



# SIERRA EV

## OWNER'S MANUAL



**TATA MOTORS**

Rev 01 / JULY 2026

## CUSTOMER ASSISTANCE

In our constant endeavour to provide assistance and complete service backup, TATA MOTORS has established an all India customer assistance centre.

In case you have a query regarding any aspect of your vehicle, our Customer Assistance Centre will be glad to assist you on our Toll Free no. **1800 209 8282**

You can also approach nearest TPEM Authorised Service Centre.

TATA MOTORS 24X7 Roadside Assistance Program offers technical help in the event of a breakdown. Call the toll-free Road-side Assistance.



For additional information, refer to "24X7 Roadside Assistance" section in the Owner's manual.

For updated Owner's Manual and Infotainment Manual refer below link

<https://ev.tatamotors.com/support.html>

Scan the below QR code for pdf file of Owner's Manual



For Dealer Network refer below link

<https://ev.tatamotors.com/dealer-locator.html>

Scan the below QR code for Dealer Network



Dear Customer,

Welcome to the TATA MOTORS family,

Thank you on the purchase of TATA MOTORS vehicle.

As a global Indian automobile manufacturer, we focus on innovation, technology and build high quality products with exceeding values of "Connecting Aspirations".

The Owner's Manual will familiarize you with the operations, equipment description, features that are either as standard or optional on your vehicle. It is requested you read this manual carefully and follow the instructions and recommendations as mentioned.

You are advised to carry out service, maintenance and repairs at TATA MOTORS EV Dealers and EV Authorised service centres through out the life of your vehicle. Always use genuine parts for continued performance of your vehicle. Avoid modification, non-genuine accessories fitment on your vehicle. TATA MOTORS does not carry any liability arising due to it. Always keep this manual in the vehicle.

You can contact our dealer or Customer Assistance toll free no. (1800 209 8282) in case of any query or support required.

Information provided in this Owner's Manual is explicit at the time of publication. However, as TATA MOTORS continues to make changes and improve products, it reserves the right to make changes in this manual or any product at any time, without notice and without any obligations.

We look forward for your continued association with us for many years to come.

Wishing you a Safe and pleasant driving experience.

### **TATA PASSENGER ELECTRIC MOBILITY LTD.**

Floor 3, 4, Plot-18, Nanavati Mahalaya, Mudhana Shetty Marg, BSE, Fort,  
Mumbai, (MH) - 400 001, India





## 01. INTRODUCTION

|                       |   |
|-----------------------|---|
| Ev Overview           | 1 |
| Know Your Vehicle     | 5 |
| Important Messages    | 9 |
| Important Information | 9 |

## 02. SAFETY

|                                                |    |
|------------------------------------------------|----|
| General Safety Tips                            | 13 |
| Seats                                          | 14 |
| Seat Belts                                     | 16 |
| Supplementary Restraint System (SRS - Airbags) | 20 |
| Children On Board                              | 27 |
| Additional Safety Features (If equipped)       | 35 |

## 03. BATTERY AND COMPONENT

|                               |    |
|-------------------------------|----|
| High Voltage Battery System   | 41 |
| Tips To Conserve Battery Life | 44 |

## 04. CHARGING

|                            |    |
|----------------------------|----|
| Instructions To Follow     | 45 |
| Important Tips             | 47 |
| Types Of Charging          | 49 |
| V2x Charging (if equipped) | 62 |

## 05. OPENING AND CLOSING

|                                              |    |
|----------------------------------------------|----|
| Keys (if equipped)                           | 71 |
| Doors                                        | 90 |
| Windows                                      | 92 |
| Bonnet And Charging Flap                     | 95 |
| Mechanically Operated Tailgate (if equipped) | 97 |
| Power Operated Tail Gate (if equipped)       | 99 |

## 06. INSTRUMENT CLUSTER

|                                        |     |
|----------------------------------------|-----|
| Digital Display (10.25" Inch)          | 105 |
| Driver Information System (DIS)        | 111 |
| Warning And Indicator                  | 126 |
| Display Messages On Instrument Cluster | 134 |

## CONTENTS

|                |     |
|----------------|-----|
| Audio Reminder | 137 |
|----------------|-----|

### 07. STARTING AND DRIVING

|                                            |     |
|--------------------------------------------|-----|
| Before You Start Your EV                   | 141 |
| Start/stop Switch                          | 143 |
| Steering System And Adjustment             | 144 |
| Electric Power Assisted Steering (EPAS)    | 146 |
| Driving Tips                               | 146 |
| Tips To Get Maximum Range While Driving EV | 150 |
| Drive And Terrain Modes                    | 154 |
| Shifter Modes                              | 157 |
| Operating Of Lights And Wipers             | 160 |
| Horn                                       | 163 |
| Seats Adjustments                          | 164 |
| Seat Zone Module (szm) (if equipped)       | 168 |
| Seat Ventilation (if equipped)             | 174 |
| Rear Seat Adjustments (If equipped)        | 175 |
| Mirror                                     | 177 |
| Driving Support System                     | 181 |

|                                                         |     |
|---------------------------------------------------------|-----|
| Advanced Driver Assistance Systems (ADAS) (if equipped) | 229 |
| Acoustic Vehicle Alerting System (AVAS) (if equipped)   | 266 |

### 08. INTERIOR FEATURES

|                                            |     |
|--------------------------------------------|-----|
| Climate Control                            | 267 |
| Fully Automatic Temperature Control (FATC) | 268 |
| Cabin Air Purification (if equipped)       | 274 |
| Fascia Switches                            | 275 |
| Power Sunroof (if equipped)                | 277 |
| Steering Mounted Controls                  | 283 |
| Regeneration Switch (if equipped)          | 285 |
| Infotainment System Display                | 286 |
| Head Up Display (HUD) (If equipped)        | 287 |
| Mic                                        | 294 |
| Speakers (if equipped)                     | 294 |
| Usb Port (if equipped)                     | 295 |
| Power Socket (if equipped)                 | 295 |
| Lamps                                      | 296 |

|                                       |     |
|---------------------------------------|-----|
| Roof Grab Handle                      | 299 |
| Mood Lights (if equipped)             | 299 |
| Vehicle Telematics (if equipped)      | 300 |
| Remote Ac Control (RAC) (if equipped) | 301 |
| Wireless Power Charging (if equipped) | 301 |
| Roof Rail (aesthetic) (if equipped)   | 307 |
| Shark Fin Antenna                     | 307 |
| Welcome And Good Bye Strategy         | 308 |

## 09. STOWAGE AREA

|                                 |     |
|---------------------------------|-----|
| Storage Compartment             | 309 |
| Glove Box                       | 310 |
| Utility Pockets On Front Doors  | 310 |
| Utility Pockets On Rear Doors   | 311 |
| Center Console                  | 311 |
| Foldable Arm Rest (if equipped) | 312 |
| Hooks (if equipped)             | 313 |
| Luggage Compartment             | 313 |
| Stowage Back Of Drive Seat      | 314 |
| Frunk                           | 314 |

## 10. EMERGENCY AND BREAKDOWN

|                            |     |
|----------------------------|-----|
| Emergency Equipment        | 317 |
| Puncture Repair Kit        | 318 |
| Towing                     | 324 |
| Fuses                      | 325 |
| Bulb Specification         | 345 |
| 24 X 7 Roadside Assistance | 346 |

## 11. TECHNICAL SPECIFICATION

|                                  |     |
|----------------------------------|-----|
| Lubricant Specifications         | 359 |
| Vehicle Specifications           | 360 |
| Vehicle Dimensions               | 363 |
| Aggregate Identification Numbers | 364 |

## 12. MAINTENANCE AND CARE

|                               |     |
|-------------------------------|-----|
| Service Instructions          | 365 |
| Service Schedule              | 366 |
| Motor Compartment             | 374 |
| Brake Fluid Level             | 375 |
| Windshield Washer Fluid Level | 375 |

## CONTENTS

---

|                                               |     |
|-----------------------------------------------|-----|
| 12V Battery                                   | 376 |
| Tyres                                         | 377 |
| Vehicle Parking For Long Duration             | 380 |
| Smart Key Battery Replacement                 | 381 |
| Remote Key Battery Re-placement (if equipped) | 382 |
| On Board Diagnostic (obd ii) System           | 383 |
| Do It Yourself                                | 383 |
| Car Care                                      | 384 |
| Value Added Services                          | 389 |
| Vehicle Interior Enrichment                   | 391 |
| Vehicle Exterior Enrichment                   | 392 |

### 13. WARRANTY

|                                                      |     |
|------------------------------------------------------|-----|
| Vehicle Warranty - Terms And Conditions              | 393 |
| High Voltage Battery Warranty - Terms And Conditions | 395 |

### 14. EXTENDED WARRANTY

|                   |     |
|-------------------|-----|
| Extended Warranty | 399 |
|-------------------|-----|

### EV OVERVIEW

#### Review and Characteristics

This is an electric vehicle. Some of the vehicle's systems operate differently and have different operating characteristics than vehicles equipped with an internal combustion engine. It is important to carefully read the entire Owner's Manual for this reason.

This Vehicle uses two types of battery systems- a high voltage system in which a high voltage battery powers the inverter and electric motor which in turn propel the vehicle and a low voltage system in which a 12-volt battery provides power to the vehicle systems and features such as the audio system, supplementary restraint systems, headlights and windshield wipers. The high voltage battery also charges the 12-volt battery.

The high voltage battery must be charged with electricity before the vehicle can be driven. As the vehicle operates, the battery gradually discharges and when completely discharged, the vehicle needs to be plugged in for charging. To increase the

range of the EV, regenerative braking has been incorporated. Basically, while the vehicle is coasting or braking, the motor works as a generator and converts the vehicle motion (kinetic energy) to electrical energy to charge the HV battery.

This vehicle is considered to be an environmentally friendly vehicle because it does not emit exhaust gases, and thus is cleaner than the conventional vehicles in terms of air pollution.

#### Main Components

##### On-Board Charger (OBC)

A device that charges the high voltage battery by converting AC power from a domestic supply into DC power and supplying it to the battery.

##### Inverter

A device that transforms direct current (DC) from the high voltage battery into alternating current (AC) to power the electric motor. It also converts braking energy into electrical energy and charges the high voltage battery while decelerating or braking.

##### DC-DC Converter

A device that converts HV DC power from the HV battery to LV DC power which is required to maintain LV battery charge, which in turn powers the LV systems like lights, wipers, infotainment, etc. in the vehicle.

##### Electric Motor

A device that converts electrical energy into rotational mechanical energy which is then transferred as rotational torque to the

## INTRODUCTION

---

wheels through the gearbox in order to drive the vehicle.

### **High Voltage Battery (Lithium-ion Phosphate)**

An onboard high-voltage electrical energy storage device

### **Power Distribution Unit**

Distributes power from the high voltage battery to the HV components like inverter, DC-DC converter, E-compressor etc.



### **General Warnings**

- Your vehicle contains a sealed Lithium Iron Phosphate high voltage battery. If the Li-ion battery is disposed of im-

properly, there is a risk of severe burns and electrical shock that may result in serious injury or death and there is also a risk of environmental damage.

- The EV system uses high voltage DC current. The system can be hot during and after starting and when the vehicle is shut off. Be careful of both the high voltage and the high temperature.
- Avoid being exposed to high-voltage components in the first place. Observe all high-voltage warning labels these indicate high-voltage components or areas. Observe all orange cables and other high voltage components, large and small these carry high voltage.
- Do not touch high-voltage components while the vehicle is in operation or cranked state.
- Do not disassemble, remove or replace high-voltage parts and cables as well as their connectors because they can cause severe burns or electric shock that may result in serious injury or death.
- The vehicle high voltage system has

no user serviceable parts. It is recommended that you take your vehicle to TPEM Authorised Service Centre for any necessary maintenance.

- Pay special attention to pedestrians. Because there is no motor noise, pedestrians may not know the vehicle is approaching, moving or about to move, and may step into the path of vehicle travel.
- When leaving the vehicle, be sure to turn off the EV system. The EV system uses high voltage current. Failure to follow the proper handling instructions may cause serious injury or death.

### Safety Of The High-voltage System

- Do not perform any modifications or work on the vehicle, especially maintenance and repair work on the high-voltage system and the body and avoid retrofitting accessories.
- If work is not carried out properly, there is the risk of fire and fatal injury from electrocution due to the high-voltage system.
- TPEM recommends to have modifica-

tions and work on the vehicle only to be carried out by TPEM Authorised Service Centre or one that operates according to TPEM specifications with personnel trained accordingly.

- Your vehicle's high-voltage system is a self-contained system. Safety is ensured as long as no unauthorised work is performed on high voltage electrical components or on the chassis.

### High-voltage System: Contact With Water

The high-voltage system is typically safe even in the following example situations:

- Water in the foot well, for instance after a rainstorm when sunroof was kept open.
- Vehicle is in water but only up to 300 mm.
- Liquid escapes in the trunk.

In these cases there is no risk of injury from electrocution. Other damage to the vehicle is possible.

## INTRODUCTION

---

### Common Terminologies And Abbreviations

EV – Electric Vehicle

HV battery – High Voltage battery

LV battery – Low Voltage (12V) battery

AC – Alternating Current

DC – Direct Current

OBC – On Board Charger

PDU – Power Distribution Unit

VCU – Vehicle Control Unit

BMS - Battery Management System

OBD - On Board Diagnostics

SoC – State of Charge

SRS – Supplementary Restraint System

CRS - Child Restraint System

DAB – Driver Airbag

PAB – Passenger Airbag

ABS - Anti-lock Braking System

EBD - Electronic Brake Force Distribution

ESC - Electronic Stability Control

PEPS - Passive Entry/Passive Start

ESCL – Electronic Column Steering Lock

EPAS – Electric Power Assisted Steering

LED – Light Emitting Diode

DRL – Daytime Running Lamp

ORVM - Outer Rear View Mirror

IRVM – Inside Rear View Mirror

EC-IRVM – Electric Chromic Inside Rear View Mirror

HVAC – Heating Ventilation and Air Conditioning

FATC – Fully Automatic Temperature Control

DIS – Driver Information System

DTE - Distance to Empty

IGN – Ignition

ACC – Accessories

EPB – Electronic Parking Brake (EPB)

CPL – Centre Positioning Light

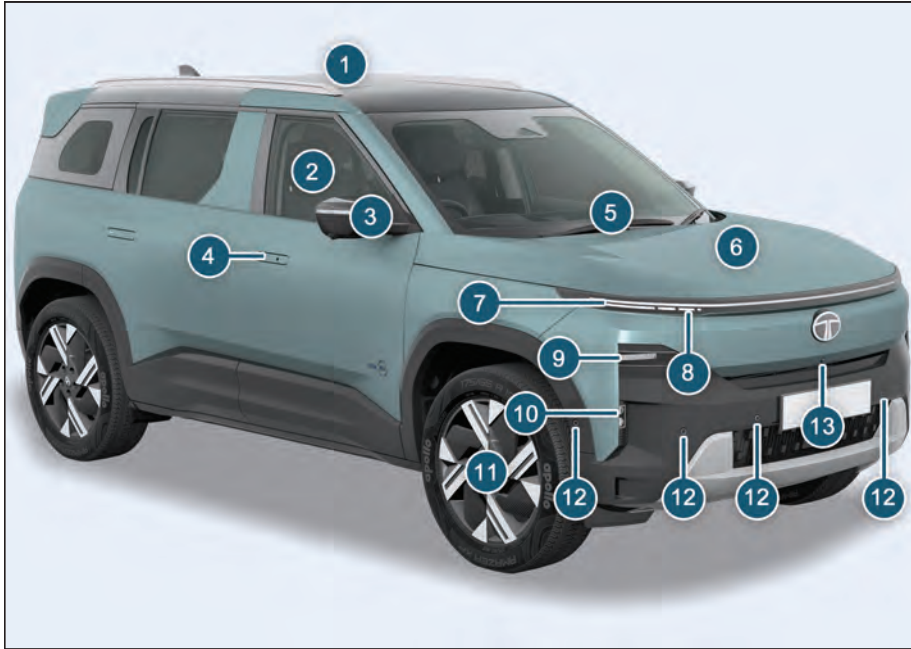
V2L – Vehicle to Load

V2V – Vehicle to Vehicle

iVBAC - Intelligent Vacuum-less Boost & Active Control



### KNOW YOUR VEHICLE



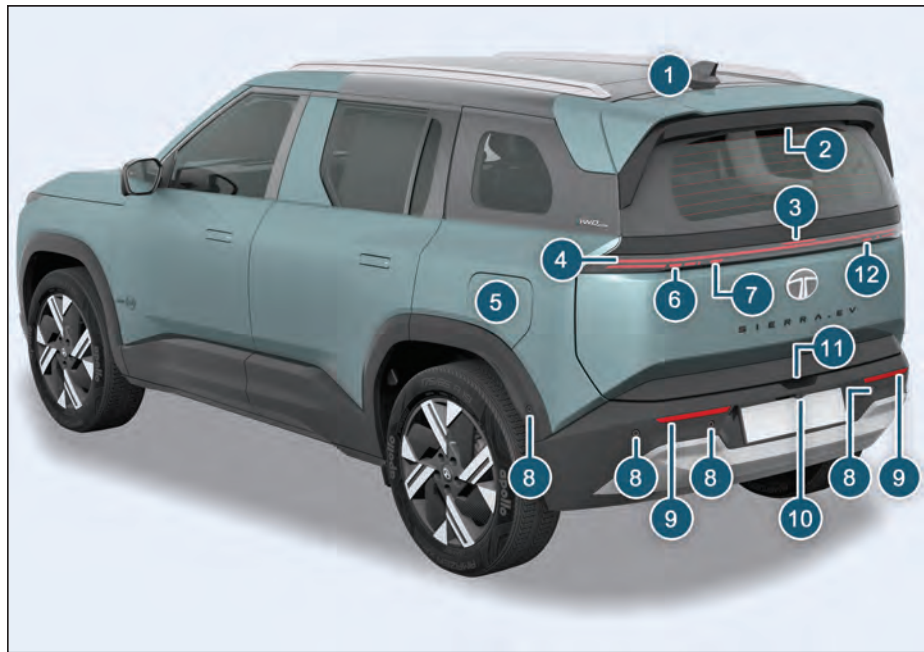
1. Sunroof
2. Windows
3. Rear View Mirror with Camera
4. Door Handle Switch (DHS)
5. Front Windshield Wipers
6. Bonnet
7. DRL/ Position Lamp
8. Turn Indicator
9. Head Lamp
10. Fog / Cornering Lamp
11. Alloy Wheels
12. Front Parking Sensors
13. Front Camera

\* Image for your reference, actual vehicle may differ.

\* Features listed above may or may not be applicable to your vehicle.

## INTRODUCTION

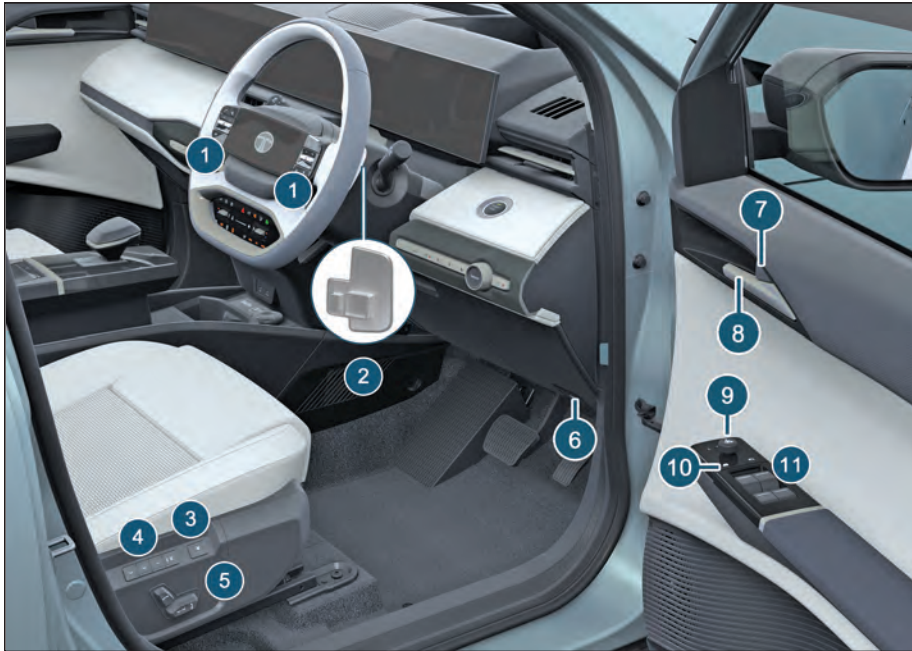
---



1. Shark Fin Antenna
2. Rear Windshield Wipers
3. CHMSL
4. Position Lamp
5. Charging Flap
6. Turn/Stop Indicator
7. Reverse Lamp
8. Rear Parking Sensors
9. Reflex Reflector
10. Rear Camera
11. Tail Gate Open Switch
12. Rear Fog Lamp

\* Image for your reference, actual vehicle may differ.

\* Features listed above may or may not be applicable to your vehicle.



1. Steering Mounted Switches
2. Regeneration Lever
3. Seat Ventilation
4. Memory Seat Adjustments
5. Power Seat Adjustments
6. Bonnet Opening Lever
7. Door Locking/Unlocking Knob
8. Door Opening Lever
9. ORVM adjusting Knob
10. Inhibit Switch
11. Power Window Switch

\* Image for your reference, actual vehicle may differ.

\* Features listed above may or may not be applicable to your vehicle.

## INTRODUCTION

---



1. Passenger Airbag
2. Infotainment Display
3. Speaker
4. Head Up Display
5. Instrument Cluster
6. Horn Pad
7. Driver Airbag
8. Start/Stop Switch
9. Drive and Terrain Modes
10. USB Charger for Rear Passenger
11. Arm Rest
12. Monostable Shifter
13. Temperature Control Panel
14. Hazard Warning Switch
15. Sound Bar
16. Glove Box

\* Image for your reference, actual vehicle may differ.

\* Features listed above may or may not be applicable to your vehicle.

## IMPORTANT MESSAGES

In this Owner's Manual, you will find the text under the heading "WARNING", "CAUTION" and "NOTE" which highlights important information. Pay particular attention to these highlighted messages. The Images / Illustrations in this owner's manual are only for reference. It may defer with actual vehicle.

### NOTE

*Indicates additional information that will assist you in gaining the optimum benefit and care for your vehicle.*

### WARNING

Indicates procedures or information that must be followed precisely in order to avoid the possibility of severe personal injury and serious damage to the vehicle.

### CAUTION

It indicates to be careful. You are capable of doing something that might result in damage to equipment.

## IMPORTANT INFORMATION

### Vehicle in water logged condition



- Before driving through water logged areas ensure that water is below running board.
- Unseen potholes could damage under surface of the vehicle.
- Drive slowly and maintain constant acceleration.
- After passing the water logged area ensure to press brakes intermittently to dry the brake disc.

### WARNING

- Do not drive through heavy water-logged area.
- Water may enter into vehicle interior and motor compartment which may

## INTRODUCTION

damage electrical, electronic circuits.

### Vehicle in flood



#### WARNING

- If your vehicle gets flooded and if you see gas bubbles around submerged EV, smoking, it may indicate a damaged battery, Immediately contact TPEM Authorised Service Centre.
- If your vehicle is flooded wait for water level to recede.
- Once the water has receded, approach the vehicle, open all doors and let the water inside the vehicle get drained completely.

- Start the ignition and check if all electronic are working. Take your vehicle to nearest TPEM Authorised Service Centre and get it inspected.
- Remove floor carpet and open the rubber/plastic grommet provided on floor board, stepney space for water to recede from interior flooring of the vehicle completely.

### Vehicle in Fire



In case of vehicle fire, immediately evacuate vehicle and contact local fire tender responder. They possess proper training and equipment to safely extinguish vehicle fire. Inform them your vehicle is EV. Contact nearest TPEM Authorised Service Centre for further assistance.

#### WARNING

- If there is damage to EV, there is risk of thermal runaway which could lead to fire.
- If the damage occurs in saltwater, the risk of battery short circuits leading to fire is increased.
- Do not modify your vehicle.
- Do not install high-wattage bulbs, non-genuine lamps, or horns.
- Do not modify suspension, wheels, tyres.
- Stick to manufacturer recommended parts.
- Avoid tampering with the wiring harness it can lead to short circuits.

#### CAUTION

Do not store or carry inflammable materials in the vehicle.

### Vehicle Scrapping

Your vehicle is equipped with SRS Air Bags and Seat Belt Pretensioner, ensure to remove and dispose it by qualified service centre or by TPEM Authorised Service Centre before scrapping your vehicle.

### Vehicle Cyber Attack

Modern vehicles are equipped with internet connectivity for navigation, entertainment and diagnostics, making them vulnerable to cyber attacks.

Vehicles collect and stores personal data, which can be targeted by hackers.

Cyber attacks can compromise infotainment system, brakes, steering and vehicle control, leading to severe cyber risk.

### Vehicle Cyber Security



- Refrain from connecting the vehicle to public or unsecured Wi-Fi networks.
- Ensure your vehicle is physically secured by locking doors and parking in safe areas to prevent from unauthorised access.
- Never leave your keys in the car, even if you are nearby.
- Keep your vehicle updated with latest software versions.
- Pay attention to any security alerts or notifications and follow the guidelines.
- Disable the bluetooth of the car where not required and vehicle is not in use.
- Keep infotainment app login creden-

tials private & do not share it with anyone.





## GENERAL SAFETY TIPS

- Safety consciousness not only ensures your safety and the safety of other road users, but it also helps to reduce the wear and tear on your vehicle.
- How quickly you make decisions to avoid an accident.
- Your ability to concentrate.
- How well you can see and judge objects.
- How well familiar you are with your vehicle controls and its capabilities.
- Turn 'ON' the side indicators at least 30 meters before taking a turn or changing the lane.
- Decelerate to a safe speed before taking turn. Do not apply brakes during cornering.
- When overtaking other vehicles, watch out for the oncoming vehicle.
- Never drive under the influence of alcohol or drugs.
- If your vehicle is equipped with an in-

formation/ navigation system, set and make changes to your travel route only when the vehicle is parked.

- Program radio presets with the vehicle parked, and use your programmed presets to make radio use quicker and simpler.
- If your car gets flooded and has soaked carpeting or water on the flooring, you should not try to start the vehicle, we recommend to kindly contact TPEM Authorised Service Centre.

### NOTE

*Do not remove the labels attached at different places on your vehicle, they include safety instructions or vehicle specifications.*

## SAFETY

### SEATS

Your vehicle is provided with good seating comfort. To make your journey more safe and enjoyable we recommend you to follow below warnings and cautions.

#### Driver's seat

##### **WARNING**

- Do not adjust seat while driving / vehicle is moving. Doing so could result in loss of control, and an accident causing death, serious injury, or property damage.
- Always sit as far back as possible from the steering wheel while maintaining comfortable control of the vehicle. Fitment of seat covers on driver seat with airbags is strictly prohibited.
- Do not keep any sitting cushion on seat. This may result in serious or fatal injury in the event of accident.
- After adjusting the seat make sure it is securely locked by pushing it forward and backward without using

lock release lever. Sudden or unexpected movement of the driver's seat could cause to lose control of the vehicle resulting in an accident.

- All passengers must be seated in seats and restrained with seats properly while riding in vehicle
- If there are occupants in the rear seats, be careful while adjusting the front seat position.

#### Front Passenger Seat

##### **WARNING**

Never ride in a vehicle with a front seat-back fully reclined. This may lead to serious injuries. Fitment of seat covers on front passenger seat with airbags is strictly prohibited.

#### Rear Seat Back

##### **WARNING**

The rear seatback must be securely latched. If not, passengers and objects

could be thrown forward resulting in serious injury in the event of a sudden stop or collision. Luggage and other objects in boot should be kept flat. If large, heavy, or piled they must be secured properly. No passenger should ride in the boot area or sit or lie on folded seat-backs while the vehicle is in motion.

#### Applicable for Hatchback/SUV

##### **WARNING**

- Under no circumstances should objects be piled higher than the seat-backs. Failure to follow these warnings could result in serious injury in the event of a sudden stop or collision. Ensure that objects are securely fastened.
- Storing items against seatback or in any other way interfering with proper locking of a seatback could result in serious or fatal injury in a sudden stop or collision. After resetting the seatback to its seating position make sure it is securely latched

by pushing it forward and backward.

### ⚠ CAUTION

Your hands might cut or be injured by the sharp edges of the seats mechanism during looking for small objects trapped under the seats or between the seat and the center console.

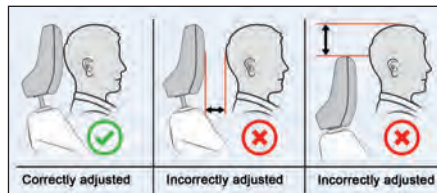
## Head Restraint

### Front Seat



Adjust the head restraint so that it is as close to the head as possible and center

of the head restraint supports the back of the head at eye level.



### ⚠ WARNING

Do not drive the vehicle without the seat head restraints. Head restraints are intended to help reduce injuries during an accident.

### Rear Seat (if Equipped)

Adjust the head restraint so that it is as close to the head as possible and the center of the head restraint supports the back of the head at eye level.



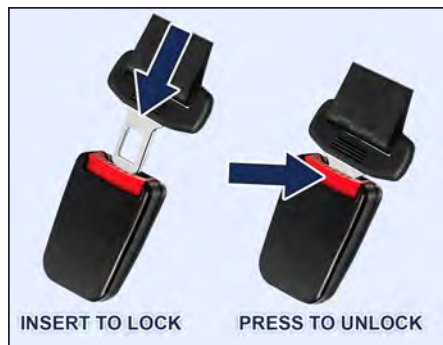
## SAFETY

### SEAT BELTS

This section describes your Vehicle's Seat belts, Airbags and Child restraints system. Please read and follow all these instructions carefully to minimize risk of severe injury or death.

- Seat belts are the primary restraint system in the vehicle. All occupants, including the driver, should always wear seat belts. Your vehicle is equipped with three point seat belts for all occupants.
- Sit back and adjust the driver seat. Make sure that your seat is adjusted to a good driving position and the back of the seat is upright.

### Buckling The Seat Belt



- Grasp the tongue then slowly pull out the seat belt over the shoulder and across the chest. When the seat belt is long enough to fit, insert the tongue into the lock buckle until you hear a “CLICK” which indicates that the seat-belt is securely locked. (Refer “INSERT TO LOCK” image)
- Position the lap portion of seat belt across your pelvic bone, below your abdomen. To remove slack, pull up a bit on the shoulder seat belt. To loosen the lap portion seat belt if it is too tight,

tilt the tongue and pull on the lap seat belt. A snug seat belt reduces the risk of sliding under the seat belt in a collision. Ensure that the seat belt running over the body (shoulder segment and lap segment) does not have any twist. Twisted seat belt may not offer effective protection when required.

- Ensure that the seat belt webbing is straight and not twisted. Twisted seat belts may not work properly in case of collision.

#### NOTE

*It is not advised by TPEM to remove the mini tongue from small buckle, located at rear middle seat.*

### Releasing The Seat Belt

To release the seat belt, push the red button on the lock buckle (refer "PRESS TO UNLOCK" image). Ensure to hold seat belt during unlocking and release it slowly towards the seat belt mounting. The seat belt will automatically retract to its stowed position. If necessary, slide the tongue down the webbing to allow the seat belt to retract fully.

#### WARNING

Due to retractor reversal action if you leave the seat belt from the unlock position it may hit you or parts like glass in the way which may cause injury to you or damage to the vehicle.

#### WARNING

- Each seating position and seat belt assembly must be used by one occupant.
- Be careful not to damage or tamper the seat belt webbing or hardware. Inspect the seat belt system period-

ically, checking for cuts, frays, or loose parts. A frayed or torn seatbelt could rip apart in a collision and leave you with no protection.

- If the seat belt webbing or hardware is damaged, get it replaced immediately at TPEM Authorised Service Centre.
- Do not insert any items such as coins, clips, etc. into the seat belt buckles, and be careful not to spill liquids into these parts. If foreign materials get into a seat belt buckle, the seat belt will not work properly.
- Do not wear seat belts over hard, sharp or fragile items in clothing, such as pens, keys, spectacles etc.
- Do not use any accessories on seat belts or modify in any way the seat-belt system. Devices claiming to improve occupant comfort or repositioning the seat belt, can reduce the protection provided by the seat belt and increase the chance of serious injury in a collision.

### Seat Belt Height Adjustments



If height adjustment is provided in the seat belt, occupant can adjust it as per their comfort.

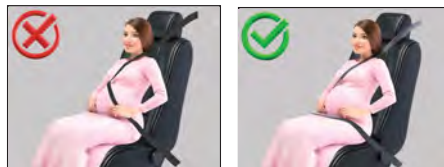
### Use of Seat Belts For Pregnant Women

#### WARNING

- Pregnant women must wear a correctly positioned seat belt. It is safer for mother as well as unborn child.
- Pregnant women should wear the lap part of the seat belt across the Pelvic Bone and as snug across the

## SAFETY

pelvic bone (hips) as possible. Keep the seat belt low so that it does not come across the abdomen. That way the strong bones of the hips will take the force if there is a collision.



### Seat Belt Warning Lamp



Your vehicle is equipped with Seat Belt Reminder (SBR) for all occupants.

#### *i* NOTE

*For Driver and CO-Driver Seat*

- Whenever an occupant is not sitting in any seating position then seat belt reminder beeping sound will not be played in the instrument cluster.
  - If any material is kept on any seat then SBR beeping sound may be played in instrument cluster. Please do not keep any material on seat.
- If the driver or any passenger do not fasten the seat belt, seat belt reminder lamp will blink and a buzzer will sound for pre-defined duration until the seat belt is buckled.

- If any passenger seat is occupied by child (without child seat), system may detect occupancy and warn with seat belt warning. It is not taken to mean child can occupy any passenger seat and use seat belt. Please refer CRS section for recommended seating position if child is sitting with child seat.

#### *i* NOTE

*Fitment of seat covers on any seating position is strictly prohibited. It may affect the function of occupant sensor.*

**Seat Belts With Pre – Tensioner (if equipped)**

The seat belt pre-tensioner system works in conjunction with the Supplementary Restraints System (SRS-Airbags). In the event of a collision, as may be necessary, pre-tensioner tightens the seat belt so that it fits the occupant's body more snugly. When pre-tensioner activates, there could be some noise and release of smoke. This is normal and there are no health hazards or fire risk.

**Seat Belt With Load Limiter (if equipped)**

The seat belt load limiter system works in conjunction with the Supplementary Restraints System (SRS-Airbags). In the event of a collision, as may be necessary, load limiter reduces the load on the rib cage region of the occupant.

If the vehicle has been involved in a collision, get it inspected immediately at TP&M Authorised Service Centre.

## SAFETY

### SUPPLEMENTARY RESTRAINT SYSTEM (SRS - AIRBAGS)



The SRS [Supplementary Restraint system] is designed to provide protection to occupants in case of collision or sudden impact, when sufficiently severe crash is detected, the SRS airbag system deploys airbags to help reduce the risk of injury to the occupant. It works in conjunction with seat belts.

There are 6 airbags provided in your car:

1. Driver Airbag
2. Front Passenger Airbag
3. Side Airbag RH
4. Side Airbag LH
5. Curtain Airbag RH
6. Curtain Airbag LH

The driver airbag is mounted in the center of the steering wheel. The front passenger airbag is located inside the dashboard in front of the passenger seat. The airbags have suitable indications on steering wheel and on dashboard.

Side airbags are mounted in front row seats.

Curtain airbags are mounted above the doors along the roof on both sides.

The word 'AIRBAG' is marked at adjacent locations of respective airbags.



The 'SRS' system also comprises of the following components depending upon the provided safety features in vehicle.

- Seat belt Pre-tensioners
- Seat belt with load limiters
- Airbag 'SRS' ECU (Electronic Control Unit)
- Collision Sensors
- SRS wiring harness
- SRS Warning lamp
- Passenger airbag switch & Indicator
- Seat belt buckle switches

The System is active when ignition switch is in the "ON" position or the ignition mode is "ON". Airbags are designed to inflate in collisions when required.

In the event of a collision, the collision sensors will detect signals, and if the Airbag ECU judges that the signals represent a severe collision, will trigger the airbags. The inflated Airbags provide a cushion to the occupants. The Airbag inflates and deflates so quickly that you may not even realize that it has activated. The Airbag will

neither hinder your view nor make it hurdle to exit the vehicle. Airbag inflation is virtually instantaneous and occurs with considerable force, accompanied by loud noise and smoke, which is normal. The inflated airbag, together with seat belts, limits the movement of an occupant, thereby reducing the risk of injury.

When an airbag inflates, you may see some smoke like particles. The particles are a normal by product of the process that generates the non-toxic gas used for airbag inflation. These airborne particles may irritate the skin, eyes, nose, or throat. If you have skin or eye irritation, rinse the area with water. For nose or throat irritation, move to fresh air. Also sometimes the smoke can cause breathing problems, in such cases get fresh air promptly.

It is not advisable to drive your vehicle after the airbags have been deployed. If you are involved in another collision, the air-bags will not be in place to protect you.

#### NOTE

- *Open your windows and doors as soon as possible after collision to reduce prolonged exposure to the smoke and powder released by the inflating Airbag.*
- *Do not touch the Airbag container's internal components immediately after an Airbag has inflated. The parts that come into contact with an inflating Airbag may be very hot.*
- *Always wash exposed skin areas thoroughly with lukewarm water and mild soap.*

#### WARNING

- Even in vehicles with airbags, all the occupants must always wear the seat belts provided. In order to minimize the risk and severity of injury in the event of a collision.
- If an occupant is out of position during collision, the rapidly deploying Airbag may forcefully contact the

## SAFETY

occupant causing serious or fatal injuries.

### WARNING

- Always use seat belts and CRS at all times. Even with airbags, you can be seriously injured or killed in a collision if you are not wearing seat belt properly or not wearing seat belt when airbag inflates.
- All occupants should never sit or lean unnecessarily close to the Airbags.
- Move your seat as far back as possible from front Airbags, while still maintaining control of the vehicle.
- All occupants should sit upright with the seatback in an upright position, centered on the seat cushion with their seat belt on, legs comfortably extended and their feet on the floor until the vehicle is parked and the IGN is turned off.

- Do not allow the front passenger to place their feet or legs on the dashboard.

### Passenger Airbag (PAB) Switch

The Passenger airbag switch is used to activate and de-activate the passenger airbag in the vehicle.

The switch is located on the left side of the dashboard and can be accessed once the co-driver side door is opened.



### PAB Switch ON

When an adult is seated in the front passenger seat, ensure that PAB switch is turned to 'ON' position. This will ensure that the passenger airbag is operational in the event of a collision.

### PAB Switch OFF

If rearward facing child seat needs to be installed on front passenger seat to carry the child then ensure PAB switch is turned OFF. This will ensure that the passenger airbag will remain de-activated in the event of a collision.

This switch can be operated by using mechanical key only. For more details refer "Keys" section in this Manual.

### Passenger Airbag (PAB) Indicator

The Passenger airbag indicator is provided to notify an occupant, whether passenger airbag is activated (ON) or deactivated (OFF) in vehicle.

The PAB indicator is positioned on the roof, close to the roof lamp.



### PAB Indicator ON:



When the PAB switch is turned to 'ON' position to activate the airbag, 'ON symbol & text' will illuminate in amber color.

### PAB Indicator OFF:



When the PAB switch is turned to 'OFF' position to deactivate the airbag, 'OFF symbol & text' will illuminate in amber color.

## SAFETY

### Wrong Seating Positions





### **i NOTE**

- Never place your arm over the airbag as a deploying airbag can result in serious arm fractures or other injuries.
- If the Airbag ECU or its mounting area is exposed to vibrations above 2.7 g, the ECU may fail and its performance is not assured.
- Do not allow the passengers to lean their heads or bodies onto doors or place objects between the doors and passengers when they are

seated on seats equipped with side and/or curtain Airbags.

- Do not place or stick any item/s in the vehicle, except at designated locations (such as utility bins, cup/bottle holders, boot space etc). Loose items may act as a projectile during a collision and cause severe to fatal injuries.
- Do not place the chest, face etc. close to the steering wheel. As there is a major risk of injury if the driver airbag was deployed.
- Please be aware that any unsecured item in your vehicle, such as your pet, unsecured CRS or a laptop, can become a potential hazard in the event of a collision or sudden stop, causing injuries to occupants in the vehicle.
- Coat hooks (if provided), must be used only for that purpose. Never hang other items on to those hooks. This could affect deployment of the Airbags, and may lead to severe to

fatal injuries.

- Always contact TPEM Authorised Service Centre if the vehicle is damaged, even if airbag has not inflated or if any part of an airbag module cover shows signs of cracking or damage.

### **⚠ WARNING**

If your SRS malfunctions, the Airbag may not inflate properly during a collision thereby increasing risk of serious injury or death. If any of the following conditions occur, your SRS is malfunctioning:

- The SRS warning lamp does not turn ON for few seconds, when the ignition switch is placed in the ON position.
- The SRS warning lamp stays 'ON' after illuminating.
- The SRS warning lamp comes 'ON'/stays 'ON' while the vehicle is in motion.

## SAFETY

- The SRS warning lamp blinks when the vehicle is running.

We recommend the customer to immediately visit TPEM Authorised Service Centre and get the SRS system inspected if any of the above conditions occur.

### WARNING

- Never make any modifications to your vehicle. The modifications carried out, but not limited to the vehicle frame, bumpers, front fenders, ride height, suspension, seat belts, interior trims, steering wheel (especially holders), and electrical system are not acceptable. This will affect the intended performance of SRS.
- Fitment of bull bars, seat covers on seats with airbags etc, is strictly prohibited.
- If you need to make any modifications to accommodate any disability you may have, please contact your

TPEM Authorised Service Centre for necessary guidance.

- Do not tamper with SRS in any way. This will lead to unexpected performance of system and may cause serious injury or death.

### Airbag Warning Sticker On Front Passenger Sun Visor



The Airbag Warning Symbol on sun visor

reminds of the extreme hazards associated with the use of a rearward-facing child restraint on front passenger seat during airbag deployment. Please refer CRS section for recommended seating position for children.

### WARNING

Never use a rearward facing child restraint on a seat protected by an active Airbag in front of it, Death or serious injury to the child can occur.

### Airbags Deployment Conditions

#### When front airbags should not deploy?

**Minor frontal collision:** Seat belt (if worn) offers adequate occupant protection in low severity collisions. The airbags are triggered only when there is a collision severe enough to trigger the airbags. Deployment of frontal airbags is not beneficial in low severity collisions.

**Side collision:** During a side collision, occupants tend to move sideways. Therefore, deploying frontal airbags in such situations will not benefit the occupants.

Side airbags and side curtain airbags are specifically designed to reduce the injuries that can occur in side collision.

**Rear collision:** During a rear collision, occupants tend to move (rearwards) away from frontal airbags. Therefore, deploying frontal airbags in such situations will not protect the occupant. Head restraints and seat belts provide occupant protection during a rear collision.

**Rollovers collision:** During a rollover collision, unbelted occupants may float inside the passenger compartment. This will increase the risk of injuries and may prove to be fatal. Wearing seat belts provide highly effective occupant protection during rollover collision. Front airbags, are not designed to deploy in a rollover as frontal airbags cannot offer any protection in rollover collision.

**When front airbags/side airbags/side curtain airbags deploy with minor or no visible damage?**

The airbags are triggered only when there is a collision severe enough to trigger the airbags. The extent of vehicle damage is

not always the correct indicator for airbag deployment. In some extreme/rare conditions of rough road driving, running into a curb or hitting other fixed objects the airbags may deploy depending upon the severity of collision. In some of these conditions, damage to the vehicle may be minor or not be readily visible.

**When front airbags/side airbags/side curtain airbags may not deploy, even with exterior visible vehicle damage?**

The airbags are triggered only when there is a collision severe enough to trigger the airbags. The amount of visible vehicle damage is not always the correct indicator for airbag deployment. Some collisions can result in visible damage but with no airbag deployment, because the airbags would not have been needed or would not have provided protection even if they had deployed. Seat belts, if worn, offer adequate occupant protection in such cases.

## CHILDREN ON BOARD

### WARNING

- Do not leave unattended children in your vehicle.
- During reversing and parking, ensure that children are far away from the vehicle.

### WARNING

- Do not put the safety seat belt under your child's arm or behind its back.
- Do not use pillows, books or towels to boost your child's height.
- Do not allow children to stand up or kneel on either the rear or the front seats. An unrestrained child could suffer serious or fatal injuries during a collision.
- Do not install a booster seat or a booster cushion with a seat belt that is slack or twisted.



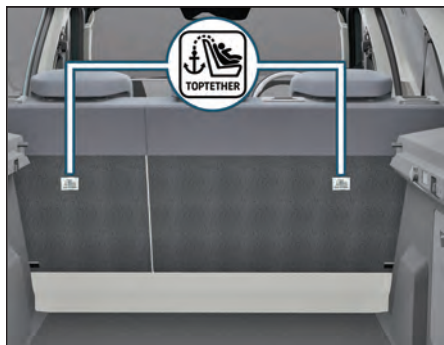
## SAFETY

### Child Restraint System (CRS)

TATA Passenger Electric Mobility Ltd. strongly recommends the use of Child Restraint Systems (CRS) for all children up to 36 Kg and to be placed at recommended positions only (Refer CRS Position table in this section).

#### Isfix

CRS can be installed in the vehicle using seat belts and/or ISOFIX with support leg (if equipped) or ISOFIX with Top Tether (if equipped). These ISOFIX attachment points are located on rear outboard seating locations which enables quick and safe child seat engagement.



*Top Tether*

Top Tether mounting anchorages are lo-

cated at backside of rear outboard seats.

The harness system of CRS holds the child in place, and in a collision, acts to keep the child positioned in the seat and reduce the risk of injuries.



Keep children in a forward-facing or rearward facing CRS with a harness until they reach the size or age or weight limit recommended by your CRS manufacturer.

#### Selection And Installation Of Crs

Always select the CRS that complies with latest safety standards (AIS 072 / ECE R44 / ECE R129). The CRS are classified



according to the child's size, height and weight. Select the appropriate CRS for your child. Ensure that the child fits properly in the CRS and it is securely installed in the vehicle.

While installing the child seats always adhere to the directions in this Owner's Manual as well as those provided by the child seat's manufacturer.

TPEM recommends Joie i-Spin Safe i-Size child seats for up to 18 Kg children. These seats are available at TPEM Authorised Service Centre.



#### NOTE

*TPEM recommends to keep the high-lighted device in close condition while using Joie i-Spin Safe i-size child seat in car.*

#### Installing The Child Seat On Front Passenger Seat

- Adjust the front passenger seat back up to its vertical position as per requirement, so that it can create adequate contact between passenger seat backrest & child seat.
- Adjust the front passenger seat forward or backward as per requirement, so that there could not be any contact between front passenger seat & child seat or child present behind it.
- If required, adjust the front passenger seat height to its suitable position.
- While installing child seat on front passenger sitting positions, adjust the buckle to its suitable position of rotation.
- While installing forward facing child

## SAFETY

seat for 15 to 18kg children on front passenger seating position, adjust the front passenger seat to its rearmost position.

### Installing The Child Seat On Rear Passenger Seats

- If required, adjust the front seat so that there could not be any contact between front seat & child seat or child present behind front seat.
- While installing child seats rear seat head restraints be removed and kept at safe location to reinstall it whenever adult passenger is sitting at that position.
- While installing child seats on rear out-board seating position, adjust their respective buckles to its required position of rotation.



### Recommended CRS Position As Per The Vehicle Matrix

The suitability of seat position for carriage of children and recommended category of CRS is shown in the table below as per the child group.

#### NOTE

*The child's life is at risk in a collision if the CRS is not properly secured in the vehicle. Be sure to secure the child in the restraint system according to the manufacturer's instructions.*

#### WARNING

Do not use an infant carrier or a child safety seat that "hooks" over a seat-back, it will not provide adequate protection in a collision.

### Recommended CRS Positions (CRS That Can Be Used With Isofix System)

| Group | Mass Group  | Front Passenger With PAB OFF | Front Passenger With PAB ON | Rear Outboard LH | Rear Outboard RH | Rear Center |
|-------|-------------|------------------------------|-----------------------------|------------------|------------------|-------------|
| 0     | Up to 10 kg | U                            | X                           | U                | U                | X           |
| 0+    | Up to 13 kg | U                            | X                           | U                | U                | X           |
| I     | 9 to 18 kg  | U<br>UF                      | UF                          | U                | U                | X           |
| II    | 15 to 25 kg | UF                           | UF                          | U                | U                | X           |
| III   | 22 to 36 kg | UF                           | UF                          | U                | U                | X           |

**X** : Seat Position not suitable for children in this mass group.

**U** : Suitable for "universal" category restraints approved for use in this mass group.

**UF** : Suitable for forward facing "universal" category restraints approved for use in this mass group.

## SAFETY

### CAUTION

A CRS in a closed vehicle can become very hot. To prevent burns, check the seating surface and buckles before placing your child in CRS.

### WARNING

Do not modify CRS in any way.

### CAUTION

- Do not install a booster seat or a booster cushion with only the lap strap of the seat belt or a seat belt that is slack or twisted.
- Do not leave any toys or other objects loose in the CRS or on the seat while the vehicle is in motion.

### Recommended CRS Positions (CRS That Can Be Used With Isofix System)

| Group | Mass Group  | Category Of Child Seat | Front Passenger | Rear Outboard LH* | Rear Outboard RH* | Rear Center |
|-------|-------------|------------------------|-----------------|-------------------|-------------------|-------------|
| 0     | Up to 10 kg | E                      | X               | IL                | IL                | X           |
| 0+    | Up to 13 kg | C,D,E                  | X               | IL                | IL                | X           |
| I     | 9 to 18 kg  | D,C,B,B1, A            | X               | IL<br>IUF         | IL<br>IUF         | X           |
| II    | 15 to 25 kg |                        | X               | IL                | IL                | X           |
| III   | 22 to 36 kg |                        | X               | IL                | IL                | X           |

**X:** The seat is not equipped for the ISOFIX system.

**IL:** The seat is suitable for the ISOFIX child seats with “Semi-Universal” approval.

**IUF:** The seat is suitable for forward facing child seats and is permitted for use in this weight category.

\*Rear outboard seating positions are suitable for ISO/R3, Class C CRS.

After a collision, we recommend to get seat belts, seats, ISOFIX and top-tether anchorages (as may be applicable) investigated at TPEM Authorised Service Centre.

Each CRS should be used for one child only.

The passenger airbag can be turned OFF manually through switch provided on side face of the dashboard at front passenger side. Visual signal of passenger airbag ON or OFF is indicated on the roof console.

When passenger airbag is ON, a rearward facing child seat shall not be installed on the front passenger seat.

When passenger airbag is OFF, a forward or rearward facing child seat can be installed on the front passenger seat.

While installing a rearward facing child seat on the front passenger seat, passenger airbag must be OFF.

Refer images in PAB Switch section.

#### WARNING

If the airbag SRS warning indicator in the instrument cluster illuminates continuously, it means that there is malfunction in the system. Remove the CRS from front passenger seat and contact your TPEM Authorised Service Centre.

## SAFETY

### Child Lock

Child lock are provided on both rear doors. It is used for safety of a child.



Child safety lever to be used for safety of children by preventing them to open rear door while sitting in passenger seat to avoid accidents while vehicle is moving.

Move the lock lever (located on vertical face of the door) as per arrow direction to activate child lock. The door which has been locked by activating the child lock cannot be opened from inside. It can be opened only from the outside.

### NOTE

*Move the lock lever in opposite direction of arrow to deactivate the child proof lock when not required*

### After Impact Safety

- In the event of a collision, the vehicle's safety system may disconnect the high-voltage (HV) battery to protect the occupants and the vehicle from potential risk caused by the HV battery. At the same time, the vehicle comes out of drive mode, during which driving is prohibited.
- Depending on the severity of the collision, drive mode can be re-activated by restarting the vehicle using the Start/Stop switch. This may reconnect the high-voltage battery and enable a limited driving "Limp Home Mode" to move the vehicle out of immediate danger
- In such cases, the vehicle should be immediately taken to TPEM Authorised

Service Centre to determine full extent of damage.

### CAUTION

It is advised not trying to drive or move the car after a severe collision. If the car poses an acute traffic hazard, an exception can be made to move it a short distance out of immediate danger, if the condition of the car allows it.

## ADDITIONAL SAFETY FEATURES (If equipped)

These are additional safety features. For tell tale related information, please check Warning and Indicator section from this Manual.

### Anti-Lock Braking System (ABS)

During Sudden/ Emergency braking or braking on slippery surfaces, when the wheels lock-up, the ABS regulates brake pressure in the individual wheels in such a way that the wheels do not lock when you brake. This allows the driver to continue steering the vehicle when braking.

The ABS warning lamp in the instrument cluster lights up when the ignition is switched on. It goes off after a few seconds if the system is healthy.

### While Braking

For Best results when using an ABS-equipped vehicle in an emergency situation take your foot off the accelerator pedal and Press the Brake pedal fully, continue pressing firmly and allow the ABS to regulate Braking for you. This allows you to

have steering control along with maximum possible braking.

### WARNING

- If ABS malfunctions, ABS may not shorten the distance in all situations.
- ABS system will have the effect of increased stopping distance due to conditions such as gravel, potholes, slippery surfaces, wet roads, ground covered with snow, etc.
- Travelling on bad road, panic braking brake pedal may become hard, this is due to ABS system taking control. ABS will not compensate for bad roads, weather conditions and poor driving judgment. Always drive carefully in adverse weather and traffic conditions.
- Always keep safe distance and adhere to speed limits.

### CAUTION

While travelling on bad road or during panic braking, the brake pedal may be

come hard and slight pulsation may be felt in the brake pedal, you may also hear motor noise from the engine compartment. You should understand this is due to ABS system taking control and you should not reduce the brake force applied in such scenarios.

### Electronic Brake Force Distribution (EBD)

EBD monitors and controls the brake pressure on the rear wheels to improve driving stability while braking. EBD provides optimal braking pressure distribution between front and rear wheels to optimize braking distance and to ensure vehicle stability by means of lowering braking pressure at rear wheels.

The ABS warning lamp along with the Low Brake fluid Malfunction lamp in the instrument cluster lights up when the ignition is switched on. It goes off after a few seconds if the system is healthy.

## SAFETY

### WARNING

- If the EBD is faulty, The ABS Warning lamp along with the Brake Malfunction lamp remains ON in the Instrument cluster.
- If EBD malfunctions, the rear wheels can lock under full braking. This increases the risk of skidding and accidents.
- If ESP, ABS and EBD malfunctions, warning lamp will glow continuously. In such cases, do not panic, reduce speed, maintain a greater following distance, and braking smoothly, drive the vehicle carefully and immediately visit TPEM Authorised Service Centre to rectify the issue.

### Electronic Stability Program (ESP)

ESP monitors stability and traction during driving. If ESP detects that the vehicle is deviating from the direction desired by the driver, the ESP automatically adjusts braking and motor power to prevent skidding or loss of control.

The ESP warning lamp on the instrument cluster lightens up when the ignition is switched on. It goes off after a few seconds if the system is healthy.

### NOTE

*When the ESP feature gets activated while driving, the ESP Malfunction lamp blinks on the cluster indicating the ESP feature is active, this is intended behaviour and is not a cause of concern.*

### WARNING

- If ESP is faulty, The ESP Warning lamp remains ON in the Instrument cluster.
- Drive carefully. Have ESP checked immediately to TPEM Authorised Service Centre as soon as possible.

### Traction Control System (TCS)

The Electronic Traction Control system function (ETC) is designed as a slip control system to prevent the driven wheels of a vehicle from excessive wheel slip.

The Traction Control system continuously monitors the front and rear wheel speeds. In event of traction loss, the system reduces wheel spin by reducing the Motor torque output and by varying the brake pressure on individual wheels.

### CAUTION

For optimal performance of the TCS feature, the tyres must be in good condition and must be inflated as per TPEM Authorised Service Centre guidelines.

### Cornering Stability Control (CSC)

Corner stability Control supports / stabilizes vehicle during partial braking on curves by reducing pressure at required inner wheel of the vehicle.

This helps to reduce probability of vehicle over steering during cornering.

### Roll over Mitigation (ROM)

The ROM function constantly monitors the vehicle behaviour with the help of the wheel speed sensors, steering angle sensor and inertial sensor and intervenes when the vehicle goes out of control and



shows tendency to roll over. Roll over Mitigation brakes individual wheels to reduce the speed of the vehicle to prevent roll over and to stabilize the vehicle.

### WARNING

- The ESP, ROM, CSC system cannot prevent all Roll overs beyond the capabilities of Physics. The capabilities of ESP, ROM, CSC equipped vehicle must never be exploited in a reckless or dangerous manner that could risk the user's safety.
- ESP, ROM, CSC systems aid in providing safety & prevent accidents. However, the Safety of the vehicle always lies in the hands of the driver.

### Electronic Brake Pre-fill (EBP)

Electronic Brake Prefill predicts a panic braking scenario based on the Accelerator Pedal release rate and prefills the Brake system so as to reduce the air gap between the brake pad and brake disc. This helps to improve the brake response time

and braking performance in case of panic situations.

### CAUTION

During EBP activation, when the driver presses the brake pedal, the brake pedal will be a bit hard this is intended behaviour as the elasticity in the brakes system is reduced by prefilling the brake system.

### Hydraulic Brake Assist (HBA)

The brake assist recognizes from the operating speed of the brake pedal whether the driver wants an emergency stop and increases the brake pressure automatically up to the ABS control range, as long as the driver keeps the brake pedal pressed.

### WARNING

- The HBA system cannot prevent collisions, the capabilities of an HBA equipped vehicle must never be exploited in a reckless or dangerous manner that could risk the user's

safety.

- The Safety of the vehicles always lies in the hands of the driver, never rely on the HBA feature to prevent accidents.

### Hydraulic Fading Compensation (HFC)

Due to frequent braking especially driving down on slopes, the brake temperature increases, which causes deterioration in the braking performance and increases the stopping distance this is called as Brake fading.

In dangerous fading situations most drivers operate the brake pedal with a small or regular braking force and they never reach to the maximum possible vehicle deceleration. The HFC function provides additional pressure to achieve the required deceleration and thereby prevent brake fading.

### CAUTION

While descending Ghat sections or hilly terrains, the driver must make use of re

## SAFETY

generative braking to reduce the usage of brakes and thus prevent chances of brake.

### WARNING

- The HFC feature is provided to assist the driver in cases of emergencies. The capabilities of an HFC equipped vehicle must never be exploited in a reckless or dangerous manner.
- The Safety of the vehicles always lies in the hands of the driver, never rely on the HFC feature to prevent accidents.

### Panic Brake Alert (PBA)

Panic brake alert warns vehicles approaching from behind when emergency or heavy braking takes place. The PBA is designed to trigger Hazard lamps automatically, which will act as an immediate warning to vehicles directly behind and nearby. PBA is an immediate and urgent warning to all vehicles around thus improv-

ing their response time to unforeseen braking situations.

The PBA feature gets activated if the vehicle speed is greater than 50 kmph and if the system detects a panic braking scenario.

### After Impact Braking (AIB)

In the event of primary collision there are chances that vehicle can no longer be safely controlled.

Accident analysis has shown that an active brake intervention post the first impact would mitigate the effect of the subsequent collisions.

After impact braking system is activated automatically and brakes the car in a safe manner to mitigate secondary collision. Hazard & brake lights are triggered to intimate surrounding users of an emergency situation.

The driver can override the AIB by depressing the Accelerator pedal if there is a risk of being hit by following traffic.

### WARNING

- In case of any failure of AIB feature, the following alert pops-up on the cluster. If the AIB alert remains on the instrument cluster drive carefully and have AIB checked immediately to TPEM Authorised Service Centre as soon as possible.
- The basic assumption for the AIB to work is that the brake system is intact after the primary impact.

### Integrated iVBAC

Introducing the Intelligent Vacuum-less Boost & Active Control (iVBAC) – a cutting-edge brake technology designed for hybrid and electric vehicles. Unlike traditional systems, the iVBAC operates vacuum-free, enhancing reliability and simplifying maintenance. This system not only supports regenerative braking but also functions as an Electronic Stability Control (ESC) unit, seamlessly integrating Anti-lock Braking System (ABS), Traction Control, and Stability Control functionalities. When you apply the brakes, the

iVBAC translates your input into precise brake pressure, ensuring a smooth and controlled experience.

### Driver Doze OFF Alert System (DDOA)

The Driver Doze off Alert system advises the driver to take a break from driving when driver is likely to doze off. This is detected based on the driver's steering behaviour.

The system evaluates the steering behaviour at speeds 50 – 200 kmph.

Driver attention Level: The Driver attention level is calculated based on driving behaviour and many external parameters and the attention level is displayed to the driver in the drive info tab.

### Hill Hold Control (HHC)

Hill Hold Control (HHC) is a comfort feature that prevents the vehicle from rolling backward when starting from a standstill on a slope or an inclined surface. It maintains brake pressure for a 2 sec while the driver moves foot from the brake pedal to the accelerator pedal, thereby avoiding rollback on an incline. Once the accelerator pedal is pressed, the ESP ECU re-

leases the brakes, allowing the vehicle to move forward smoothly.

The HHC will function on both vehicle facing uphill and downhill if the following conditions are met.

1. Vehicle must be equipped with HHC.
2. Vehicle must be on sufficient gradient.
3. Parking brake must not be engaged.
4. Correct gear engaged
  - a. Vehicle facing uphill – Gear should be in forward
  - b. Vehicle facing downhill – Gear should be in reverse

### WARNING

The HHC feature cannot prevent vehicle roll back in all scenarios, the safety of the vehicle lies in the hands of the driver.

The HHC warning lamp on the instrument cluster lights up when the ignition is switched on. It goes off after 3 seconds if the system is healthy.

### WARNING

If HHC is faulty, The HHC Warning lamp remains ON on the Instrument cluster. Drive carefully. Please contact immediately to the TPEM Authorised Service Centre as soon as possible.

### Dynamic Wheel Torque by Brake (DWT-B)

DWT-B, or Dynamic Wheel Torque Vectoring by Braking, enhances vehicle agility and steering responsiveness by selectively applying braking force to the inner wheels during turns. This intervention reduces understeer, the tendency for the vehicle to push wide in corners, enabling drivers to maintain higher speeds through curves while maintaining control and stability. This technology improves the overall driving experience, especially in dynamic situations where precise handling is crucial.

### Dynamic Braking Support (DBS)

Dynamic Braking support feature makes use of the data from ADAS module to detect a crash imminent scenario, and if the

## SAFETY

driver doesn't brake hard enough to avoid a crash, the brake control system supplements the braking by driver. By providing additional braking assistance, DBA helps to reduce stopping distances and improve overall vehicle safety.

### WARNING

- The DBS feature is provided to assist the driver in cases of emergencies. The capabilities of DBS equipped vehicle must never be exploited in a reckless or dangerous manner.
- The Safety of the vehicles always lies in the hands of the driver, never rely on the DBS feature to prevent accidents.
- Ensure that Front Collision Warning and Auto Emergency Braking functions are activated from Infotainment system to keep DBS functional.

### Roll Away Re-clamp

If vehicle rolling observed by EPB system

in EPB applied condition, motor automatically re-clamp to prevent rolling.

To avoid potential roll-away caused by a reduction in clamp force due to temperature drop, the motor re-applies the clamping force as a preventive measure.

### Dynamic Steering Torque (DST)

Dynamic Steering Torque Control (DST) function provides stability to the vehicle on different surfaces during braking. The ESP systems calculates and provides the steering direction request to correct the steering which in turn assists the driver in achieving stability during critical braking situations on different road surfaces.

### WARNING

While DST function intervention is ongoing, do not remove hands from steering wheel.

### CAUTION

- Drive the Vehicle cautiously on Slippery road surfaces.

- DST Function controls the steering and make it slightly rotate either left or right side depending upon road surface while braking. Ensure to follow the steering input.
- User can rotate the steering Wheel past the DST control as per wish but with a slight more force.

### HIGH VOLTAGE BATTERY SYSTEM

#### Temperature Limits

Battery pack and vehicle can operate safely in limits from -5°C to 55°C.

#### NOTE

*To control the battery temperature of the high voltage battery the air conditioner is used to cool down the battery and may switch on automatically without request from control panel which may generate noise from operation of the air conditioner compressor and cooling fan.*

#### HV Battery Life & Maintenance

This Vehicle comes with a standard battery warranty as mentioned in warranty section. Regular service of the vehicle and charging protocol to be followed to maximize the battery life.

#### Energy Information

The vehicle battery pack has a maximum energy as specified in Technical Specification. Energy retention capacity deteriorates

over several cycles of usage and hence range deterioration happens over time.

This decrease in range during the end of life of battery is expected and is not considered as a malfunction of the battery pack. During these conditions, it is recommended to contact TPEM Authorised Service Centre.

#### Brake Energy Recovery System

The vehicle features energy regeneration system, which regenerates expended energy during coasting or braking during the drive. This system allows the battery to be recharged under the above mentioned conditions.

Please note, the regeneration system does not fully recharge the battery, it only provides a chance to recover a portion of energy that would be lost during braking. When you release the accelerator or press the brake pedal, energy flows from wheels to high voltage battery, thereby charging it.

Regeneration is done by converting driving force (kinetic energy) into electrical energy that is stored in the Lithium Iron Phosphate battery while the vehicle is decelerating or being driven downhill. This is called regenerative braking

#### Heavily Discharged High Voltage System

In the case of a heavily discharged HV battery, there is a chance that the low voltage battery is discharged as well. In this case, please contact your nearest TPEM Authorised Service Centre.

#### WARNING

Do not try to jump start the vehicle or tow the vehicle without guidance from the service assistant.

#### Heated High Voltage Battery

In such a condition, the battery has safety logics to limit the performance or disconnect by itself with prior warning. The vehicle should be stopped and allowed to cool down and contact TPEM Authorised Service Centre.

## BATTERY AND COMPONENT

### Long Storage of HV Battery Pack

The HV battery undergoes discharge at a rate of approximately 3% over a period of 30 days in storage. Do not allow the vehicle to be discharged to 0% in storage. It is recommended that the vehicle must be charged to a charge level in between 30% to 50% before leaving the vehicle for long time storage. After this time period the vehicle must be charged to 100% using Normal Charging before use.

### High Voltage System Failure

In the case of high voltage system failure, which may arise due to various reasons, contact TPEM Authorised Service Centre for further assistance.

### Predicting Energy Usage

The vehicle battery energy usage is displayed in the instrument cluster in the form of estimated range. This range is updated by the system algorithm, depending on the driving conditions.

Displayed range in the instrument cluster is a tentative number based on drive route, driving pattern and usage pattern history

over the past drives. It is recommended keep a 20km buffer in estimated range before planning the trip

### Disposal

The disposal of an HV Battery must be done with utmost care and will be carried out by TPEM Authorised Service Centre after sales service at the end of the battery life time or if the battery pack has passed its warranty period.

Refer EPR no.as given below-



### *i* NOTE

- It is advised to contact TPEM Authorised Service Centre which shall guide & help in dis-mantling, han

*dling and disposal through agencies who are certified by central/state pollution control board & obtain certificate of disposal from these agencies as proof of sustainable disposal.*

- If you decide not to use the recommended TPEM Authorised Service Centre or TPEM Authorised Workshop to dispose of your high voltage battery, the responsibility of the consequences of environmental pollution or accidents must be borne solely by you, the owner of the vehicle.*
- Customers who wish to dispose of battery by themselves shall deal only with registered entities (list of these is available on the CPCB/SPCB website) after duly verifying validity of necessary registration documents. After disposal the EPR certificate has to be obtained by the customer from the entity. It is request that customer shall provide this certificate to nearest*

*TPEM Authorised Service Centre which is to be kept for records & submitted to central pollution control board as proof of disposal according to policy guidelines.*

### **WARNING**

EV battery contains materials like Lithium Iron, graphite, plastic & steel etc which can have impact on environment and are harmful if not handled/disposed of carefully. There is a risk of severe burns and electrical shock that may result in serious injury while additionally posing a risk of environmental damage.

### **CAUTION**

Keep the following things in mind to prevent damage of the battery pack:

- Avoid exposure of the vehicle to extreme environmental temperature for extended period of time.
- Do not leave the vehicle for longer period when High voltage battery is

near zero SOC. Follow the long duration storage guideline.

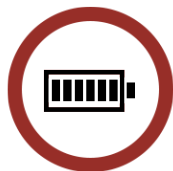
- Do not use the High voltage battery for any other purpose.
- The capacity of the High voltage battery will decrease with time and usage to hold charge like all such batteries. As the battery ages and capacity decreases, this will result in a decrease from the vehicle's initial mileage range. This is normal, expected, and not indicative of any defect in your High voltage battery.
- When the HV Battery reaches the end of its life cycle, the charging capacity or State of health (SoH) reduces. During such decrease in charge capacity and range of vehicle, contact TPEM Authorised Service Centre and get the SoH of Battery pack get inspected.
- One full slow charging after every 4 Fast charging cycle is recommended for regular charging.

- If over time the maximum charge capacity and the maximum electric range begin to degrade, contact TPEM Authorised Service Centre for inspection and maintenance.
- If the vehicle is involved in a collision, we recommend that you contact to TPEM Authorised Service Centre to inspect if the high voltage battery is still connected or damaged in any way.
- The HV battery capacity of the high voltage battery may decrease when the vehicle is stored in high/low temperatures.
- Electric range may vary depending on the driving conditions, even if the charge amount is the same. The high voltage battery may expend more energy when driving at high speed or uphill. These actions may reduce the vehicle electric range.

## BATTERY AND COMPONENT

### TIPS TO CONSERVE BATTERY LIFE

#### 1. Battery Charging



It is advisable to charge the vehicle upto 100%. This ensures accurate SoC calculation and consistent energy content intake during charging. Charging to 100% also maintains battery health for longer time and ensures better range throughout the life cycle of the vehicle.

It is also advisable to slow or home charge the vehicle to 100% as much as possible. Slow Charging and full charging is the best way to charge your vehicle to maintain the health of the high voltage battery.

After every 4 fast charging cycles, it is advisable to slow or home charge the vehicle to 100% SoC

#### 2. Cell Balancing or Equalization



Cell balancing or equalization and SoC calibration happens during charging, especially at higher SoC, > 90%. This causes charging to take longer time above 90%. Allow the vehicle to reach 100% SoC before terminating the charging session, as both cell balancing and SoC calibration are essential steps for any battery to perform optimally for a prolonged time.

#### 3. Extreme Temperatures



Extreme temperatures impact upon how well a battery can maintain its level of charge. Avoid charging the high voltage

battery when it is hot and vehicle is driven for long distances especially in summers. Allow vehicle temperature to cool down before charging. It is always better to charge EV when ambient temperature is low especially at night.

#### 4. Vehicle Parking



Avoid parking vehicle in direct sunlight for long duration. Try to park it under shade, tree etc. This helps in keeping battery temperature low resulting enhance battery life over the life cycle.



## INSTRUCTIONS TO FOLLOW

- It is recommended to charge the vehicle to 100% every time, whenever vehicle is being charged.
- Avoid charging vehicle under heavy rain / thunderstorms.
- Avoid driving vehicle below 10% SoC.
- Make sure the charge station's supply cable is positioned so it will not best stepped on, tripped over, or otherwise subjected to damage or stress.
- There are no user serviceable parts inside the charging gun. Contact TP&M Authorised Service Centre should you experience any problems with the charging gun. Do not attempt to repair or service the charge station or charging gun yourself may result in injury.
- Do not operate your charge station and gun if it or the supply cable is visibly damaged. Contact your Service Representative for service immediately. Refer to the 'Emergency and Break-down' section in this manual for information on the Service Representative in your area.
- Do not place fingers inside the coupler on either end of the charging gun.
- Do not allow children to operate this device. Adult supervision is mandatory when children are in proximity to a charge station that is in use.
- Not for use in commercial garages.
- Slow Charging only happens in park brake engaged condition. So always keep the park brake engaged during a charging session.
- Change of vehicle state (Ignition OFF to Ignition ON or vice-versa) should be avoided while charging
- Post switch off the charger, provide min 5 seconds for touching and pulling out the gun.
- If the charging gun removed and reinsertion required it could be done after at least 10 seconds of removal of the charging gun from Socket.
- Do not disengage/play around with the Park brake/hand brake while vehicle is in fast charging condition.

- Overcurrent and leakage current protections are given in the home charging box and charging gun. The RCBO should always be in ON state during normal charging use-case and there should be no error (Red) LEDs on the charging gun. In case any tripping of RCBO is observed or error LEDs start blinking on the Charging gun, please contact TP&M Authorised Service Centre.
- Home charging box comes with a key and lock. It is recommended to lock the box during overnight charge or when the charging box is not in use to avoid misuse of charging point.

 **WARNING**

Unplug both couplers of your Portable Charging Gun before cleaning.

 **NOTE**

*During normal operation, the charging gun or couplers may feel warm. If either coupler or the charging gun feels hot*

## CHARGING

*during charging, unplug the gun and have a qualified electrician inspect the connections before you continue charging.*

### CAUTION

Do not use a damaged charging station, plug point or charging port. Using the charger with a worn or damaged port may result in unanticipated consequences.

### WARNING

The charger generates electromagnetic waves that can seriously impact medical electric devices such as an implantable cardiac pacemaker in a person. When a person has an implant like the one mentioned above, make sure to ask the medical team and the manufacturer whether charging your EV will impact the operation of the medical electric device implant. In such case, do not go near the vehicle when it is charged.

ing.

### WARNING

- Ensure that the charging gun is always stored in a safe place. Do not expose it to rain or wet conditions. Avoid pouring or dripping water or other liquids over it.
- If water penetrates the electrical devices, the risk of electric shock increases. Ensure that all plugs and cables are free of moisture before using the charging gun. Never connect the charging gun to the mains with wet or moist hands or when the charging gun is wet.

### NOTE

*Charging station and domestic plug point must be approved/certified by a qualified electrician before using the charging gun. Coupler Receptacle has to have proper Grounding, electrical connection and has to contain a Residual-Current Circuit Device (RCD).*

*ual-Current Circuit Device (RCD).*

### CAUTION

- Make sure that the device is always stored in a safe place. Do not expose the device to rain or wet conditions.
- DO NOT use this product if the EV charge connector/cable is damaged.
- During charging the vehicle must not be exposed to rain, lightning and snow.

### NOTE

*Charging should be done in Ignition OFF state.*

---

**IMPORTANT TIPS****Do's and Don'ts**

- Do not direct high pressure washer fluid/ water jets (Pressure above 0.5 bar) at electrical devices and connector during washing. This is to prevent malfunction/failure of electrical system due to water ingress. No High pressure washing in motor compartment, Under-floor battery pack and CCS Charging port.
- Always check the SoC level before start of journey & ensure car is adequately charged. You may check the SoC level on the mobile app also.
- Remote AC command not to be executed through mobile app while/during the charge initiation process.
- If AC is switched ON remotely using Zconnect, it is required to switch it off using the Zconnect app before unlocking the vehicle. If it is not followed, the vehicle requires two ignition ON cycle to move as it will not move in the first ignition ON cycle.
- The charging gun provided for home charging has to be stored safely and securely in the trunk of the vehicle or has to be plugged on to the Home Charging Box in locked condition.
- The wall box charging unit is also used for slow or home charging. It comes with a key and lock. It is recommended to lock the home charging box when the vehicle is kept for overnight charging or when nobody is around while the vehicle is being slow charged. This ensures that the charging unit along with the charging gun cannot be misused or stolen.
- Wet surfaces are good conductors of electricity. Though the vehicle is equipped with safety mechanisms to protect users, it is advisable to take a few precaution while plugging in for charging. Hence, before charging, ensure that the power source socket, the charging gun and the charging port (CCS2) port in the vehicle are dry. Also ensure that you are standing on dry ground and your hands are dry as well while using the high voltage charging equipment.
- Usage of damaged cables, Power Source socket and vehicle side CCS2 port must be avoided as they may result in electrical hazard and inconsistent charging experience.
- While plugging in for home charging, ensure power source is off. Subsequently ensure charging gun is connected at both ends – One at power source and the other at vehicle's CCS2 port. Then switch ON the power source switch to commence charging. Confirm that the vehicle is charging from the green charging telltale displayed on the instrument cluster. The cluster remains ON to display charging status for 60 sec after the start of charging.
- If charging gun is removed before 100% charging and again needs charging up to 100%, it is advisable to wait for at least 10 seconds before reinserting the gun in the charging port.
- Once charging is complete and gun is removed from the charging port, it is

## CHARGING

---

advisable to pause for 30 sec before switching on the car to start driving.

- When the vehicle is shut off after drive, it is advisable to pause for at least 10-15 sec before charging. It allows the vehicle's electrical system time to de-energize and stabilize before the charging commences.
- Comply with the following in order to prevent electrical shock when charging:-
- Use a waterproof charger.
- Do not touch the charging connector and charging plug with your hands wet, or do not stand in water or snow while connecting the charging cable.
- Be careful when there is lightning.
- Be careful when the charging connector and plug are wet.
- Replace the charging cable if the cable coating is damaged to prevent electrical shock. When connecting or removing the charging cable, make sure to hold the charging connector handle.
- Only use the certified charging cable.

If you use a separate extension cable such as a reel or use an uncertified cable, it may cause abnormalities of electrical outlets, leading to fire or explosion.

- If you pull the cable itself (without using the handle), the internal wires may be disconnected or get damaged. This may lead to electric shock or fire.
- Do not leave the vehicle with the charging door open. An open charging door may indicate that the vehicle door has been unlocked and may be subject to vehicle theft.
- Always keep the charging connector and charging plug in clean and dry condition. Be sure to keep the charging cable in a condition where there is no water or moisture.
- Make sure to use the designated charger for charging the electric vehicle. Using any other charger may cause failure.
- Make sure to use the designated charger for charging the electric vehicle. Using any other charger may

cause failure.

- Before charging the battery, turn OFF the vehicle.
- When the vehicle is switched OFF while charging, the cooling fan inside the motor compartment may automatically operate. Do not touch the cooling fan while charging.
- Be careful not to drop the charging connector. The charging connector can be damaged.
- Do NOT use an extension cord, when using the L1-Trickle charger, as this may overheat and/or cause damage.
- When charging or right after charging the high voltage battery, the cooling will be made using air conditioner system in order to control the high voltage battery temperature. At this time, the noise might occur by the air conditioner compressor and cooling fan, which is normal.

## TYPES OF CHARGING

| SN | Types Of Charging  | Charging Component Specification                                                                                                                                        | Charging Time In Hrs                        | Charge Time In SoC Band | Charge Gun                                                                          | Power Source                                                                        |
|----|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1  | Normal/AC Charging | <ul style="list-style-type: none"> <li>Nominal Voltage ICCPD: 230V</li> </ul>                                                                                           | ~25 (Option I)<br>~29.1 (Option II)         | 10-100 %                |  |  |
| 2  |                    | <ul style="list-style-type: none"> <li>Nominal Voltage: 230V AC RMS single Phase 50Hz</li> <li>Power Rating: 3.3Kw AC RMS</li> <li>Rated Current 13A AC RMS</li> </ul>  |                                             | 10-100 %                |                                                                                     |                                                                                     |
| 3  | AC Charging (WMU)  | <ul style="list-style-type: none"> <li>Nominal Voltage: 230 V AC RMS single Phase 50Hz</li> <li>Power Rating: 7.2Kw AC RMS</li> <li>Rated Current 32A AC RMS</li> </ul> | ~9.3 (Option I)<br>~10.7 (Option II)        | 10-100 %                |  |  |
| 4  | Fast/DC Charging   | <ul style="list-style-type: none"> <li>Charging station voltage capability should be greater than or equal high voltage battery pack nominal voltage.</li> </ul>        | ~26.5 Min (Option I)<br>~27 Min (Option II) | 20-80 %                 |  |  |
|    |                    |                                                                                                                                                                         | ~35 Min                                     | 80-100%                 |                                                                                     |                                                                                     |

## CHARGING

### Normal/ac Charging

In electricity grid, electric power is AC (alternating current) by nature. However, electric power in battery is DC (Direct Current) by nature. Hence, to charge an electric car by AC grid, power has to be converted from AC to DC. And to convert AC power to DC power On-board Charger is used. This type of charging is called Normal charging/AC charging.

Normal charging is recommended for usual charging of the vehicle. This charging method is most suitable for parking spots where the car will stay parked for longer duration of time

### Precautions For Normal Charging

1. Proper maintenance of earthing pit is must. Add water & add salts at regular intervals into the earth pits in order to maintain the value of earth resistance. Check annually the condition of the electrodes so as to add or replace electrodes.
2. The electrical socket used for EV charging and its associated wiring should be able to supply 15A dedi-

cated load continuously.

3. Vehicle charging port must be free of dust, water or snow while connecting the charging gun; if not proper cleaning method must be used to remove dust, water and ice.
4. Don't try to pull off the charging gun during charging.
5. Don't pull out the charging gun if it is in locked condition as excess force can break or damage the locking mechanism.

### Normal Charging Procedure

1. Engage the Automatic Parking Brake. (Charging won't start if EPB is not engaged).
2. Connect the plug to AC power socket.



3. DO NOT plug into a power strip.



4. Open the protective cap on Charging Gun.



5. Open the protective cap on Charging Inlet.



6. Before connecting the charging gun to vehicle charging socket, make sure the gun lock is released.

### ⚠ CAUTION

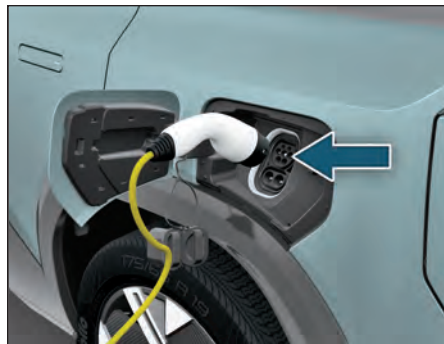
- If the Gun Lock is not released please don't insert the Charging Gun forcefully into the socket. It may damage the Charging Socket.
- Don't use the electric connection with extension or power strip for the slow charging or AC charging of the vehicle, this will lead to heat up the cables and charging gun. Prolong

charging IN such condition may lead to melting of wire and charging gun.

7. If the actuator is engaged and the gun is not getting inserted properly, contact TPEM Authorised Service Centre.
8. Remove any dust on the Charging Gun and Charging Inlet. Connect the charging gun to vehicle AC Charging Inlet.



## CHARGING



9. Switch on the AC supply.
10. Charging Gun will be locked after switching on the AC supply. You will hear a “click” sound, when the gun is connected correctly.



### *i* NOTE

*When vehicle is in Charging Mode, it will not go in Drive (D), Sport (S) or Reverse(R).*

11. Normally the car starts automatically charging. If not, please refer ‘Troubleshooting Guide for Normal Charging’ table.
12. Open the door and see instrument cluster for State of Charge, Time to Charge and Gun connection status.
13. In case the park brake is not engaged

or partially engaged the charging won’t start and ‘Engage Park Brake to start charging’ message will be displayed on instrument cluster.

### *i* NOTE

*Infotainment and cabin cooling can be used during charging of the vehicle by putting the vehicle in Ignition.*

14. To stop the charging, switch off the AC power supply.
15. The charging gun will be unlocked after switching off the AC supply and pressing charging gun unlock switch.

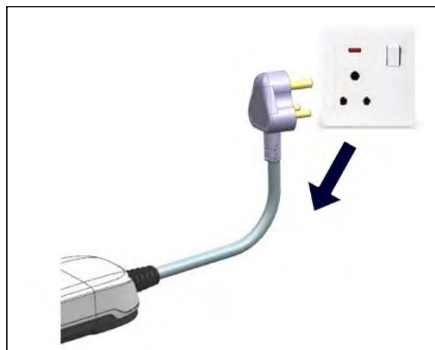




### *i* NOTE

*If the gun does not get unlocked in first attempt of pressing fascia switch, repeat the operation.*

16. Pull out the plug.



17. Put on the protective caps on both Charging Gun and Vehicle Inlet.

### *i* NOTE

- *Once Normal/Fast charging is completed, 90 seconds of time gap is required before the vehicle can be started. After turning off the vehicle, wait for four seconds if you want to start the vehicle again.*
- *In emergency charging shutdown conditions, Gun won't be unlocked. Please turn the ignition ON-OFF via*

*PEPS switch. If the issue still persists contact TPEM Authorised Service Centre.*

- *In unforeseeable circumstances if the charging gun is stuck to the socket after charging is done, Please turn the ignition ON-OFF via PEPS switch. If the issue still persists contact TPEM Authorised Service Centre.*

### Emergency charging Gun unlocking

In case of electrical malfunction, you can unlock the Charging Gun from inside of luggage compartment as per procedure given below.

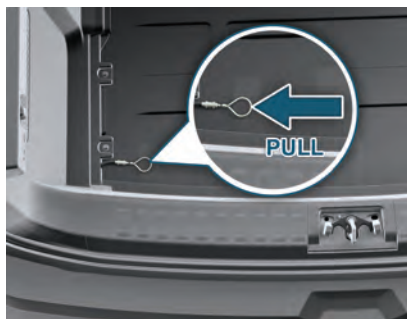
1. Open the tailgate
2. Remove the luggage cover

## CHARGING

---



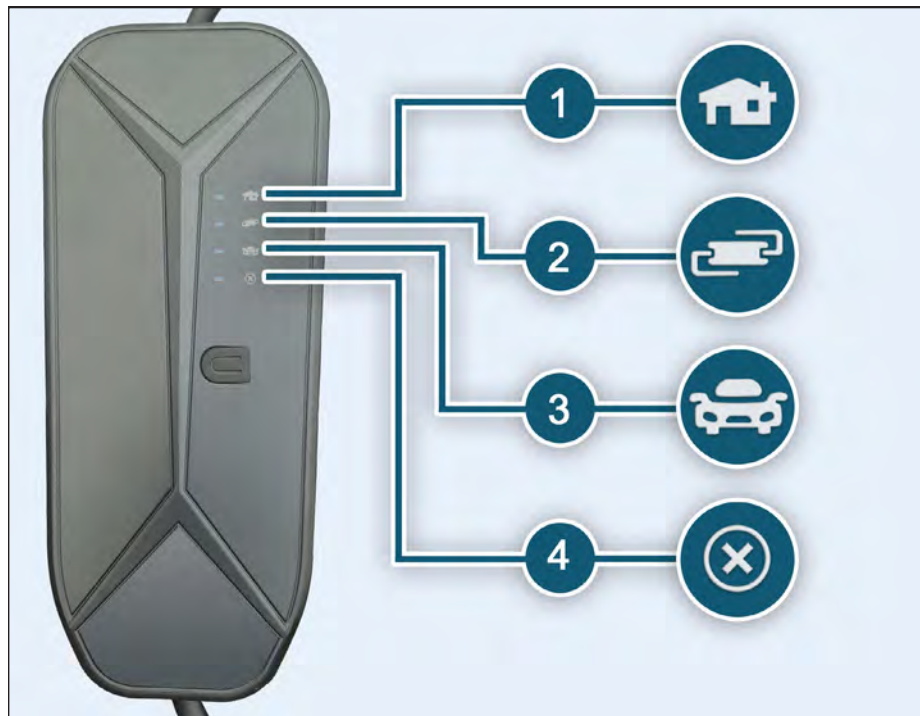
3. Pull the cable of charging gun unlocking as shown in below picture.



4. Then pull out the charging gun.

**Normal Charging Control Box Indications:**

1. Home
2. Control Box
3. Vehicle
4. Fault



## CHARGING

| Working State         | Home  | Control Box | Vehicle | Fault | Example                                                                            | Description                                    |
|-----------------------|-------|-------------|---------|-------|------------------------------------------------------------------------------------|------------------------------------------------|
| Self-inspection state | Blink | Blink       | Blink   | Blink |  | Self-inspection for system                     |
| Standby state         | On    | Off         | Off     | Off   |  | No fault<br>Check for engagement of park brake |
|                       |       |             |         | Blink |  | Plug temperature is high                       |
| Charging state        | On    | On          | Blink   | Off   |  | No fault                                       |
|                       |       |             |         | Blink |  | Plug temperature is high                       |
| Charging stopped      | On    | On          | On      | Off   |  | No fault                                       |
|                       |       |             |         | Blink |  | Plug temperature is high                       |

## Troubleshooting Guide For Normal Charging

Refer the below table if charging is not starting or if it stops abruptly. The below symbols of 'House', 'Control Box', 'Vehicle' and 'Fault' can be seen on the control box beside the respective LEDs.

| Fault Category          | Indication                                                                        | Home                                                                              | Control Box                                                                       | Vehicle                                                                           | Fault                                                                               | Recommended Action                                                                                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                                                                   |  |  |  |  |                                                                                                                                                                                                               |
| Interface fault in home |  | Blink                                                                             | Off                                                                               | Off                                                                               | On                                                                                  | Improper earth connection. Check the earth pit.<br>Short circuit between PE and phase. Error in domestic supply side. "Stop Charging"                                                                         |
|                         |  | Blink                                                                             | Off                                                                               | Off                                                                               | Blink                                                                               | AC voltage is either less than 190V or more than 250V. Error in domestic supply side. "Stop Charging"<br>Proper connection of plug and socket should be ensured. Also, check socket rating and use 15A socket |
| Control box fault       |  | Off                                                                               | Blink                                                                             | Off                                                                               | On                                                                                  | Contact TPEM Authorised Service Centre                                                                                                                                                                        |
|                         |  | Off                                                                               | Blink                                                                             | Off                                                                               | Blink                                                                               |                                                                                                                                                                                                               |

CHARGING

| Fault Category | Indication                                                                        | Home | Control Box | Vehicle | Fault | Recommended Action                           |
|----------------|-----------------------------------------------------------------------------------|------|-------------|---------|-------|----------------------------------------------|
| Vehicle box    |  | Off  | Off         | Blink   | On    | Go to nearest TPEM Authorised Service Centre |
|                |  | Off  | Off         | Blink   | Blink |                                              |

Legend

 -Blink

 - Off

 - On

### AC Charging (Wall Mount Unit)

This type of charging will help customer to improve the charging time for vehicle charging.

#### AC (WMU) Procedure

1. Parking brake should be in engage condition (Charging won't start if parking brake is not engaged).
2. Open the protective cap on Charging Inlet.
3. Remove the charging gun from the WMU. (WMU will be separately installed)



4. Open the protective cap on WM Charging Gun.



5. Before connecting the WMU charging gun to vehicle charging socket, make sure the gun lock is released.
6. If the actuator is engaged and the gun is not getting inserted properly, contact TPEM Authorised Service Centre.
7. Remove any dust on the Charging Gun and Charging Inlet. Connect the WMU charging gun to vehicle AC WMU Charging Inlet.



8. Scan the RFID provided, on the WMU to start charging.
9. Charging gun will be locked automatically. You will hear a "click" sound, when the gun is connected correctly.
10. Normally the charging gets started automatically. If not, please refer 'Troubleshooting Guide in WMU's owners' manual.

## CHARGING



11. Open the door and see instrument cluster for State of Charge, Time to Charge and Gun connection status.
12. To stop the charging, scan the RFID on WMU.
13. The charging gun will be unlocked after pressing the fascia switch on the dashboard panel. Pull out the gun.
14. Put on the protective caps on both Charging Gun and Vehicle Inlet. Place the Charging gun back and close the charging inlet flap.

### Fast/dc Charging

- Fast charging of electric vehicle is achieved by using Fast/DC charging stations, they convert the AC power from the grid to DC power and can directly charge the HV battery pack thus by passing the On-Board Charger.
- Fast charging can be done wherever Fast/DC charging station is available. User can charge at high speeds at public charging stations.

#### NOTE

- *Battery performance and durability can deteriorate if the fast charger is used constantly. Use of Fast Charging should be minimized in order to help prolong high voltage battery life.*
  - *After a maximum of four fast charging cycles, you must use Normal charging to 100% State of Charge for the optimum performance of the high voltage battery pack.*
- Electric vehicle can be fast charged

using any fast charging station or equipment compliant to Combined Charging System standard having Type 2 connector (CCS Type 2).

### Fast Charging Procedure

1. Engage the Automatic parking brake (EPB) switch. (Charging won't start if EPB is not engaged).



2. Pull out the Charging Gun from DC/Fast Charging Station.
3. Open the protective cap on Charging Gun.





4. Pull the 'Charging-inlet Flap Open Lever' to open the charging door.
5. Open the charger-inlet flap.
6. Before connecting the charging gun to vehicle charging socket, make sure the gun lock is released.

### CAUTION

If the Gun Lock is not released please don't insert the Charging Gun forcefully into the socket. It may damage the Charging Socket.

7. Remove any dust on the Charging Gun

and Charging Inlet.

8. Connect the charging gun to vehicle AC and DC Charging Inlet.
9. Switch on the DC charging station supply.



10. Charging Gun will be locked after switching on the DC charging station.
11. You hear a "click", when the Gun is connected correctly, Click sound is because of Gun locking after supply is switched on.



12. Normally the car starts automatically charging. If not, please refer Charging Gun's Fault Indication & Indication Priority Table on the charging station.

### NOTE

*When vehicle is in Charging Mode, it will not go in Drive (D), Sport (S) or Reverse (R).*

13. To know the State of Charge, Time to Charge and Gun connection status please see instrument cluster. This status is displayed only when driver door is opened.

## CHARGING

14. In case the park brake is not engaged or partially engaged the charging won't start and 'Engage Park Brake to start charging' message will be displayed on instrument cluster.

### *i* NOTE

*Infotainment and cabin cooling can be used during charging of the vehicle by putting the vehicle in Ignition.*

15. To stop the charging, switch off DC charging station.
16. The charging gun will be unlocked 15 seconds after switching off the supply from DC charging station. For fast charging no fascia switch input is required. It unlocks automatically.
17. Put on the protective caps on both Charging Gun and Vehicle Inlet.

### *i* NOTE

*If you remove the charging gun from the vehicle and if you wish to reinsert the gun to recharge the vehicle, please wait for at least 10 seconds before charging*

*gun is plugged again.*

### *i* NOTE

*In emergency charging shutdown conditions, Gun won't be unlocked. Contact TPEM Authorised Service Centre.*

## V2X CHARGING (if equipped)

The V2X charging system offers a flexible and energy exchange method for charging electric vehicles (EVs) without the need for charging stations. A new framework for vehicle-to-vehicle charging technology is introduced that can work plug-in electric cars.



V2X switch is located below parcel tray at RH side of luggage compartment.

### Vehicle Ignition OFF / ON Conditions

The transponder in the ignition key carries a Unique Identification Code (UID). The

vehicle unlocks when the code on the key matches with the code on the Vehicle Control Unit (VCU). In case of PEPS variant, Immobilizer function is provided by PEPS ECU.

### Vehicle to vehicle charging (V2V)



V2V Gun

#### Vehicle Ignition OFF state

1. User can decide the discharge SoC limit of the source vehicle (donor) and can preset the SoC limit in infotainment before starting the V2V function.
  - Since infotainment system won't be awake in Ignition OFF state, Limit can be set only in ignition ON or Crank on condition.
  - Discharge SoC Limit cannot be changed during the V2V function.
  - Incase if no limit is set by the user, 30% SoC will be consider as default discharge SoC limit.
2. Keep both sink (recipient) and source (donor) vehicle nearby and ensure maximum up to 3m distance between the charging inlets of the vehicles.
3. Make sure to keep the vehicles and the V2V charger in dry environment/in shade
4. If the Source/sink vehicle is in OFF state, Open/close the door or if vehicle is in locked state Press RKE unlock Button and open the door to wake up

the vehicle control unit. Once wakes up, Vehicle control unit will remains in wakeup state for 120sec.

5. Remove the caps & connect the V2V gun to the source and sink vehicle within 120sec of step 4. Check the labels on the V2V gun and make sure that the V2Vgun side labeled as "source" should connect to the source vehicle & the other end labeled as "sink" should connect to the sink vehicle. If user couldn't connect the V2V gun within 120sec of step 4. Remove the Gun completely and repeat from step 4.
6. Make sure park-brake/EPB is engaged.
7. Ensure V2V gun is connected to both vehicles and press the V2X switch of source vehicle within 120 sec of step 5 to start the 220V AC power supply and to charge the sink vehicle. If user couldn't press V2X switch within 120sec of successful step 5. Remove the Gun completely and repeat from step 4 to initiate the V2V function

## CHARGING

---

- again.
8. Check for the telltale indication frequently when the V2V function is in progress to ensure there is no interruption/fault.
  9. V2V function will stop from source vehicle side, if there is no energy dissipation has happened for 100secs from successful step 7. In this case, remove the gun completely and start from step 4 to reinitiate the V2V function
  10. Charging will automatically stop if the source vehicle SoC goes below the user set SoC limit or at 30%. In that case follow the procedure from step 16.
  11. Also monitor the SoC of both vehicles frequently to check if sink vehicle is charged enough/ source vehicle SoC is reached the minimum required SoC level.
  12. To stop the V2V charging from source vehicle side, press the V2X switch of source vehicle to disable the 220V AC supply and follow from step 16 after 5sec.
  13. To stop the V2V charging from sink vehicle side, press its fascia switch and then ensure to press the V2X switch of source vehicle to stop the AC power supply. & follow step 14.
  14. After 5 sec delay press source vehicle fascia switch and then remove both the guns.
  15. In case V2V charging gets interrupted from source/sink vehicle side due to any fault, Press the Gun unlock fascia switch in both vehicle and completely remove the gun from both side and repeat from step 4. If the issue still remains, Press the Gun unlock fascia of both vehicle and completely remove the gun. After that keep the vehicles in Ignition OFF state for 120 seconds. Then repeat from Step 4.If the charging get interrupted again from vehicle side go to step 16 and contact nearby TPPEM Authorised Service Centre.
  16. To remove the gun, Press Fascia switch of both vehicles. Close the V2V charging gun caps to ensure that it is not exposed to mud/metal particle etc.

## DON'TS

1. Don't shift the gear
2. Don't crank source or sink vehicle during V2V charging
3. Don't use un authorized V2V gun
4. Don't press the V2X switch of sink vehicle throughout the V2V function
5. Don't keep the V2V charging gun open at sink vehicle side while pressing the V2X switch in the source vehicle
6. Don't perform V2V charging in other vehicles which is not recommended by the TPPEM ( list to be published later) as source and sink vehicle
7. Don't keep the V2V gun caps open after the use to ensure it is not exposed to mud/metal particle etc.
8. Don't start V2V charging function if source Vehicle SoC is lesser than or equal to 30%.
9. Do not perform V2V charging during rain or in any situation where it might be exposed to water
10. Don't press V2X switch and fascia

- 
- switch together to stop the function( keep minimum 5sec delay )
11. Don't keep both vehicle at far distance where the gun has to stretch maximum to connect between the vehicles.
  12. Don't disengage the park brake in either of the vehicles during the charging.
  13. Do not insert the V2L connector when vehicle is in Crank or EV ready condition.
  14. Don't disassemble or remodel the V2V adaptor
  15. Don't Drop the V2V gun to cause high impact and don't keep heavy objects over the gun
- Vehicle Ignition ON state**
1. Keep both sink (recipient) and source (donor) vehicle nearby and ensure maximum up to 3m distance between the charging inlets of the vehicles.
  2. Turn ON the ignition of both sink (recipient)/source (donor) vehicle. Make sure to keep the vehicles in Ignition ON mode throughout the procedure
  3. Engage the park-break OR EPB of both vehicles.
  4. Remove the caps & connect the V2V gun to the source and sink vehicle. Check the labels on the V2V gun and make sure that the V2Vgun side labeled as "source" should connect to the source vehicle & the other end labeled as "sink" should connect to the sink vehicle.
  5. Ensure V2V gun is connected to both vehicles and press the V2X switch of source vehicle to start the 220V AC power supply and to charge the sink vehicle. ( TBC with VCU team-V2V status if V2X switch enabled in sink vehicle also)
  6. Check telltale indication in instrument cluster of vehicles to ensure V2V charging is on.( check for the HMI)
  7. Check for the telltale indication frequently when the V2V charging is in progress to ensure there is no interruption/fault.
  8. Charging will automatically stop if the source vehicle SoC goes below 30%.
- In that case follow the procedure from step 14.
9. Also monitor the SoC of both vehicles frequently to check if sink vehicle is charged enough/ source vehicle SoC is reached the minimum required SoC level
  10. If SoC of source vehicle reaches the agreed Limit and to stop the V2V charging from source vehicle side, press the V2X switch of source vehicle to disable the 220V AC supply and follow from step 14 after 5sec
  11. If Sink vehicle is charged up to the agreed SoC level and to stop the V2V charging from sink vehicle side, press its fascia switch to stop charging and then ensure to press the V2X switch of source vehicle to stop the AC power supply. & follow step 12
  12. After 5 sec delay press source vehicle fascia switch and then remove both the guns & follow step 15
  13. In case V2V charging gets interrupted from source/sink vehicle side due to any fault, Press the Gun unlock fascia
-

## CHARGING

switch in both vehicle and completely remove the gun from both side while keeping the vehicles in Ignition ON and repeat from step 1. If the issue still remains, Press the Gun unlock fascia switch in both vehicle and completely remove the gun from both side both. After that both vehicle should brought in to ignition OFF state. Keep the vehicles in Ignition OFF state for 30sec. Then repeat from Step 1. If the charging get interrupted again from vehicle side go to step 14 to 15 and contact TPEM Authorised Service Centre.

14. To remove the gun, Press Fascia switch of both vehicles. Close the V2V adaptor caps to ensure that it is not exposed to mud/metal particle etc.

15. Turn off the Ignition in both vehicles

### DON'T'S

1. Don't switch off the Ignition in source or sink vehicle during V2V charging
2. Don't crank source or sink vehicle during V2V charging
3. Don't use un authorized V2V gun

4. Don't press the V2X switch of sink vehicle throughout the sink vehicle charging
5. Don't keep the V2V charging gun open at sink vehicle side while pressing the v2x switch in the source vehicle after connecting the source side gun.
6. Don't perform V2V charging in other vehicles which is not recommended by the TPEM ( list to be published later) as source and sink vehicle
7. Don't keep the V2V gun caps open after the use to ensure it is not exposed to mud/metal particle etc.
8. Don't start V2V charging function if source Vehicle SoC is lesser than or equal to 30%.
9. Do not perform V2V charging during rain or in any situation where it might be exposed to water
10. Don't press V2X switch and fascia switch together to stop the function(keep minimum 5sec delay )
11. Don't keep both vehicle at far distance where the gun has to stretch maximum

to connect between the vehicles.

12. Don't disengage the park brake in either of the vehicles during the charging.
13. Don't insert the V2L connector when vehicle is in Crank or EV ready condition.

### Vehicle to load charging (V2L)



### Vehicle Ignition OFF state

1. User can decide the discharge SoC limit of the vehicle and can preset the SoC limit in infotainment before starting the V2L function.

- 
- Since infotainment system won't be awake in Ignition OFF state, Limit can be set only in ignition ON or Crank ON condition in the previous cycle.
  - Discharge SoC Limit cannot be changed during the V2L function.
  - In case if no limit is set by the user, 30% SoC will be considered as default discharge SoC limit
2. Make sure to keep the vehicle and the V2L adaptor in dry environment/in shade.
  3. If the vehicle is in OFF state, Open/close the door or if vehicle is in locked state Press RKE unlock button and open the door to wake up the vehicle control unit. Once wakes up, Vehicle control unit will remain in wakeup state for 120sec.
  4. Remove the V2L adaptor caps & connect the V2L adaptor gun to the source vehicle within 120sec of step 3.
  5. Make sure park-brake/EPB is engaged.
  6. If user couldn't connect the V2L gun within 120sec of step 3, then remove the Gun completely and repeat from step 3.
  7. Connect external load to the V2L adaptor 3pin socket. Make sure that Electrical loads are healthy/in good condition and ensure the relevant cables and plugs of loads are connected properly.
    - When using multiple loads make sure Total power consumption of the loads should be lesser than 3.3Kw or total current demand of loads should be lesser than 16A.
    - If the electric appliances demand exceed the maximum power and current capacity that the vehicle can provide, V2L function will stop from vehicle side
    - It is recommended to use home appliances with Power factor greater than 0.85
    - If multiple loads are connected on extension box, it is recommended to use extension box with MCB (16A) & with M Type plug and socket. The MCB switch needs to be in switched off/ turned off condition.
  8. Press V2X switch within 120sec of step 4 to start 220V AC power supply from vehicle. If user couldn't press V2X switch within 120sec of step 3. Remove the Gun completely and repeat from step 3 to initiate the V2L function.
  9. Ensure safety precaution against the live 220VAC voltage in the V2L adaptor.
  10. Turn on the loads.
  11. Check telltale indication in instrument cluster to ensure V2L discharging is initiated/in progress.
  12. Check for the telltale indication frequently when the V2L discharge function is in progress to ensure there is no interruption/fault.
  13. V2L function will stop if there is no energy dissipation has happened/if there is no load is connected for 100secs from successful step 8. In this case, remove the gun completely and start from step 3 to reinitiate the V2L dis-
-

## CHARGING

---

charge function

14. To stop the V2L discharge function, First switch off the external loads and then press V2X switch second time to disable the 220V AC power supply and Go to step 17.
15. V2L function will automatically stop if the source vehicle SoC goes below the user set value or at the default value of 30%. In that case go to step 17
16. In case V2L function gets interrupted from vehicle side due to any fault, Press the Gun unlock fascia switch in vehicle and completely remove the gun repeat from step 3.
  - If the issue still remains, Press the Gun unlock fascia and completely remove the gun. Keep the vehicles in Ignition OFF state for 120sec. Then repeat from Step 3.
  - If the charging get interrupted again from vehicle side go to step 17 and vehicle along with the V2L adaptor need to be taken to nearby TPEM Authorised Service Centre.
17. Switch off and remove the Connected

Loads then Press Fascia switch and remove the V2L adaptor. (Close the V2L adaptor caps to ensure it is not exposed to mud/metal particle etc).

### DON'TS

1. Don't crank the vehicle during V2L charging
2. Don't use unauthorized V2L adaptor
3. Don't use High power home appliances like air conditioner, dryer having power consumption more than 3.3Kw and current requirement more than 16A
4. Don't hang the appliances on the V2L adaptor
5. Don't use appliances or extension box which is not having national safety certificate. Refer each device manual to know the usage and precautions to be taken.
6. Don't use any unhealthy or improper electrical connection/ apparatus for V2L discharge function like loads with insulation failure, short circuit, improper 3pin/2pin plug, open cables without plug)
7. Don't allow the appliances/extension box cables to twist or overlapped.
8. Don't use the appliances if its cable



- sheath is damaged.
9. Don't use the electric devices which require continuous power supply like medical equipment. since AC power supply may get interrupt based on the vehicle condition
  10. Don't use the load which required high power at the starting/initial operation
  11. Don't use the loads which is sensitive to Inverter type AC supply
  12. Don't touch V2L adaptor socket side during V2L function
  13. Don't keep the V2L adaptor caps open after the use to ensure it is not exposed to mud/metal particle etc.
  14. Don't start V2L charging function if Vehicle SoC is lesser than 30%.
  15. Don't perform V2L charging during rain or in any situation where it might be exposed to water
  16. Don't press V2X switch and fascia switch together to stop the function.( keep minimum 5sec delay)
  17. Don't disengage the park brake during V2L charging

18. Don't mishandle the V2L adaptor.
19. Do not insert the V2L connector when vehicle is in Crank or EV ready condition.
20. Don't shift the gear
21. Don't disassemble or remodel the V2L adaptor
22. Don't Drop the V2L adaptor to cause high impact and don't keep heavy objects over the adaptor

### Vehicle Ignition ON state

1. Turn ON the source vehicle ignition. Make sure to keep the vehicle in Ignition ON mode throughout the procedure.
2. Make sure to engage the EPB.
3. Make sure to keep the vehicle and the V2L adaptor in dry environment/in shade.
4. Remove the caps & connect the V2L adaptor gun to the source vehicle.
5. Connect external load to the V2L adaptor 3pin socket.
  - Make sure that Electrical loads to

be used for V2L charging are healthy/in good condition and ensure the relevant cables and plugs of loads are connected properly.

- Total power consumption of the loads used should be lesser than 3.3Kw and it is recommended to use the loads with Power factor greater than 0.85
  - In case of multiple loads, it is recommended to use extension box with MCB (16A) & with M Type plug and socket.
6. Press the V2X switch to start the 220V AC power supply.( check for the HMI).
  7. Ensure safety precaution against the live 220VAC voltage in the V2L adaptor
  8. Turn on the loads.
  9. Check telltale indication in instrument cluster to ensure discharging is initiated/in progress.( check for the HMI)
  10. Check for the telltale indication frequently when the V2L is in progress to ensure there is no interruption/fault.

## CHARGING

---

11. To stop the V2L charging, first switch off the External loads and then Press V2X switch in the vehicle to stop the AC power supply and Go to step 14 .( check for the HMI)
  12. V2L charging function will automatically stop if the source vehicle SOC goes below 30% (Configurable). In that case go to step 14.( check for the HMI & telltale indication on instrument cluster)
  13. In case V2L charging gets interrupted from vehicle side due to any fault, Press the Gun unlock fascia switch in vehicle and completely remove the gun while keeping the vehicles in Ignition ON and repeat from step 1
    - If the issue still remains, Press the Gun unlock fascia and completely remove the gun. After that vehicle should brought in to ignition OFF state. Keep the vehicles in Ignition OFF state for 30sec. Then repeat from Step 1.
    - If the charging get interrupted again from vehicle side go to step 14 to 15 and vehicle along with the V2L adaptor need to be taken to TPEM Authorised Service Centre.
  14. Switch off the Connected Loads then Press Fascia switch and remove the V2L adaptor and connected loads. (Close the V2L adaptor caps to ensure it is not exposed to mud/metal particle etc.)
  15. Turn OFF the vehicle ignition.
- DON'T'S**
1. Don't switch off the vehicle Ignition during V2L charging
  2. Don't crank the vehicle during V2L charging
  3. Don't use unauthorized V2L adaptor
  4. Don't use any unhealthy or improper electrical connection/ apparatus for V2L charging. ( e.g.: loads having power consumption>3.3Kw ,insulation failure, short circuit, improper 3pin/2pin plug, open cables without plug)
  5. Don't keep the V2L adaptor caps open after the use to ensure it is not exposed to mud/metal particle etc.
  6. Don't start V2L charging function if Vehicle SoC is lesser than 30%.
  7. Don't perform V2L charging during rain or in any situation where it might be exposed to water
  8. Don't press V2X switch and fascia switch together to stop the function.( keep minimum 5sec delay)
  9. Don't disengage the EPB during V2L charging
  10. Don't mishandle the V2L adaptor.
  11. Do not insert the V2L connector when vehicle is in Crank or EV ready condition.

### KEYS (if equipped)

A key is an electronic access and authorization system available as a standard feature with your vehicle.

#### Unlocking Principle

The transponder in the ignition key carries a Unique Identification Code (UID). The vehicle unlocks when the code on the key matches with the code on the Body Control Module (BCM).

In case of PEPS variant, Immobilizer function is provided by PEPS ECU.

#### Loss of Keys

If one of the keys is lost, Contact the TPEM Authorised Service Centre immediately.

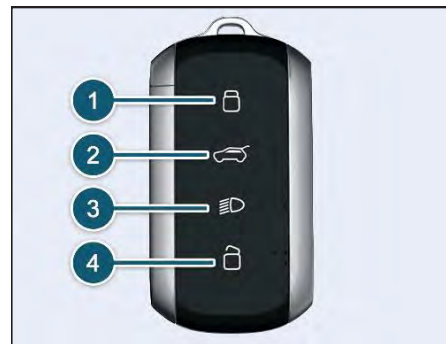
#### WARNING

- Do not turn 'ON' ignition switch by using key with any type of metal wound around its grip or in contact with it. This may be detected as abnormal condition by immobilizer and prevent vehicle from starting.
- Do not leave the key in high temperature areas. The transponder in it will behave abnormally when reused.

#### Types Of Keys (if equipped)

1. Smart Key (PEPS)
2. Remote Key (SASS)
3. Smart Card
4. Digital Key (Mobile App)
5. Mechanical Key

### 1. Smart Key (peps)



Keep the smart key with user to perform the passive access. It is used for locking, unlocking and starting the vehicle.

1. Locking all doors
2. Tailgate unlatch /Power Operated Tailgate Open and Close (if equipped)
3. Short press for Approach Light Long press Approach light button for SoC state
4. Unlocking all doors

## OPENING AND CLOSING

### 1. Locking All Door

Press the lock button once (1) to lock all the doors of the vehicle. Successful lock will be indicated by two flashes of turn signal indicators. If lock button is pressed on the key with the any door open, locking-unlocking takes place with audible warning indicators do not flash.

### 2. Tail Gate Unlatch

Press the Tailgate opening button once (2) to unlock the tailgate with in authentication range of Smart key i.e. within 60 meters.

If vehicle in lock state and user press the smart key tailgate unlock button then he / she should press tailgate DHS within 30sec to open the tailgate.

If vehicle in unlock state, user press the smart key tailgate unlock button then he / she should press tailgate DHS at any point of time to open the tailgate.

### Power Operated Tailgate Open and Close (if equipped)

To open and close tailgate press the button. If button is long pressed for 3 seconds then the tailgate will open or close.

### 3. Short Press For Approach Light

This feature helps to find and reach the parked vehicle. When you press approach light button (3) once, low beam, roof lamp and position lamps will turn 'ON'. This feature helps to find and reach the parked vehicle or to reach home in dark/ cloudy condition after parking. To switch 'OFF' the approach lights, press and release the same button or it automatically turns 'OFF' after approx. 30 sec.

### Long press Approach lamp button for SoC state (If equipped)

Long press the approach light button (3) more than 3 second, Vehicle display the Battery SoC on CPL for few sec.

### 4. Unlocking All Doors

Press the unlock button once (4) to unlock all the doors. Successful unlock will be indicated by one flashes of turn signal indicators.

#### NOTE

*If smart key battery is low/drained or vehicle battery is low/drained, user can*

*unlock and enter into vehicle by using mechanical key blade, which is present inside the smart key.*

### Emergency Key Blade In / Out



Press the knob (1) to release the key. Pull the key blade (2) out.

### Smart Key Features

#### Vehicle Search

In vehicle locked condition, if lock button on smart key is pressed, the turn indicators of vehicle flash 4 times.

#### Automatic Activation of Immobilizer

If smart key is not found within the passenger compartment, engine will be immobilized and vehicle cannot start

#### Auto Locking / Unlocking Of Doors / Auto Relock

Vehicle doors get automatically locked when all doors are closed and the vehicle speed crosses 10 kmph.

Also, when unlocked with remote key and if no door is opened within 30 seconds, vehicle doors get automatically locked.

When vehicle power is in OFF mode and vehicle all doors are in closed condition, vehicle doors get automatically locked after 180 sec.

In case of PEPS variants, door will get unlocked when ignition is OFF by pressing start/stop button.

### Anti-grab / Anti-scan Coding

The remote control set of this security system is protected against the use of devices called 'scanners' and 'grabbers' which can record and reproduce some types of remote codes.

#### Important Tips

- Don't operate Unlock button of remote in the vicinity of your vehicle, as it could lead to an unintentional unlocking your vehicle.
- For battery, replacement procedure refer 'MAINTENANCE' section.
- Do not remove the battery connection of the vehicle while the vehicle has been locked by remote.

#### Smart Key Precautions

1. If smart key is close to radio transmitter such as radio station or an airport which can interfere with normal operation of the transmitter.
2. If smart key is near a mobile two way radio system or a cellular phone, then it will not work properly.
3. If another vehicle's smart key is being

operated close to your vehicle, signal will fluctuate.

#### NOTE

- *Keep smart key away from electromagnetic materials that blocks electromagnetic waves to the key surface.*

#### Force Panic ON Operation

When vehicle is in OFF condition, if we press lock button and unlock button simultaneously, Force panic operation gets activated. In this case, turn indicators of vehicle start flashing and horn will blow automatically.

#### Force Panic OFF Operation

By pressing any button of smart key, Force panic operation gets deactivated.

## OPENING AND CLOSING

### 2. Remote Key (SASS)

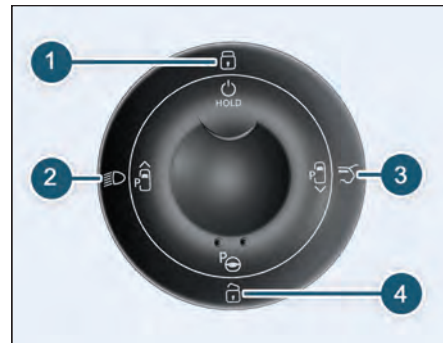
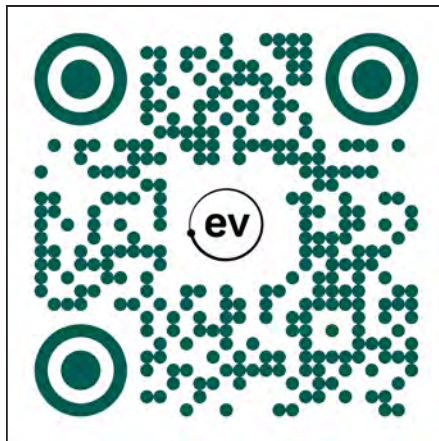


*Preferred Remote Key Location*

Remote key is used to perform passive access: locking, unlocking and vehicle starting.

It is also used for effortless parking. Park In and Park Out features in your key fob can be used to move your car in or out of tight parking spaces without you stepping inside.

To view related videos, scan the QR code below using your mobile device.



*Remote Key*

#### 1. Short press for locking all doors

To lock all the doors of the vehicle, press the lock button once (1). Successful locking will be indicated by two flashes of the turn signal indicators. If the lock button is pressed while any door is open, the vehicle will lock and then immediately unlock, accompanied by an audible warning, and the turn signal indicators will not flash.



### Long press to start the vehicle.

To start the vehicle, long press and hold the lock button (1). Vehicle will enter into Ready state. For Detailed procedure please refer 'Remote Parking (Driver outside the vehicle)' under AUTOMATIC PARKING ASSIST (APA) section.

#### NOTE

*No function is associated for double press.*

### 2. Short press for approach lamp

When you press the approach light button (2) once, the low beam, roof lamp, and position lamps will turn 'ON'. This is particularly useful for finding your vehicle or reaching home in dark or cloudy conditions after parking. To switch 'OFF' the approach lights, press and release the same button, or they will automatically turn 'OFF' after approximately 30 seconds. User can change the auto turn OFF time to 30sec, 60sec and 90sec through infotainment screen.



### Double short press for battery SoC state

Double short press the approach light button (2) to display the Battery State of Charge (SoC) on the CPL for a few seconds.

### Long press the button for vehicle forward movement (if equipped).

To move the vehicle forward, long press the button (2). Ensure the vehicle is al-

ready in 'Ready' state. For detailed procedure please refer AUTO-MATIC PARKING ASSIST (APA) section.

## OPENING AND CLOSING

### 3. Short press the button for unlocking all doors

Press the unlock button once (3) to unlock all the doors. Successful unlock will be indicated by one flashes of turn signal indicators.



### Long press the button to Auto Park the vehicle

To Automatic Park the vehicle, long press the button (3). Ensure vehicle is already in 'Ready' state. For de-tailed procedure please refer 'Remote Parking (Driver outside the vehicle)' under AUTOMATIC PARKING ASSIST (APA) section.

#### NOTE

*No function is associated for double press*

### 4. Short press the button to unlock the trunk

Press the tailgate opening button (4) once to unlock the tailgate within the authentication range of the remote key (Approx. 50 meters). If the vehicle is locked and the Remote key tail-gate unlock button is pressed, the tailgate DHS must be pressed within 30 seconds to open the tailgate.



If the vehicle is unlocked and the Remote key tailgate unlock button is pressed, the tailgate DHS can be pressed at any time to open the tail-gate.

### Double short press to open/close tailgate (if power operated tailgate available)

To open and close tailgate double short press the button. Then the tail-gate will open or close.

### Long press the button for vehicle backward movement (if equipped)

To move the vehicle backward, long press

the button (4). Ensure vehicle is already in 'Ready' state. For detailed procedure please refer AUTO-MATIC PARKING ASSIST (APA) section.

### Lock + Unlock

If Lock and Unlock Button (1)+(3) is pressed simultaneously. Force panic feature will be enabled. Press it again simultaneously to disable the feature.

### Remote Key Features

#### Vehicle Search

If lock button on Remote key is pressed when the vehicle is locked, the turn indicators of vehicle flashes four times for the driver to locate the vehicle.

#### Automatic Activation of Immobilizer

If the user tries to start the vehicle using start/stop switch without a valid Remote key or Digital key inside the vehicle's passenger compartment or Smart card on center console [SASS variant], the vehicle will be immobilized and will not start.

Also, vehicle will not start remotely using invalid Remote key.



### Auto Locking / Unlocking of Doors / Auto Relock

The vehicle doors automatically lock when all doors are closed and the vehicle speed exceeds 10 kmph.

If the vehicle is unlocked with the re-mote key and no door is opened within 30 seconds, the doors will automatically lock.

When vehicle power is in OFF mode and all doors are closed, the doors will automatically lock after 180 seconds with key inside the vehicle.

For SASS variants, the doors will unlock when ignition is turned OFF by pressing the start/stop button.

### Anti-grab / Anti-scan Coding

The remote control set of this security system is protected against the use of devices called 'scanners' and 'grabbers' which can record and reproduce some types of remote codes.

### Important Tips

Don't operate Unlock button of re-mote in the vicinity of your vehicle, as it could lead to an unintentional un-locking your vehicle.

For battery, replacement procedure refer 'MAINTENANCE AND CARE' section.

Do not remove the battery connection of the vehicle while the vehicle has been locked by remote.

### Remote Key Precautions

If remote key is close to radio transmitter such as radio station or an air-port which can interfere with normal operation of the transmitter.

If remote key is near a mobile two way radio system or a cellular phone, then it will not work properly.

If another vehicle's remote key is being operated close to your vehicle, signal will fluctuate.

### NOTE

- Remote key button to be pressed at center line for better functioning, no need to press at symbols.
- Keep Remote key away from electromagnetic materials that blocks electromagnetic waves to the key surface.

- *In case any button of the key is accidentally pressed for more than 240 seconds, the remote stops functioning till the time the button is pressed. The LED on the remote also stops glowing. The function of the remote gets reinstated immediately when the user stops pressing the push button of remote.*

### Force Panic ON Operation

When vehicle is in OFF condition, if we press lock button and unlock button simultaneously, Force panic operation gets activated. In this case, turn indicators of vehicle start flashing and horn will blow automatically.

### Force Panic OFF Operation

Press Lock/Unlock button of remote key to deactivate force panic operation.

### Digital Access Settings

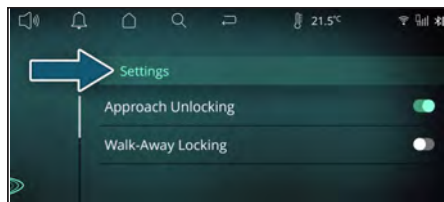
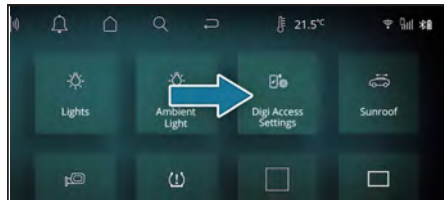
In Digi Access settings, user can enable or disable 'Approach Unlocking' and 'Walk Away Locking'.

If any of the feature is disabled then that

## OPENING AND CLOSING

feature will not work even if the key is in proximity area.

Go to> App Drawer > Settings > Vehicle > Digi Access Settings



### Approach Unlocking

This feature automatically unlocks the vehicle when the Digital Key is detected nearby via Bluetooth. It allows you to enter your vehicle effortlessly without taking the key out of your pocket or bag.

This feature activates when the user approaches the vehicle from the driver or co-driver side with the key, within an approx. range of 1.5 meters.

### Walk-Away Locking

The vehicle will automatically lock itself when the Digital Key moves out of the locking range. This feature ensures your vehicle is secure without the need to manually lock it, offering peace of mind as you walk away.

This feature activates when the user moves more than 3 meters away from the vehicle with the key.

### *i* NOTE

- *One remote key is provided with vehicle.*
- *Key blade is provided separately along with remote key. It can be used to unlock the vehicle in case of emergency when vehicle battery is completely drained.*
- *Key range is 50m for short or double short press operations. Refer APA section for key range in long press operations.*
- *Keep the remote key away from electromagnetic materials that blocks/interferes electromagnetic waves to the card surface.*
- *It is recommended to keep key fob stationary when not in use. Key fob has got inbuilt motion sensor to save battery power when not in motion. Frequent activation of motion sensor can drain key fob battery early.*

### *i* NOTE

*If vehicle is not used frequently, disable approach unlocking and walk-away locking features to save battery.*

### NOTE

- The key range is around 50m in line of sight for short or double short press operations. Refer **AUTOMATIC PARK ASSIST** section under **STARTING AND DRIVING** for key range in long press operations.
- **Key Fob Sleep Mode:** If the remote key remains stationary and no buttons are pressed for more than 3 minutes, the key fob will enter sleep mode to conserve battery. While in sleep mode, the key fob will not be detected. For cases, where key fob is not detected inside , key not detected indicator will show up on instrument cluster.
- To wake up the key fob, simply lift and shake it. The key fob will be activated and ready for use again. ✓ When any button is pressed on Remote key, a green light will illuminate inside the key.
- If there is any fault or the function terminates, a red light will illuminate

*inside the key.*

- As new technology is used to provide new features like Approach unlock/Walkaway lock, key fob battery consumption is increased. Ensure to keep key fob stationary when not in use.

## OPENING AND CLOSING

### Remote Key Functions

| Button No/symbol                                                                  | Button Name           | Press Type   | Function                            |
|-----------------------------------------------------------------------------------|-----------------------|--------------|-------------------------------------|
|  | Lock                  | Short        | Lock                                |
|                                                                                   | -                     | Double Short | Reserved for future use             |
|                                                                                   | Power Button          | Long         | Remotely turn ON vehicle            |
|  | Approach Lamp         | Short        | Approach Light ON/OFF               |
|                                                                                   | -                     | Double Short | Vehicle Battery SOC                 |
|                                                                                   | Vehicle Forward Move  | Long         | Vehicle Forward Move (if equipped)  |
|  | Unlock                | Short        | Unlock                              |
|                                                                                   | -                     | Double Short | Reserved for future use             |
|                                                                                   | Automatic Park        | Long         | Automatic Park                      |
|  | Tailgate Unlock       | Short        | Tailgate Unlock                     |
|                                                                                   | Tailgate Open/Close   | Double Short | Tailgate Open/Close                 |
|                                                                                   | Vehicle Backward Move | Long         | Vehicle Backward Move (if equipped) |
| Button 1 + Button 3                                                               | Lock and Unlock       | Simultaneous | Force Panic                         |

### 3. Smart Card



This is a smarter and more convenient way to access your car.

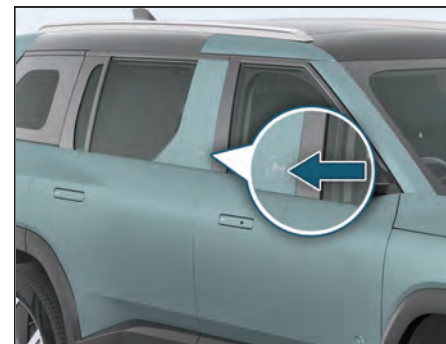
The vehicle can be accessed using Smart Card, which is especially useful when the mobile or remote key's battery is drained.

#### NOTE

- Smart card works exclusively for vehicle lock/unlock and vehicle starting operations.
- For vehicle access, NFC reader is present on driver's door side B-pillar.

- For vehicle starting, NFC reader is present on center console. Vehicle starting procedure is covered under 'STARTING AND DRIVING' section.
- Please ensure you always carry the smart card with you for emergency use.
- The smart card can be customized. If you wish to customize it, kindly contact a TPEM Authorised Service Centre.
- One smart card is provided with the vehicle. If you require an additional smart card, please reach out to a TPEM Authorised Service Centre.
- Keep the smart card away from electromagnetic materials that may block or interfere with the electromagnetic waves affecting the card's functionality
- Do not bend or twist the smart card, as it may stop functioning. In case of loss or damage, please contact TPEM Authorised Service Centre.

### Vehicle Access Using Smart Card



To unlock your vehicle using smart card, look for NFC reader symbol on the B-Pillar [Driver side door]. Then simply tap the card at the NFC reader symbol on the B-pillar and hold at least for 2 seconds. Successful unlock will be indicated by one flashes of turn signal indicators.

To lock your vehicle, tap and hold at least for 2 seconds on the B-pillar at the NFC reader symbol [Driver side door]. Successful locking will be indicated by two flashes of the turn signal indicators. If tap-ping Smart card for locking while any door is

## OPENING AND CLOSING

open, the vehicle will lock and then immediately unlock, accompanied by an audible warning, and the turn signal indicators will not flash.

### 4.digital Key (mobile App) (if equipped)

With the mobile app, you can access your car seamlessly.

#### 1. Setting Up Your Digital Key Through Smartphone

##### Complete KYC:

Ensure user have completed the Know Your Customer (KYC) process at the dealership to register user mobile number with TPEM.

##### Download the Wallet App:

The app is available for download on respective de-vice manufacturer supported digital app store.

User can set digital key through TPEM iRA.ev App as well.

##### Compatibility Check:

Make sure device is updated with latest software revision.

Refer TATA digital web page to find out compatible device list.

<https://ev.tatamotors.com/sierra/ev/digital-access.html>

##### Create an Account:

Sign up and verify your account within the app.

##### Activate Connected Car Subscription:

Make sure user have an active connected car subscription.

Connected car subscription is mandatory while adding /deleting owner key from mobile app/ iRA.ev app.

##### In-Car Setup:

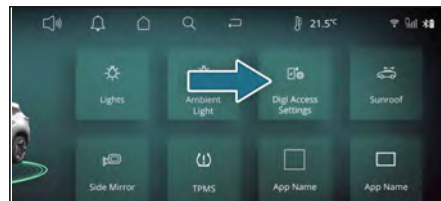
Enter the Vehicle: Use the remote key or Smart Card to access the vehicle.

##### Access Settings:

Navigate to the set-tings menu on the in-foatainment screen.

##### Select “Digi Access”:

In this menu, user can find options to activate and reset all digital keys.



### NOTE

*In this section device refers to Smart-phone or Smart watch (if equipped). Preferred location of smart phone is center console.*

### WARNING

Device name will be reflected after pairing, please choose your preferred name before pairing as it is not editable once paired. Please check with your device manufacturer for location/menu to edit device name.

### 2. Pairing Your Smartphone (owner Device)

#### Pairing initiation using Infotainment Screen:

Following are the major steps involved:

#### Availability Check

If no Digital Keys are paired to the vehicle, then the Infotainment screen will prompt a message to start the Owner device pairing process. Steps to follow: App drawer > Digi Access > Digital Keys.



#### Initiation

Select "Yes" displayed on the Infotainment screen.

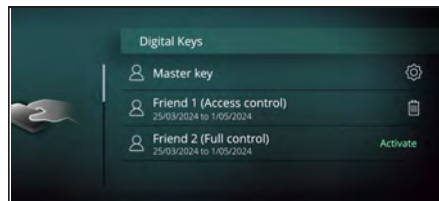
Follow the instructions displayed on the infotainment screen for digital key pairing.

#### App Instruction

Follow on-screen instructions on your Smartphone to pair it with the vehicle.

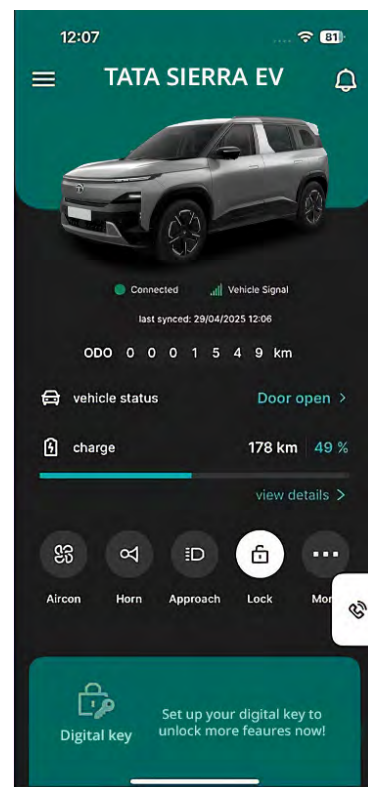
After successful pairing, the owner key profile will be available under the "Digital Keys" section.

Your added digital key will also be visible inside your device's wallet. Once pairing is completed, device name inside smartphone will be reflected as Owner name in infotainment screen.



#### Pairing initiation using iRA.ev app:

Alternatively, owner device can be paired using iRA.ev app. In order to pair device Follow on-screen instructions to pair your smartphone with the vehicle.



## OPENING AND CLOSING

### NOTE

- *Smartphone pairing could fail in case KYC is not completed, Phone is not compatible, Bluetooth/UWB is not turned ON, NFC not turned ON, connected car subscription not available etc.*
- *If your mobile device doesn't get detected during pairing, contact TPDM Customer care or dealer.*
- *If owner mobile number is changed, user need to do KYC again to register new mobile number.*
- *TPDM prioritizes your privacy by encrypting all personal information and usage data. Our robust privacy policy ensures that your data is securely stored and managed.*

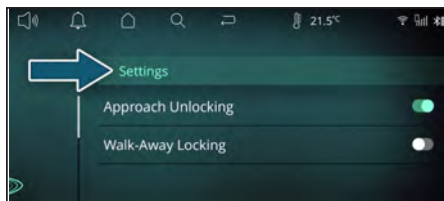
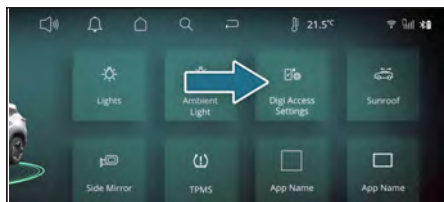
### 3. Digital Key Features

If any of the feature is disabled then that feature will not work even if the key is in proximity area.

In Digi Access settings, user can enable or disable 'Approach Unlocking' and 'Walk

Away Locking'.

Go to> App Drawer > Settings > Vehicle > Digi Access Settings



### Approach Unlocking

This feature automatically unlocks the vehicle when the Digital Key is detected nearby via Bluetooth. It allows you to enter your vehicle effortlessly without taking the key out of your pocket or bag.

### Walk-Away Locking

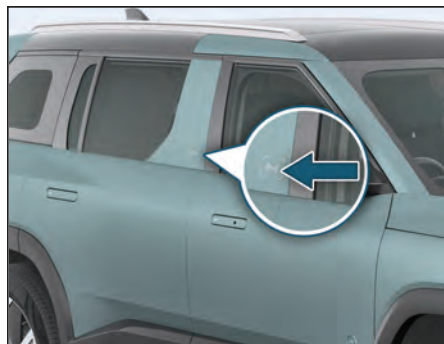
The vehicle will automatically lock itself when the Digital Key moves out of the locking range. This feature ensures your vehicle is secure without the need to manually lock it, offering peace of mind as you walk away.

### NOTE

- *If vehicle is not used frequently, disable approach unlocking and walk-away locking features to save vehicle's battery.*
- *To use Approach unlock and walk away locking features, ensure to turn on Bluetooth and UWB in device.*
- *The device must be compatible with Bluetooth and UWB communication.*
- *The digital key will work properly as long as your phone is within the Bluetooth range of the car, typically around 7 m for passive use and 50 m or beyond for active use.*



### Vehicle Access using Device NFC



To unlock your vehicle using device's NFC, look for NFC reader symbol on the B-Pillar [Driver's door side]. Ensure that the device's NFC should already be turned ON. Then simply tap the device NFC at the NFC reader symbol on the B-pillar and hold at least for 2 seconds. Successful unlock will be indicated by one flashes of turn signal indicators. To lock your vehicle, ensure that the device's NFC is turned ON, tap and hold NFC device at least for 2 seconds on the B-pillar at the NFC reader symbol [Driver side door]. Successful lock-

ing will be indicated by two flashes of the turn signal indicators.

If tapping NFC device for locking while any door is open, the vehicle will lock and then immediately unlock, accompanied by an audible warning, and the turn signal indicators will not flash.

#### *i* NOTE

- *Keep the digital key away from electromagnetic materials that block or interfere with electromagnetic waves to the card surface.*
- *NFC should be turned ON in device in order to use it as a key.*
- *Please check location on phone for NFC from your device manufacturer and place that portion of device on vehicle driver's side B-pillar NFC marked location for vehicle access.*
- *When device battery gets discharged, device's NFC function will still be supported for few hours.*

#### *i* NOTE

*If vehicle is not used frequently, disable approach unlocking, walk-away locking features to save battery.*

### 4. Managing Digital Keys

#### Manage Shared Keys:

Owner can view all shared keys, along with the names of friends, their control levels, and options to delete these keys.

#### Sharing the digital key

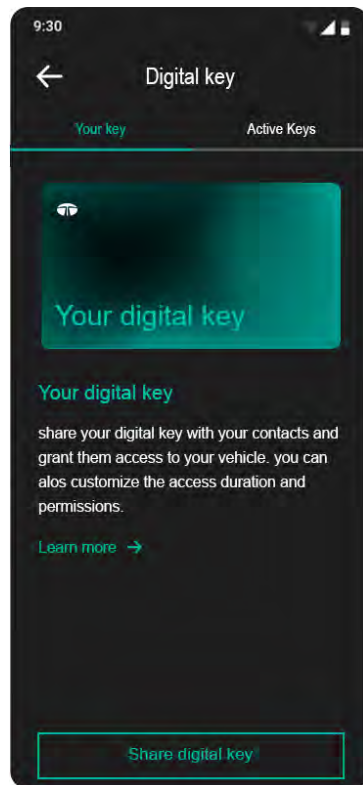
The vehicle owner can share the digital key with up to seven devices with unique digital keys.

User can share digital Key by following below steps:

- Open the Wallet App.
- Navigate to the Digital Key section.
- Select "Invite for sharing"
- Choose the level of access and tap invite
- Access shared by owner via third party messaging app/SMS.

## OPENING AND CLOSING

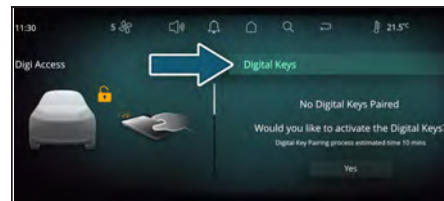
- Digital key sharing can also be initiated using iRA.ev app.



### Shared Key Profile Activation

Shared Key can be activated by navigating to vehicle's infotainment "App Drawer > Digi Access > In the Digital Keys section".

Except the owner all other devices (phone, watch) will be called as friend key. Friend key can be generated and provided by the owner only. Owner can then define the control profile and key duration for each friend key before generating the key.



- The owner can generate friend key from wallet app.
- The shared key from Owner to friend will be displayed with their name, access control level, and an activation icon.
- Follow the on screen instructions to activate the shared key.

- After successful activation, the infotainment screen will display a “Shared key activation completed” confirmation message.
- A friend's name, control profile, and the start and end dates (in dd/mm/yyyy format) will be shown after key activation.
- After successful activation, a friend can view their access control in their wallet app.
- Owner can view all paired friend's profile in their wallet App.

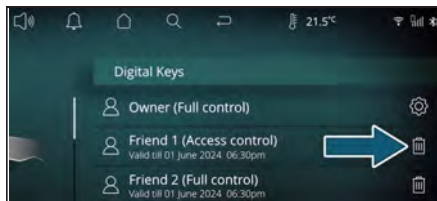
### NOTE

- *Access types: A Friend key with Full control provides full vehicle access and the ability to start the vehicle until the specified valid date and time. A Friend key with Access control will only allow access to the vehicle until the specified valid date and time. The vehicle cannot be started using Access control.*
- *Owner can share keys for minimum one day.*

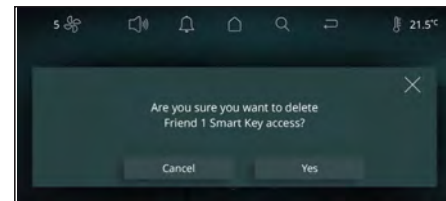
### Deleting the smartphone (Friend device)

The Owner can delete the shared key from the vehicle infotainment by following the steps below. The owner will require the owner device to delete the shared key/s from the vehicle infotainment.

- Go to App Drawer > Digi Access > Digital Keys > Tap on delete Icon next to the Friends Name.



- Confirm by selecting “Yes” to delete or “Cancel” to abort the process.



- The friend key can be deleted from the owner's wallet or the iRA.ev app. It can also be deleted by the friend from their wallet app.

### NOTE

- *Only Owner is authorized to delete the Owner's and friend's device from the infotainment screen/apps. Friend can delete his/ her profile from his/her app only.*
- *Owner device should be available in the vehicle to delete the friend's key.*
- *Friend's key deletion notification will be sent to respective friend's device and owner's device.*
- *Owner can delete all the keys.*
-

## OPENING AND CLOSING

- If the owner deletes a digital key from the infotainment system, the selected key(s) will be immediately revoked.
- *Pre-Deletion Mode:* For security reasons, when the owner stops sharing or removes a digital key, the deleted shared key will enter a “pre-deletion” state. The key will re-main in this state until another active key uses the vehicle or the vehicle is locked.
- If the owner deletes a friend's digital key from their app, the shared key will initially enter pre-deletion mode before being fully revoked.
- If the friend deletes their own digital key from the mobile app, the shared key will initially enter pre-deletion mode before being fully revoked.
- If the owner suspends their digital key using the “Find My Device” webpage, the owner key will initially enter pre-deletion mode before being fully deleted.

- In the event of vehicle is stolen and the owner contacts TPEM customer support for key deletion (with all required prerequisites met), TPEM customer support can delete all digital keys.

### Changing the Owner Mobile Device

The owner can change the owner device from the vehicle's infotainment system by following the steps below. To change the owner device, the user will need a Smart card or Remote Key.

- Go to App Drawer > “Digi Access” > digital keys > Tap the settings icon>Select “Change Owner's De-vice”
- Follow the provided instructions.
- Keep the Smart card in center console or remote key inside the vehicle.
- Confirm by selecting “Yes” to change or “Cancel” to abort the process.

### NOTE

*If the device is stolen and you have deactivated the SIM. Please ensure to*

*delete the digital key associated with the lost device. Options for deleting owner's or Friend device is present under “Deleting the Owner key” and “Deleting the smartphone (Friend device)” section.*

### Deleting the Owner key

The owner can delete the owner key from the vehicle infotainment by following the steps below. The owner will require a Smart card or Remote Key to delete the owner device.

- Navigate to App Drawer > Digi Access > Digital Keys > tap the settings icon next to Owner's name or Master Key > Select Delete Owner's Digital Key
- Follow the instructions displayed on the infotainment screen.
- Owner can also delete the owner key from owner's wallet app or iRA.ev app.
- For more details on other options, check the device manufacturer's “Find My Device” webpage or contact TPEM customer care.

### Delete All Digital Keys

- Go to App Drawer > “Digi Access”>Digital keys > Tap on settings icon>Select “Delete all Digital Keys”
- Keep the Smart card in center con-sole or Remote key inside the vehicle.
- Confirm by selecting “Yes” to delete all digital keys or “Cancel” to abort the process.



#### *i* NOTE

- *TPEM prioritize your privacy. In case any of the above feature doesn't work properly, kindly contact TPED Authorised Service Centre.*
- *When pairing the owner key or changing the owner device, place the device on the center console.*

- *The user should delete all digital keys before selling the car to ensure security and prevent unauthorized access.*
- *Selecting “Delete All Digital Keys” will remove all added keys, including the owner's key. Please ensure that this action is done carefully.*

### 5. Mechanical Keys



Mechanical key can be used to lock and unlock the vehicle in case Digital key/Smart Card/Remote Key is not work-

ing.

This key serves as an emergency backup, and its usage is described in the Smart Key section.

#### *i* NOTE

- *Mechanical Key is provided separately along with remote key.*
- *It can be used to unlock the vehicle in the event of an emergency such as when vehicle's battery is completely drained.*

## OPENING AND CLOSING

### DOORS

#### Option 1- Door Locking / Unlocking With Key From Outside

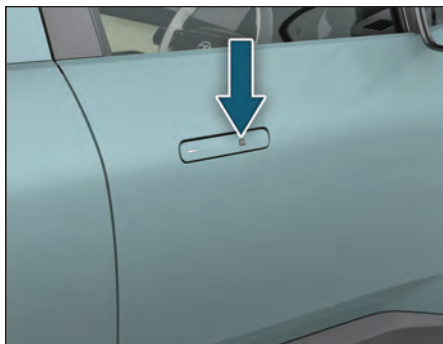


Driver door can be locked or unlocked from outside using the key blade.

Insert the key and turn it clockwise to lock and anticlockwise to unlock the door.

#### Option 2- Door Locking / Unlocking Using Door Handle Switch (DHS) From Outside

To lock / unlock all the doors without operating smart key button / key blade.



Press the door handle switch (DHS) provided on the driver door to lock/unlock all the four doors except Tailgate.

#### *i* NOTE

- *Authentication range for smart key shall be 1 to 1.5 meters from outside the respective door or tail gate.*
- *Passive entry only works during ignition off.*

#### Option -3 Door Locking / Unlocking Using Door Handle Switch (DHS) (SASS Variant)

If Walk away lock / Approach unlock is disabled by user and user wants to lock / unlock all the doors without operating Remote Key button / Smart Card / Digital key iRA.ev app / Mechanical key, Press the door handle switch (DHS) provided on the driver or Co-driver side door with remote key/Digital key present within authentication range.

It will lock/unlock all the four doors except tailgate.

#### *i* NOTE

- *Authentication range for smart key shall be 1 to 1.5 meters from outside the respective door or tail gate.*
- *Passive entry only works during ignition off.*

### Horn Honking When Door Locking Using Door Handle Switch (DHS)

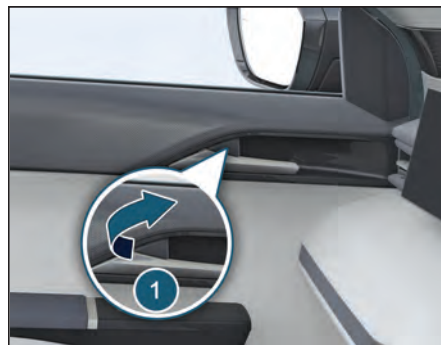
If vehicle is in unlock condition and smart key is not available, (i.e. Smart Key is away from authentication range) and if you try to lock the vehicle through door handle switch then vehicle horn honking gets activated for 9 sec.

If vehicle is in unlock condition and Smart key is present inside the vehicle. If you try to press the door handle switch then vehicle horn honking gets activated for 9 sec.

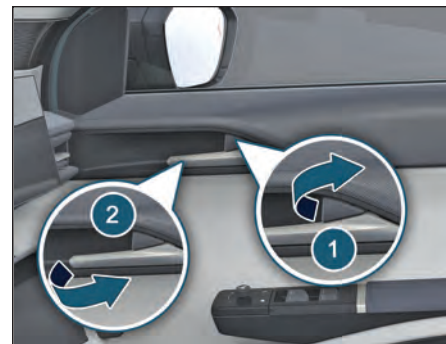
### Horn Honking When Door Locking Using Door Handle Switch (DHS) (SASS Variant)

If vehicle is in unlock condition and Digital key/Remote key is present inside the vehicle. If you try to press the door handle switch (DHS) on driver or co-driver side then vehicle horn will honk once.

### Locking / Unlocking The Doors From Inside



All doors can be opened from inside by pressing knob on driver door and independently on other doors.



Pull the door opening knob (1) and then opening lever (2).

#### **NOTE**

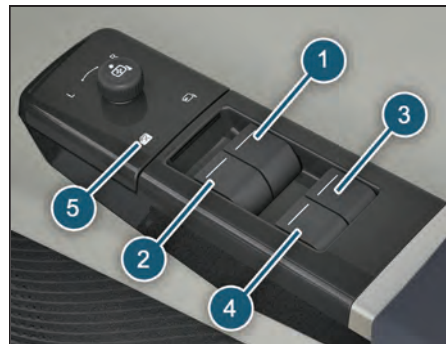
*There is a single pull override feature on driver door. All door can be unlocked by inner handle without operating lock knob of inner handle.*



## OPENING AND CLOSING

### WINDOWS

#### Power Windows



1. Front RH Switch
2. Front LH Switch
3. Rear RH Switch
4. Rear LH Switch
5. Inhibit Switch

Window glasses 1-2-3-4 on all four doors can be operated by switches provided on the main control panel located on the driver's arm rest. They work only when the key is in the 'IGN ON' position.

#### NOTE

*Power windows can be operated for 30 seconds in 'IGN OFF' and 'KEY OUT' positions, provided the doors are closed.*

#### Express Up

Window glass can be closed by single pull of the switch. Express up feature is provided for the driver's door only.

#### Anti-pinch Function (if Equipped)



The Anti-pinch function will stop window upward movement if any obstruction or re-

sistance detected.

Thus, it gives full and reliable protection for hand, neck etc. Anti-pinch function.

#### Anti-pinch Inhibition

After 3 successive anti-pinch reversals with less than 5 seconds delay between each reversal in switch operated mode, Anti-pinch function shall be de-activated until complete closed condition is detected.

#### Express Down

Window glass can be opened by a single press of the switch (1). Express down feature is provided for the driver's door only.



### Inhibit Switch



When switch is pressed, amber light turns 'OFF'. The individual switches provided on other doors are not functional. It can be only operated by driver side switch.

As the switch is depressed amber light turns ON and individual switches provided on rear and front passenger door cannot be operated. Still it can be only operated by driver side switch.

### WARNING

- If children operate the windows they could get trapped, particularly if they are left unsupervised. There is also a risk of injury.
- Activate the window inhibit feature when children are travelling. When leaving the vehicle, always take the key with you and lock the vehicle. Never leave children unsupervised in the vehicle.

### Individual Switches

Individual switch has been provided on all doors.

Glasses are wound up or down by pulling or pressing the switch.

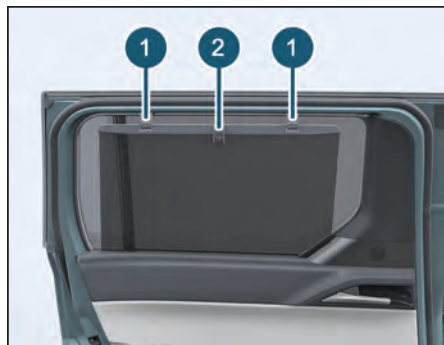


To close the glass pull the switch in upward direction.

To open the glass press the switch in downward direction.

## OPENING AND CLOSING

### Lateral Sunshade Curtain (if equipped)



The purpose of sunshade is to prevent discomfort caused by direct sunlight coming into the vehicle.

#### Sunshade operating procedure

**Pull Up:** Lift the sunshade fabric by using pull handle (2) and engage the pull bar onto both the hooks (1) simultaneously. After engagement, move the pull handle in downward direction.

**Pull Down:** Hold the pull handle (2) and remove the pull bar from both the hooks (1)

by raising it slightly. Slowly pull down the sunshade fabric and place it over the door trim.

#### CAUTION

- Do not hang any other objects except the shade curtain on the hooks.
- Always ensure that the sunshade pull bar is engaged in both the hooks at the same time. There is a risk of fabric damage, if it is engaged in only one hook.
- Sunshade to be opened and closed by using pull handle only.
- Do not hold or fold the stiffener rods on the sunshade.
- Sunshade fabric may get damaged due to head rested on it/contact with any pointed item like ornaments, nails etc/pulling of cloth/ fabric abruptly.
- When window is open, avoid using sunshade to prevent any risk of damage to sunshade fabric. Risk of fabric damage is higher, especially

in vehicle running conditions.

- Sunshade to be operated only from inside the car in door closed condition.

### BONNET AND CHARGING FLAP

#### Bonnet Opening

1. Make sure to engage the parking brake for your safety.
2. To open the bonnet, pull the bonnet release lever two times. It will pop up slightly.



3. Raise the bonnet slightly and with your finger slide the secondary lock lever.



#### NOTE

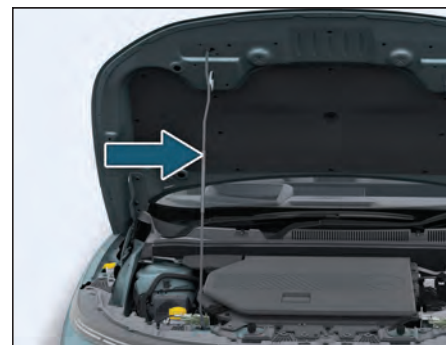
- Make sure that the wiper arms are not raised before you lift up the bonnet to avoid damaging the wiper arms and the bonnet.
- Do not operate wipers when bonnet is open.

4. Lift the bonnet up. Pull the bonnet stay rod from its clip and insert the free end into the slot provided on frame.

#### CAUTION

Insert the stay rod into the hole securely. If the rod drops off, your body

may be caught below the bonnet.



#### Bonnet Closing

1. To close the bonnet, hold the bonnet by one hand, disengage the stay rod and clamp it back properly.
2. Lower the bonnet close to the bumper, then let it drop down.

#### WARNING

Ensure that the bonnet is properly locked before driving or it can fly up unexpectedly during driving.

## OPENING AND CLOSING

### Charging Flap

1. Make sure to engage the parking brake for your safety.
2. To open the charging flap, press the switch located on fascia switch.



3. The cover will fully open.



### Closing charging flap

- For closing charging flap cover, gently push it till it gets locked.

#### NOTE

*If charging cap needs replacement, make sure that it is replaced by a genuine cap at TPEM Authorised Service Centre only.*

## MECHANICALLY OPERATED TAILGATE (if equipped)

| Option                               | Image                                                                             | Operation                                                                                                                                                                                                                                      |
|--------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option I<br>Using Smart Key          |  | Press a tail gate opening button on remote and release. To close, slam the tail gate to latch and it gets locked.<br>Note: vehicle to be in authentication range.                                                                              |
| Option II<br>Through Fascia switch   |  | To open the tail gate, press the switch located on fascia switch.<br>Note: If vehicle is in locked condition then tailgate unlatch via fascia switch will work only in ignition ON condition.                                                  |
| Option III<br>Using DHS on tail gate |  | DHS on tail gate is pressed with valid key in the authentication range, the tail gate gets unlatched. To close, slam the tail gate to latch then it gets locked. If the valid smart key is left inside the trunk then tail gate gets unlocked. |

## OPENING AND CLOSING

---

### NOTE

- *During closing tailgate if doors are in locked condition and valid smart key is inside the trunk, then tailgate can be unlocked by pressing tailgate switch.*
- *For tailgate unlatch remote operation, vehicle shall be in OFF mode.*
- *Tailgate once unlatch it will not get locked automatically with other doors.*
- *If doors are in unlocked condition, tailgate can be unlocked via tailgate handle switch independent of smart key.*
- *Avoid keeping smart key inside the boot space area while closing tailgate.*

### WARNING

- In case of manual operated tailgate, after unlatching, tailgate doesn't get opened automatically.

- Tailgate should not be opened holding tailgate garnish, this might damage the part. It is suggested to open through tailgate edge.
- Tailgate can't be locked using mechanical key/ flip key/ smart key. It can be locked by slamming it.

### WARNING

In case if tailgate gets stuck or unable to operate. Please contact the nearby TPEM Authorised Service Centre.

### POWER OPERATED TAIL GATE (if equipped)

The Power Operated Tailgate has capability to automatically open and close when the trigger is provided by the user. It consist of a motor which will drive the tailgate to open or close. A hazard flashes and buzzer indicate the customer while opening and closing of power tailgate.

#### NOTE

*Before opening the tailgate, make sure that the vehicle stops, the gear is in parking gear, and the handbrake is pulled to avoid any damage.*

To open, close or stop the tailgate automatically one of the triggers from the following will be used

1. PEPS Key
2. DHS Tailgate Switch
3. Tailgate Fascia Switch
4. Tail gate Kick Sensor [hands free access]
5. Tailgate Closure Switch (Inside Tail-

gate, Close Only)

#### 1. PEPS Key



1. To open and close tailgate press the button. If button is long pressed for 3 seconds then the tailgate will open or close.
2. To stop the tailgate while opening and closing is in progress, short press tailgate button. Automatic tailgate will stop in open or close position and will work only when PEPS key is near to vehicle vicinity and this feature is enabled from infotainment.

#### 2. DHS Tailgate Switch



1. Press the tailgate open switch (which is located externally on tailgate door) to access the vehicle trunk, then after 2 sec delay, tailgate start auto open / close based on the vehicle lock / unlock state and with valid smart key within vicinity area.
2. If any interrupt (Trigger) given during the tailgate auto operation, then it will stop the operation. It will resume the operation after next interrupt (trigger).

## OPENING AND CLOSING

### 3. Tailgate Fascia Switch



1. To open and close tailgate, press the tailgate fascia switch.
2. To stop tailgate, press fascia switch while tailgate is closing or opening.

### 4. Tailgate Kick Sensor (Hands Free Access)



1. To open and close tailgate automatically, kick near to rear bumper area as shown in picture.
2. To stop the tailgate while opening and closing is in progress, kick near to rear bumper area.
3. Automatic tailgate opening, closing or stop will work only when PEPS key is near vicinity area and gesture feature is enabled from infotainment.

### Things to follow using Tailgate kick sensor (Hands Free Access)

1. Do not kick on the bumper i.e. while doing kick do not hit/touch to the bumper inside/outside. Kick at the center of bumper in line to TATA logo and nearby are (proximity calibrated to 1.5 feet)
2. A quick kick will only work, all other forms of gesture will not work e.g. continuous kick, slow kick, not close to the bumper, kick at left or right bumper vehicle edge etc.
3. After applying kick move back to a sufficient distance such that not to cause any obstruction to tailgate opening.
4. Do not kick or keep leg near to HFA area for long time. It will not be considered as valid kick and tailgate will not open / close in such operation.
5. While cleaning the bumper area with cloth/sponge/ high pressure water pump if PEPS key is within the vicinity area/ with user then it be considered as valid kick to open/ close tailgate. So, user is advised to not keep PEPS



key within the vicinity area while cleaning.

6. In the rainy season if the mud is accumulated at HFA area, then HFA sensor working performance may degrade. So, user is advised to keep kick sensor area clean for better performance.

### 5. Tailgate Closure Switch (inside Tailgate)



1. Inside tailgate switch can be used to close, stop tailgate and to store user intended maximum tailgate height when it's open.

2. To close or stop tailgate, press inside tailgate closing switch.

### Height Setting Through Tailgate Closure Switch:

User can limit tailgate opening by setting the customized tailgate height. Below is the procedure.

1. Open the Tailgate to the position required as the maximum height. Press any Tailgate control to stop movement at the required position. The final position can be achieved manually, if required.
2. Make sure the Tailgate is stationary for at least 3 seconds.
3. Press and hold the Tailgate close button for 5 seconds to set the maximum opening height. You will hear a long beep from POT ECU to confirm height set.
4. After step 3 is executed, if Tailgate initiates closing operation, then height will not be set.
5. If the Height is not set successfully user will hear 3 beep sound from

buzzer

6. Close the Tailgate, then open again to check that it opens to the set height

### Buzzer Indication For Opening/closing/fully Closed (if equipped)

1. Buzzer sound is provided to indicate the user about current tailgate operation when buzzer setting from infotainment is enabled.
2. When tailgate opening initiated, buzzer and hazard flashes will drive twice.
3. When tailgate closing initiated, buzzer and hazard flashes will drive twice.
4. When tailgate is fully closed, buzzer and hazard flashes will drive once.

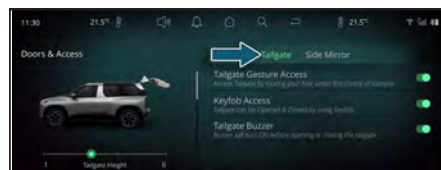
### Obstacle Detection And Anti-pinch

1. It is advisable to check if there is obstruction in the tailgate travel while closing or opening.
2. If a tailgate close operation is initiated and an obstacle is detected during ongoing movement, then the current tailgate close operation will be aborted and a rewind tailgate open operation will start.

## OPENING AND CLOSING

3. If tailgate open operation is initiated and an obstacle is detected during on-going movement, then the current tailgate open operation will be aborted, and tailgate will stop.
4. In manual operation obstacle detection will not work.
5. Hardware based anti-pinch is not available, however software-based obstacle/anti-pinch is available.
6. If an obstacle is detected close to the edge of vehicle, then due to time and force required for obstacle detection mechanism, that obstacle may not get detected and will create pinch to that obstacle. So, it is always advised to check if there is any obstruction between closing tailgate and vehicle closing edge to avoid any injury/ damage.

### Tailgate Infotainment Setting



1. Enable/Disable Gesture Feature
2. Enable/Disable RKE Feature
3. Enable /Disable Buzzer Feature
4. Tailgate Height Setting

### WARNING

1. Obstructing the tailgate operation intentionally may cause serious injury to the person or damage to vehicle though it is equipped with obstacle detection feature.
2. Power strut equipped tailgate shall not be operated manually. However in the exceptional case like battery discharge it has to be operated gently. While operating manually (opening/closing) recommended to hold

tailgate at the center portion.

3. Do not grab or hold Power struts or try to disassemble them.
4. Don't drive the vehicle with Tailgate open, doing so will allow exhaust fumes/dust to enter the vehicle cabin, will obstruct rear approaching vehicles visibility and also may damage power struts.
5. Don't apply forces on power struts (e.g. pushing vehicles, tightening straps or applying other fasteners). Doing so will damage the struts and may lead to injuries.
6. Don't allow anyone to occupy luggage compartment as it is highly dangerous location in case of crash event.
7. Don't modify or repair any part of power tailgate. Reach out to the nearest TPEM Authorised Service Centre in case of any functional failures.

8. Never leave children or animals unattended in your vehicle. Children or animals might operate the power tailgate that could result in injury to themselves or others or damage to the vehicle.
9. Make sure objects in the rear cargo area do not come out when opening the tailgate on the slope way. It may cause serious injury.

### NOTE

1. Make sure no object, fingers or person etc. is obstructing tail gate operation while opening and closing, if so it can get trapped with damage or serious injury.
2. Driving vehicle with tail gate open or partially open is not recommended.
3. Accessories other than OEM approved, not to be fitted on tailgate. It may result into improper operation of power tailgate or damage the power strut.

4. Multiple operation of power tailgate can drain the battery, if ignition is off.
5. Do not leave power tailgate open for a long period of time. This may drain the battery.
6. Power tailgate can be operated when the engine is not running. However the power operation consumes large amounts of vehicle electric power. To prevent the battery from being discharged, do not operate them excessively.
7. When jacking up the vehicle to change a tyre or repair the vehicle, do not operate the power tailgate. This could cause power tailgate to operate improperly.
8. The outside power tailgate switch may not function properly in extreme cold or wet climatic conditions. In such cases it is operate through remote key or from center console switch.
9. In cold and wet climates, door lock and door mechanisms may not work

- properly due to freezing conditions.
10. When power tailgate is opened manually (without electrical operation), more effort will be required to open and close compared to the non-powered tailgate.
  11. Play protection feature is available to prevent overheating. This feature prevents the power tailgate to operate and allow the power struts to cool down. The power tailgate system can be operated again after 1-2 minute time period.
  12. It is recommended to wait until the power tailgate fully closed before starting the vehicle. The power tailgate may not close fully if the vehicle is started during automatic closing.



## DIGITAL DISPLAY (10.25" INCH)



NOTE: All indicators and values are for illustration purpose only.

## INSTRUMENT CLUSTER

| Gauge         | Information                                                                                                                                                                                                                                                                                                                                                          | Note/Warning                                                                                                                                                                                                                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speedometer   | The Speedometer Indicates the actual vehicle speed in km/h                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>At every key IN and Ignition ON, the speedometer Bar moves to MAX and return to '0' position.</li> <li>This is welcome strategy and self-check feature.</li> <li>In vehicle running condition if the Speedometer is not showing the Vehicle speed, then take the vehicle to TPEM Authorised Service Centre.</li> </ul>  |
| Odometer      | Odometer Indicates distance travelled by vehicle.                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>The odometer reading does not return to 0 when maximum value is reached, the display will freeze to maximum value.</li> </ul>                                                                                                                                                                                           |
| SoC Gauge     | SoC (State of Charge) gauge indicates the battery state of charge to user in percentage                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>When battery SoC goes below 5%, first Bar in gauge will start blinking.</li> <li>Do not drive the vehicle with low SoC.</li> <li>When the battery is low or near to empty position, low battery warning tell-tale turns Red. Pop up message is displayed to connect the charger to the vehicle for charging.</li> </ul> |
| EV mode Gauge | <ul style="list-style-type: none"> <li>This function provides instantaneous power consumption mode of vehicle during driving and displayed in the instrument cluster.</li> <li>During the IGN ON of the vehicle, EV mode gauge will starts sweep from REGEN mode to DRIVE mode and then back to the REGEN mode to indicate the welcome strategy behavior.</li> </ul> | When all functional modes are activated, then take the vehicle to TPEM Authorised Service Centre.                                                                                                                                                                                                                                                              |

| Gauge                        | Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Note/Warning                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distance To Empty (DTE)      | <ul style="list-style-type: none"> <li>For DRIVE mode, LED BARs will be ON as per the power consume.</li> <li>Range indicates approximate distance (km) that the vehicle can travel with current battery charge.</li> <li>Range shall be indicated both in IGN ON &amp; IGN OFF conditions.</li> <li>In IGN OFF when charger is connected and charging is happening then Range value will display as long as screen is activated in the cluster.</li> <li>'RECHARGE' shall be displayed which indicates that it's the time to take your vehicle to the nearest charging station and the distance that a vehicle can travel with current charge is 20 Kms.</li> </ul> | <ul style="list-style-type: none"> <li>If Range is displayed as '—', take vehicle to TPEM Authorised Service Centre.</li> <li>The Range values may vary significantly based on driving conditions, driving habits, and condition of the vehicle.</li> <li>The Range value is an estimate of the available driving distance.</li> <li>Change of distance unit is not applicable.</li> </ul>                                       |
| Regeneration Level Selection | <p>This function provides Regeneration Level settings to user from Minimum to Maximum in steps of Level 0, Level 1, Level 2 and Level 3 of vehicle during driving and displayed in the instrument cluster.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>The "Maximum" Regen Level setting provides the maximum amount of regenerative braking power &amp; it recaptures the most energy and reduces wear &amp; tear on the brakes.</li> <li>The "Minimum" Regen Level setting incorporates a reduced regenerative braking force that recaptures less energy but allows the vehicle to coast further than in the "Maximum" Regen Level.</li> </ul> |

## INSTRUMENT CLUSTER

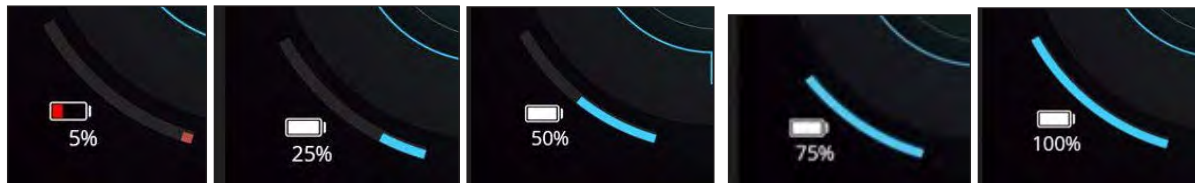
---

### State of Charge (SoC) Gauge For High Voltage Battery

Provided in the instrument cluster as a tell-tale. It shows the charging status of the high voltage battery. Low charge or minimum position on the indicator indicates that there is not enough energy in the high voltage battery.

Full charge or max position indicates that the HV battery is fully charged.

- When driving on highways, make sure to check in advance if the HV battery is charged enough.
- When the bar turns red on the high voltage charge indicator, the low charge warning lamp turns ON to alert you of the battery level.
- At <25% SoC, Sports mode cannot be selected. Max speed can be attained. At <10% SoC, limp mode gets activated and speed limits are triggered.



### Action to be Taken When Charging Stops Abruptly

- Check the reason for interruption of charging.(Refer 'Troubleshooting guide for Normal Charging' table).
- Switch off the AC supply.
- Remove the charging gun from the charging inlet.
- Wait for 5 minutes.
- Restart the charging. (Refer charging procedure).

### Cleaning of Charging Inlet



### Cleaning of Charging Inlet

Covering the charging gun and charging inlet by dust cap will ensure protection from water and dust.

### Precautions to be Taken While Cleaning the Charging Inlet

- Keep the vehicle lid always closed
- When the lid is open ensure that dust caps are in closed position
- During normal charging, make sure that DC charging cap is closed
- In case of any dust/mud/snow accumulation in the charging port and also on CCS2 especially actuator area, it can be cleaned with blowing air before charging.
- Allow the water to drain completely through drain holes.
- Allow the charging port to dry completely.

#### NOTE

*Water entering into the charging port will always be drained through the drain system. If water is stagnant in charging port area call TP&M Authorised Service Centre to rectify the issue.*

## INSTRUMENT CLUSTER

### Illuminated Charging Bowl

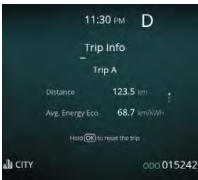
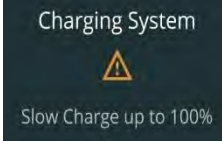
Provided in the charging flap as a tell-tale. It shows the charging status of the high voltage battery.

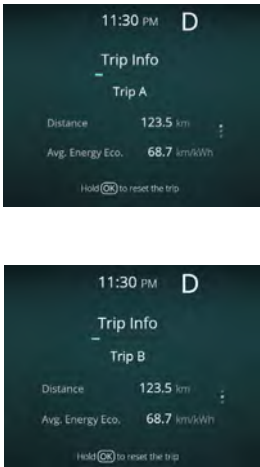
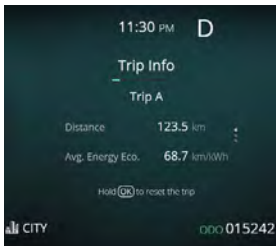
| S.N. | Color       | Illuminated Charging Bowl Status                                                  | Condition                                   | Criteria                                                                                                                  |
|------|-------------|-----------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | White Blink |  | Charging under initiation or authentication | Illumination ring to blink gradually in white colour. Blink duration till "charging initiation" in progress or max 150sec |
| 2    | Blue Blink  |  | Charging started and under progress         | Illumination ring to blink gradually in blue colour. Charging blink duration to be 60sec                                  |
| 3    | Green Solid |  | Charging completed to 100%                  | Illumination ring to solid ON in Green colour. Charge completion solid ON duration to be 60sec                            |
| 4    | Red Blink   |  | Charging error - failure                    | Illumination ring to blink gradually in Red colour. Error blink duration to be 60sec                                      |

**DRIVER INFORMATION SYSTEM (DIS)**

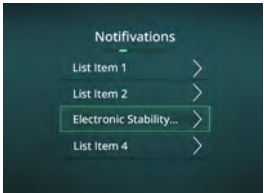
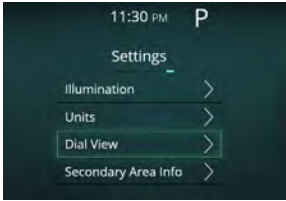
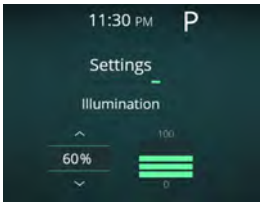
| Driver Information          | System Image                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service reminder            |  | User can select Service Reminder Screen using controls on steering wheel to navigate & by pressing Set Button in Settings Screen. Service reminder is a feature to alert the user for service action. When distance since last service meet the maximum distance criteria, a service screen will get activated for the user to indicate the service of the vehicle has to be done along with the tell-tale indication. User can reset the Service Reminder Symbol by right/left & SET buttons on the steering wheel. |
| Outside Ambient Temperature |  | Displays outside ambient temperature in °C.<br><b>NOTE:</b> The temperature sensor is in the front bumper of the vehicle, therefore the temperature reading can be affected by heat reflection from the road surface. This can cause an incorrect temperature reading when speed is under low speeds or when stopped.<br>If display shows '- - ', take your car to TPDM Authorised Service Centre.                                                                                                                   |
| Door Ajar (if equipped)     |  | This feature monitors the Door Input and warns Driver if any Door is Open<br><b>NOTE:</b> If any other door is open roof lamp will be 'ON' provided that roof lamp switch is in ON position.                                                                                                                                                                                                                                                                                                                         |

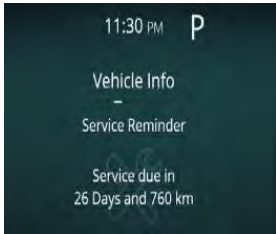
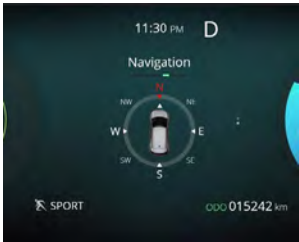
## INSTRUMENT CLUSTER

| Driver Information                             | System Image                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current Gear Indication                        |  | <p>Current gear engaged by the transmission shall be displayed on DIS.</p> <p><b>Note:</b> If is displayed, it means 'Fault' condition. In such case, take vehicle to TPEM Authorised Service Centre.</p>                                                                                                                                                                                                                                             |
| Seat Belt Reminder                             |  | <p>The seatbelt warning indicator remains ON for 4 seconds, when ignition is turned ON. The warning lamp remains ON till all occupied seats belts are buckled.</p> <p>If seatbelt remains unbuckled and vehicle speed goes beyond 15 kmph, then final audio warning will go more than 90 seconds</p> <p>Note: Once the seatbelts are fastened, the buzzer and warning lamp turns OFF. Seatbelt reminder remains OFF when reverse gear is engaged.</p> |
| Critical Battery (Thermal Runaway) Malfunction |  | <p>This message will appear in cluster along with audio warning to "Evacuate the vehicle". In case there is a critical malfunction in high voltage battery which can lead to thermal damage, this message will appear in cluster. Customer is recommended to park vehicle in safe zone and evacuate to avoid injury.</p>                                                                                                                              |
| Slow charge up to 100%                         |  | <p>It is recommended to slow charge vehicle to 100% SOC to ensure consistent performance and better health of high voltage battery. Once this message appears in cluster, it is recommended that user slow charges vehicle's high voltage battery to 100% SOC.</p>                                                                                                                                                                                    |

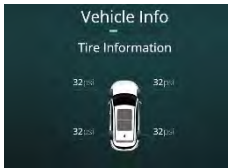
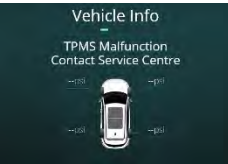
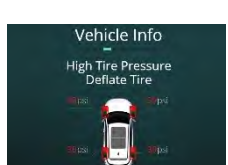
| Driver Information                           | System Image                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Average Energy Economy for Trip A and Trip B |  | <p>Displays “Average Energy consumption” for trip A or B since it was reset<br/>Resolution: 0.1 Wh/Km<br/>Average Energy Consumption shall Reset to 0 when respective Trip meter is reset.<br/>Average Energy Economy shall be displayed as ‘—.’ for initial 0.5 km of respective trip. Once 0.5 km distance is covered, Average Economy shall be displayed.<br/>Even after 0.5 km distance covered for particular trip, Average Energy economy is displayed as ‘—.’ take vehicle to TPEM Authorised Service Centre.<br/>Note:</p> <ul style="list-style-type: none"> <li>AEE value is estimate of Energy economy. It may vary significantly based upon driving conditions, driving habits and condition of vehicle.</li> <li>Average Energy Consumption shall get reset to 0 when Battery is removed and refitted.</li> </ul> |
| Distance To Empty                            |  | <p>DTE indicates approximate distance (km) that the vehicle can travel with current battery charge.<br/>DTE shall be indicated both in IGN ON &amp; IGN OFF conditions.<br/>In IGN OFF when charger is connected and charging is happening then DTE value will display as long as screen is active in the cluster.<br/>‘RECHARGE’ shall be displayed which indicates that it’s the time to take your vehicle to the nearest charging station and the distance that a vehicle can travel with current charge is 20 Kms.</p>                                                                                                                                                                                                                                                                                                     |

## INSTRUMENT CLUSTER

| Driver Information                                                | System Image                                                                      | Description                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infotainment Information<br>On Instrument Cluster<br>Display Unit |  | The instrument cluster will display information like media, navigation and FM.                                                                                                                                                                                                                                                              |
| Settings Screen                                                   |  | User can enter into setting screen by pressing select button while being in setting screen.                                                                                                                                                                                                                                                 |
| Illumination Setting                                              |  | <p>User can select Illumination Setting by Scroll down &amp; pressing Set Button in Setting Screen provided park lamp ON.</p> <p>User can increase the illumination from (20% to 100%) in 5 steps by using UP &amp; SET Button.</p> <p>User can decrease the illumination from (100% to 20%) in 5 steps by using DOWN &amp; SET Button.</p> |

| Driver Information     | System Image                                                                      | Description                                                                                                                                                                                                                                                                                                                |
|------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service Reminder Reset |  | <p>User can select Service Reminder Screen by Scroll down &amp; pressing Set Button in Setting Screen.</p> <p>User can reset (Yes / Cancel) the Service Reminder by UP / DOWN &amp; SET Buttons.</p> <p>Note:</p> <p>In the Setting menu if there is no user input for 10 secs the previous screen shall be displayed.</p> |
| Compass screen         |  | <p>Compass Feature shall be used for navigation and orientation that shows direction relative to the geographic cardinal directions</p>                                                                                                                                                                                    |
| Charging Limits        |  | <p>Set charging limit of the battery. The target charging level can be changed by 10%. Once the charging is completed as per set limit, the message is displayed</p>                                                                                                                                                       |

## INSTRUMENT CLUSTER

| Driver Information                     | System Image                                                                                                                                                                                                                                                                                               | Description                                                                                                                                                                 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TPMS (TYRE PRESSURE MONITORING SYSTEM) |  <p>The image shows a top-down view of a car with four tires. Each tire has a green pressure gauge icon and the text '32 psi' next to it. Above the car, the text 'Vehicle Info' and 'Tire Information' is displayed.</p> | User can select TPMS screen by scrolling up or down. TPMS system tell-tale comes ON with Tire pressure value near the respective Tire with Recommended level" Text Message. |
| TPMS - MALFUNCTION                     |  <p>The image shows a top-down view of a car with four tires. Each tire has a red exclamation mark icon. Above the car, the text 'Vehicle Info' and 'TPMS Malfunction Contact Service Centre' is displayed.</p>           | TPMS MALFUNCTION CONTACT SERV CENT" text warning comes 'ON' for 4 seconds when TPMS system malfunction. Take your vehicle to TPMS Authorised Service Centre.                |
| TPMS - LOW TYRE PRESSURE               |  <p>The image shows a top-down view of a car with four tires. Each tire has a red exclamation mark icon. Above the car, the text 'Vehicle Info' and 'Low Tire Pressure Inflate Tire' is displayed.</p>                    | When any of the Tire Pressure drops significantly below Recommended levels then TPMS system tell-tale comes ON with "Low Tire pressure Inflate Tire" Text Message.          |
| TPMS - HIGH TYRE PRESSURE              |  <p>The image shows a top-down view of a car with four tires. Each tire has a red exclamation mark icon. Above the car, the text 'Vehicle Info' and 'High Tire Pressure Deflate Tire' is displayed.</p>                   | When any of the Tire Pressure increase significantly above Recommended levels then TPMS system tell-tale comes ON with "High Tire pressure Deflate Tire" Text Message.      |



| Driver Information               | System Image                                                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TPMS - AIR LEAKAGE               |  <p>The image shows a top-down view of a vehicle with red exclamation marks on the front and rear left tires. Text on the screen reads: "Vehicle Info", "Leakage Detected", and "Check Tires".</p>                                                         | <p>When any of the Tire Pressure drops significantly below Recommended levels then TPMS system tell-tale comes ON with "Leakage Detected Check Tires" Text Message.</p> |
| TPMS - TIRE OVER-LOAD            |  <p>The image shows a top-down view of a vehicle with red exclamation marks on all four tires. Text on the screen reads: "Vehicle Info", "Tire Overheated", and "Slow Down".</p>                                                                           | <p>When any of the Tire Pressure increase significantly above Recommended levels then TPMS system tell-tale comes ON with "Tire Overheated slow down" Text Message.</p> |
| DDOA SCREEN<br>ATTENTION LEVEL-5 |  <p>The image shows the Driver Assist screen with the text "Driver Attention Assist STAY ALERT!". Below the text is a green bar representing the attention level, with "Low" on the left and "High" on the right. A green eye icon is in the center.</p>   | <p>User can select DDOA screen by scrolling up or down switch on steering wheel.<br/>DDOA screen with Driver Attention Level 5 (Attention Level in Green Color)</p>     |
| DDOA SCREEN<br>ATTENTION LEVEL-4 |  <p>The image shows the Driver Assist screen with the text "Driver Attention Assist STAY ALERT!". Below the text is a yellow bar representing the attention level, with "Low" on the left and "High" on the right. A yellow eye icon is in the center.</p> | <p>DDOA screen with Driver Attention Level 4(Attention Level in Yellow color)</p>                                                                                       |

## INSTRUMENT CLUSTER

| Driver Information               | System Image                                                                      | Description                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| DDOA SCREEN<br>ATTENTION LEVEL-3 |  | DDOA screen with Driver Attention Level 3 (Attention level In Amber Color) & with interrupt message for 4 seconds       |
| DDOA SCREEN<br>ATTENTION LEVEL-2 |  | DDOA screen with Driver Attention Level 2 (Attention level In A light RED color) & with interrupt message for 4 seconds |
| DDOA SCREEN<br>ATTENTION LEVEL-1 |  | DDOA screen with Driver Attention Level 1 (Attention Level in RED color) & with interrupt message for 4 seconds         |

### Driver Information System (DIS) Setting

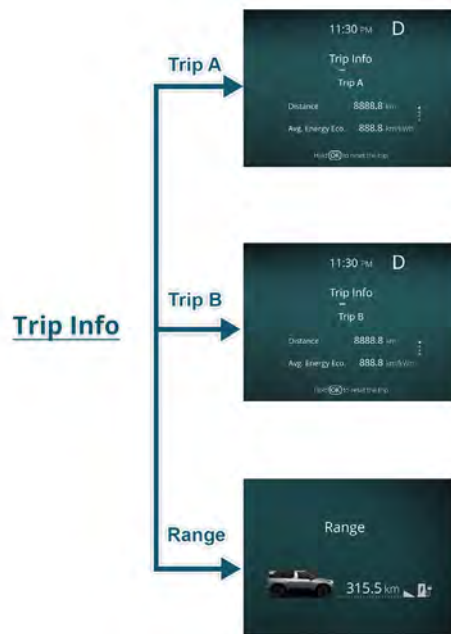
Operate the Up & down and Set Switch on steering wheel to see the Trip Info, Drive Assist, Vehicle Info, Notification, Navigation, Layout and Settings Window.

Operate the Set Switch on steering wheel to reset TRIP A, AEE A, Average Speed A, Trip Time A (When TRIP A is displayed) and reset TRIP B, AEE B, Average Speed B and Trip Time B (When TRIP B is displayed).

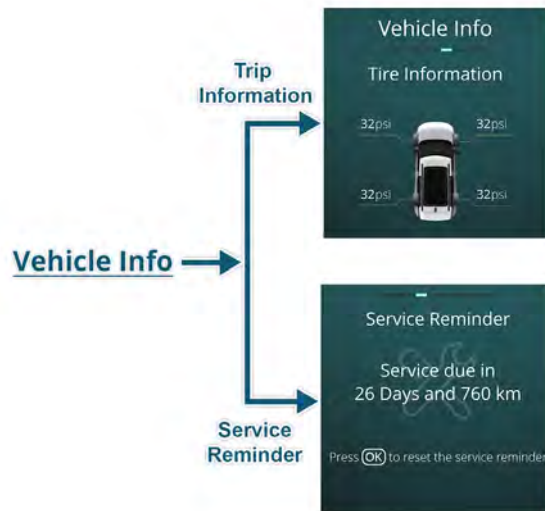


## INSTRUMENT CLUSTER

### Trip Information



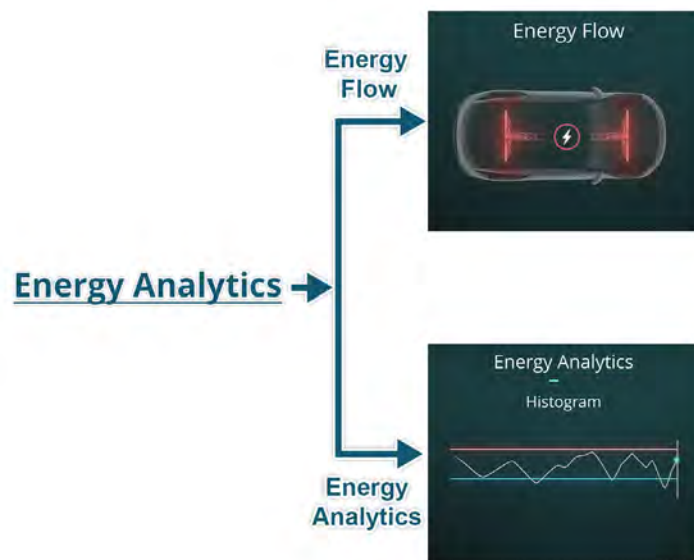
## Vehicle information



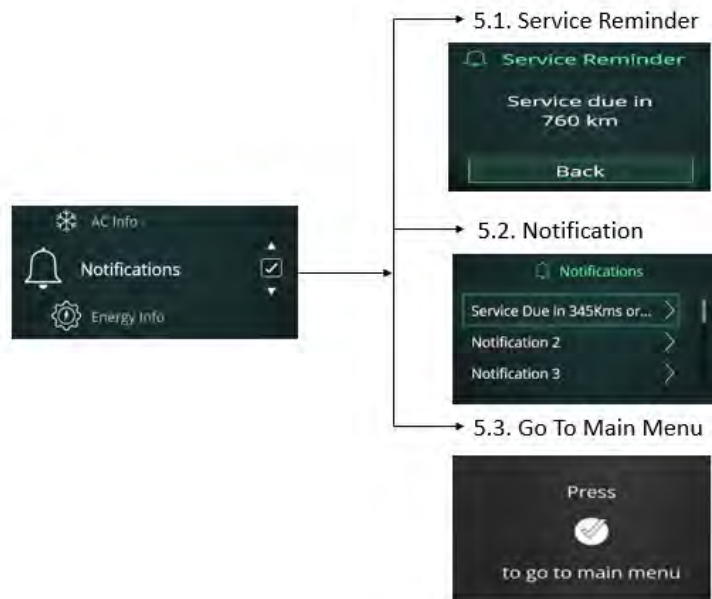
## INSTRUMENT CLUSTER

---

### Energy Analytics



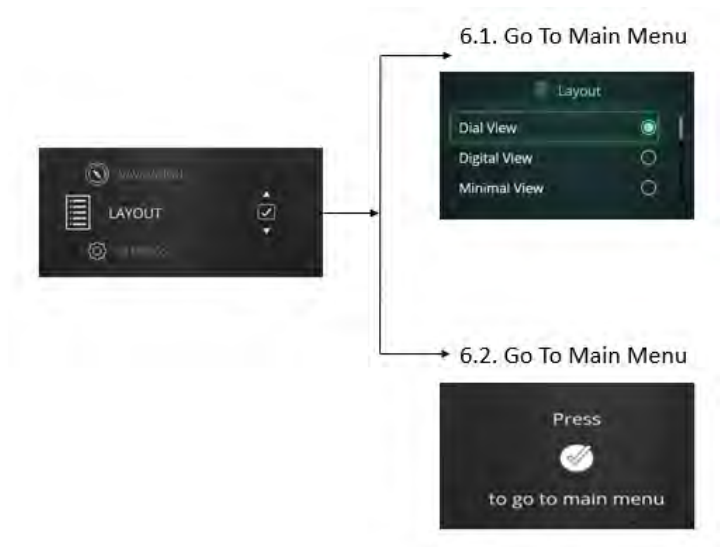
## Notification



## INSTRUMENT CLUSTER

---

### Layout





## Setting Screen



## INSTRUMENT CLUSTER

### WARNING AND INDICATOR

| Warning Lamps                | Color         | Indicator                                                                                                                                                              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Service Indicator            | Amber/<br>Red | <br> | In case of Power Train Sensors (AC Charging Inlet Temperature Sensor, AC Linear Pressure Sensor etc.) & Actuators (Fan, Pump, Regen Switch, Eco / Sport Switch etc.) failure then this Amber indicator will glow. When there is high severity then Red indicator will glow. Please take your vehicle to nearest TPEM Authorised Service Centre at the earliest.                                                                                                    |
| Immobilizer<br>(if equipped) | Red           |                                                                                       | <ol style="list-style-type: none"><li>1. This lamp comes on when the system disables vehicle start if the original key is not used.</li><li>2. Lamp blinks: Vehicle is in immobilized condition when key is not inserted.</li><li>3. Lamp ON: Problem with key/system. Contact a TPEM Authorised Service Centre.</li><li>4. Lamp OFF: Normal condition (Authenticated user) and vehicle will start.</li></ol>                                                      |
| Turn Signal                  | Green         |                                                                                       | Indicates direction indicated by the turn signal.<br>Blinks along with buzzer while operating left/right turn indicator only when ignition is switched 'ON'. The direction indicator arrow on Instrument Cluster flashes along with external indicator lights as selected. Both tell tales shall blink simultaneously when Hazard switch is pressed irrespective of Ignition ON and the Tick-Tock sound shall be given when any one or both the tell tales are ON. |

| Warning Lamps                                  | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------|-------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High Beam                                      | Blue  |  | This lamp comes on when the high beam headlamps are switched 'ON' or flashed.                                                                                                                                                                                                                                                               |
| LV Battery charging                            | Red   |  | This symbol lights up when the 'IGN' is turned 'ON' and should go 'OFF' after the vehicle starts.<br><b>NOTE:</b><br>If it remains 'ON' while the vehicle is running, it indicates that the battery is not getting charged. Switch off all unnecessary electrical equipment and get the problem attended at TPEM Authorised Service Centre. |
| Airbag status                                  | Red   |  | This lamp comes on when ignition is switched 'ON' and goes 'OFF' in approx. 4 seconds. If it continuously remains on or blinks then contact the TPEM Authorised Service Centre immediately.                                                                                                                                                 |
| Park Brake / Brake Fluid Low / EBD malfunction | Red   |  | Illuminates momentarily when ignition is switched 'ON'. Once parking brake is re-released, it turns 'OFF'. If it remains 'ON', it indicates.<br>1. Brake fluid level is low.<br>2. Park brake is applied & turns 'OFF' when it is released.<br>3. ABS/EBD system has a fault.                                                               |
| Cruise Control lamp (if equipped)              | Green |  | This symbol lights up when the 'IGN' is turned 'ON' and shall go 'OFF' after 4 sec. The Cruise Control is used to indicate the status of cruise control system to the driver. Lamp ON indicates cruise control feature is present and it is activated.                                                                                      |
| EPAS                                           | Amber |  | Illuminates momentarily when ignition is switched 'ON'.<br>Illuminates when there is a fault in the EPAS. Contact the TPEM Authorised Service Centre immediately.                                                                                                                                                                           |

## INSTRUMENT CLUSTER

| Warning Lamps                                      | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Driver Seat Belt Indicator                         | Red   |  | <p>Seat belt warning indicator comes 'ON' for 4 seconds, when ignition is turned 'ON' irrespective of seat belt buckle status.</p> <p>If seat belt is not fastened then tell tale will be ON as initial warning with No audio warning i.e. chime. If seat belt remains unbuckled &amp; vehicle speed goes above 15 kmph, Final Warning will start with audio chime for 93 seconds continuously.</p> <p>NOTE:</p> <ul style="list-style-type: none"> <li>Once the seat belt is fastened, the buzzer &amp; warning indicator will go 'OFF'.</li> <li>Seat belt indicator &amp; audio alarm remains OFF when reverse gear is engaged.</li> </ul> |
| AVH Indicator (If equipped )                       | Green |  | <p>AVH Indicator turns 'ON' for 4 seconds, when ignition is turned 'ON' irrespective of input state.</p> <p>This feature monitors AVH function in ESP system and warns the driver in case of AVH function malfunction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ABS                                                | Amber |  | <p>Illuminates when ignition is switched 'ON' and goes 'OFF' in 3 seconds. Illuminates continuously if there is any malfunction in ABS. Normal braking system will be operational without assistance of ABS. Contact TPEM Authorised Service Centre immediately.</p>                                                                                                                                                                                                                                                                                                                                                                          |
| Key Not Detected (if equipped)                     | Amber |  | <p>This lamp comes on when the Valid Smart key is not detected inside the vehicle.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Press / Brake Pedal to Start vehicle (if equipped) | Amber |  | <p>This lamp comes on with IGN ON till user presses the brake pedal to start the vehicle.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Warning Lamps                              | Color       | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|-------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daytime running lamps<br>DRL (if equipped) | Green       |  | This lamp comes on when the Daytime Running lamp is 'ON'.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Door Ajar lamp (if equipped)               | White / Red |  | All four door and Tail gate are indicated independently when the respective door or tail gate is open.                                                                                                                                                                                                                                                                                                                                                                                       |
| ECO                                        | Green       |  | Illuminates momentarily when ignition is switched 'ON'.<br>When ECO lamp is ON, it indicates the car is in 'Economy' drive mode.                                                                                                                                                                                                                                                                                                                                                             |
| CITY                                       | Blue        |  | Illuminates momentarily when ignition is switched 'ON'.<br>If CITY lamp is ON, it indicates 'City' drive mode, which is default mode.                                                                                                                                                                                                                                                                                                                                                        |
| SPORT                                      | Amber       |  | This symbol comes ON when SPORT driving mode is activated.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Speed limit warning indicator              | Amber       |  | When the vehicle speed crosses 80 kmph, then speed limit warning indicator turns 'ON' along with an audio chime for every two minutes (audible warning).<br>When the vehicle speed is reduced below 75 kmph, then the speed limit warning indicator and the audio warning will turn off.<br>If vehicle speed crosses 120 kmph, the speed limit warning indicator flashes along with an audio warning for every two sec one beep (audible warning) until the vehicle speed is above 120 kmph. |

## INSTRUMENT CLUSTER

| Warning Lamps                     | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|-------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   |       |                                                                                   | When the vehicle speed is reduced below 115 kmph, then speed limit warning indicator turns 'ON' along with an audio chime for every two minutes one beep (audible warning)                                                                                                                                                                                                                                                                                                |
| TPMS<br>(Isolated/Non-Isolated)   | Amber |  | 1. This symbol comes ON and blink for 4 second if Tyre Pressure is LOW/HIGH, Tyre temperature is HIGH, Tyre air pressure leakage. After 4 second symbol will continuously ON till warning is present.<br>2. This symbol comes on and blink for 10 second if TPMS system has fault and TPMS Sensor fault / missing. After 10 second symbol will continuously ON till fault is present, Please take your vehicle to nearest TPEM Authorised Service Centre at the earliest. |
| HHC warning lamp<br>(if equipped) | Amber |  | Illuminates momentarily when ignition is switched 'ON'.<br>If continuously on then HHC, system is in fault condition. Please take your vehicle to TPEM Authorised Service Centre at the earliest.                                                                                                                                                                                                                                                                         |
| Park Lamp Indicator               | Green |  | Park Lamp Indicators used to display/Indicate the Position Lamp to Driver.                                                                                                                                                                                                                                                                                                                                                                                                |
| Charging Fail Indicator           | Red   |  | This symbol is displayed when the vehicle is not getting charged even if the charger is connected. Contact the TPEM Authorised Service Centre to get the charging fail issue resolved                                                                                                                                                                                                                                                                                     |
| Charger Connected                 | Blue  |  | This symbol lights up as soon as the charger is connected for charging the battery                                                                                                                                                                                                                                                                                                                                                                                        |

| Warning Lamps            | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Charging Indicator       | Green |  | This symbol is displayed when your vehicle is getting charged.                                                                                                                                                                                                                                                          |
| Motor High Temperature   | Red   |  | This symbol lights up when the temperature of the motor is higher, and motor becomes hot.<br>Park your vehicle safely and wait for the temperature to become normal. If the problem persists, contact the TPEM Authorised Service Centre.                                                                               |
| Battery High Temperature | Red   |  | This symbol lights up when the temperature of the battery is higher, and battery becomes hot. Contact the TPEM Authorised Service Centre if this indicator is getting on frequently.                                                                                                                                    |
| Limp Home Mode           | Amber |  | This symbol indicates the vehicle gone into limited performance mode. This usually happens when the battery reaches 10% threshold or if there is any minor fault in power transmission or electrical components.                                                                                                        |
| High Voltage (HV) Alert  | Red   |  | This symbol lights up the voltage of the battery is too high and cause damage. Park your vehicle safely and contact the TPEM Authorised Service Centre.                                                                                                                                                                 |
| Drive Ready              | Green |  | This symbol indicated that your vehicle is ready to drive                                                                                                                                                                                                                                                               |
| zero charge/Low Charge   | Red   |  | This feature provides the HV battery Low/Zero Charge Status to the user. A bulb check shall be performed for this TT at every IGN ON for 4 seconds. The TT shall remain ON irrespective of the Input state during these 4 seconds. This tell tale shall be controlled turned ON/OFF by receiving the SoC input when the |

## INSTRUMENT CLUSTER

| Warning Lamps                          | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|-------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        |       |                                                                                   | SoC level is low TT will turn ON to indicate charging system battery low to the user.                                                                                                                                                                                                                                                                                                                                                                                                                |
| AC ON                                  | Blue  |  | This feature provides AC status to user. A bulb check shall be performed for this TT at every IGN ON for 4 seconds. The TT shall remain ON irrespective of the Input state during these 4 seconds. This tell tale is ON when AC is turn ON by user.                                                                                                                                                                                                                                                  |
| Co-Driver Seat Belt Indicator          | Red   |  | If Co-Driver (Passenger) is present and its seat belt is not buckled and IGN is ON then tell tale will be ON as initial warning with No audio chime.<br>If Co-Driver (Passenger) seat belt remains unbuckled and vehicle speed goes above 15 km/ hr, Then Seat belt tell tale will also start flashing along with audio alarm during Final Warning.<br>Note: Buckle the Co-Driver (Passenger) seat belt to stop Audible warning and tell tale OFF.                                                   |
| Rear Seat Belt Indicator (If equipped) | Red   |  | If Rear Passenger (Right / Middle / Left) is present and its seat belt is not buckled and IGN is ON then tell tale will be ON as initial warning with No audio chime.<br>If Rear Passenger (Right / Middle / Left) seat belt remains unbuckled and vehicle speed goes above 15 km/ hr, Then Seat belt tell tale will also start flashing along with audio alarm during Final Warning.<br>Note: Buckle the Rear Passenger (Right / Middle / Left) seat belt to stop Audible warning and tell tale OFF |
| EPB MIL Fault                          | Amber |  | In case of malfunction in EBP MIL function in ESP system this indicator will glow. Please take your vehicle to nearest TPDM Authorised Service Centre at the earliest                                                                                                                                                                                                                                                                                                                                |



| Warning Lamps           | Color | Indicator                                                                         | Remarks                                                                                                                                                                                                  |
|-------------------------|-------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AUTO VEHICLE HOLD (AVH) | Amber |  | In case of malfunction in AVH function in ESP system this amber indicator will glow. Please take your vehicle to nearest TPEM Authorised Service Centre at the earliest                                  |
| ESP (if equipped)       | Amber |  | Illuminates momentarily when ignition is switched 'ON'.<br>If continuously ON then ESP system is at fault condition, Please take your vehicle to nearest TPEM Authorised Service Centre at the earliest. |

## INSTRUMENT CLUSTER

---

### DISPLAY MESSAGES ON INSTRUMENT CLUSTER

#### Warning Messages

| SN | Warning / Information Title                                   | Warning Message Title  | Action To Be Taken                                          |
|----|---------------------------------------------------------------|------------------------|-------------------------------------------------------------|
| 1  | Fasten Seat Belt - Driver                                     | Seat Belt Reminder     | Fasten Driver Seat Belt                                     |
| 2  | Speed Limit Warning                                           | Speed Limit Warning    | Over Speeding Detected Slow Down                            |
| 3  | Drive Control Shift Denied                                    | Drive Mode Warning     | Drive Control Shift Denied                                  |
| 4  | Hill Hold Control Failure                                     | Hill Hold Control      | Malfunction Detected Contact TPEM Authorised Service Centre |
| 5  | Hill Decent Control Failure                                   | Hill Decent Control    | Malfunction Detected Contact TPEM Authorised Service Centre |
| 6  | Charging Level Low State                                      | Charging Level Warning | Charge the vehicle                                          |
| 7  | Fasten seat belt front passenger                              | Seat Belt Reminder     | Fasten Front passenger Seat Belt                            |
| 8  | Transmission Failure Limp home Activated Visit Service Centre | Transmission System    | Malfunction Detected Contact TPEM Authorised Service Centre |

**Alert Messages**

| SN | Alert / Information Title            | Alert Message Title                            | Action To Be Taken                                                                   |
|----|--------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------|
| 1  | Service Reminder Days                | Service Due in/Service Overdue by "value" days | Contact to TPEM Authorised Service Centre                                            |
| 2  | Park Brake Engaged                   | Brake Alert                                    | Push EPB switch down to Release Park Brake                                           |
| 3  | Charging Full                        | Battery Fully Charged                          | Remove Charger Safely                                                                |
| 4  | Service Reminder Kms                 | Service Due in/Service Overdue by "value" km   | Contact to TPEM Authorised Service Centre                                            |
| 5  | Charging below 100%                  | Battery XX% Charged Range YYY kms              | Remove Charger Safely                                                                |
| 6  | Auto Headlamp                        | Lamp Alert                                     | Headlamps will get activated automatically                                           |
| 7  | Battery Low & user changes gear to S | Sport Mode Not Recommended                     | Change gear to any Eco or City mode                                                  |
| 8  | Charging ON Park brake OFF           | Engage Park Brake to Start Charging            | Engage Park Brake to Start Charging                                                  |
| 9  | HV Critical alert                    | Critical Alert Contact Service Centre          | Contact TPEM Authorised Service Centre                                               |
| 10 | Slow Down Vehicle Speed              | Slow Down to Turn Off Vehicle                  | Press the brake pedal to slow down vehicle                                           |
| 11 | Slow Charge up to 100%               | Slow Charge up to 100%                         | It is recommended that user slow charges vehicle's high voltage battery to 100% SOC. |
| 12 | Critical Battery Malfunction         | Park vehicle safely and Evacuate Immediately   | It is recommended to park vehicle in safe zone and evacuate to avoid injury.         |

## INSTRUMENT CLUSTER

---

### Interrupt Messages

| SN | Alert / Information Title                     | Action To Be Taken                                               |
|----|-----------------------------------------------|------------------------------------------------------------------|
| 1  | Rotate steering wheel (In ESCL jam condition) | Press Start Button while Turning Wheel                           |
| 2  | Drive Alert - Tea Break                       | Take a Break                                                     |
| 3  | Steering Failure-Visit Garage                 | Steering Failure Contact TPEM Authorised Service Centre          |
| 4  | Steering Failure-Stop Driving                 | Steering Failure Stop the Vehicle Safely                         |
| 5  | Door Ajar                                     | Close the door                                                   |
| 6  | ESCL                                          | Press Start Button While Turning Wheel                           |
| 7  | No Key                                        | Smart Key Out of Range                                           |
| 8  | Low Key Battery                               | Smart Key Battery Low Replace Battery                            |
| 9  | Press Brake Pedal                             | Press Brake Pedal to Start Engine                                |
| 10 | Drive Modes                                   | Respective drive mode                                            |
| 11 | Release Park Brake                            | Press Brake Pedal and push EPB switch down to Release Park Brake |

## AUDIO REMINDER

| Sr. No | Feature                            | Condition                                                                                                                                                       | Reminder                                                                                                                                        |
|--------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Parking Lamp 'ON' Reminder         | If you forget to turn OFF the park lights and driver door is open                                                                                               | An audio warning will be started. Do not forget to turn OFF your park lights as it may drain the vehicle's battery.                             |
| 2      | Parking brake 'ON' reminder        | If Park Brake is applied and vehicle is driven above 5 Km/h, tell-tale shall blink along with chime continuously. Disengage the park brake to stop the warning. | Tell-tale will turn 'ON' and buzzer will provide audio warning continuously. Disengage the park brake to stop audio warning.                    |
| 3      | Reverse Gear reminder              | If reverse gear is engaged                                                                                                                                      | The buzzer sound will alert you for 1 second.                                                                                                   |
| 4      | Driver Seat Belt reminder          | If seatbelt is not fastened and vehicle goes above 15 kmph                                                                                                      | Then final audio warning will go on for more than 90 seconds. Seat belt tell-tale light will remain continuously ON when audio alarm is active. |
| 5      | Front passenger Seat Belt reminder | If front passenger has not fastened seatbelt and if vehicle speed goes above 15 kmph, then final audio warning will go on for more than 90 seconds.             | Seat belt tell-tale light will remain continuously ON when audio alarm is active.<br>Note: Fasten the seatbelt to stop audio warning.           |
| 6      | Drive mode chime                   | When user switches drive mode from city to eco or city to sport (if equipped)                                                                                   | Sound warning for 1 second will be given to alert user.                                                                                         |

## INSTRUMENT CLUSTER

| Sr. No | Feature                                      | Condition                                                                                                                                                                                  | Reminder                                                                                                                                                                                                              |
|--------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7      | Electronic Steering Column Lock (ESCL) chime | This feature informs the driver to rotate steering wheel when ESCL gets engaged inadvertently.                                                                                             | This chime is sounded in IGN OFF mode for 3 secs.                                                                                                                                                                     |
| 8      | High Temperature alert for Motor             | When machine and inverter temperature cross the max limit from the BMS                                                                                                                     | Buzzer will start along with the motor high temperature warning lamp blinking to indicate the user to contact TPEM Authorised Service Centre. TT and buzzer will be in sync continuously till the state remains TRUE. |
| 9      | TPMS chime                                   | 1) If,<br>Tyre Pressure is low<br>Tyre Pressure is high<br>Tyre temperature is high<br>Tyre air pressure leakage<br>2) If,<br>TPMS system has fault<br>TPMS Sensor has fault or missing    | TPMS chimes shall sound for 4 secs and for TPMS fault conditions TPMS chime shall sound for 10 sec.                                                                                                                   |
| 10     | PEPS Key not detected chime                  | If PEPS key is not detected in the vehicle                                                                                                                                                 | Sound warning will be given to alert User                                                                                                                                                                             |
| 11     | Low battery charging Chime                   | When Auxiliary battery charging fault occurs with high criticality, buzzer will start along with the warning lamp blinking to indicate the user to contact TPEM Authorised Service Centre. | Warning lamp and buzzer will be in alert continuously till the state remains TRUE.                                                                                                                                    |

| Sr. No | Feature                            | Condition                                                                                                                        | Reminder                                                                                                                                                                                                                         |
|--------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12     | Charging ON & Park Brake OFF Chime | When charger is connected & Park brake is dis-engaged.                                                                           | Sound will be given to alert User                                                                                                                                                                                                |
| 13     | Rear Seat Belt reminder            | If Rear Passenger is present & its seat belt remains unbuckled and vehicle speed goes above 15 km/ hr, Final Warning will start. | Audio Chime will continue for 93 seconds. Seat belt tell-tale will remain continuously ON when audio alarm is active.                                                                                                            |
| 14     | High Temperature alert for Battery | When maximum battery cell temperature crosses the limit specified                                                                | Sound will start along with the battery high temperature warning lamp blinking to indicate the user to contact TPEM Authorised Service Centre. Warning lamp and buzzer will be in sync continuously till the state remains TRUE. |





### BEFORE YOU START YOUR EV

- Make sure that the area around the vehicle is clear.
- Do a check of the fluid levels - coolant, brake fluid, and windshield washer fluid as frequently as possible.
- Make sure that all windows and lights are clean.
- Examine the tyres for their appearance, inflation pressure and condition.
- Make sure that all doors are closed.
- Position the seat and adjust the headrests.
- Adjust the inside and outside mirrors.
- Fasten seat belts and ask all passengers to do likewise.
- Do a check of the operation of the warning lights when the power switch is pushed to the ON position. For additional information, refer to "Warning lamps and audible reminders" in the "Instrument Cluster" section of this manual.

### Procedure To Stop Your EV

1. Hold down the brake pedal until the vehicle comes to halt.
2. Continue to press the brake pedal, first shift to N mode and further shift to P mode/engage parking brake.
3. Press the start/stop button to OFF position to stop the vehicle.

## STARTING AND DRIVING

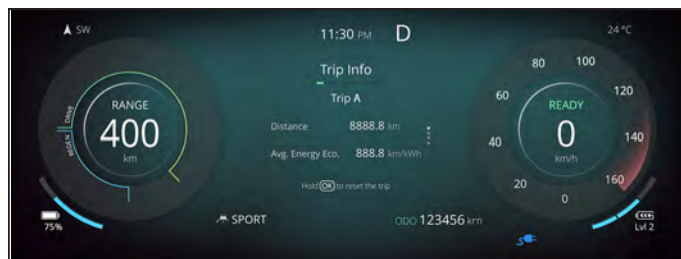
### Range Of Your EV

Your EV can drive as per given range in below table, when the high voltage battery is 100 % charged. However, in certain situations like driving at high speed or when the air conditioner/heater is ON, the distance to empty can reduce significantly, as the high voltage battery consumes more electricity.

| S.n. | Range                                                                                 | Battery State |
|------|---------------------------------------------------------------------------------------|---------------|
| 1    | Option I : RWD - 566 Km* (P1+P2)<br>Option II : RWD - 655 Km* , AWD - 625 Km* (P1+P2) | 100%          |
| 2    | Option II : RWD - 533 Km (P1+P2)                                                      | 100%          |

P1+P2 Cycle - Urban and Extra- Urban Driving Cycle (max speed 90kmph)

\*Range mentioned under standard test condition.



If the “—” symbol is displayed, charge the vehicle immediately. After you charge your vehicle, the distance to empty reading may vary significantly depending on previous operating patterns. When previous driving patterns include high speed driving, resulting in the driving battery using more electricity than usual, the estimated distance to empty is reduced.

When the high voltage battery uses a little electricity in ECO mode, the estimated distance to empty increases. Distance to empty may depend on many factors such as the charge available in the high voltage battery, weather, temperature, durability of the battery, geographical features, and driving style. Natural degradation may occur with the high voltage battery depending on the number of years the vehicle is used. This may reduce the distance to empty. Contact your nearest TPEM Authorised Service Centre to replace the battery in that case.

#### NOTE

*Any additional load in the car drains the battery or it may effect on range of vehicle.*

### START/STOP SWITCH



#### Vehicle Passive Start - Condition

##### Single Press Start

1. Bring the key with you and place at its location mentioned in "procedure to start EV", sit in the driver seat.
2. Press the brake pedal and then press the start/stop switch.
3. Green colour LED on start/stop button will turn ON.
4. Once vehicle is started successfully, the green colour LED on start/stop button stays ON.

#### Two Step Start

##### Step 1

1. Bring the key with you and place at its location mentioned in "procedure to start EV", sit in the driver seat.
2. Press the start/stop button without pressing brake pedal.
3. Amber colour LED on start/stop switch turns ON.
4. Vehicle will remain OFF and Steering is unlocked.

##### Step 2

1. Press the brake pedal and then press start/stop button to start the vehicle.
2. Green colour LED on start/stop button will turn ON.
3. Once vehicle start successfully, green colour LED on start/stop switch will remain ON.

#### Three Step Start

1. Bring the key with you and place at its location mentioned in "procedure to start EV", sit in the driver seat.
2. Press the start/stop button without

pressing brake pedal.

3. Amber colour LED on start/stop button will turn ON.
4. Limited information will be displayed on instrument cluster and steering will be unlocked. Vehicle remains OFF.

##### Step 2

1. Press the start/stop button without pressing brake pedal again.
2. Green colour LED on start/stop button will turn ON.
3. Vehicle will remain OFF but all electrical equipment and infotainment system can be used.

##### Step 3

1. Press the brake pedal and then press start/stop button to start the vehicle.
2. Green colour LED on start/stop button will turn ON.
3. Once the vehicle is started successfully, the green colour LED on start/stop button stays ON.

## STARTING AND DRIVING

### Vehicle Passive Stop - Stationary Condition

- Press the start/stop button with or without brake.
- ACC and IGN turns OFF.
- LED on start/stop switch turns OFF.

#### WARNING

When vehicle is in OFF mode, if user tries to lock the vehicle from outside by pressing door handle switch and PEPS detects that the smart key is left inside the vehicle, an audio warning/ chime is sounded and doors will not get locked.

### Backup Start

To start the vehicle when smart key battery voltage is low, the user needs to press start/stop button two times with an interval of 2.5 seconds after pressing the brake with valid smart key at its location (in Center Console).



#### NOTE

- If smart key is inside the vehicle and on pressing start/stop switch, if start/stop switch green LED blinks more than 10 sec. duration then contact TPEM Authorised Service Centre.
- If ESCL (Electronic Column Steering Lock) is not unlocked properly, then vehicle doesn't go into ACC mode.

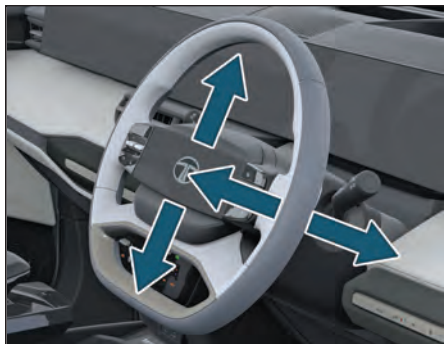
## STEERING SYSTEM AND ADJUSTMENT

### Steering Lock and Unlock



### To adjust the Steering Wheel

1. Adjust the seat to a comfortable position.
2. Push the tilt lever completely down to unlock the steering column.
3. Adjust the steering wheel to the desired position by moving the steering wheel in axial and radial direction (Telescopic & Tilt).



4. Pull the tilt lever up completely to lock the steering column.
5. Make sure that steering wheel is securely locked by checking up and down direction.

### NOTE

*When adjusting the steering wheel, make sure that:*

- *You can see control pedals without any obstacles.*
- *You can see all the displays in the instrument cluster clearly.*

- *Hold the steering wheel such that both hands are on the outer edge at the 3'o clock and 9'o clock positions.*

### WARNING

- Before you start the car, make sure the steering wheel position is locked. Do not unlock or adjust the steering wheel while the vehicle is in motion.
- Do not put your hand through the steering wheel to operate, this may result in an accident.
- Do not alter the steering wheel in any way. Avoid painting it or installing stickers, decorative items, covers, or other accessories. Such modifications may interfere with proper airbag deployment.
- Repair or replacement of the steering wheel should be done exclusively at TMPVL Authorized Service Centre.

### WARNING

You could lose control of your vehicle if you try to adjust the steering wheel while driving. There is a risk of an accident.

## STARTING AND DRIVING

### ELECTRIC POWER ASSISTED STEERING (EPAS)

Your vehicle is equipped with electric power assisted steering system. The EPAS system makes steering the vehicle easier with less effort.

In the EPAS system, steering effort increases with vehicle speed for improved control at higher speeds and becomes lighter at lower speeds to enhance maneuverability.

If the vehicle is 'OFF' or if the EPAS system becomes inoperative, the vehicle still can be steered with more steering effort however ignition switch should be in ON position else steering wheel will get locked.

This EPAS system is available with the following assist features

1. Speed sensitive assist control
2. Active return control

#### NOTE

- A click noise may be heard from the EPAS relay after the ignition switch

is turned ON or OFF position.

- The steering wheel may not unlock normally in some cases when ESS button is pressed. If this happens, turn the steering wheel to the right or left slightly to unlock the steering wheel while pressing ESS button.
- Contact the nearest TPEM Authorised Service Centre if in case of the above scenarios.

#### WARNING

In case of below malfunction conditions, then, take your vehicle to the nearest TPEM Authorised Service Centre and have the EPAS system checked as soon as possible.

- Vehicle noise may be heard when the vehicle is driven at low speeds.
- If the EPAS system does not operate normally, the warning light will illuminate on the instrument cluster. The steering wheel rotation may become difficult to control or operate.

### DRIVING TIPS

- Plan your tour in advance get help from google maps, check for road conditions, kms to be covered in a day, halting destination, fuel station, food station, hospitals and roadside assistance in case of emergency.
- Always wear your seat belt and ensure all passengers so the same.
- Follow speed limits and adjust your speed according to road condition.
- Keep safe following distance from the vehicle in front of you.
- Obey traffic rules and sign at all times.
- Avoid distraction like mobile phone, texting while driving.
- Check your mirror frequently and be aware of surrounding.
- Never drive under the influence of alcohol or drugs.
- Avoid aggressive driving behaviour like tailgating or excessive over speeding.
- Regularly maintain your vehicle to ensure good working condition.

- Take adequate break and rest every after 200 kms.
- Check tyre pressure, coolant, oil indication, any leakage under the vehicle regularly during travelling.
- Use horn, light and indicators as per condition.

### WARNING

We strongly advise you not to drive in late hours/sleep hours to avoid fatality because body response is slow to respond to any situation in this time period.

### NOTE

*Remember, safe driving is crucial for your safety and safety of others on the road, drive responsibly following above driving tip may help is good and enjoyable travelling experience.*

## Driving Through Water / Flooded Water

- Drive through calm water only if it is not

deeper than 300mm and at this depth, the vehicle speed to be maintained at creep speed.

- If car gets completely or partially submerged in water, switch OFF the ignition, evacuate the car and call RSA (Roadside Assistance) at 18002098282 for assistance
- We do not recommend you to drive through flooded water as it may enter the vehicle interior and motor compartment which could damage power electronic, electronic & electrical systems. Judge the depth of water before driving through it.

### WARNING

If vehicle is flooded with water, do not attempt to start the vehicle. Tow the vehicle to a safe place. Contact a nearest TPEM Authorised Service Centre.

## Driving on wet roads

- Check wiper blades, lights and brakes for proper functioning and condition.
- Check the tyre treads depth, the con-

dition of the tread and tyre.

- Avoid harsh braking and sharp turns. It may cause loss of control and lead to skid.
- Keep lights 'ON' if visibility is poor.

### WARNING

On wet road or during light showers, "Aquaplaning" can occur. "Aquaplaning" is the loss of direct contact between the road surface and the vehicle's tyres due to a water film forming between them. Steering or braking the vehicle can be very difficult, and loss of control can occur.

There is no hard and fast rule about aquaplaning. The best advice is to slow down when the road is wet.

### NOTE

*If you have driven for a long time in heavy rain without braking, there may be a delayed reaction from the brakes when braking for the first time. You have to press the brake pedal more firmly.*

## STARTING AND DRIVING

*Maintain a greater distance from the vehicle in front.*

### Driving on snowy roads

While driving on snow, it is advisable to use the snow chain on roads. Follow assembly and safety instruction provided by the snow chain manufacturers.

#### WARNING

Wet ice (0°C and freezing rain), snow or ice can be slick and very hard to drive on. The vehicle will have much less traction or “grip” under these conditions. Try to avoid driving on wet ice until the road is salted or sanded.

- Whatever the conditions, drive with caution. Accelerate and slow down with care. If accelerating too fast, the drive wheels will lose even more traction.
- Allow more stopping distance under these conditions. Braking should be started sooner than on dry pavement.

- Allow greater following distances on slippery roads.
- Watch for slippery spots (glare ice).
- These may appear on an otherwise clear road in shaded areas. If a patch of ice is seen ahead, brake before reaching it. Try not to brake while on the ice, and avoid any sudden steering maneuvers.
- Do not use cruise control on slippery roads.

### Driving at Night

- Ensure that all lights are working and windshield, window glasses are clean.
- Drive more slowly at night than in the daytime, as the visual range is restricted at night. Maintain a speed such that you can stop within illuminated distance of headlamps.
- Do not use the high beam unless inevitable. It may dazzle the driver of the oncoming vehicle, thus causing an accident.
- Use headlamp main/dip beam to alert

other road users on turns/ cross roads etc.

- Use side indicators for lane change or turning

### Driving on Gradients

When climbing gradient, plan it in advance so that the vehicle speed is maintained.

When driving down a hill, the regenerative braking should be used. Do not switch OFF the vehicle.

#### WARNING

On long and steep gradients you must reduce the load on the brakes by taking your foot off the accelerator pedal. This allows you to take advantage of regenerative braking effect and helps avoid overheating of service brakes resulting in reduced braking efficiency.

### Driving in Heat and cold weather

The heating and cooling on the car uses energy from the battery. Set temperatures to a comfortable 24° C - 26° C with Auto mode and Econ activated, and see the comfort as well as the range go up signifi-



cantly

### CAUTION

Do not park the vehicle in temperatures below -22°C for more than seven days. If the outside temperature is -22°C or less, the Lithium Iron Phosphate battery may freeze and it cannot be charged or provide power to run the vehicle. Move the vehicle to a warm location

### Driving on Highway

Do not change the accelerator pedal inputs rapidly. Go as smooth as possible. EV's being instantaneous torque and power – there is very little lag in translating the pedal input to vehicle response.

Stopping distance progressively, increases with vehicle speed. Maintain a sufficient distance between your vehicle and the vehicle ahead.

For long distance driving, perform safety checks before starting a trip and take rest at certain intervals to prevent fatigue.

EV's give best range between speeds of 40 – 60 kmph. Therefore they are ideal for

city applications. Driving in this range along with following of other points here will add your mileage significantly. A driving speed band of 60 to 80 kmph is recommended on highway. At high speed, the range may result in significant drop in range.

## STARTING AND DRIVING

---

### TIPS TO GET MAXIMUM RANGE WHILE DRIVING EV

#### Acceleration, Braking and Coasting

**Acceleration:** This vehicle has a single speed automated gearbox. In accelerating mode, the torque supplied by the motor via the gearbox to the front wheels is linear in nature.

#### 1. Regenerative Braking

- This vehicle is equipped with a regenerative brake system. The primary purpose of the regenerative brake system is to provide some power to help recharge the Lithium Iron Phosphate battery and extend driving range.
- The electric motor when decelerating and braking transforms kinetic energy to electrical energy in order to charge the high voltage battery. (Torque is applied in the opposite direction when decelerating to generate



braking force and electricity).

- A secondary benefit is an effect similar to “vehicle braking” seen in IC vehicle cars. Here, it depends on HV battery condition.
- In the Drive mode, when the accelerator is released, the regenerative brake system provides some deceleration and generates power for the high voltage battery.
- Power is also generated when the brake pedal is applied.
- The calibration on the regen is done in such a way that most people can experience a “single pedal” drive at most times, just lift your feet of the accelerator pedal to slow the vehicle down and gain range. Brake lightly if required
- When you brake and take your foot off the accelerator pedal, more regenerative brake is applied than in the drive mode. However, during high-speed driving you may feel that regenerative brake provides less deceleration than the motor braking in an ordinary vehicle. This is normal.
- Less deceleration is provided by the regenerative brake system when the Lithium Iron Phosphate battery is fully charged. Regenerative brake is automatically reduced when the high voltage battery is fully charged to prevent it from overcharging.
- Regenerative brake is also automatically reduced when the battery temperature is high/low to prevent battery damage.
- The brake pedal should be used to slow or stop the vehicle depending on traffic or road conditions. The vehicle brakes are not affected by regenerative brake system operation.
- Use correct Regen Selection for better range.
- It is recommended to use below mentioned Regen level for better mileage:
  - Regen level “0” or “1” in case of plain highway.
  - Regen Level “1” & “2” In city drive with normal traffic.

- Regen Level “2” & “3” In city drive with heavy traffic.

### 2. Interior Climate

Heating and Air conditioning system uses energy from the high voltage battery and this reduces range. For maximizing range during air-conditioning on driving, it is recommended to set the air conditioning in Auto mode with Econ activated. Also the set temperature should be set between 24-26 deg C. EV is equipped with Remote Air conditioning. You can remotely start Air Conditioning system. However, this also consumes energy from the high voltage battery and reduces the driving range.



### 3. Driving Speeds

At high speeds, greater than 80kmph, high amount of energy is spent in propelling the vehicle and hence reduces range. Similarly, idling for long duration



also reduces range significantly

### 4. Driving Style

Driving behavior has a significant influence on the driving range of an electric vehicle. Frequent and heavy accelerations will have a detrimental impact on car's driving range whereas travelling at a steady pace, in between 40-80 km/hr will help an electric car to maximize range. Predictive driving with gradual acceleration reduces your reliance on hard braking. This helps conserve energy during acceleration and regenerate optimally during deceleration. However, brake should be applied as necessary to avoid hazards to occupants and surrounding.



### 5. Tyre Pressure

Maintaining specified Tyre pressure only ensures comfortable ride comfort but also maximizes range by minimizing rolling



losses of the vehicle. It is advisable to regularly monitor and maintain the tyre pressure within specified limits.

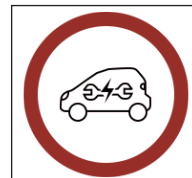
### 6. Unauthorized Electrical Accessories

Unauthorized aftermarket electrical accessories can potentially consume higher energy than factory fitted ones and may affect range directly. They can also lead to functional complications and lower component life in the long run. It is recommended to fit only TATA Genuine Accessories at TPEM Authorised Service Centre.



### 7. Vehicle Maintenance

Vehicle to be serviced regularly as per service schedule in authorized service stations. It helps in maximizing vehicle performance and component life.

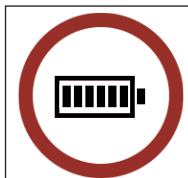


## STARTING AND DRIVING

---

### 8. Battery Charging

For every 4 consecutive fast charges, 1 complete slow charge is recommended (<20% to 100% SoC) to ensure consistent performance and better health of the high voltage battery. This improves charge balancing during charging and deliver better driving range.



### 9. Dead Weight

Dead weight like bags, boxes etc. which is not required should be removed from the car. This lowers the energy required to propel the vehicle and improves driving range.



### Limp Home Strategy

| Soc Intervention |                                                                                                                                            |             |              |            |               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|------------|---------------|
| SoC Zone         | IC message and state                                                                                                                       | Performance | Gradeability | Max. Speed | Cabin cooling |
| SoC <=25%        | If SoC drops to 25% when the vehicle is in Sports mode, it will automatically shift to City mode and the same will be displayed on cluster | No change   | 18%- 29%     | No change  | No change     |
| SoC <=10%        | SoC gauge 1 <sup>st</sup> bar ON with single chime. Limp Home tell tale on cluster. "RECHARGE" will display.                               | Reduced     | 18 %         | 80 kmph    | Reduced       |
| SoC <=5%         | SoC gauge 1 <sup>st</sup> bar will blink. Blinking Limp Home with continuous chime.                                                        | Reduced     | 18 %         | 60 kmph    | No cooling    |

| Fault Intervention |                                                             |             |              |            |               |
|--------------------|-------------------------------------------------------------|-------------|--------------|------------|---------------|
| Sr. No.            | Telltale Indication                                         | Performance | Gradeability | Max. Speed | Cabin cooling |
| 1                  | HV critical ON + Single Chime. Limp home Tell-tale blinking | Reduced     | 18 %         | 60 kmph    | No change     |
| 2                  | Limp home Telltale blinking + single chime                  | Reduced     | 18 %         | 60 kmph    | No change     |

# STARTING AND DRIVING

## Limp Home Condition of EV

In situations when certain conditions in the vehicle are not met or when some fault arises in the vehicle, the vehicle control unit intervenes and puts the vehicle into Limp Home Mode. The vehicle will give reduced performance in these situations. These limp home interventions are defined on two levels which are provided in the table.

### NOTE

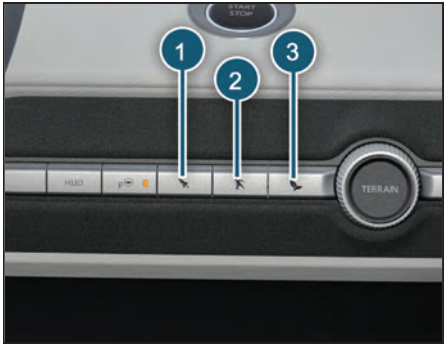
*There will be a single audio chime whenever the vehicle goes into Limp Home Intervention along with the IPC message. Sports mode cannot be selected if the vehicle SoC is below 25% or the vehicle is in limp home mode. If the vehicle is already in sports mode, it will automatically switch to drive mode. A message 'Gear Shift not allowed' will be displayed with an audio warning. When level 1 intervention takes place, the vehicle will not start in the next ignition cycle.*

## Limp Home Mode Telltale Warnings

|  | Blinking             | HV Critical Fault<br>Contact TPEM Au-<br>thorised Service<br>Centre.                                 |
|-----------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------|
|  | Continu-<br>ously ON | HV Critical Fault<br>Mobility is not al-<br>lowed<br>Contact TPEM Au-<br>thorised Service<br>Centre. |
|  | Continu-<br>ously ON | Vehicle has entered<br>in<br>Limp Home mode                                                          |

## DRIVE AND TERRAIN MODES

### Drive Mode



1. Boost Mode
2. Sport Mode
3. ECO Mode

### 1. Boost Mode

Press the switches to select / deselect the respective drive mode.

Under specific conditions, when activated boost mode, increases motor Output for a Specific period of time beyond normal driving levels.

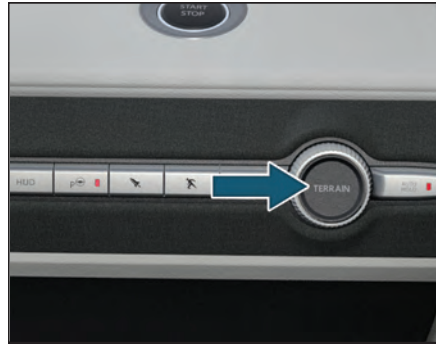
### 2. SPORT Mode

Press this switch for more torque from Motor.

### 3. ECO Mode

Press this switch for Optimum Motor Torque and Power output for efficient performance.

### Terrain Mode Rotary Switch



Rotate the knob clockwise to select the terrain mode. Selected mode will be displayed in cluster.

**In 4x4 AWD vehicle there are following modes :**

- Normal Road Mode
- Grass / Snow
- Mud Ruts / Gravel Mode
- Sand Mode
- Rock / Crawl Mode
- Custom Mode

- Wet Mode
- Rough mode

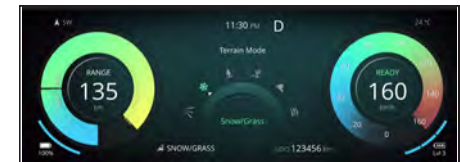
### Normal Road Mode

In this Mode System will adapt to the prevailing surface conditions automatically and enables control settings based on the condition sensed.



### Grass / Snow

This mode allows you to have greater stability under conditions of bad weather. For use on and off-road on surfaces with poor traction, such as roads covered with snow.



## STARTING AND DRIVING

### Mud Ruts / Gravel Mode

This mode allows you to use on surfaces with poor traction, such as roads covered by mud or gravel. The transmission is set to provide maximum traction.



### Sand Mode

This mode allows you to use on surfaces with poor traction, such as roads covered by Sand. The transmission is set to provide maximum traction.



### Rock / Crawl Mode

This mode allows you for off-road driving. This mode used such as roads covered

rocks. The transmission is set to provide maximum traction.



### Custom Mode:

As user select 'Custom' it will take to HMI screen for customized selection of different feature like Drive mode, Regen level, AVAS – Sound level, mood lighting options etc.



### Wet Mode (if equipped)

The mode is designed for driving in the rain. It is characterized by having better traction and handling so you can effectively and safely drive during these condi-

tions.



### Rough Road Mode (if equipped)

Rough Road ore optimizes the vehicle behavior for driving over rough road or other unyielding obstacles.

**In 4x2 2WD vehicle there are following modes :**

- Normal Road Mode
- Wet Mode (if equipped)
- Rough Road Mode (if equipped)
- Custom Mode

### Normal Road Mode

In this Mode System will adapt to the prevailing surface conditions automatically and enables control settings based on the condition sensed.



## SHIFTER MODES



### Neutral (N)

The monoshifter is in 'N' mode position and 'N' will be indicated in Instrument Cluster.

### Drive Mode (D)

The monoshifter is in 'D' mode position and 'D' will be indicated in Instrument Cluster.

### Park (P)

The monoshifter is in 'P' mode position and 'P' will be indicated in Instrument Cluster. When starting the vehicle or parking



### Wet Mode (if equipped)

The mode is designed for driving in the rain. It is characterized by having better traction and handling so you can effectively and safely drive during these conditions.



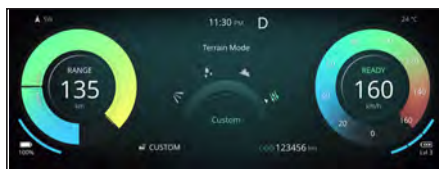
### Rough Road Mode (if equipped)

Rough Road mode optimizes the vehicle behavior for driving over rough road or other unyielding obstacles.



### Custom Mode:

As user select 'Custom' it will take to HMI screen for customized selection of different feature like Drive mode, Regen level, AVAS – Sound level, mood lighting options etc.



## STARTING AND DRIVING

---

the vehicle. Apply the parking brake whenever the vehicle is to be parked.

### **Reverse (R)**

The shifter knob is in 'R' shifter position and 'R' will be indicated in Instrument Cluster.

## Monostable Shifter

Your vehicle is equipped with Monostable shifter, where the shifter returns to its stable position the moment it is released.

It is provided with 2 UP positions and 2 DOWN positions: UP2-UP1-Stable-DOWN1-DOWN2.

The user shall release the shifter after each shift, as any new shift will be possible from stable position only.

The user has to confirm the desired shifting position on display.

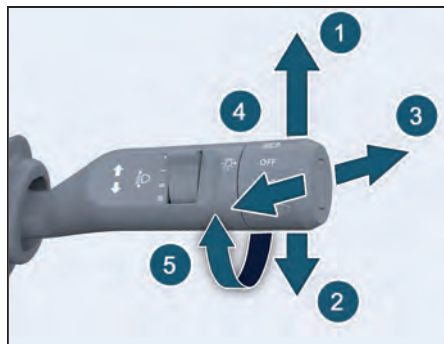
Shifter shall not be attempted from non-stable position by continuously holding the shifter in the previous shift position.

| Gear shift Position |        | Mono-stable shifter movement         | Unlock Button Press | Brake Pedal Press |
|---------------------|--------|--------------------------------------|---------------------|-------------------|
| UP2                 | P to R | Stable position -> Up2 (2nd detent)  | Yes                 | Yes               |
|                     | D to R | Stable position -> Up2 (2nd detent)  | Yes                 | Yes               |
|                     | P to N | Stable position -> Up1 (1st detent)  | Yes                 | Yes               |
| UP1                 | N to R | Stable position -> Up1/Up2           | Yes                 | Yes               |
|                     | D to N | Stable position -> Up1 (1st detent)  | Yes                 | No                |
| Stable Position     |        |                                      |                     |                   |
| Down 1              | P to D | Stable position -> Down1/Down2       | Yes                 | Yes               |
|                     | R to N | Stable position -> Down1(1st detent) | No                  | No                |
|                     | R to D | Stable position -> Down2(2nd detent) | Yes                 | Yes               |
| Down 2              | N to D | Stable position -> Down1/Down2       | Yes                 | Yes               |

## STARTING AND DRIVING

### OPERATING OF LIGHTS AND WIPERS

#### Combi-switch (RH Stalk)



#### 1. Left Turn Signal

Move the lever to horizontally forward.

#### 2. Right Turn Signal

Move the lever to horizontally downward.

#### NOTE

*When the turn is completed, the signal will cancel and the lever will return to its normal position.*

#### 3. High Beam

Move the lever forward to select the high beam function. Pull the lever back to normal for low beam.

#### 4. High Beam Flash (Spring Return)

To flash the high beam, pull the lever towards you from the normal position. It will return to its normal position when you release it.

#### 5. Headlamp Rotary Switch

OFF Position



All lamps will remain 'OFF.'

Parking Lamp



Rotate stalk to turn 'ON' the Parking lamps.

Auto Light



The headlights will be automatically switched ON depending on ambient light

conditions (while entering a tunnel or when it is twilight).

#### Day Time Running Lamps (DRL)



Day time Running Lamps (DRL) are used to increase the visibility of the vehicle to other drivers during daytime.

To activate and deactivate DRL, keep the ignition switch is 'ON' position and switch the parking lamp ON-OFF twice within approx. three seconds.

Activation and Deactivation of DRL can be done by DRL soft switch, which is available on the Head Unit Display.

#### Low Beam



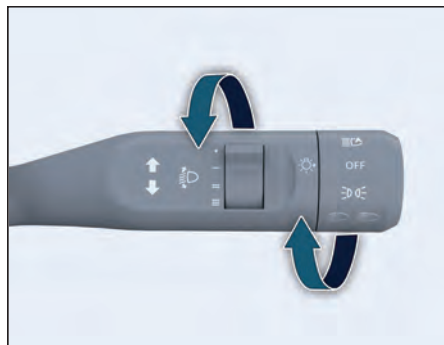
Rotate stalk to turn 'ON' the Low Beam function.

#### 5. Lane Change Signal

To signal a lane change, move the lever slightly up or down to the point where the turn signal light begins to flash, but the lever does not latch. The turn signal will

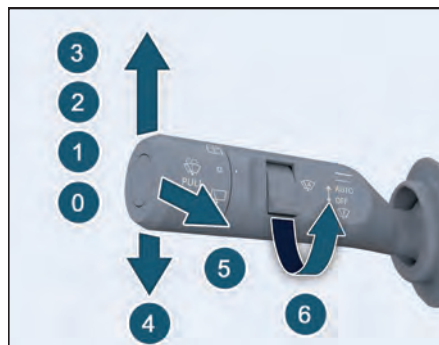
flash six times automatically.

### Head Lamp Leveling Rotary Switch



Inner rotary switch on right hand stalk is provided for head lamp leveling. With the inner rotary switch, Head lamp leveling can be done with head lamp in Low Beam and in 'ON' position. Select correct position before start of trip, when the vehicle is stationary. Depending on the number of passengers and luggage in the vehicle headlamp focus may change. This can be adjusted by rotating the knob to one of the three level positions.

### Combi-switch (LH Stalk)



#### 0. Off" Position

The wiper is switched 'OFF'.

#### 1. Intermittent Wipe

Push the stalk upwards to operate intermittent wipe.

Inner rotary switch on left hand stalk is provided for intermittent front wiper delay. The switch has five delay timers.

#### 2. Slow Wipe

Push the stalk towards position (2) for continuous slow wipe.

#### 3. Fast Wipe

Push the stalk towards position (3) for continuous slow wipe.

#### 4. Flick Wipe (Spring Return)

Pull the stalk downwards and hold it for continuous wipe, the wiper continuously wipes across the windshield at low speed till the stalk is released.



#### 5. Front Windshield Washer

- Pull the lever little longer, to spray the washer fluid on the windshield.
- The windshield wipers will operate for three cycles after the lever is released and for one more cycle after five seconds.



## STARTING AND DRIVING

### Manual Mode (if Equipped)

- Pull the lever little longer, to spray the washer fluid on the windshield.
- The windshield wipers will operate for three cycles after the lever is released and for one more cycle after five seconds.

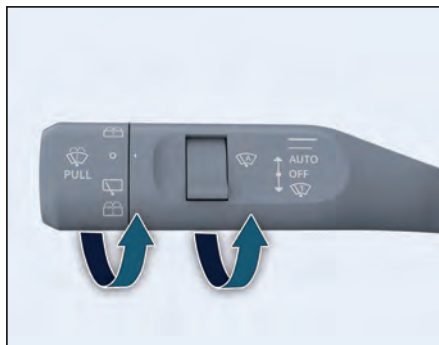
### Auto Mode (Rain / Light Sensor (if equipped))

- If your vehicle is fitted with rain and light sensor, the wipers will automatically wipe the windscreen, if it senses rainfall. Make sure that the wiper stalk is in Auto position.
- The integrated rain and light sensor is mounted on front windshield glass to sense rain and light. As per the input from sensor, the wipe and light functions will work automatically.

#### NOTE

*When you crank the engine, the supply to washer motor is momentarily cut off.*

### Rear Wash And Wipe



### Rear Wipe

Turn the rotary knob counter clockwise such that it aligns its positions with arrow mark to operate rear windshield wiper continuously.



#### NOTE

*Rear wiper will not work as long as tail-gate is open.*

### Rear Windshield / Wiper and Washer Switch

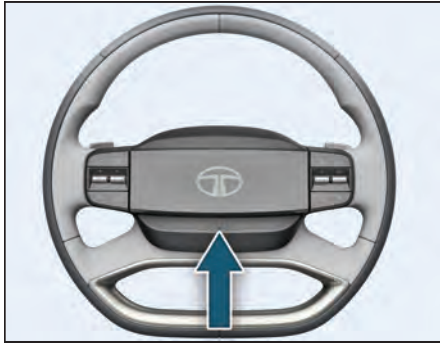
Turn the rotary knob counter clockwise such that it aligns its positions with arrow mark and hold it to operate rear windshield wash and wipe function. It will return to 'Rear wipe' position as soon as it released and continue to wipe.



#### WARNING

If you operate wash and wipe function for more than 15 seconds the controller cuts off the supply to the washer motors to avoid overheating.

### HORN



Horn is located on steering wheel. Use it whenever required.

#### **WARNING**

- Check out for No Horn zone, where use of horn is prohibited.
- Avoid using sharp objects which can create scratch on illuminated emblem.
- Do not use sharp objects to clean the gap between horn pad and steering wheel.

- Do not use carwax, window cleaner, glass coat and chemical solvents such as strong detergents containing high alkaline or volatile organic solvents (gasoline, acetone etc). This may damage the driver airbag deco piece.
- Please use a soft, clean cloth dampened with normal water to clean the steering wheel.
- Avoid using extremely hot or cold water, as rapid temperature changes can cause the plastic components to become brittle. Do not use high-pressure water for cleaning.
- Wipe the steering wheel with soft cloth only.
- Do not insert any coins, clips, foreign materials etc in the gap between the Driver Airbag and Steering Wheel. This may lead to no horn condition and horn will not work properly.

- Do not use any unauthorized accessories on steering. Devices that claim to offer improved grip or enhanced surface finish may actually compromise functionality and safety.
- Please refer to the car care section from “MAINTENANCE AND CARE” section for the steering wheel cleaning precautions.

## STARTING AND DRIVING

### SEATS ADJUSTMENTS

#### Front Seat Manual

##### Driver Seat



Following seat adjustments can be carried out manually.

1. Driver Seat Backrest Angle Adjustment
2. Driver Seat forward/backward adjustment
3. Driver Seat height adjustment

#### WARNING

Do not adjust the driver's seat while driving. Adjusting the seat while driving could cause the driver to lose control of the vehicle.

#### 1. Driver Seat Backrest Angle Adjustment

To change the seat back rest angle, lean forward slightly and pull up the lever (1). Adjust seat backrest until it reaches desired comfortable position. Make sure that lever returns to its original position and seat is securely latched.

#### NOTE

*Adjust the seat backrest until your arms are slightly angled when holding the steering wheel.*

#### WARNING

Never travel in a moving vehicle with the seat backrest in an excessively reclined position as this can be danger

ous.

You could slide under the seat belt in a collision.

#### 2. Driver forward / backward adjustment

Lift lever (2) and slide the seat forwards or rearwards. Release lever and make sure that seat is securely latched.

#### NOTE

*Adjust the driver seat position in such a way that the driver will be able to operate the control pedals conformably.*

#### 3. Driver Seat height adjustment

To raise the seat, pull and continue pumping the lever (3) in the upward direction until the seat is at the desired height. To lower the seat, pump the lever downward until the seat is at desired height.



### Co-driver Seat



Following seat adjustments can be carried out manually on Co-Driver seat.

1. Co-Driver Seat forward/backward adjustment
2. Co-Driver Seat Backrest Angle Adjustment

#### **WARNING**

Do not adjust the driver's seat while driving. Adjusting the seat while driving could cause the driver to lose control of the vehicle.

### 1. Co-Driver seat Backrest Angle Adjustment

Similar to driver seat, to change the seat back rest angle, lean forward slightly and pull up the lever (1). Adjust seat backrest until it reaches desired comfortable position. Make sure that lever returns to its original position and seat is securely latched.

### 2. Co-Driver forward / backward adjustment

Similar to driver seat, lift lever (2) and slide the seat forwards or rearwards. Release lever and make sure that seat is securely latched.

### Seat Cushion- Thigh Extension

User can extend the driver and co-driver seat cushion- thigh extension as per their comfort.

- Slide your legs forward to ensure your thighs have minimal contact with thigh extender front area.



- Grab the lever on the front of the seat and pull upwards.
- Adjust the length of the “thigh extension” by pulling in forward direction until desired position is achieved.

## STARTING AND DRIVING



- Ensure the cushion thigh extender is properly locked in position
- Release the handle and make sure that the seat cushion thigh extender has reached the correct position

### *i* NOTE

- *Make sure the cushion thigh extender is fully closed during entry and exit.*
- *Do not apply excessive or abusive loads on the cushion thigh extender, the maximum load should not exceed 10 kg.*

ceed 10 kg.

### Front Power Seats

#### Driver Seat



1. Seat Backrest Angle Adjustment
2. Forward / backward adjustment
3. Seat height adjustment

#### 1. Seat Backrest Angle Adjustment

To change the seat back rest angle, lean forward slightly and operate the button slightly (1) as per arrow mark either forward or rearward as required. Adjust seat

backrest until it reaches desired comfortable position and then release the button.

### *i* NOTE

*Adjust the seat backrest until your arms are slightly angled when holding the steering wheel.*

#### 2. Forward / backward adjustment

To move seat forward push the button (2) slightly forward as per the arrow mark, to move seat rearward pull the button in rearward direction make sure that seat is securely latched.

### *i* NOTE

- *Adjust the driver seat position in such a way that the driver will be able to operate the control pedals comfortably.*
- *Do not apply over force on power operated buttons as it may damage them. Button operates with slight finger force as they are electronically controlled.*

- *Avoid unnecessary operation of power seat adjustment buttons as it consumes power from vehicle battery.*
- *Do not operate more than one operation button simultaneously.*

### 3. Driver Seat height adjustment

To raise the seat, pull and continue pumping the button (3) in the upward direction until the seat is at the desired height. To lower the seat, pump the button downward until the seat is at desired height.

### Co-driver Seat



1. Seat Backrest Angle Adjustment
2. Forward / backward adjustment

#### 1. Co-Driver seat Backrest Angle Adjustment

To move seat forward push the button (2) slightly forward as per the arrow mark, to move seat rearward pull the button in rearward direction make sure that seat is securely latched

To change the seat back rest angle, lean forward slightly and operate the button slightly (1) as per arrow mark either for-

ward or rearward as required. Adjust seat backrest until it reaches desired comfortable position and then release the button.

#### 2. Co-Driver forward / backward adjustment

Similar to driver seat, lift button (2) and slide the seat forwards or rearwards. Release button and make sure that seat is securely latched.

## STARTING AND DRIVING

### Manual Boss Mode (if equipped)



Manual Boss mode is a feature that can make comfort to the rear seat passenger by increasing the leg room.

Rotate the boss mode button clockwise and push / pull the front seat to slide forward / rearward to create more leg room space for rear sea occupant.

#### NOTE

- *User is advised to use Manual Boss mode when there is no passenger on the Co-driver Seat for maximum*

*com-fort and safety.*

- *If there is Passenger on the Co-driver seat and Rear Seat Passenger presses the Boss Mode switch then Seats will adjust as per the buttons pressed which may injure the Co-driver Passenger.*
- *Co-driver Seat position obscures visibility of co-driver ORVM, when seat is reclined to forward most position and seat few notches ahead on track. Driver has to cautiously position co-driver seat, such that the ORVM vision is unobscured.*

### SEAT ZONE MODULE (SZM) (if equipped)

The Seat Zone Module (SZM) is ECU used to move the seat of the driver to tailor the shape or position of the seats for maximum comfort at the push of a button and then these seat positions can be stored and recalled as per user preference.

Along with Seats the driver and co-driver Mirror positions can also be adjusted and Mirror positions can be stored and recalled as per user preference in forward as well as reverse shift.

The Memory seats shall be adjusted in three axis directions, allowing adjustments for the seat (e.g. forward or back-ward, height up or down, and Recline or incline). Along with this Easy Entry Exit feature is also provided.

The Memory ORVM shall be adjusted in Vertical and Horizontal Direction using joystick available at driver door.

### SZM – Switch Location On Driver Seat



### SZM Features Primary Movements



By pressing the Switches, seat can be adjusted at comfort level in 3 directions as above. By pressing the Switches, seat can be adjusted at comfort level in 3 directions as above.



| Sr. no. | Mechanism   | Direction          |
|---------|-------------|--------------------|
| 1       | Squab       | Incline/Recline    |
| 2       | Slide       | Forward / Rearward |
| 3       | Lift/Height | Up/Down            |

## STARTING AND DRIVING

### Seat And ORVM Memory Store Operation



To store a seat position user shall first bring a seat to intended position by pressing primary switches (Lift/Slide/Squab) and to store the ORVM position user shall bring the ORVM in intended position by using the Joystick.

User can store maximum 3 memory seat and ORVM positions as follow:

#### I. To store seat and ORVM position in M1 button:

Press the SET button, and within 5 sec-

onds (or until the SET button LED lights up), press the M1 button to save the seat and ORVM positions at M1.

#### II. To store seat and ORVM position in M2 button:

Press the SET button, and within 5 seconds (or until the SET button LED lights up), press the M2 button to save the seat and ORVM positions to M2.

#### III. To store seat and ORVM position in M3 button:

Press the SET button, and within 5 seconds (or while the SET button LED is illuminated), press the M3 button to store the seat and ORVM positions at M3.

- When the Seat and ORVM position is stored, user can see pop up for respective Memory Button on IPC and a chime sounds to confirm the settings have been stored.
- The seat and ORVM memory storage function operates in the Ignition OFF (Awake, Accessory, and Accessory Delay), Ignition ON, and Run power modes.



### Seat And ORVM Memory Store Operation

- To recall the Stored Seat and ORVM Position, Press M1/M2/M3 as per user choice.
- After Pressing M1/M2/M3 seat and ORVM will automatically adjust to previously stored seat and ORVM position.
- When the M1/M2/M3 position is recalled, user can see pop up for respective Memory Button on IPC and a chime sounds to confirm the settings have been recalled.

### NOTE

- When the vehicle speed > 6 kmph ( $\pm 1$ kmph), to recall the stored seat position, long press M1/M2/M3.
- If user want to recall seat position stored beyond Pre stop, then Seat will travel till Pre stop position.
- If the speed of the vehicle is > 6kmph ( $\pm 1$ kmph), & switch M1/M2/M3 is one touch/short pressed, memory seat position will not be recalled.

### Cancel Ongoing Seat And ORVM Recall Operation

Ongoing Seat Recall operation can be cancelled by following ways:

- During ongoing recall operation any switch is pressed (For E.g. Lift/ Slide/ Squab)
- During ongoing recall operation any Memory Button (M1/M2/M3) or SET button is pressed.
- During ongoing recall, vehicle speed is increased above 6 km/h ( $\pm 1$ kmph).

Ongoing OVM Recall operation can be cancelled by following ways:

- During ongoing recall operation joystick is pressed.
- During ongoing recall operation any Memory Button (M1/M2/M3) or SET button is pressed.
- During ongoing recall, vehicle speed is increased above 6 km/h ( $\pm 1$ kmph).

When the Seat or ORVM recall cancelled, user can see pop up for respective Memory Button on IPC and a chime sounds to confirm the settings have been stored.

### NOTE

- The stopped seat and ORVM movement will resume once user presses again any Memory Button.
- While recalling the positions, if seat switch is pressed then only seat recall operation will cancel while the ORVM position recall operation will continue.
- While recalling the positions, if Joystick is pressed then only ORVM re-

*call operation will cancel while the Seat position recall operation will continue.*

### Easy Entry Operation

By opening a driver door in IGN OFF state(driver entry), Seat will adjust height and Slide axis and will travel to backwards for predefined time and height axis will go to lowest height position for pre calibrated time.

- When the Driver door is closed, Seat will adjust slide and height and will go to previous seat position.
- If user close door before completion of height down motion, height up motion will start.
- If user close door before completion of slide rearward motion, slide for-ward motion will start.

### Easy Exit Operation

- By opening a driver door in IGN OFF state (driver exit), Seat will adjust height and Slide axis and will travel to backwards for predefined time and



## STARTING AND DRIVING

height axis will go to lowest height position for pre calibrated time.

- When the driver door is closed, after driver gets out of the vehicle, seat will adjust slide and height and will go to previous seat position.
- If user close door before completion of height down motion height up motion will start.
- If user close door before completion of slide rearward motion, slide forward motion will start.

### ORVM Memory Store Operation In Reverse Mode

When the IVI setting is enabled for ORVM recall operation in Reverse Mode, to store the ORVM position in Reverse Mode, the user shall engage reverse Mode and adjust the ORVM to the intended position using the joystick.

Users can store ORVM positions as follows:

#### I. To store ORVM position in the M1 button:

Press the SET button and within 5 seconds

or until the SET button LED glows, press the M1 button to store the ORVM position at M1.

#### II. To store ORVM position in the M2 button:

Press the SET button and within 5 seconds or until the SET button LED glows, press the M2 button to store the ORVM position at M2.

#### III. To store ORVM position in the M3 button:

Press the SET button and within 5 seconds or until the SET button LED glows, press the M3 button to store the ORVM position at M3.

#### NOTE

*Any existing ORVM memory positions are overwritten when storing a new memory position.*

### ORVM Memory Recall Operation In Reverse Mode

- To recall the stored ORVM position, press M1/M2/M3 as per the user's choice.

- When the IVI setting is enabled for ORVM recall operation in Reverse Mode, after pressing M1/M2/M3, the ORVM will automatically adjust to the previously stored position.
- When the M1/M2/M3 position is recalled, the user can see a pop-up for the respective memory button on the IPC, and a chime sounds to confirm that the settings have been recalled.

#### NOTE

*When the vehicle speed is > 10 kmph ( $\pm 1$  kmph) in reverse, and the ORVM position is in the process of recall, the ORVM will automatically adjust itself to the last recalled normal ORVM memory position.*

#### NOTE

*After returning to the normal ORVM position (> 10 kmph in reverse), if the reverse speed decreases below 10 kmph, the ORVM position will remain in the normal position unless the user moves*



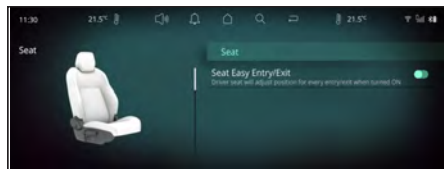
*out of reverse mode and engages reverse gear again.*

### Infotainment Setting

Easy Entry Exit Settings:

By pressing Enable Easy Entry operation from infotainment Screen, EEE function will be enabled and from next door open / close, Seat will adjust height as per Easy Entry/ Easy Exit operation.

By pressing Disable Easy Entry operation from infotainment Screen, EEE function will be disabled and from next door open / close, Seat will not adjust height as per Easy Entry/ Easy Exit operation.



### NOTE

*Easy Entry/Exit function once enabled from Infotainment, will remain active till it is disabled from Infotainment.*

ORVM Settings in Reverse Mode:

By pressing Enable ORVM in Reverse Gear from infotainment Screen, ORVM Memory Store and Recall in Reverse Gear function will be enabled.

By pressing Disable ORVM in Reverse Gear from infotainment Screen, ORVM Memory Store and Recall in Reverse Gear function will be disabled.

### NOTE

- *User is advised to use primary switches (height/Slide/Squab) in Run mode as in other power modes, vehicle battery may get drained.*
- *User is advised to not use primary movement/ recall / EEE for 1 min continuously for individual motor (Height/Slide/ Squab) - for seat*

*movement. There is compulsory rest time added of 60 sec. by software for protection of motor (for each motor) i.e. TON Time < TOFF Time.*

### WARNING

*Do not use memory functions while the vehicle is moving. Doing so could result in loss of control, and an accident causing death or serious injury.*

## STARTING AND DRIVING

### SEAT VENTILATION (if equipped)

To start ventilation, press button once.



It has 3 ventilation adjustment in decreasing order and LED glows for each press.

To stop the ventilation long press the button for few seconds.

Default setting is highest speed on first press.

The ventilated seat by default is set to OFF whenever the vehicle START/STOP button is turned on.



#### *i* NOTE

- Do not apply excessive force on ventilation button as it may get damaged.
- Button operates with slight finger force as they are electronically controlled.

#### To protect ventilated seats-

- Use the air ventilation seat ONLY when the vehicle HVAC system is on.
- Never use a liquids like alcohol, high viscosity oils or other to spill on ventilated seats.
- Avoid spillage of liquids on the ventilated seats surface this may lead to blockage of ventilated seat system and may not function properly.
- Do not add seat covers, as it will not allow ventilated seats to function prop-

erly.

- Do not keep plastic covers of seat as it is, as it will not allow ventilated seats to function properly.

#### *i* NOTE

*Ventilated seats to be vacuum cleaned regularly as there are chances of air vent hole blockage after usage.*

### REAR SEAT ADJUSTMENTS (If equipped)



Following seat adjustments can be carried out manually.

1. Seat Backrest Angle Adjustment/ Seat folding knob
2. Seat Armrest Adjustment

### Seat Backrest Angle Adjustment / Seat Folding Knob



*Angle Adjustment for 60-40% only*

To change the seat back rest angle, lean forward slightly and pull backrest release knob (1). Adjust seat backrest until the seat is securely locked and lever returns to the original position

This knob is used for seat folding as well. (Refer Seat folding section)

### Seat Armrest Adjustment

A foldable arm rest (2) is available in seat. When not required, fold the armrest back into the seat.

#### **NOTE**

- Remove all items and cups before folding the cup holders.
- Use cups, containers, bottles of right size and which have lids.
- The content could otherwise spill.

### Rear Seats Folding Option I (60-40%) (If equipped)

You can increase the luggage capacity by folding the respective rear seats as required using seat folding knob.

**To fold the seat follow given procedure:**

- Pull the backrest release knob to fold the seat forward.

## STARTING AND DRIVING



- Fold the backseat as shown in the figure.



- Fold another seat as shown in figure.



### NOTE

*Ensure that 'foldable arm rest' is close before seat folding. Fold both seats if required.*

### WARNING

- You should always engage the rear seat if you do not need the through-loading feature.
- If the rear bench seat and seat backrest are not engaged they

could fold forwards, e.g. when braking suddenly or in the event of an accident.

- The vehicle occupant would thereby be pushed into the seat belt by the rear bench seat or by the seat backrest. The seat belt can no longer offer the intended level of protection and could even cause injuries.
- Objects or loads in the trunk can not be restrained by the seat backrest. There is an increased risk of injury.
- Before every trip, make sure that the seat backrests and the rear bench seat/rear seat are engaged and securely latched.

### Rear Seats Folding (100%)

You can increase the trunk capacity by folding the rear seat. For folding:

Pull the backrest release knob provided on both side simultaneously.

Fold the backseat as shown in the figure.



### **WARNING**

- You should always engage the rear seat if you do not need the through loading feature.
- If the rear bench seat and seat backrest are not engaged they could fold forwards, e.g. when braking suddenly or in the event of an accident.
- The occupant would thereby be pushed into the seat belt by the rear bench seat or by the seat backrest. The seat belt can no longer offer the

intended level of protection and could even cause injuries.

- Objects or loads in the trunk cannot be restrained by the seat backrest. There is an increased risk of injury.
- Before every trip, make sure that the seat backrests and the rear bench seat/rear seat are engaged and securely latched.

## MIRROR

**Internal Rear View Mirror (IRVM) (if equipped)**



To adjust the mirror move the mirror up, down or sideways manually to obtain the best rear view.

When you drive at night, set the selector tab to select anti-glare mode (if equipped) to reduce glare from the headlights of vehicles behind you.

## STARTING AND DRIVING

### *i* NOTE

*Use antiglare position only when necessary, as it reduces rear view clarity.*

### Automatic Dimming IRVM (if equipped)



The LED indicator on the IRVM (1) shows the active status of auto dimming function. The auto dimming IRVM defaults to the ON position whenever the ignition switch is turned ON and it is switched OFF whenever reverse gear is engaged.

Automatic dimming (3) rear view mirror au-

tomatically controls the glare from the headlights of the car behind you in the night time or low light driving conditions. Press ON/OFF button (2) to turn ON the automatic dimming function.

### *i* NOTE

*For proper operation, keep the photocell sensors (1) & (3) clean and do not cover the area between the IRVM and the windshield.*

### Outer Rear View Mirrors (ORVM)

You can adjust the outer rear view mirrors by switch to a desired position.

### *i* NOTE

*Objects visible in mirror are actually closer than they appear. Always make sure of the actual distance from the road users traveling behind by glancing over your shoulder.*

### Motorized ORVM Adjustment (if equipped)

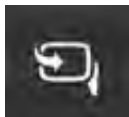


The switch to adjust the motorized mirrors is located on the driver's door. You can adjust the mirrors when the ignition switch is in the "ACC" or "ON" position.

### To adjust the Mirror



Move the mirror selection switch to L (for left side) and R (for right side) to select the mirror you wish to adjust.



Use the four positions of the knob to adjust the rear view mirrors to the required position.

### ORVM Folding

#### Option 1: Auto folding by Smart Key



When you lock the vehicle, ORVMs will be folded automatically.

When you unlock the vehicle, ORVMs will be unfolded automatically.

In case to repeated usage, Mirror Folding/Un-folding will stop functioning and will be re-activated after delay of 2 mins. During that period avoid repeated pressing of Switch.

### NOTE

*ORVM auto unfold not work, in case of user select the ORVM auto unfold option disabled through infotainment screen.*



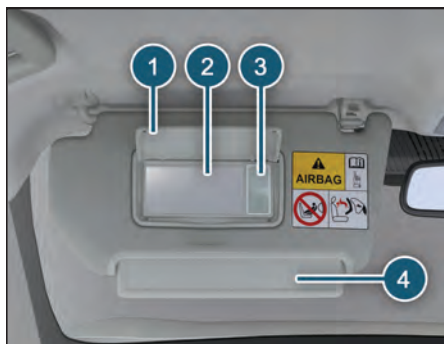
## STARTING AND DRIVING

### Option 2: Auto Folding by Knob



To fold / unfold the ORVMs, keep the Selector switch in center position (i.e. neither 'L' nor 'R, position) and then toggle down. This will operate when the ignition switch is in the "ACC" or "ON" position.

### Sun Visors



The main purpose of the sun visor is to protect the eyes of the driver and passengers from sun glare. The sun visor is installed on the roof just above the head. Users can easily position the sun visor blade according to their convenience on the windshield or the door side by rotating it with one hand. In addition to protecting eyes from sun glare, the sun visor also includes features that contribute to the comfort of the driver and passengers.

1. Mirror Flap
2. Vanity Mirror

3. Light for Vanity Mirror
4. Sun Visor Extender

#### 1. Mirror Flap (if equipped):

The sun visors can be pulled down to block the glare coming through the windshield. To block the glare from side windows, pull down the sun visor and release it from retainer. Swing the sun visor to the side.

#### 2. Vanity Mirror (if equipped)

Vanity mirror is provided on the back of the front passenger side sun visor.

#### 3. Light for Vanity Mirror (if equipped)

It is provided beside the vanity mirror. Light glows 'ON' as soon as flap of vanity mirror is open.

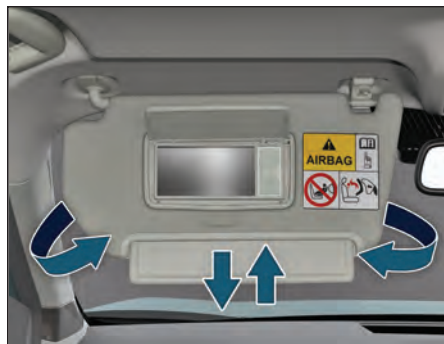
#### 4. Sun Visor Extender

The main purpose of the sun visor is to protect the eyes of the driver and passengers from sun glare. User can extend the sun visor to avoid sun glare.

The sun visors can be pulled down to block the glare coming through the windshield. To block the glare from side window,



dows, pull down the sun visor and release it from retainer. Swing the sun visor to the side.



### Paper Holder

It is provided to keep tickets, papers, cards, visiting cards etc.

### Sliding Blade

This feature allows you to place the sun visor at the desired position on the door side by sliding the blade to block sun glare.

## DRIVING SUPPORT SYSTEM

**Electronic Parking Brake (if equipped)**



Always ensure parking brake is released and parking brake warning lamp is OFF before start of the drive. Park brake warning lamp in cluster at vehicle running condition indicates failure in brake system needs to be checked with TPEM Authorised Service Centre and vehicle needs extreme precaution until you reach TPEM Authorised Service Centre.

**P switch:** P switch, located beside the gear shift lever, replaces the conventional EPB switch. To apply the Electronic Parking Brake (EPB), press the P switch for short duration until gets applied. To release it, Long press and hold the P switch until gets released while pressing the brake pedal.

### NOTE

*Apply the parking brake properly before leaving the vehicle and release it before moving.*

### WARNING

- If the EPB is faulty, the EPB malfunction lamp will remain ON in the instrument cluster. In such a case, park the vehicle safely on a flat surface and shift the gear selector lever to the 'P' position.
- Have iVBAC ECU checked immediately at a TPEM Authorised Service Centre.

## STARTING AND DRIVING

### How to Apply and Release

#### Manually

To apply the Electronic Parking Brake (EPB), short press the P switch. To release it, long press the P switch while pressing the brake pedal.

#### NOTE

- *If the EPB fails to apply or in degraded state, prevent vehicle movement by blocking the diagonal wheels.*
- *In the event of vehicle Communication bus failure: All iVBAC and other tell tales are ON in cluster, EPB will not release by P Switch.*
- *For such situation to release EPB, Keep AVH Button pressed for short duration until EPB gets released and to apply EPB, press AVH switch for short duration. If the EPB fails to apply or in degraded state, prevent vehicle movement by blocking the diagonal wheels.*

#### Auto Apply

EPB gets automatically applied for added driver comfort if any of the following conditions is met:

- If Ignition is turned off.
- Gear lever is engaged to park.
- If driver opens the driver door and unbuckles the driver seat belt. (When vehicle is in drive/reverse/neutral gear mode)
- If vehicle is in AVH activation mode, then after 3 minutes of AVH activation vehicle will switch to EPB apply.

#### WARNING

- Ensure EPB to be engaged state before leaving the vehicle.
- In case of vehicle break down with 12V battery availability, user can release EPB by pressing AVH switch for long duration when vehicle is in ready state.

#### CAUTION

If the EPB fails to apply or in degraded state, prevent vehicle movement by blocking the diagonal wheels.

#### Auto Release

- Keep the vehicle in Ready mode.
- Close the driver door and fasten the driver seat belt
- Shift the gear lever from Park (P) or Neutral (N) to Drive (D) or reverse (R)
- Press the accelerator pedal to automatically release the EPB.

Once all above conditions met, EPB will release automatically. Make sure park brake indication in cluster is turned off after automatic release of EPB.

#### NOTE

*When user is engaging or releasing EPB, please note that noticeable electric motor sound is normal and indicates EPB is working as intended.*

### CAUTION

- For Smooth Automatic release of EPB, please press accelerator pedal slowly
- It is recommended to always park the vehicle with parking brake engaged to avoid any vehicle movement which can damage vehicle or injure pedestrians.
- You should not allow anyone who is not familiar with vehicle to touch the EPB button, unintentional park brake release can lead to injure someone.
- If any abnormal noise or burning smell is noticed, kindly check with the TPMS Authorised Service Centre.

### EPB Malfunction Indicator

Both EPB malfunction lamp and parking brake lamp glow on cluster for 3 seconds during ignition ON then disappears. If there is any abnormality in the system the malfunction indicator stays continuously

ON.

In the case of any fault in the ESP system, this lamp may or may not appear.

### Controlled Dynamic Deceleration (CDP) (for Parking)

CDP function can be utilized in cases where sufficient braking force cannot be achieved through the brake pedal, such as a failure in the brake pedal, brake booster, or master cylinder.

Under such circumstances, when the P switch is long pressed, IVBAC control unit builds hydraulic pressure in the brake system, generating braking force at individual wheels to safely bring the vehicle to an emergency stop.

### WARNING

- It is strongly advised that no one other than the driver should operate P switch while the vehicle is in motion as CDP function offers high rate of deceleration.
- CDP function is designed as a safety feature to ensure controlled

vehicle deceleration in emergency situations.

### CAUTION

In case of unintentional activation, if the co-driver or any other occupant press P switch while the vehicle is in motion without the driver's awareness, the driver can override the CDP function by pressing the accelerator pedal.

### Precautions during Vehicle Towing With P Switch based EPB

Before towing please ensure EPB is not engaged as it can damage brake pads and brake components during vehicle towing.

1. EPB should be manually released if battery of the vehicle is healthy during towing and Ignition should be kept in ON state till vehicle reaches to service Centre.
2. If vehicle battery is not in healthy state during vehicle towing connect the service center for the EPB release, external power for manual release of EPB

## STARTING AND DRIVING

is not recommended.'

3. If it is not possible to keep the ignition ON till vehicle reaches to service Centre then keep EPB button pressed in release position, Brake paddle pressed and turn off the ignition this will avoid auto engagement of EPB during switching off the ignition.

### NOTE

- In case of vehicle is power down EPB cannot be released, connect the service center.
- It is not recommended to use external supply to disengage the EPB as it will damage the caliper components permanently.

### WARNING

EPB and AVH will get auto released when a drive off situation is detected This is applicable even when vehicle is in neutral gear and due care needs to be taken especially when the vehicle is parked in nose down condition.

### NOTE

- Make sure park brake indication in cluster is turned off after automatic release of EPB.
- For safety you can engage EPB only once even though ignition is turned off in case vehicle starts to roll away but you cannot release EPB in ignition off condition.
- Also it is recommended to press and hold the brake pedal and release the EPB manually when driving downhill or backing up the vehicle.

### WARNING

- If parking brake warning is still ON in the cluster even though EPB is getting applied and released, the system needs to be checked by TPEM Authorised Service Centre.
- Don't drive your vehicle with EPB engaged at any time it will cause excessive brake pad wear and it will

damage the brake rotors.

### Automatic Vehicle Hold (if equipped)



AVH holds the brakes once vehicle speed reaches zero at traffic lights or a junction, thus avoids unintended vehicle rolling. Once activated through AVH switch Automatic Vehicle Hold maintains brake force even after you release the Brake Pedal. You can then release the brake pedal and remain stopped, even on a hill.

To disengage AVH, press the accelerator pedal.

### How To Apply

1. Vehicle is in ready condition
2. Press and hold the Brake pedal.
3. Ensure seatbelt is fasten and driver's door is closed.
4. Press AVH switch.
5. Auto Hold indication turns on in the Cluster which indicates AVH is turned ON and in Standby mode.
6. When the vehicle reaches the standstill condition though brake pedal is released AVH holds the vehicle and AVH indication changes the color from white to green.
7. Once 3 min of AVH activation completes, EPB gets applied automatically.

### NOTE

- *When the vehicle is turned off keeping the Auto Vehicle Hold in the ON condition, Auto Vehicle Hold will get re-leased and EPB will get automatically applied.*
- *For safety and smooth take off, press the accelerator pedal slowly when the AVH is active.*

### Vehicle Hold Warning Indicator

AVH indication and warning lamps turns ON (white which will appear on the cluster is provided below.

AUTO  
HOLD

AVH indication ON

AUTO  
HOLD

AVH active indication (Green color)

AUTO  
HOLD

AVH failure indication (Amber color)

### Auto Hold Deactivation

In Ignition ON condition depress the Auto hold switch the Auto Hold indication in white color will disappear from the cluster indicating AVH is turned OFF.

AVH once turned ON will not be turned OFF automatically until it is deselected by switch input from user.

### NOTE

*Auto hold function will not become active if:*

- *Driver Seat belt is not buckled.*
- *Driver Door is not closed properly.*
- *EPB is in applied condition.*

For end user safety Auto hold will shift automatically to EPB in below conditions:

## STARTING AND DRIVING

1. Vehicle is in standstill for more than 3 minutes.
2. Monoshifter is shifted from any of Drive (D), Reverse(R) to Park (P) Position.
3. If you turn off the vehicle/Ignition in standstill condition.
4. Vehicle is standing on steep slope.

In above conditions AVH indication will change from Green to white and EPB indication will turn on in the cluster.

### WARNING

If any abnormality is present in the system, AVH malfunction lamp in amber colour will glow. Switch OFF ignition for 30 seconds and check if the same behavior is there. If the Malfunction lamp is still there, get your parking brake system checked with the TPEM Authorised Service Centre.

### Ensure Before You Park

- Park the vehicle in a safe place. Switch on the indicator signal before turning to park.

- Apply the parking brake.
- Make sure that all window glasses are closed and all lamps are turned 'OFF'.
- At night, put on the parking lights if required.
- Remove the key from the ignition switch and lock the vehicle.
- Use wheel chocks if the vehicle is parked on a slope.

### NOTE

*When parking on a downhill gradient, place the gear lever in 'Reverse' position. While parking on uphill gradient, place the gear lever in the '1st' position.*

### NOTE

*Do not use parking brake for braking unless unavoidable circumstances like when service brake is not working properly. The braking distance is considerably longer and the wheels could lock. There is an increased danger of skidding and accidents.*

## Surround View System (SVS) (if equipped)

Surround view system displays the surroundings around the vehicle to the driver for safe and comfortable drive.

SVS assists the driver while reversing and maneuvering the vehicle at lower speeds.

## Camera Location As Shown In The Images



Front Camera

## STARTING AND DRIVING



Left side Camera



Rear Camera



Right side Camera

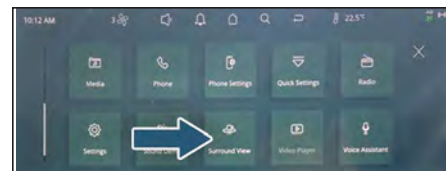
### Activation Of SVS

The function is activated when:

1. Surround view soft switch is pressed on Fascia switch



2. Surround view soft switch is pressed on Head unit.





## STARTING AND DRIVING

3. The shift lever is in D (Drive), N (Neutral) or R (Reverse) and vehicle speed is under 17 kmph and surround view soft switch is pressed.
4. Engage the reverse gear and vehicle speed is below 17 kmph.

### Deactivation Of SVS

SVS function is deactivated when one of the following step is performed.

1. Surround view soft switch is pressed again
2. Vehicle speed is more than 15 kmph
3. Disengage the reverse gear

### NOTE

- When vehicle speed is more than 15 kmph, the SVS function will turn off. The function will not automatically turn on again, even though vehicle speed gets below 17 kmph. Press the switch again, to turn on the function.
- When vehicle speed is more than 15 Kmph SVS screen will be switch

*to only rear view during reverse gear.*

- During vehicle speed is more than 15 Kmph and driver activate through soft switch/hard switch rear view shall display to user.

### Surround View System Features

The Surround view system has the following features

1. 2D View
2. 3D View
3. Front Corner View
4. Rear Corner View
5. Full View
6. Settings
7. Cancel Icon

#### 1. 2D View

By selecting 2D Icon which is available on the right corner side of the infotainment screen, cameras provide about 360 degree 2D top view of vehicle's surrounding. In 2D top view mode 4 camera icons will

be present around the model car image to switch to different sides of view. The different 2D views are as follows.

- I. 2D Top view + Front view
- II. 2D Top view + Rear view
- III. 2D Top view + Left view
- IV. 2D Top view + Right view



2D Top + Front view



2D Top + Rear view





2D Top + Left view



2D Top + Right view

### 2. 3D View



3D view with 8 different views

By selecting 3D Icon, cameras provide about 360 degree 3D view of vehicle's sur-

rounding on the Infotainment screen

In 3D mode view 8 camera icons will be present around the model car image to switch to different angle of view.

### 3. Front Corner View

If driver wants to focus on the front corner view, then the icon can be pressed to select the view.

By selecting front corner view icon, camera provides a focused view on the front left and right corners to provide a better visibility for safe maneuver.



Fig 3. Front corner view

If driver wants to focus on the rear corner view, then the icon can be pressed to select the view.

By selecting rear corner view icon, camera provides a focused view on the rear left and right corners to provide a better visi-

bility for safe maneuver.

### 4. Rear Corner View

If driver wants to focus on the rear corner view, then the icon can be pressed to select the view.

By selecting rear corner view icon, camera provides a focused view on the rear left and right corners to provide a better visibility for safe maneuver.

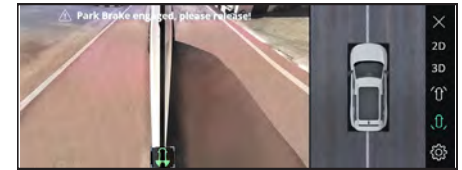


Fig 4. Rear corner view

### 5. Full View

#### 1. 2D Full front view

By selecting 2D front view Icon which is available on the model car image, camera provides about wide 2D front view of vehicle's surrounding on the Infotainment screen.

Press full view button to view front objects

## STARTING AND DRIVING

closer and press the same button to go back to the normal 2D front view



2D Full Front view

### II. 2D Full Rear view

By selecting 2D rear view Icon which is available on the model car image, cameras provides about wide 2D view of vehicle's surrounding on the Infotainment screen.

Press full view button to view rear objects closer and press the same button to go back to the normal 2D rear view



2D Full Rear View

### III. 2D Full Left view

By selecting 2D front view Icon which is available on the model car image, cameras provides about wide 2D front view of vehicle's surrounding on the Infotainment screen.

Press full view button to view front objects closer and press the same button to go back to the normal 2D front view.



2D Full left view

### IV. 2D Full Right view

By selecting 2D right view Icon which is available on the model car image, cameras provides about wide 2D view of vehicle's surrounding on the Infotainment screen.

Press full view button to view right objects closer and press the same button to go back to the normal 2D right view.



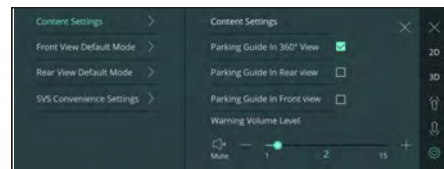
2D Full Right view

### Settings

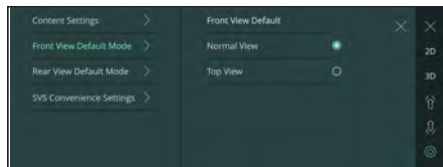
By selecting the settings icon available on the infotainment screen, driver can change the settings as required.

User can change the content settings based on the user choice.

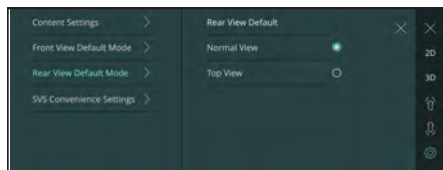
User can change the front and rear default view to any of the view i.e. either normal view or top view



SVS content settings



SVS front view default mode settings



SVS rear view default mode settings

### Cancel Icon

By selecting the cancel icon which is available on the top right corner of the infotainment screen, user can exit from the surround view system function.

It can be used for all the SVS features such as 2D, 3D, front & rear corner views.

It cannot be visible when vehicle gear state is reverse gear.



## Understanding Guidelines Indication Static Guidelines



## Dynamic Guidelines



## Red Line

Indicates, if rear objects are in this colored zone, you have to stop the vehicle and not allowed to go backward. If you still go backward, your vehicle will hit the object.

## STARTING AND DRIVING



### Yellow Line

Indicates, if rear objects are in this colored zone, you have to take utmost care. However, objects fall in this zone, may not hit vehicle.



### Green Line

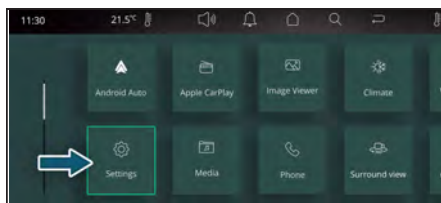
Indicates, if rear object is in this colored zone, you have to be cautious. Still you can go backward safely.



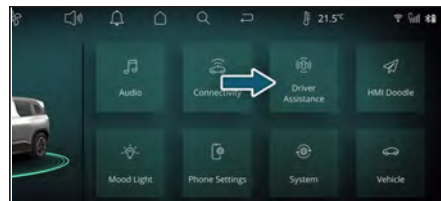
### PDC Guidelines Settings

User can change the timer settings for PDC guidelines which is available on the infotainment display

By selecting the infotainment settings icon available on the infotainment screen use able to open the settings options available in the system.

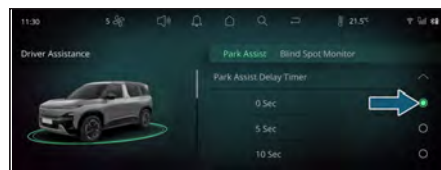


The system will display the below screen when user select the settings icon. Select the driver assistance icon which is available on the screen.



In driver assistance system will provide many other options in that user should select the park assist delay timer. System will provide three different option such as 0sec, 5 sec and 10sec.

Based on the user choice he/she can select any option from the three.

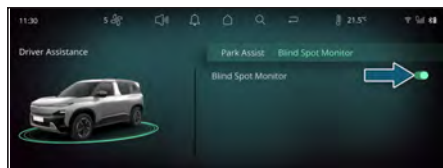


### Blind View Monitor

Blind view monitor will help to reduce the crashes that happen when driver is being overtaken or changing the lanes.

This system should work in ignition on and run condition irrespective of the vehicle speed.

We can enable/disable the blind view monitor in HMI settings based on the user choice.



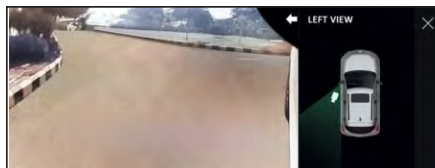
This feature is activated when user turn on the left/right turn indicator. This feature is activated when user turn on the left/right turn indicator.

### Activation Of Blind View Monitor

- On activating the right turn indicator, right side rear view should be displayed on the infotainment along with static overlays.



Right rear side view when turn on the right indicator



Left rear side view when turn on the left indicator

### Deactivation Of Blind View Monitor

This feature is deactivated when user turn off the left/right turn indicator.

### Understanding Static Overlays Indication

**Red Line:** Indicates, if rear objects are behind this colored line, you are not allowed to change the lane. If you still change the lane, your vehicle will hit the object.

**Yellow Line:** Indicates, if rear objects are behind this colored line, you have to take utmost care. However, objects fall in this zone, may not hit vehicle.

**Green Line:** Indicates, if rear object is behind this colored line, you have to be cautious. Still you can safely change the lane.

### NOTE

- When SVS is in active condition then user turn on the left/right turn indicator then system should display the blind view monitoring and if user turn off the turn indicator then system return back to the SVS screen.
- When SVS is not in active condition, user turn on the left/right turn indicator then system should display the blind view monitoring. Once user

## STARTING AND DRIVING

*turn off the turn indicator then system return back to infotainment home screen.*

### Camera Precaution

- As the camera is, IP protected, do not detach, disassemble or modify in any manner from the actual position. This will show required visual information in display.
- Do not use camera when tailgate is open. If tailgate is open, visual information may not be the actual rear view of the vehicle & system will warn with message 'Tail Gate Open, Please close.'
- Do not use camera when driver/passenger door is open. If any one of the door is open, visual information may not be the actual view of the vehicle & system will warn with message 'Door Open, Please close'. And also corresponding door side display shall be in dark image.
- Do not use camera when ORVM is folded. If ORVM is folded, visual infor-

mation may not be the actual view of the vehicle & system will warn with message 'ORVM Folded'.

- When the camera is operated under fluorescent lights, sodium light or mercury light etc., illuminated areas on the lens may appear to flicker in the display.
- Do not attach any advertisement or styling or any kind of stickers on top of camera. If this happens, camera cannot provide you the visual image and may damage camera.
- Do not add any accessory, which will obstruct camera field of view.

### Cleaning Camera

1. Due to environmental reasons, dust, mud or fog may accumulate on the camera lens. So regularly clean the camera lens.
2. Use water to clean the camera lens. Do not use extreme cold or hot water. Rapid changes in temperature may brittle the camera lens. Do not apply High Pressure water for cleaning.

3. Wipe the camera lens with soft cloth.
4. Do not use hard cloth or material to wipe the camera lens. This will cause scratches on the camera, and leads to deteriorated visual image on the display.
5. Do not apply organic solvent, car wax, window cleaner or glass coat to clean the camera. This may damage the lens
6. Do not use chemical solvents such as strong detergents containing high alkaline or volatile organic solvents (gasoline, acetone etc). This may damage the camera lens
7. Do not apply heavy force on lens, while cleaning.
8. Do not remove mud, snow on the camera lens using stick or hard material. Use normal water and soft cloth.

### WARNING

- The camera uses fish eye lens. So the size of the objects or in the display may differ from the actual size and distances in low light condi



tions, the screen may darken or image may appear faint.

- If the tyre sizes are changed, the position of the fixed guidelines displayed on the screen may change.
- During rainy conditions, image may get obscured. In such conditions, do not depend on camera view.
- The camera used in the vehicle, may not reproduce the same color of the real object.
- Do not apply any kind of force on the camera.
- High humidity and variation in ambient temperature may result into condensation inside the camera lens, which may further result into degradation of camera video feed on the screen. It is recommended that not to rely on camera video feed for parking assistance in such scenario. This phenomenon is temporary and will be automatically recovered with reduction in humidity and less variation in ambient temperature.

- The area displayed by the camera is limited. The camera does not display objects that are close to or below the bumper, underneath the vehicle, or objects out of the camera's field of view. The area displayed on the screen may vary according to vehicle orientation or road conditions.
- Surround View System (SVS) is an aid only. User need to check the surrounding for safety.

### Rear View Camera



Rear View Camera is a visual reverse guiding system. When reversing or parking, make sure that there are no persons, animals or objects in the area where you are reversing.

The display will be shown on the infotainment screen.

## STARTING AND DRIVING



### Activation

#### Reverse gear

This system will start, if reverse gear is engaged, or park assist button (if equipped) is pressed or manual activation is done through Infotainment screen.

#### Deactivation

System will stop, if reverse gear is disengaged, or park assist button (if equipped) is pressed.

If started through infotainment, the system can be stopped using a cross button on infotainment screen.

### Understanding Guidelines Indication



*Static guidelines*



*Dynamic guidelines*

#### Green Line

You can safely reverse the vehicle, but be cautious if objects fall in this zone.

#### Yellow Line

You have to take utmost care if objects fall in this zone. However, the objects may not hit vehicle.

#### Red Line

Red line indicates that you have to stop reversing the vehicle. If you still go backwards, the car will hit the obstacle.

#### Do's And Don't

- Do not use camera when tailgate is open. If tailgate is open, visual information may not be the actual rear view of the vehicle & system will warn with message 'Tail Gate Open, Please close.'
- When the camera is operated under fluorescent lights, sodium light or mercury light etc., illuminated areas on the lens may appear to flicker in the display.
- Do not attach any advertisement or styling or any kind of stickers on top of camera. If this happens, camera cannot provide you the visual image and it may damage the camera.
- Do not add any accessory, which will



cause blockage to the camera's field of view.

### Cleaning Camera

1. Due to environmental reasons, dust, mud or fog may accumulate on the camera lens. So regularly clean the camera lens.
2. Use water to clean the camera lens. Do not use extreme cold or hot water. Rapid changes in temperature may brittle the camera lens. Do not apply High Pressure water for cleaning.
3. Wipe the camera lens with soft cloth.
4. Do not use hard cloth or material to wipe the camera lens. This will cause scratches on the camera, and leads to deteriorated visual image on the display.
5. Do not apply organic solvent, car wax, window cleaner or glass coat to clean the camera. If this is applied, wipe it off as soon as possible.
6. Do not apply heavy force on lens, while cleaning.
7. Do not remove mud, snow on the cam-

era lens using stick or hard material. Use normal water and soft cloth.

### WARNING

- The camera uses fish eye lens. So the size of the objects or in the display may differ from the actual size and distance. In low light conditions, the screen may darken or image may appear faint.
- If the tire sizes are changed, the position of the fixed guidelines displayed on the screen may change.
- During rainy conditions, image may get obscured. In such conditions, do not depend on camera view. The camera used in the vehicle, may not reproduce the same color of the real object.
- In case of damage of the rear portion of the vehicle, camera position may change. Which causes wrong visual information on display. In case of damage, make sure that, camera is fitted properly at the intended location.

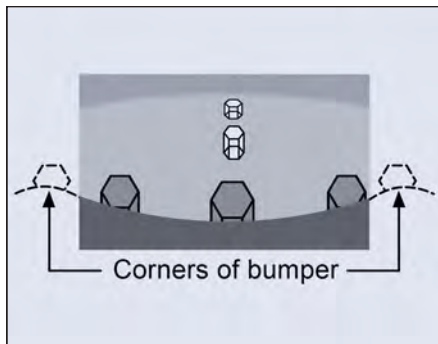
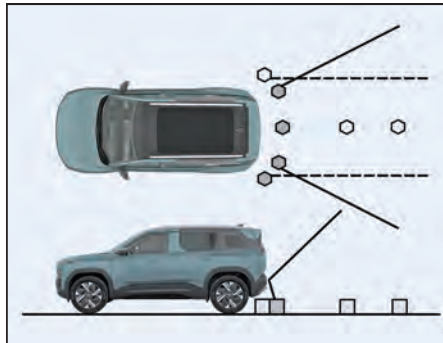
- In case of uneven road conditions or up-hill or downhill conditions, do not depend on rear view camera park aid.
- Do not apply any kind of force on the camera.
- Always use rear View mirrors along with Rear View Camera for confirming the safety of the rear and the surrounding conditions.
- High humidity and variation in ambient temperature may result into condensation inside the camera lens, which may further result into degradation of camera video feed on the screen. It is recommended that not to rely on camera video feed for parking assistance in such scenario. This phenomenon is temporary and will be automatically recovered with reduction in humidity and less variation in ambient temperature.
- The area displayed by the rear view camera is limited. The camera does not display objects that are close to

## STARTING AND DRIVING

or below the bumper, underneath the vehicle, or objects out of the camera's field of view. The area displayed on the screen may vary according to vehicle orientation or road conditions.

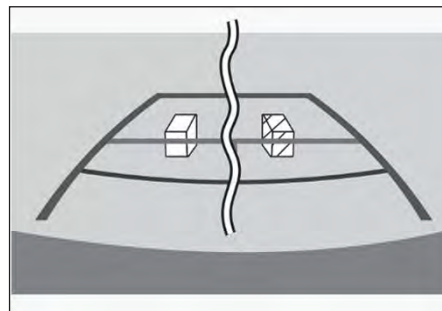
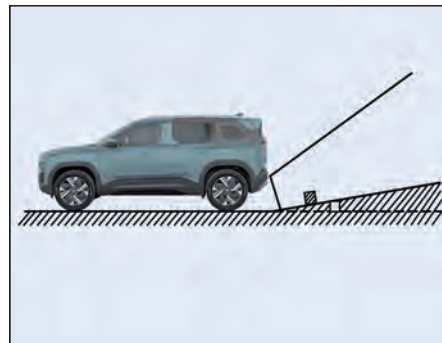
### Rear View Camera System Precautions Area Displayed on Screen

The rear view camera system displays an image of the view from the bumper of the rear area of the vehicle.



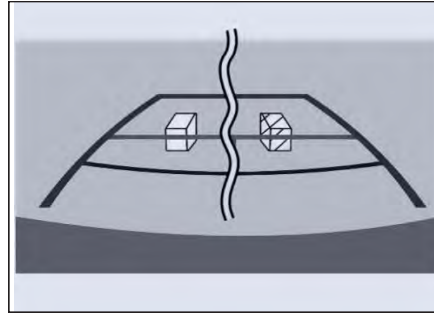
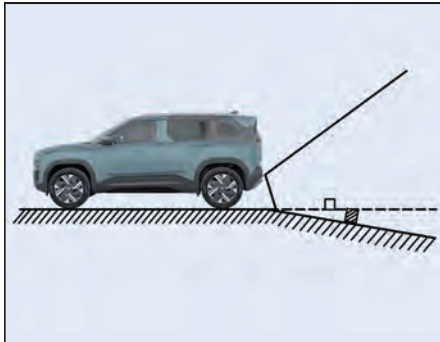
- The area displayed on the screen may vary according to vehicle orientation conditions.
- Objects, which are close to either corner of the bumper or under the bumper, cannot be seen on the screen.
- The camera uses a special lens. The distance of the image that appears on the screen differs from the actual distance. The camera may not display items that are located higher than the camera's field of view.

### When sharp up gradient behind the vehicle



- The distance guidelines will appear to be closer to the vehicle than the actual distance.
- Because of this, objects will appear to be farther away than they actually are.
- In the same way, there will be a margin of error between the guidelines and the actual distance/course on the road.

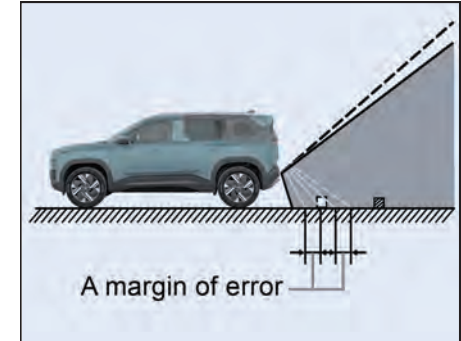
### When sharp down gradient behind the vehicle



### *i* NOTE

*The distance guidelines will appear to be further from the vehicle than the actual distance. Because of this, objects will appear to be closer than they actually are. In the same way, there will be a margin of error between the guidelines and the actual distance/course on the road.*

### When any part of the vehicle sags



When any part of the vehicle sags due to the number of passengers or the distribution of the load, there is a margin of error between the fixed guide lines on the screen and the actual distance/course on the road.

### When approaching three-dimensional objects

The distance guidelines are displayed according to flat surfaced objects (such as the road). It is not possible to determine the position of three-dimensional objects

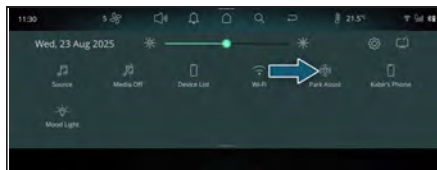


at rear bumper area, leading to improper operation.

### Park Assist System (Front and Rear)

Park Assist system is an electronic parking aid that will assist you to park your vehicle safely when in reverse gear mode. It provides audio and visual information through the vehicles infotainment system.

The user can view the Park assist screen by selecting this feature in Infotainment display to see any obstacle behind the vehicle while parking.



The system also displays the Park assist screen when the reverse gear is engaged. It provides audio and visual information through the vehicles infotainment system / Buzzer.

### WARNING

0 to 25 cm obstacle detection performance is not guaranteed due to ultrasonic sensor technology limitation.

Variant where infotainment display is not present and audio warning is given through a buzzer, on activating the Park Assist system, a tone will be played within first two seconds to indicate the proper functioning of the system. After these two seconds, normal functioning of the system will continue. If no tone is heard for first two seconds, it shall mean that Park Assist System is faulty. The owner should, in that case, go to the nearest dealer for rectification.

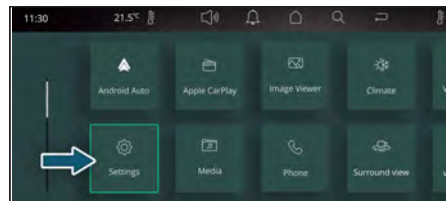
### Front Park Assist System (FPAS) (if equipped)



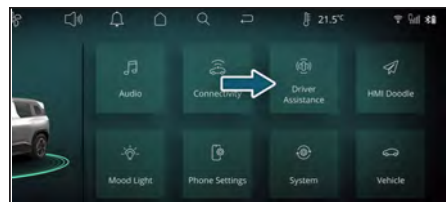
## STARTING AND DRIVING

### Activation Conditions

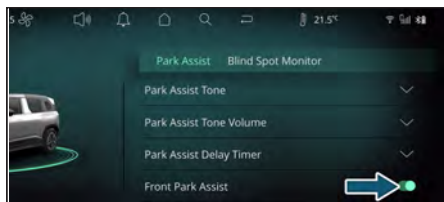
1. Front park assist option can be enabled through Infotainment screen. Go to settings



Select Driver Assistance



Enable front park assist option



2. If reverse gear is engaged and Front Park assist option is enabled through infotainment screen.
3. If user has turned ON Low speed activation from user settings menu and vehicle speed is below 10 kmph (Forward Direction) and some object is detected in the front of the vehicle then audio warning for 10 sec will sound, while the visual warning will continue to be shown till the object is present.

### Deactivation Condition

1. If user has turned OFF low speed activation from user settings, then warning will come as above point no 3 only for the first time after IGN ON.
2. If vehicle speed is above 10Kmph in forward gear.

3. If started through infotainment screen button, the system can be stopped by clicking a cross button on infotainment screen.

### Audio Warning Option I

| Approx. Distance Range From Bumper (in cm) | Visual Warning | Audible Information |
|--------------------------------------------|----------------|---------------------|
| 25 – 30                                    | Red Zone       | Continuous Beep     |
| 31 – 60                                    | Yellow Zone    | Fast Beep           |
| 61 – 100                                   | Green Zone     | Slow Beep           |

### Audio Warning Option II (for APA)

| Approx. Distance Range From Bumper (in cm) | Visual Warning | Audible Information |
|--------------------------------------------|----------------|---------------------|
| 0 – 40                                     | Red Zone       |                     |

### Audio Warning Option II (APA Variant)

| Approx. Distance Range From Bumper (in cm) | Visual Warning | Audible Information |
|--------------------------------------------|----------------|---------------------|
| 0 – 50                                     | Red Zone       | Continuous Beep     |
| 50 – 100                                   | Yellow Zone    | Fast Beep           |
| 100 – 150                                  | Green Zone     | Slow Beep           |

#### NOTE

Audio warning may come from Infotainment system speaker or through Buzzer, depending on vehicle model and configuration.

### Park Assist System Limitations

Reverse Park Assist system is not a collision avoiding system. It is solely the driver's responsibility to park the vehicle safely.

Reverse Park Assist feature works on ultra

| Approx. Distance Range From Bumper (in cm) | Visual Warning | Audible Information |
|--------------------------------------------|----------------|---------------------|
|                                            |                | Continuous Beep     |
| 40 – 80                                    | Yellow Zone    | Fast Beep           |
| 80 – 120                                   | Green Zone     | Slow Beep           |

### Reverse Park Assist System (RPAS)



#### Activation Conditions

This system will start, if reverse gear is engaged, or park assist button (if equipped) is pressed or manual activation is done through Infotainment screen.

#### Deactivation Condition

System will stop, if reverse gear is disengaged, or park assist button (if equipped) is pressed.

If started through infotainment, the system can be stopped using a cross button on infotainment screen. System will stop, if reverse gear is disengaged, or park assist button (if equipped) is pressed. If started through infotainment, the system can be stopped using a cross button on infotainment screen.

#### Audio Warning Option I

| Approx. Distance Range From Bumper (in cm) | Audible Information |
|--------------------------------------------|---------------------|
| 25– 40                                     | Continuous Beep     |
| 41 – 80                                    | Fast Beep           |
| 81 – 120                                   | Slow Beep           |



## STARTING AND DRIVING

---

sound echo technology, due to which performance is not guaranteed in following scenarios:

- If the object has a sharp edge surface or a surface that causes ultrasonic echoes to disperse, the object may not be detected properly.
- If the object is a mesh fence or made of thin wires where ultrasonic echoes cannot be sufficiently reflected, detection may be limited.
- Fast-moving objects passing through the sensor's detection field may not be detected because the echoes are not processed by the system in time.
- If the object is made of or covered by foam, sponge, snow, ice, mud, dirt, slush, frost, or water droplets (for example, after washing or during rain), ultrasonic signals may be absorbed or distorted, resulting in reduced detection performance.
- If the sensor surface or the area around the sensor is covered by a hand, sticker, accessory, protective film, or aftermarket coating, the system

may not operate properly.

- Objects close to the rear bumper may go undetected depending on the detection field. Drivers should always exercise caution while parking.
- Detection performance may be affected if the vehicle ride height changes due to heavy cargo or load, abnormal tire pressure, suspension alteration, or other vehicle modifications.
- If the vehicle sustains an impact and the sensor is damaged or misaligned, the system may provide inaccurate distance information or false alerts. Inspection by an authorized dealer is recommended.
- If the sensor area is exposed to extreme temperatures due to direct sunlight, freezing weather, or rapid temperature changes, detection performance may be reduced.
- Ultrasonic noise or interference from nearby sources (such as other vehicle sensors, horns, engines, air braking systems, exhaust fans, presence of external noise, wireless transmitters,

or mobile devices) may lead to false warnings or detections..

- When the vehicle speed exceeds approximately 10 km/h, the system will not warn the driver even if an object is detected, and a message such as "Vehicle speed is high, drive slowly" may be displayed.
- Driving on uneven road surfaces such as gravel, unpaved roads, artificial speed breakers, steep gradients, or rough terrain may affect detection accuracy.
- Poles or objects with square or rectangular cross-sections, especially when detected at the vehicle edge, may not be recognized correctly.
- Accessories installed near the bumper area (such as larger trailer hitches, bike racks, carriers, bumper guards, or aftermarket bumpers) may reduce system performance or cause false alerts.



### WARNING

Due to any reason, if the sensor gets misaligned or loses its intended fitment position, contact your dealer for refitment.

### NOTE

*Parking sensor performance may affect in following cases:*

- *Use of unauthorized registration plate. Use RTO authorized size registration plate only. High security registration plate dimension in mm – 500 x 120 (Approx).*
- *If external accessories are installed at rear bumper area, leading to improper operation.*

### Park Assist System Preventive Maintenance / Cleaning

1. Regularly clean the sensors and keep them free from dust, ice, mud, water, chewing gum etc. for proper working of the system. Use a smooth cloth for cleaning.

2. Do not use water at high pressure for cleaning the sensor.
3. Do not cover the sensors. This will interrupt park assist performance.
4. Do not remove mud, snow on the sensors using stick or hard material. Use normal water and soft cloth.

### Park Assist Malfunction Indications

In case of park assist system malfunctions, fault screen may appear on the infotainment system.

Reason for this fault may be

1. Body Control Module Failure
2. Sensor Malfunction
3. Partner components such as Infotainment music system, Instrument Cluster failure

### Automatic Parking Assist (APA) (If equipped)

Automatic Parking Assist allows your vehicle to park in an identified slot, and handles steering, braking and accelerating.

The APA system uses 12 ultrasonic sensors and 4 cameras to assist the driver in

parking the vehicle.

### CAUTION

- The Park Assist system is designed to support parking in many scenarios, however its performance may vary depending on specific environmental and vehicle conditions.
- Driver should always be attentive while parking, check surroundings and should intervene when necessary to avoid injuries/damages.
- In case of remote parking also, driver should always pay attention while vehicle is moving and take control whenever necessary.
- It is recommended to read this section carefully before using APA feature.
- Please note that the system may not detect all types of obstacles, such as small children, pedestrians, cyclists, animals, or objects that are moving, protruding, hanging, unusually positioned, outside the effective

## STARTING AND DRIVING

detection range or thin objects less than 12mm, objects which doesn't reflect USS waves like fabrics, foam blocks, curbs covered in grass. Environmental factors can also affect sensor performance, occasionally resulting in missed or inaccurate readings.

- Although the APA feature is designed to account for unexpected events, the driver remains responsible for ensuring safe maneuvering.

### *i* NOTE

*All time-related parameters in this section are subject to change by the manufacturer.*

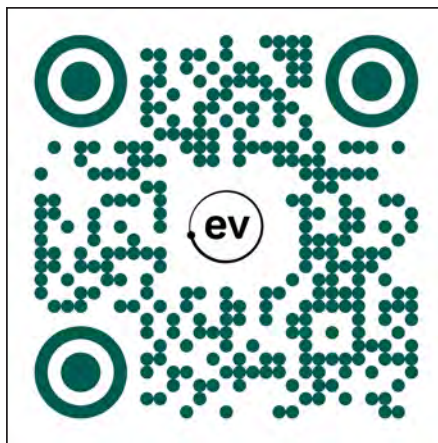
Automatic parking can be done in two ways:

**Option I :** Auto Parking (Driver Present inside the vehicle)

**Option II :** Remote Parking (Driver Outside the Vehicle)

To view related videos, scan the QR code

below using your mobile device.

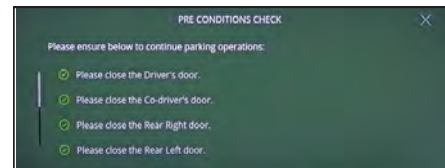


### Option I: Auto Parking (driver Present Inside)

Preconditions for Activation

- Vehicle in D gear
- Driver seat belt is fastened
- Vehicle all doors are properly closed
- Vehicle bonnet is closed
- Vehicle tailgate is closed

- ORVMs are unfolded.
- Vehicle charging above SoC level (min 5% charged).
- Vehicle charger not connected



### *i* NOTE

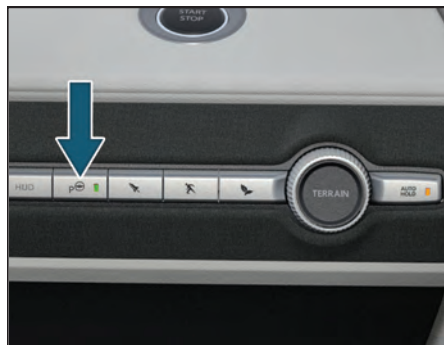
- All sensors should be operational, no warning telltale should be ON.
- Parking feature will not work until all the preconditions are met.

In case of sensor failure given tell-tale will be shown on instrument cluster.

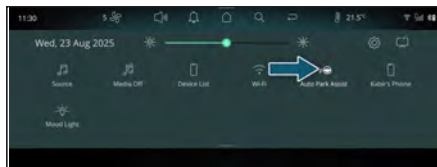


### Activation of APA

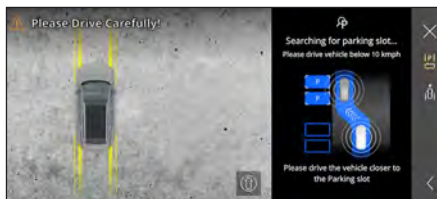
- Start the vehicle through Push button switch
- Activate the parking features through infotainment screen or through APA switch near centre console.



To activate through infotainment screen  
Go to Apps> Auto Park Assist

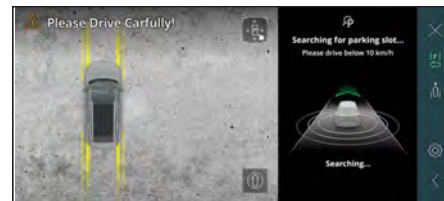


Press the soft keys on the touch screen to switch between the park in or park out parking modes. Park-in is selected by default.



### Park In:

- Once park-in is selected, below screen appears on infotainment screen. The parking assistance system starts searching for the appropriate parking space in case of park in.



### NOTE

*During searching, vehicle speed should be less than 10kmph.*

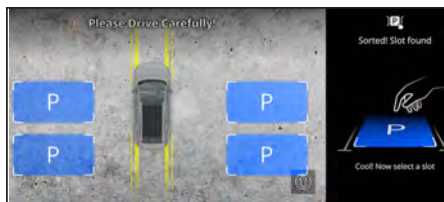
*User will be given warning if speed exceeds 10kmph till 15kmph. If speed exceeds 15kmph APA will get terminated.*

- During slot searching all available parking slots on both right and left sides will be visible to the driver.*
- However if user wants to park on particular side (either left or right) driver can provide the direction for search by giving left or right indicator. If right indicator is turned ON, then parking slots only on the right side will be shown by the parking system.*

## STARTING AND DRIVING

- During slot, searching if driver exceeds the required speed limit or any precondition of APA is not met, then visual warning will be given.
- For optimal slot searching and APA performance, well-marked parking lines (preferably in yellow or white), good road conditions, and clear visibility are recommended. The system performs best when parking slots are free of obstructions, potholes, waterlogging, or unclear markings etc.
- System analyses available slots considering internal, external & surrounding conditions and then suitable slot is offered, if found unsuitable, the slot may not be offered.
- If the driver does not select a slot or release the brake pedal within 30 seconds of bringing the vehicle to a standstill, the APA will exit, and the home screen will be displayed on the IVI.

- When slot searching is in process, parking slots will be displayed on infotainment screen. If slot is being offered is of your choice Stop the vehicle by pressing the brake pedal
- Once you stop the vehicle nearby a particular slot with the intention of parking-in to the slot, select the slot by touching on it.
- Selectable parking slot will be highlighted in blue color. System detects minimum 1 and maximum 4 slots.

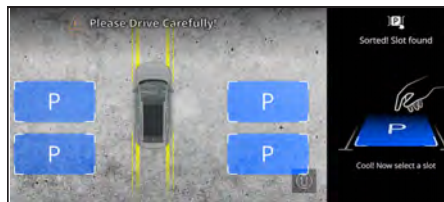


### NOTE

- Parking slot detection will not happen if vehicle is driven in reverse direction.
- Parking slot detection will not work if lateral distance between vehicle and parking slot is greater than 2 meters or less than 0.5 meters.
- Parking Slots will get detected and offered as you drive across the slot at designed speed. Slots may get offered as and when they are deemed to be park able.
- Auto Parking and remote parking options will be displayed on screen.



- By clicking on rotate arrow, user can select the parking direction of vehicle (Front / Rear facing) (if equipped).



- Once the above screen appears on infotainment screen, release the Brake Pedal, Remove hands from the steering wheel.

### *i* NOTE

- Before selecting the parking slot, driver should check surrounding

and whether available slot is suitable and safe for the vehicle.

- During automated parking, the vehicle speed will be between 1-5kmph.
- Auto parking will be paused if any pre-conditions are not met.
- If driver door or driver's seat belt is opened during parking, APA feature will get terminated.
- For other pause conditions, the system will wait for 30secs, if the conditions are met, APA will resume; otherwise APA will terminate.

- After successful parking, Park-In completed message will be displayed on the screen and park brake will be applied automatically.



### *i* NOTE

User should check parking brake is applied before leaving the vehicle.

### Park Out:

- Driving out of a parking space is available if parking-in is done by the parking assistance system.
- Start the vehicle through Push button switch when driver inside.
- Activate the parking features by selecting Auto Park Assist through infotainment screen.
- Select the Park out option on screen.



## STARTING AND DRIVING

### NOTE

- If park in is done through Auto Park Assist then only Auto park out is possible.
- The vehicle shall not be moved manually once auto parking is done.
- In case park-in is not done through Auto Park Assist, system will prompt “Park-out is not possible, parking in is done manually”.
- If user doesn't provide park-out direction through side indicators then home screen will be displayed after defined time.
- If the surroundings are changed post parking-in is completed, system may not perform park-out.



- Select the desired park-out direction

through right or left indicators in case of perpendicular parking only.



### NOTE

*Depending on the surrounding environment, if the auto park out is not possible in either direction, then the choice of direction selection via side indicator is nullified.*

- Once below screen appears on infotainment screen then only release the brake pedal and remove hands from steering wheel.



- Park out maneuvering is done by the parking assistance system, only after driver releases the brake pedal.
- Park-Out successful message will be displayed on the screen at the end of successful Park-out.

### Parking Pause

Parking assistance system once live, will pause for 30 seconds in situations such as the following:

- If brake pedal is pressed
- If passenger, or rear doors are open.
- If side ORVMs are folded
- If bonnet, tailgate is not closed.

If there are an obstacles present on the maneuvering path.



### **i NOTE**

- To resume parking, remove all pause conditions within 30 sec of the pause.
- Acceleration input will not be taken by system during parking maneuver.
- At the end of any APA maneuver or in case of APA termination due to any reason, the vehicle will be shifted to P mode for safety.

### **Parking Termination**

Parking assistance system will terminate in situations such as the following:

- If driver door is opened
- If gear is changed
- If EPB pressed
- If APA button is pressed again
- If driver seat belt is removed during maneuvering.
- If Pause conditions are not removed within 30sec of the pause
- If driver holds the steering wheel

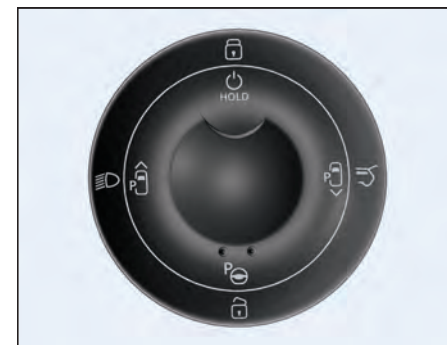
### **i NOTE**

- If automatic parking is terminated, it has to be started again from slot searching.
- Auto Parking will get terminated if parking is not completed within 180sec or took more than 8 maneuvers to park.
- During parking maneuvering, hazard light will be continuously ON.

### **Option II: Remote Parking (driver Outside The Vehicle)**

Remote Parking enables the driver to perform parking in or parking out maneuvering using Remote device.

During remote parking driver needs to press APA switch on the remote key until the parking is completed while standing within 5 meters periphery of the vehicle.



## STARTING AND DRIVING



### ⚠ CAUTION

- It is a parking assist system and may not be applicable in all cases / all environmental conditions. Driver should always be attentive while parking and should intervene when necessary.
- Although it is an automatic feature, driver should always pay attention while vehicle is moving and take control whenever necessary.
- While parking vehicle remotely make sure to keep safe distance from vehicle.

- Please ensure the windows, sunroof and doors are closed & no occupant present inside the vehicle before leaving the vehicle for Remote Parking.
- Ensure parking brake is applied and vehicle

### Preconditions for Park-In activation

- Driver seat belt is fastened.
- Vehicle doors are properly closed.
- Tailgate, Bonnet is closed.
- ORVMs are unfolded.
- Vehicle into "D" gear
- Vehicle charging above SoC level (Min 5% charged).
- Vehicle charger not connected

### Activation of Remote Parking

#### *i* NOTE

- *Remote control parking-in is always initiated from inside the vehicle. Once free slot is selected and re*

*mote parking is selected from info-tainment screen, park-in maneuvering will be done remotely from outside of the vehicle.*

- *Frequent use of remote key will discharge the key battery.*
- *Driver should continuously keep watch on surroundings while parking remotely and should immediately stop the vehicle when needed by releasing the APA button on remote key.*

Start the vehicle through Push button switch

Activate the APA feature through infotainment screen.

To activate Go to Apps> Auto park Assist

Select the parking modes from the screen: park in



## STARTING AND DRIVING



### Park In:

The parking assistance system starts searching for the appropriate parking space in case of park in.

#### NOTE

- During searching, vehicle speed should be less than 10kmph.
- User will be given warning if speed exceeds 10kmph till 15kmph. If speed exceeds 15kmph APA will get terminated.
- During slot searching all available parking slots on both right and left sides will be visible to the driver.
- However if user wants to park any one particular side (either left or right) driver can provide the direc

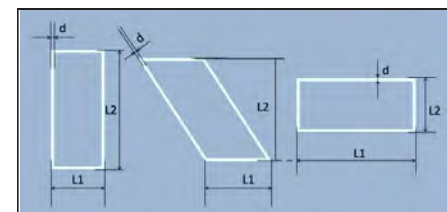
tion for search by giving left or right indicator. If right indicator is turned ON, then parking slots only on the right side will be shown by the parking system.

- During slot searching, if driver exceeds the required speed limit or any precondition of APA is not met, then visual warning will be given.
- For optimal slot searching and APA performance, well-marked parking lines (preferably in yellow or white), good road conditions, and clear visibility are recommended. The system performs best when parking slots are free of obstructions, potholes, waterlogging, or unclear markings etc.
- For the ease of remote parking please select spacious parking slot to ensure smooth and effortless vehicle movement.
- Ensure parking type and direction are safe.

When slot searching is completed, parking

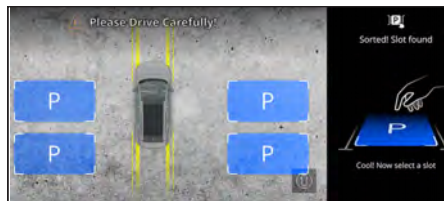
slots will be displayed on infotainment screen. When slot searching is completed, parking slots will be displayed on infotainment screen.

### Slot Dimensions:



| Parking Type | Perpendicular | Parallel | Angular  |
|--------------|---------------|----------|----------|
| L1           | ~2.7- 3m      | 6m       | ~2.7- 3m |
| L2           | 6m            | 3m       | 6m       |
| d            | 0.1m          | 0.1m     | 0.1m     |

## STARTING AND DRIVING



### *i* NOTE

*Parking slot detection will not happen if vehicle is driven in reverse direction.*

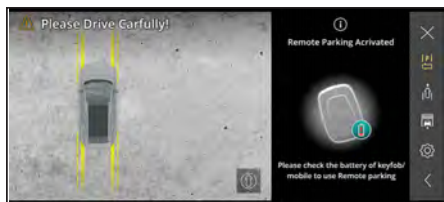
If slot is being offered is of your choice.

Stop the vehicle

Once you stop nearby a particular slot with the intention of parking-in to the slot, ensure selecting the slot by touching on it.

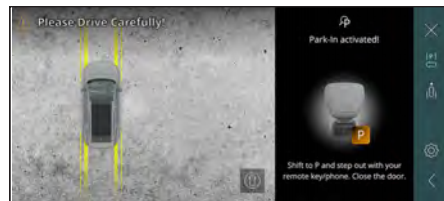
Selectable slot will be highlighted in blue color.

Once parking slot is selected then driver can tap on Remote Parking on the display screen to enable remote maneuvering.



Engage the park gear and then release the brake pedal.

Driver has to unbuckle the seat belt and leave the vehicle by closing the driver's door to meet the below pre-conditions.



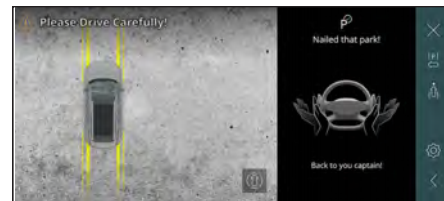
### Preconditions for Remote Park In:

- Park gear is engaged.
- Vehicle is started.
- Vehicle doors are properly closed.
- Tailgate, Bonnet is closed.
- ORVMs are unfolded.
- Smart phone or key is connected with the vehicle via Bluetooth.
- Remote key has sufficient battery.
- All Keys are outside of the vehicle
- Vehicle charging above SoC level (Min 5% charged).
- Charger is not connected.
- Distance between the vehicle and smart phone / key should not be more than 6m.

### *i* NOTE

- If remote parking is cancelled then the hazard light will be continuously ON for 5sec.
- Remote Parking system continues with parking, only when driver press and hold the APA switch present on the Key.
- During remote maneuvering, the vehicle speed will be between 1-3kmph. During remote maneuvering, the vehicle speed will be between 1-3kmph.
- If driver release APA switch on the key then the parking will pause.
- During R-APA if there is EPB failure detected then hazard light will glow for 120sec (Fast blinking) and honking will be ON for 30sec. User need to attend the vehicle immediately and apply the primary brake immediately to avoid the roll back and to turn off the warning.

- During Remote APA, hazard light will be continuously ON (Normal blinking) and will turned off when parking is finished, ensure safe movement of the vehicle.
- Successful Park IN/OUT: Vehicle will turn OFF and Lock after 5sec/30sec resp. if not attended.
- Unsuccessful Park IN/OUT: Vehicle will turn OFF and Lock after 30sec. if not attended. If attended the vehicle will remain ON.
- The driver shall continuously supervise the low-speed maneuvering process and shall immediately terminate the process when his/her situational awareness is degraded.
- After successful parking, Park-In message will be displayed on infotainment screen. Vehicle will be turned OFF and locked.
- Ensure vehicle is parked safely before leaving.



### Switching Vehicle ON and OFF Procedure from remote Key

- Press “lock button” on the remote key
- Press and hold the “Power” button to remotely turn ON the vehicle.
- To turn OFF the vehicle, long press the “Power” button for 2 seconds until vehicle is turned OFF.
- Upon successful OFF, a Good bye animation will be displayed.

### *i* NOTE

Once vehicle is turned ON remotely, if no interactions done within 30 seconds, vehicle will be turned OFF automatically.

## STARTING AND DRIVING

### Park Out:

- Turn ON the vehicle ignition through remote key.
- In case of park-out through Remote key, park-out direction is always opposite of park-in direction irrespective of parking-in scenarios.

### Preconditions to activate Remote Park Out:

- Vehicle doors are closed
- Tail gate is closed
- Bonnet is closed
- ORVM is unfolded
- Vehicle need to be turned ON
- Vehicle charging above SoC level (Min 5% charged).
- Charger not connected
- Smart phone or key is connected with the vehicle via Bluetooth.
- Remote key has sufficient battery.
- All Keys are outside of the vehicle
- Distance between the vehicle and smart phone / key should not be more

than 6m.

- Phone is sufficiently charged or in specific modes ( Low Power, Power saving mode , airplane mode, Setting related to network, BLE, UWB etc.)

### NOTE

- *An automated parking out maneuver via Remote key can be initiated from outside the vehicle, after the vehicle is started by key.*
- *Children are not recommended to use remote key.*
- *Objects in the blind spot areas may or may not be detected by sensors, driver should always pay attention while selecting parking space.*
- *User should ensure vehicle is locked after parking it remotely.*
- *The phone settings do not impede/ hamper remote APA functionality.*

### Remote Parking Pause

During parking if the driver exit the APA mode or if he leaves a radius around the

vehicle (Approximately 5m), the vehicle movement is immediately stopped, until driver comes within radius and presses the APA switch within 30 seconds. APA will pause if key fob is out of range.

With no action from driver within 30 seconds, the vehicle will be secured by applying parking brake and parking will be terminated.

### Remote Parking Termination conditions:

- Driver door open
- Passengers or rear doors are open
- Tailgate is open
- There is obstacle present in maneuver path
- If pause condition are not removed within 30sec of the pause.
- If no input received from remote devices within the pause timer.
- Low battery of remote key.

### Disclaimer:

- Driver should intervene to override the parking assistance system in case of:

Comfort steering & comfort braking, imminent collisions, malfunctions, driver take over requests.

- If obstacle are detected in maneuvering path then vehicle will go into pause. Under certain conditions system may terminate the maneuver to prevent collision.
- In case one or more malfunctions or sensor damage, relevant notifications will be sent to the driver and the affected vehicle functions may not be available

### Automatic Park Assist Limitations:

- **Parking on inclines:** If slope is above 8%, Automatic Parking Assist Feature will be terminated.
- **Angle of angular parking slot in which vehicle can be parked:** In the Automatic Parking assist system, system is only able to park the vehicle in the angular parking slot of angle in between 60 to 120 deg.
- **Selected Slot gets occupied:** If while maneuvering APA, the selected parking slot gets occupied (by another ve-

hicle or pedestrian). The APA terminates the current maneuver. The driver must select a new parking slot to continue.

- **The vehicle is installed with a snow chain, spare tire, or different size of wheel:** If any wheel of the vehicle is tied with snow chain or installed with spare tyre / different size of wheel then parking will not find the valid parking slot or parking will be fail or parking performance is low.
- **Wheel Alignment and Tyre pressure is not balanced:** APA feature may not work properly if tyre pressure is low or higher than the standard tyre pressure or wheel alignment is not accurate.
- **Detection and tracking by Auto Park Assist:** Fast moving objects may not be detected by system, user need to pay attention while parking-in and out.
- **Attachments on rear bumper:** If vehicle is loaded with cargo longer or wider than vehicle or a trailer is connected, the valid parking slot will not be detected by system or parking will fail.

- **Sensor Position Change:** If sensor position is shifted by an impact to the bumper, APA may not work properly.
- **An object attached to steering wheel:** If a stop lock is attached to steering wheel can cause APA failure.
- **ORVM Fold:** APA is not possible when ORVMs folded manually.

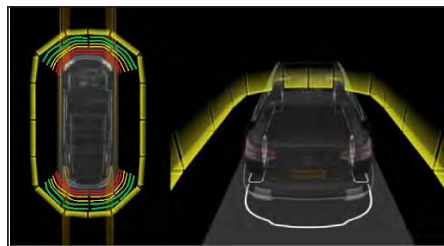
## STARTING AND DRIVING

### Features of APA (If equipped)

#### 1. Parking Distance Warning

The maneuver distance warning warns the driver against collisions with stationary and dynamic obstacles around the car during low-speed maneuvering.

Warning will be provided to the driver about critical objects via visually and acoustically.



#### Visual Warning -

Green, Yellow and Red is used to represent obstacle proximity.

For closer proximity red is used and for farther proximity green is used.

#### Acoustic Warning -

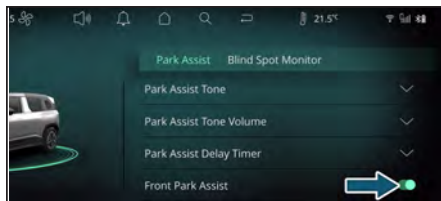
Intermittent tones with increasing fre-

quency from green zone till the last two red lines

Continuous tone for highest criticality.

#### Low Speed Activation -

If Driver has turned ON Low Speed activation from user settings menu and vehicle speed is below 10 kmph (Forward Direction) and some object is detected in the front of the vehicle then audio warning will be there for few seconds, while the audio and visual warning shall continue to be shown till the object is present.



#### *i* NOTE

- *Low Speed Activation can be enabled or disabled provided, if Front Park Assist is enabled.*

- *To activate Maneuver Distance Warning, vehicle speed should be less than 10kmph.*

#### Activation

- Front manoeuvre distance warning can be activated through user setting menu
- It gets activated if Automatic Park Assist is activated.
- Rear Manoeuvre distance warning is automatically activated after every vehicle start.
- Rear Manoeuvre distance warning is automatically activated if reverse gear is engaged.

#### *i* NOTE

*Feature gets deactivated if vehicle speed crosses 15kmph speed*

#### Partial availability of Manoeuvre Distance Warning

- The Maneuver distance warning function deactivate the warnings for the

corresponding side (left, right) when a door is open.

- It deactivates the warning for the front side when bonnet is open.
- It deactivates the warning for the rear side when the tailgate is open.
- It deactivates the warning for the rear side when a trailer is attached.

### NOTE

*A visual warning is displayed on the display if there are any malfunction in Maneuver distance warning due to sensor issue.*

### NOTE

- *The Maneuver distance warning function gradually reduces acoustic warning to muted or directly muted tone when in standstill for more than a given time with an obstacle in intermittent or continuous tone range.*
- *In cases when the obstacle becomes dynamic the audible signals*

*are reinitiated automatically.*

- *Obstacles which are in the predicted driving path will be treated as obstacles that may cause collision.*

### WARNING

- The driver remains responsible at any time for controlling the vehicle
- Driver to watch the surrounding traffic situation closely, be ready to take over steering and braking at any time, and actively intervene where appropriate.
- The system may not detect vehicles moving in a different direction or animals. Apply the brakes when necessary.
- Maneuver distance warning cannot independently react to all traffic scenarios. Driver is responsible for controlling, supervising, and intervening if required during cold and inclement weather conditions.

- Avoid approaching an object too fast. Avoid driving off quickly while Maneuver distance warning is not yet active. There is a risk of injury or risk of damage to property.

## 2. Low Speed AEB and Reverse AEB

Low Speed Autonomous Emergency Braking is a safety function that aims at avoiding or mitigating collisions during forward driving with low speed. Similarly Reverse AEB is used to avoid or mitigate collisions during reversing with low speed. Cameras and ultrasonic sensor fusion is used to detect the obstacle and potential hazards. Low speed or Reverse AEB assists the driver by -

- Emergency braking on static objects
- Emergency braking on dynamic objects

### NOTE

- *The emergency braking function is designed to provide a brake intervention at a late point in time so that*



## STARTING AND DRIVING

*the driver experiences maximum freedom during maneuvering.*

- *The emergency braking function also provides the provision of manual overriding option, if driver depresses the brake pedal fully.*
- *Brake support and emergency braking is active up-to 8kmph of the vehicle speed.*

### Preconditions to activate emergency braking:

- Front seat belts are fastened.
- Vehicle doors are properly closed.
- ORVM should be unfolded

#### NOTE

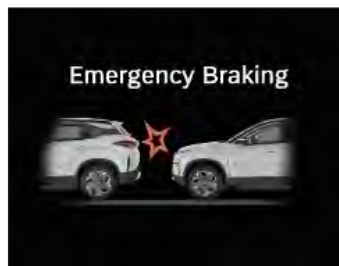
*Emergency braking will not actuate if the preconditions are not satisfied. Also, in case where driver is steering or accelerating*

### Activation / Deactivation

- Low speed AEB is OFF by default on every vehicle start and does not retain

the previous ON/OFF setting from the last vehicle start.

- To activate go to setting> Driver Assistance> Auto park Assist> Low Speed AEB
- Low Speed AEB is designed to safely assist you in low speed maneuvering when exiting parking lots and confined spaces.
- This feature remains active for 500 meters of distance after the feature is turned ON. After 500m distance the feature will be deactivated automatically.



When emergency braking is actuated, Visual warning appears on the screen.



When there is failure or malfunction in the emergency braking function then Tell-Tale symbol will appear on the instrument panel cluster.

#### NOTE

- *Low Speed AEB is not designed or intended to function in stop & go moving traffic.*
- *Once Emergency braking is actuated, and collision is prevented the vehicle will remain stationary for about 5 secs and overrides the driver acceleration or any maneuvering attempt.*
- *Since safety is a priority, the Low Speed AEB system detects and may apply brakes for shallow objects and uneven surfaces.*



### WARNING

- Due to system limits, a collision cannot be prevented under all circumstances. The function is available below 8kmph. The system cannot serve as a substitute for the driver's personal judgment in assessing the traffic situation.
- The driver remains responsible to perform the driving task and monitor the environment. The emergency braking is aid to the driver and does not relieve from the driver's responsibilities to drive with attention and due care.
- With folded ORVMs, trailer hitch, Sensor blockage, the emergency braking function cannot work properly as sensors are concealed.

### 3. Reversing Assist

Reversing Assist (RA) supports the driver to drive backward out of narrow roads (e.g., narrow dead-end road or confusing parking or street situations) by following back the already driven path. Upon activa-

tion of the function the vehicle follows the already driven path by automated lateral and longitudinal control. In case an obstacle appeared in the tracked path, the function brings the vehicle to standstill in order to avoid a collision.

#### Preconditions for Activation:

- Driver has to drive the vehicle in forward direction at least for 5m to enable Reversing Assist.
- The driver's seat belt should be fastened.
- All the doors and trunks of vehicle should be closed.



RA can be enabled in the background by driving the vehicle forward for more than 5 meters at speed less than 30 Kmph and wheel turning at less than 22 Deg/sec.

Once the vehicle is brought to stop after driving, user needs to activate RA by selecting it from the infotainment screen. Once Notification observed on the screen that RA is active user needs to release the break paddle to enable vehicle movement. Gear will be changed automatically to support the reverse assist.

Remaining distance to stop is displayed on the screen.

Once after reaching the desired destination, the driver will be intimated acoustically and visually.

### NOTE

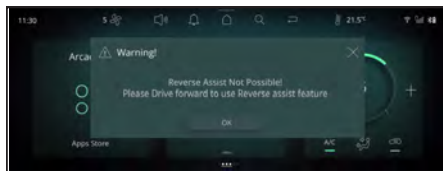
- *Vehicle speed should be less than 30kmph while recording the RA forward path.*
- *Vehicle speed will be less than 5kmph while reverse maneuvering and upto 50 meter distance.*
- *The system remembers the last 50 meters of travel, as long as the vehicle has moved only in the forward direction. Any movement in reverse*

## STARTING AND DRIVING

*resets the memory related to distance traveled*

### Reversing Assist will pause:

- If RA is active and vehicle is idle for duration of 30 sec.
- If brake pedal is pressed
- Any Door/Bonnet/Tailgate is open
- Seatbelt unbuckled
- ORVM folded
- Dynamic/Static object in trajectory.



### WARNING

- Driver should always be attentive while reversing and should intervene when necessary.
- Driver is solely responsible for any damages resulting from:
  - Improper use of system
  - Driving in adverse environmental conditions
  - Driver's negligence
  - Modification/damage to cameras and sensors
- The rear-view camera is used as a detecting sensor to detect pedestrians. If the camera lens is covered with foreign material, such as snow or rain, it may adversely affect camera performance and Reverse Assist may not operate normally.
- Reverse Assist will stop if camera detects any object /pedestrian.
- Reverse Assist may or may not operate properly on slope, driver should always pay attention while reversing.

### Reversing Assist will terminate:

- If driver takes control of steering wheel
- Changes gear
- Driver door is opened
- If vehicle speed increases more than 5kmph.
- If parking brake is applied by the driver.
- If reverse assist is canceled through infotainment screen.
- Any permanent obstacle is detected in the path of parking.
- The gradient of the surface is greater than required limit.
- If it reaches maximum maneuver time i.e., 180s.
- Any internal system functional fault occurs.

### WARNING

Reverse Assist may not work properly:

- Stray animals (dogs, cows, etc.) pedestrian or any other object unexpectedly entering reverse path.

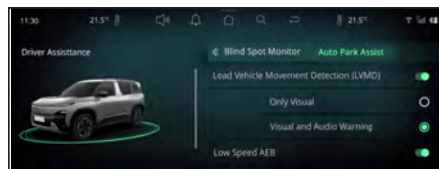
- Street vendors, carts, or two-wheelers moving closely behind the vehicle.
- Vehicle reversing on unpaved roads or gravel where sensor data may be unreliable.
- If reversing path is highly curved
- Reverse Assist feature may not drive back S pattern due to steering angle at wheel level may not meet to offer path in automated parking
- The stored path will not be retained over driving cycle therefore after coming to standstill driver does ignition off then path will be cleared.

#### 4. Lead Vehicle Movement Detection (LVMD)

The Lead Vehicle Movement Detection function monitors when the lead vehicle starts moving from a standstill and notifies the driver with audio and visual alerts about the available space ahead in case the driver is not attentive.

To enable LVMD Go to > Driver Assis-

tance>Auto Park Assist>LVMD



User can select either “Visual only” or “Visual and audio” warning alerts.

The driver is alerted when the lead vehicle starts moving from stationary and the distance between lead and subject vehicle is greater than 3m.

#### Preconditions for Activation:

- The LVMD feature is enabled and active.
- Any gear other than Reverse is engaged.
- There are no detected system faults or failure conditions that would inhibit feature operation.

#### Lead Vehicle selection:

- Lead vehicle is in region of interest.
- Lead vehicle stationary for more than

10 seconds.

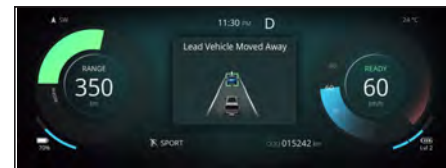
- Lead vehicle has longitudinal distance > 0.3m and <5m from ego vehicle.
- Lead vehicle shall be Car, Truck, Motorcycle and Auto.

#### Lead Vehicle Alert Conditions:

The ego vehicle has remained stationary for more than 5 seconds.

The selected lead vehicle has moved forward by more than 3 meters.

No other lead vehicles or Vulnerable Road Users (VRUs) are detected directly ahead.



#### NOTE

- The system identifies and selects the closest vehicle ahead as the lead vehicle.

## STARTING AND DRIVING

- *If another vehicle cuts in front, it may become the new lead vehicle if it meets the selection criteria.*
- *Two-wheelers are also tracked and can be designated as lead vehicles when they satisfy the selection criteria.*
- *If the lead vehicle leaves the detection zone and no other vehicle is detected, an alert may still be triggered.*
- *Alerts are suppressed if a pedestrian or cyclist enters the gap between vehicles.*
- *Before issuing an alert, the system verifies the presence of other vehicles and may update the lead vehicle selection accordingly.*

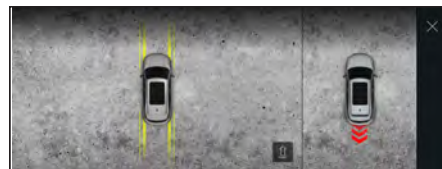
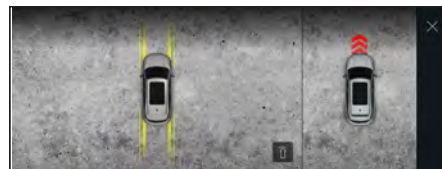
### WARNING

- Default warning is visual warning only.
- Driver should always be attentive in all traffic conditions.

- Cameras, ultrasonic sensors performance may be affected adversely when sensors are covered with foreign material such as snow or rain, or affected by extreme hot or cold weather, or windy environment. In these cases, LVMD may not operate normally.
- LVMD function may not work properly when there is System failure/Faults/ sensor malfunctions are detected.

## 5. Summon Mode

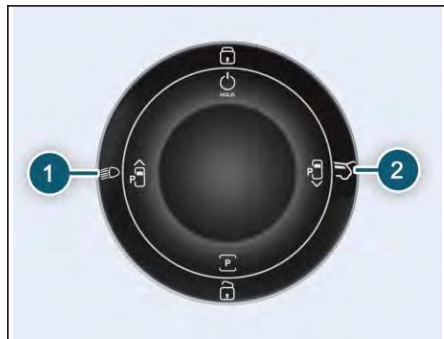
With Summon Mode, you can move the vehicle forward or backward in a straight line from 12 meters to 24 meters using a Remote Key.



### Preconditions for Activation:

- Driver seat belt is not fastened.
- Vehicle doors, tailgate, bonnet are properly closed.
- Vehicle charging above SoC level (Min 5% charged).
- Vehicle charger not connected.

- Vehicle is turned ON through Start/Stop Switch or Remote Key.
- (Remotely turning ON vehicle is already mentioned in remote parking section).
- Smart phone or key is connected with the vehicle via Bluetooth.
- Remote key has sufficient battery.
- All Keys are outside of the vehicle
- Distance between the vehicle and smart phone / key should not be more than 6m



Once the vehicle is powered on:

Press and hold the Forward (1) or Reverse (2) button to move the vehicle in the desired direction, while standing within range, change of direction will terminate the Summon mode.

Releasing the button will pause the vehicle movement for up to 30 seconds. Press and hold the button again to resume.

For pause, cancel, and termination conditions, please refer to the Remote Parking section.

### *i* NOTE

- *User must stay within 6 meters of the vehicle at all the times.*
- *Always monitor the vehicles path to avoid collisions.*
- *Do not operate the vehicle over roadside ditches, potholes, or uneven surfaces.*
- *Visually monitor the surroundings, especially for small animals or objects that may not be detected by the system.*

- *Avoid using Summon Mode on slopes exceeding the specified gradient.*
- *Ensure Remote Key has a sufficient battery.*

## 6. Anywhere Parking

With Anywhere Parking, you can choose exactly where you want to park your vehicle and customize the slot's position and direction.

### Activation:

- Launch APA through infotainment screen/ Press APA switch on console.
- Click on Anywhere Parking icon on screen.



- Stop the vehicle to define parking slot

## STARTING AND DRIVING



- Carefully read the instructions and click on “OK”.



- Click on the screen where you wish to park your vehicle.



- Upon clicking the screen, slot will be

displayed as shown in the image



- Move the slot in all four directions using the arrow marks on the right side of the screen.
- Adjust the orientation of the slot clockwise or anticlockwise. Adjust the slot till it turns to blue in colour.

### *i* NOTE

*Valid slot will be shown in blue colour and invalid slot will be shown in red colour.*



- After finalising the slot position and it is blue, click on the P icon over the slot to confirm the slot.
- Now user can select Auto Parking or Remote Parking.



- Follow all the steps according to previous sections of Auto Park or Remote Park Assist.

### NOTE

- *It is the user's sole responsibility to ensure the selected slot is safe and free of risk.*
- *Do not select the slot over footpath or slot on an elevated plane.*
- *Do not select slot over a drain, road-side ditches, potholes or uneven grounds.*
- *Refer APA/RAPA warning/limitations/disclaimer section before using any sub feature of APA.*

### WARNING

- Auto parking will be stopped if any of the pre-condition of APA is not met.
- In parallel park-out, direction will be opposite of park-in. For angular parking direction will be by default. For perpendicular parking user can select the direction and if park out is not possible in user provided direction then default direction will be se

lected.

- The driver remains responsible at any time for maneuver compliance with road safety and specific road traffic regulations.
- The parking assistance system cannot independently react to all traffic scenarios. Driver is responsible for controlling, supervising, and intervening if required.
- With folded ORVMs, trailer hitch, Sensor blockage, the parking assistance system cannot work properly as sensors are concealed.
- APA system is trained to identify the objects those are connected to the ground.
- In case the parking assistance system is not performing up to the mark (specifically: Slots are not being offered) user should check the ultrasonic sensors are free of any dust, mud, ice or snow accumulation, foreign object present before approaching service Centre.

- The driver must follow all traffic rules and ensure the safety of own vehicle as well as other vehicles, people, animals, and obstacles around the vehicle.
- In case the cameras and sensors are not operational or faulty, it is recommended to not use them and get them immediately inspected at TPEM Authorised Service Centre.

### WARNING

- Automatic Parking Assist will not work properly in case following road conditions:
- Icy roads: Risk of vehicle getting stuck OR risk of uncontrollable vehicle behavior.
- Snow-covered roads: Risk of vehicle getting stuck OR risk of uncontrollable vehicle behavior. Also in snowy weather the sensor cannot detect the obstacle, the performance of obstacle detection will be



## STARTING AND DRIVING

low/ fail. Parking will not find the valid parking slot.

- Muddy & slippery undergrounds: Risk of vehicle getting stuck OR risk of un-controllable vehicle behavior.
- Ice, mud, snow, foreign object accumulation on sensors can affect the functionality of APA. It can lead to blocking, reduced sensing accuracy and reduce system responsiveness making it difficult for sensors to recognize the lane markings, objects, and other vehicles.
- Lighting: Insufficient lighting can hinder the performance of cameras, making it difficult to detect objects.
- Heavy rain: In case of heavy rain, sensor detection may not work properly.
- Close to bank slopes and abysses: Risk for a vehicle crash.
- In cold and severe weather conditions such as heavy rainfall road, heavy snowfall road, foggy condi-

tions, icy road, wet conditions, dusty roads, etc. the parking assistance system may not function also can limit the sensor performance.

- The system might not function properly on steep hills and poor illuminations.
- Parking slots with curbs, parking stoppers, small animals.

### WARNING

Automatic Parking Assist will not work properly in following cases:

- The slot is unmarked or partially occupied.
- The system may not accurately identify specific driving surfaces or detect unsafe driving areas and parking spots; therefore driver should always be attentive.
- The slot dimensions are not as per requirement.
- The surroundings are cluttered or parking slot is very tight.
- If the vehicle's speed is too high.
- Objects present in slot like shopping cart, pipes etc.
- If glare reflection because of Sun or shadow is present in parking slot.
- If unmarked or irregular parking spaces, faded markings present.
- Open manholes or potholes.
-



### ADVANCED DRIVER ASSISTANCE SYSTEMS (ADAS) (if equipped)

The Advanced Driver Assistance System (ADAS) provides the alerts and controls to improve the overall safety of vehicle and increases the driver comfort.

#### WARNING

- If elevations or speed breakers present in parking slot.
- The system performance may vary on uneven roads, gravel roads, or broken roads.
- The system will not function properly if the sensor or sensor assembly is disassembled, sensor is blocked, scratched, or damaged.
- If there are stickers on the cameras or sensors.
- The system performance will be affected in construction sites, diverging/merging roads, near garbage filled roads/areas, road under construction etc.
- The system performance will be affected when driving very close to the other vehicle.
- The objects that are sharp or slim, such as chains, ropes, small poles, cones, blades, etc. might not be detected by the system.

- The system performance may be limited while parking near pillars, between vehicle & pillars.
- The system performance may be limited in underground parking and similar areas with low light intensity.
- System may not detect the overhead objects like low-hanging signs or tree branches, truck rear hanging guards etc.
- Single reference and curb side parking offering may not happen behind wide objects such as Buses , Trucks, Bullock carts, ETC., slot offering behind narrow objects like cycles, scooters, etc. may depend on their manner of parking
- Driver is solely responsible for any damages occur through:
  - Improper use of the system
  - Driver negligence or inattention
  - Feature used in unsupported environments.

- ADAS is only an aid system, it is NOT a substitute for the driver's attention. The driver must always remain in control of the vehicle, observe the surroundings and drive safely.
- ADAS system is assist system for driver for comfort driving. The driver should observe surroundings environment obstacles in vehicle path and judge distance of vehicle from obstacle and apply enough braking. Driver should also observe traffic signs on road, lane marking always and act accordingly.
- The correct operation of the ADAS sensors will be compromised if they

## STARTING AND DRIVING

---

are misaligned due to accident damage at the windshield/bumper area of the vehicle.

### Limitations of Advanced Driver Assistance System

There might be degraded/no/unexpected functionality of ADAS for following cases:

- There is adverse environmental conditions such as heavy snow, heavy rain, etc.
- The ADAS sensors are covered/blocked with snow, dirt, mud, etc.
- There is splash on ADAS sensors due to water logging on the road.
- The temperature around the ADAS sensors is too high or too low.
- A trailer, carrier or other attachment is installed in your vehicle.
- The bumper around the radars is covered with objects such as a stickers, guards, paint, bike rack etc.
- The area around the radars is impacted, damaged or the radars are out of position.
- Windshield area around the ADAS camera is damaged or camera is out of position.
- Your vehicle height is low due to heavy loads, abnormal tyre pressure (tyre pressure is low, uneven) or a tyre is damaged, etc.
- The road contains multiple metallic components (for example, metallic bridges, metal construction poles etc).
- The vehicle drives on a curved road.
- The vehicle severely vibrates while driving on bumpy/uneven paths.
- When the ADAS sensors are blocked by other vehicles, walls or parking-lot pillars.
- The vehicle is driven through a toll-gate.
- Driving on a road where trees or grass are overgrown.
- Driving in areas where the sensor does not detect another vehicle or structure (like bullock cart, horse cart etc) for an extended period of time.
- When an object or vehicle makes sharp lane changes or driving direction changes.
- ADAS features may not operate properly when driving where the heights of the adjacent lanes are different. The function may not detect the vehicle on a road with different lane heights (underpass joining section, grade separated intersections, etc.).
- ADAS features may not operate properly on sloped road conditions.
- ADAS features may not operate properly if interfered by strong electromagnetic waves.
- ADAS features may not operate properly when there are structures beside the driving road. In certain instances, the system may degrade while recognizing the structures (guardrails, street light, road sign, tunnel wall, overhead structures etc.) beside the road.
- Driving on narrow roads.
- A motorcycle or bicycle is near or any vehicle approaches too close.

- In certain complex environmental situations or sensors misaligned, ADAS system may provide false warning or braking or may not provide warning or braking for objects.

### **i NOTE**

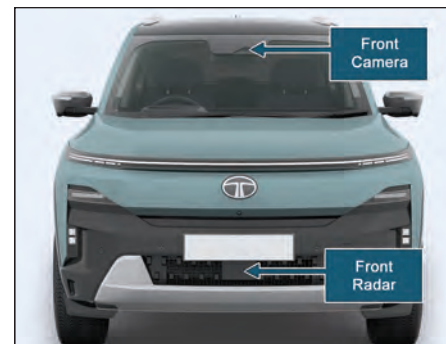
- *Vehicles bumper and windshield should be kept free of snow, ice, mud and must not be covered by any material so as to ensure proper working of ADAS features.*
- *Keep the bumpers and windshield clean.*
- *Radar based functionalities and warnings are given only for moving objects.*

If any damage to bumper/windshield of the vehicle, it is recommended to get the vehicle inspected by TPEM Authorised Service Centre.

### **Following are the features of ADAS**

1. Forward Collision Warning (FCW) and Automatic Emergency Braking (AEB)
2. Lane Keep Assist (LKA) and Lane Depart Warning (LDW)
3. Adaptive Steering Assist (ASA)
4. High Beam Assist (HBA)
5. Traffic Sign Recognition (TSR)
6. Adaptive Cruise Control (ACC) (if equipped)
7. Blind Spot Detection (BSD) and Lane Change Alert (LCA)
8. Rear Cross Traffic Alert (RCTA)
9. Door Open Alert (DOA)
10. Rear Collision Warning (RCW)

### **Front Advanced Driver Assistance System (front ADAS)**



Front ADAS features uses front radar and front windshield camera, which are tentatively placed as below,

#### **1. Forward Collision Warning and Automatic Emergency Braking (FCW & AEB)**

Forward Collision Warning System warns the driver by providing audio and visual warning when there is a possibility of collision with the preceding 4 wheelers, cyclists or pedestrians detected ahead. Automatic Emergency Braking system

## STARTING AND DRIVING

avoids accident or mitigates collision impact by applying brake whenever driver applied brake is not sufficient or not applied.

### Prerequisite for Activation

The following condition shall be satisfied for FCW and AEB to activate.

1. Front Camera and front RADAR are fault/blockage free.
2. Front windshield should be clean.
3. Front bumper should be clean.
4. Vehicle speed should be 7~180 kmph.
5. Gear shall be in Drive or Neutral Mode.
6. Steering wheel position should be stable and not to be turned suddenly.
7. No other system failures or degradation of functions related to AEB systems like Braking system, Steering system etc.

### User Settings

FCW and AEB shall be default ON during start of Vehicle.

When vehicle is at standstill user can select FCW-AEB features settings from infotainment as below

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist

1. User can turn OFF FCW-AEB feature.
2. User can turn ON only FCW or both FCW and AEB.
3. Select FCW sensitivity (low, medium and high) to adjust the distance at which collision warnings are provided:
  - Low - FCW is provided at a lower than normal distance to collision risk
  - Medium - FCW is provided at normal distance to collision risk
  - High - FCW is provided at a higher than normal distance to collision risk

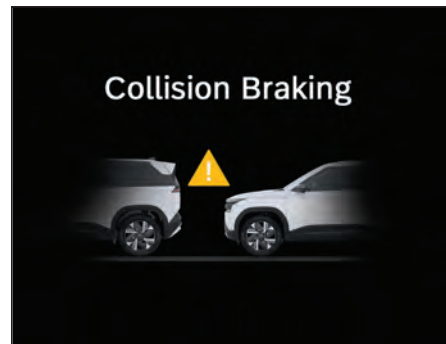
### Feature Operation

Warning Behavior (When FCW and AEB Both OFF)

When both FCW and AEB are disabled, the system does not respond to collision risks and the following tell-tale appears on the instrument cluster panel.



**Warning Behavior (When Only FCW is ON)**



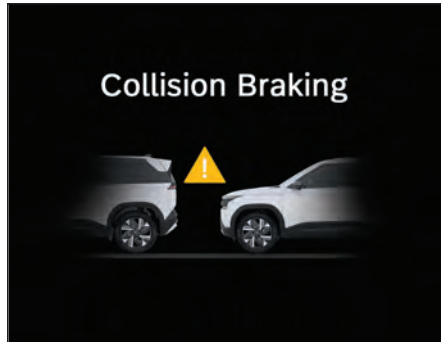
When Only FCW is ON, the system gives only one level of warning. The driver is

alerted to the risk of an imminent collision by visual and audible warning which may be accompanied by a short brake pulse.

### **Warning Behavior (When FCW and AEB Both ON)**

When both FCW and AEB are ON, the system gives two level of warning along with brake intervention as described below.

1st Warning - The driver is alerted to the risk of an imminent collision by visual and audible warning which may be accompanied by a short brake pulse.



2nd Warning - If the driver does not take action or the driver braking force is not enough to mitigate collision, system issues the second level of visual and audible warning and applies emergency braking.



After the vehicle has come to a complete halt post emergency braking, the system holds the vehicle stationary for around 2 seconds. After these 2 seconds it is driver responsibility to control the vehicle and to prevent it from moving forward or creeping.

### **Feature Failure**

#### **Feature Failure Behavior:**

When system fails due to some ob-



struction/blockage of front camera or radar then FCW-AEB feature works in degraded mode. In this case tell-tale may appear on instrument cluster and visual warning pop up will appear along with audio warning.

To resolve this failure the user should clean the front camera or radar for any obstruction/blockage. If the issue still persists then turn Ignition OFF to ON. If the issue is still there then visit TPEM Authorised Service Centre.

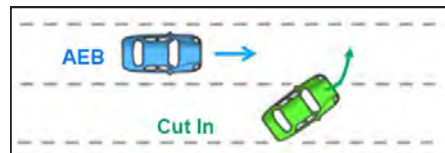
## STARTING AND DRIVING



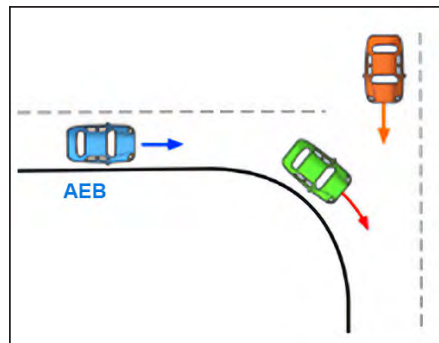
### System Limitation Scenarios

FCW-AEB system is subject to system limitations and may be unavailable or degraded performance will be there in following situations.

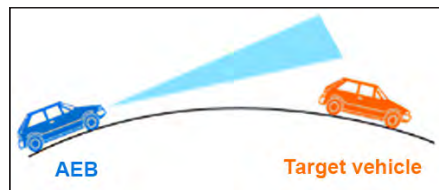
- Close cut-in situations, where another vehicle cuts-in from an adjacent lane at close range



- Intersection crossing scenarios, where other vehicles may approach at an angle to driving path



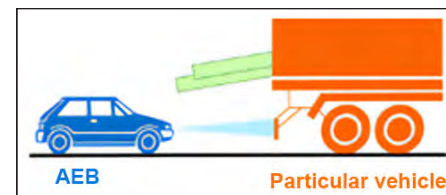
- Hill Situation



Target vehicle is not detected due to road inclination. It is not possible to detect tar-

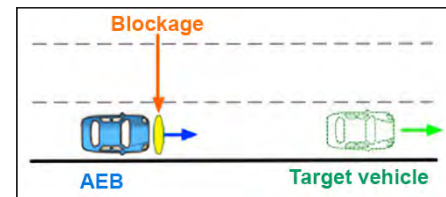
get in uphill and downhill situation.

- Special Vehicles / Vehicle with specific load



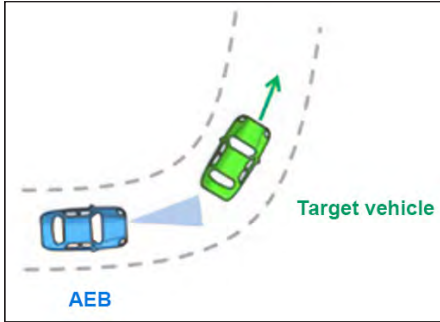
When a specific type of load is loaded or a special vehicle, it can collide due to the sensor detection limitation.

- Sensor Blockage



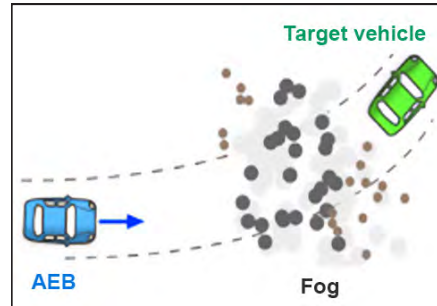
In case of front radar or camera contamination / blockage, collision with the front vehicles or other objects may occur.

- Large Curve Road



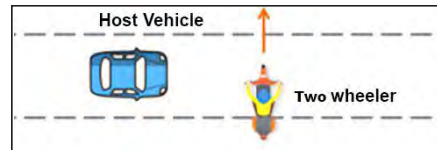
- When driving on a large curve road, due to the sensor limitation, collision with the front vehicles or other objects may occur.

- Driver field of vision



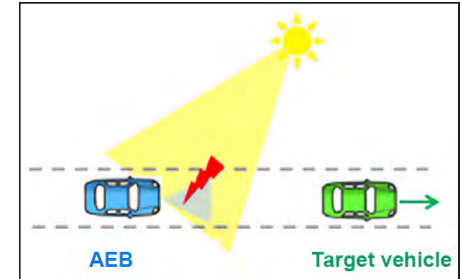
In extreme weather condition (snow, rain, fog, etc.), the sensor limit may cause a collision between the vehicle and other objects ahead.

- Two-Wheeler movement



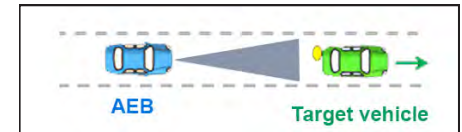
If there is collision danger with two wheeler that moves laterally, collision is possible due to limitations of sensors

- Low Camera visibility



Possible collision with forward vehicles and other objects in situations where the camera's visibility, due to backlight, reflected light, direct sunlight glare, darkness, is lesser.

- Front vehicle asymmetrical Tail-light at night

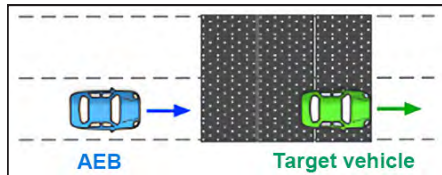


Possible collision due to camera limitations if there is no tail light of the front vehicle

## STARTING AND DRIVING

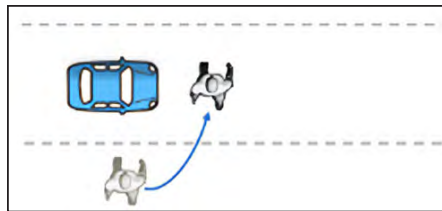
hicle at night or if the mounting position is asymmetrical.

- Detection problem due to complicated structure around the road.



If a construction section, railroad track, or other metallic object is on the road (eg, tollgate, subway construction site, tunnel, lane-proximity guardrail, etc.) then there can be possible collision.

- Pedestrians suddenly jumping in front of the vehicle



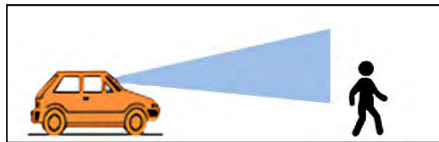
Possible collision if pedestrian suddenly jumps in area where sensor cannot detect.

- If a part of the body of the pedestrian is a covered or standing pedestrian



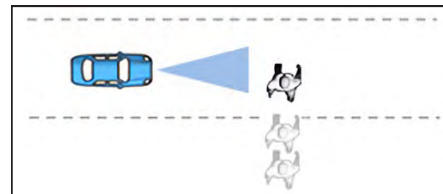
In the case of pedestrians whose body part is obscured by umbrellas, bags, special clothes, etc., they cannot be detected and collision may happen. A pedestrian sitting / lying down may be collided with due to no detection.

- Low illumination



If the illumination is low, possible collision may happen with objects due to no detection by camera.

- Fast moving pedestrians



Possible collision with fast moving pedestrians due to sensor limitations.

- Small pedestrians



Possible collision with small pedestrians due to no detection in camera FOV.

- Performance degradation of radar and camera in extreme weather condition (rain, fog, snow, etc.) and during metallic poles, fences etc. In these scenarios there can be false warning and braking observed by the system, hence, the driver is advised to not rely on the ADAS system in such scenarios.



ios.

- When approaching forward vehicles, pedestrians or bicyclists with low overlap.
- The FCW-AEB system is a collision mitigation system. It does not always guarantee collision avoidance. Driver has to keep attention on road always for any possible collision.
- When driving above 85 kmph, where only partial braking is applied by AEB.
- When approaching pedestrians or bicyclists at above 65 kmph, where the system does not intervene.
- AEB for Vehicles - Junction Turning

To ensure that the FCW-AEB system detects an approaching vehicle in Junction Turning case, the vehicle must be driven at a low speed and oncoming vehicle should be clearly visible. The subject vehicle should be in left lane making a right turn with turn indicator ON.

In junction turning scenarios, FCW-AEB system applies brakes and there may be no warning. Depending on the situation,

the system may be able to mitigate the collision but not avoid it completely. The driver is solely responsible for always driving while maintaining a safe speed and safe distance.

- AEB will not work under following conditions:

I. When driver applies Steering in order to avoid a collision

II. When the driver presses the accelerator pedal beyond a certain limit

III. When target vehicle or object ahead is not detected anymore.

- For some special obstacles like barricades, cows and other animals, there is a possibility that FCW warning and Braking may not be given.

### Disclaimer

- FCW-AEB system may not work in all-weather / Traffic conditions.
- FCW-AEB system is able to detect 4 wheelers, pedestrian, and cyclists but not in all conditions.
- The driver is solely responsible for driving carefully and safely at safe

speed and safe distance in all conditions

- The driver has to follow warnings and instructions before use of FCW-AEB system to avoid serious injury or death.
- It should be noted that AEB system cannot be guaranteed to work 100% effectively in all situations.
- FCW-AEB system should never be tested by driving toward a person or object or animal. This may result in serious injury or loss of life.
- FCW-AEB system may not provide warning and braking for objects in crossing and sometimes in stationary position
- FCW-AEB system works by predicting the path and direction of travel of obstacles. In certain complex environment situations or sensors misaligned. The system may provide false warning or braking or may not provide warning or braking for front object.
- During AEB collision the passengers can experience great discomfort and high jerk while the system applies

## STARTING AND DRIVING

maximum possible brake to avoid/mitigate the collision. This can cause minor to medium level injuries to the passenger. It is advisable that to avoid major injuries the passengers should always wear seatbelt.

- In situations where there is water on the road, there may be false detections and false warning and braking may be given.

### Lane Keep Assist (LKA) And Lane Depart Warning (LDW)

It is a safety feature which helps the driver to keep the vehicle in the lane by providing warning and steering inputs to the user when vehicle departs the lane without driver's intention.

Whenever vehicle moves towards lane edge, driver is alerted with and LKA brings the vehicle back into the lane.

#### Prerequisite for activation

The following condition shall be satisfied to activate LKA-LDW

1. Front Camera should be fault free and blockage free.
2. Front windshield should be clean.
3. Vehicle speed is more than 60kmph.
4. Steering wheel position should be stable and not to be turned suddenly.
5. Lane markings are properly detected and a lane is properly recognized.
6. Turn Indicator of the side of departure is OFF.
7. In case turn signal is given, LKA will re-

sume after few sec when Turn signal is turned OFF.

8. No other system failures or malfunction related to Steering system.

#### WARNING

This is a driving assist system for comfort driving, however driver should always be attentive while driving.

#### User Settings

LKA-LDW shall be default ON at start of vehicle.

User can select this feature through infotainment system user interface:

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist

1. User can turn ON/OFF LKA-LDW feature.
2. User can turn ON only LKA or both LKA-LDW feature.
3. When both LDW & LKA are ON, below points user shall refer:
  - If LKA-LDW is activated through infotainment setting screen and user

presses steering wheel switch, LKA-LDW gets activated.

- If LKA-LDW is activated through infotainment setting screen and user presses steering wheel switch, LKA only will get deactivated and LDW will be activated.
- If LKA-LDW is activated from steering wheel switch and user deactivates it from infotainment setting screen, it will get deactivated.
- If LKA is deactivated from steering wheel switch and user activates it from infotainment setting screen, LKA+LDW will get activated.

### Feature Operation

#### When LDW and LKA both ON

When both lanes not recognized / Vehicle speed <60kmph. Feature will be in standby mode and tell-tale will appear in white color on Instrument cluster.



When one of operating condition is not satisfied (either one lane detected, or both lanes detected).



Feature will be in standby state and tell-tale will appear in instrument cluster. (Lane will be in white color and vehicle in green color).

When both lanes recognized & vehicle speed > 60 Kmph then tell-tale will blink for few sec on instrument cluster. If either right or left lane detected then only that lane will be in green, vehicle and other lane will be in white color.



When vehicle departed either left or right lane then tell-tale will blink for few sec with 3 times audio alert and vehicle and crossed lane will be in red color.



Visual warning in case of system failure "Lane Departure Warning - Lane keep Assist System Failure Please contact service centre" will appear in amber color.



Audio warning for few sec in case of calibration issue "Lane Departure Warning - Lane Keep Assist System Not calibrated. Please contact service centre".

When LKA feature is active steering correction tell-tale will appear in Instrument cluster in green color.



LKA feature gives audio and visual warnings in case of hands OFF situation and next intervention occurring within 180 sec, however driver should always be attentive while driving.



### NOTE

If steering wheel is hold with very light grip, system will prompt user to keep hands on steering wheel.

### WARNING

Warning will be deactivated once driver keep the hands on steering wheel.

## STARTING AND DRIVING

### NOTE

*LKA and LDW will not work for the side where vehicle is drifting when same side of indicator signal is ON.*

### Feature Failure

#### Warning Indicator (When only LDW ON)

When LKA-LDW feature fails or system is not calibrated then tell-tale will appear on instrument cluster in amber color.



Visual warning in case of system failure “Lane Departure Warning System Failure Please contact service centre”.



Audio warning for few sec in case of calibration issue with visual warning “Lane Departure Warning System Not calibrated. Please contact service centre”.



### System Limitation Scenarios

LKA-LDW system is subject to system limitations and may be unavailable or degraded performance or false warnings may be there in following situations. In these situations, it is advisable to turn the LKA-LDW function OFF and driver should take full control of the vehicle:

- Road Edges are not detected if vehicle is departing lane
- Road under construction

- Reduced visibility due to bad weather on account of rain, fog, snow etc
- Roads with indistinct/faded lane markings
- Road sections with sharp curves
- Narrow roads
- High speed driving especially in sharp curves
- Road sections where there are turn lanes or when the original lane merges or separates
- Passing through tunnels when brightness changes suddenly.
- Horizontal or vertical slope
- Preceding vehicle obstructing lane visibility
- Improper headlight aiming/headlight covered with dirt
- Any other environmental conditions affecting camera vision
- Poor visibility due to sun glare or blockage in lens
- Non Standard Lane Marking (e.g. Rural road lane marking, cat eye lane

marking, width of lane marking is not as per IRC standard etc)

- When driving on a road with reflective material on the road surface that may interfere with lane marking detection.
- All other painted signs on a road such as a road sign, an arrow, a zebra crossing, a figure, a letter, tyre mark, etc.
- When lane like pattern is formed on the road due to any spillage or grass, tiles/road, system may provide unexpected lane departure warning.
- All artificial structures near a lane such as a guardrail, an outer wall of a tunnel, a sidewalk, curb and a lane change prohibition bar, etc.
- Shadow of all artificial structure and other vehicles.
- Preceding vehicle and a cut-in/out vehicle.
- A boundary between asphalt road and concrete road.
- Erased, faded, or damaged lane.
- An edge occurred by color discontinu-

ity of asphalt.

- When there are duplicate lane markings in close proximity.
- When there are hatch and hazard markings in between lane markings.

### WARNING

LKA-LDW is intended to assist the driver and thus driver should pay attention to the road all the times, taking primary responsibility of driving. This feature cannot function as intended in all kinds of driving, weather, traffic and road conditions.

### **Adaptive Steering Assist (ASA)**

Adaptive Steering Assist (ASA) maintains the vehicle's position at the centre of a lane.

The system works on the basis of detected lane markings and helps reduce the steering effort required of the driver, by providing continuous steering corrections.

When conditions are met, in the absence of adequate lane markings, the system can also follow the path of a preceding vehicle, increasing the availability of the system. The driver can override the system at any time by applying positive steering inputs or accelerator pedal input.

The system requires the driver to hold the steering wheel at all times and pay attention to the road.

Adaptive Steering Assist works along with Adaptive Cruise Control (ACC). Thus, Adaptive Steering Assist requires ACC to be activated.

### **Prerequisite for activation**

The following conditions shall be satisfied to activate ASA:

## STARTING AND DRIVING

---

- ASA is enabled through driver assist settings.
- Adaptive Cruise Control is active and not overridden by driver accelerator pedal input.
- All prerequisites of ACC are met.
- Standard lane markings are detected clearly on any one side.
- Lane width is as per standard road building norms applicable to the country (typically 2.5-4.5 meters)
- The vehicle position is stable, entirely within lane boundaries and the lane is recognized by the system.
- Turn indicators are OFF.
- Steering wheel position should be stable and not to be turned suddenly.
- Vehicle speed should be within 30 to 165 kmph.
- No malfunctions of Adaptive Steering Assist & Electronic Stability Control systems

Once activated, the system can continue operation when lane markings are de-

tected on only one side, or if lane markings are not detected on either side but a preceding vehicle is being followed.

### User Settings

Adaptive Steering Assist shall be by default, ON with each new start cycle of the vehicle.

Adaptive Steering Assist can be selected from the driver assist settings from Infotainment screen.

Follow the following sequence:

Go to Home page » All Apps » Settings » Driver Assist

### Following the Preceding Vehicle Path

The system can follow the path of a preceding vehicle (excluding 2-wheeled vehicles such as bicycles, motorcycles, scooters etc.) and continue offering assistance even when lane marking detection is not clear, when the following conditions are met:

Vehicle speed is lower than 75 km/hr (once activated, this mode of control shall be released when the vehicle speed exceeds 80 km/hr).

Distance between driven vehicle and preceding vehicle is within 50 meters (once activated, this mode of control shall be released when the distance exceeds 70 meters).

The preceding vehicle's path has not deviated too much with respect to driven vehicle, or with respect to the path predicted for the preceding vehicle by the system.

### System Limitation Scenarios

ASA system is subject to system limitations and may be unavailable or degraded performance or give false warnings in following situations.

- Inclement weather (rain, snow, fog, dust, haze, mist etc.)
- Camera blockage (direct sunlight or headlamp glare on camera, mud/dirt/moisture/ stickers/ debris on windshield surface in front of camera etc.)
- Low ambient light (headlamps of driven vehicle/ tail lamps of preceding vehicle not working or providing insufficient illumination, no/ inadequate street lighting, loss of lane marking retro reflectivity etc.)

- Strong shadows falling across road or vehicles.
- Faded lane markings or obscuration of markings due to barricading, dirt, debris, water, oil, traffic, poor road surface, construction activity etc.
- Non Standard Lane Marking Like Rural road lane marking, cat eye lane marking, width of lane marking is not as per IRC standard etc.
- Low contrast of vehicles with horizon, lane markings with road surface
- Passing through a tunnel (sudden brightness change).
- Road curves that are very sharp or vehicle speed that is too high for a curve (typically, standard highway curves can be managed by the system at applicable highway speed limits)
- Merging or diverging lane markings, duplicate lane markings, intersections where lane markings disappear.
- Road seam lines that create a contrast similar to lane markings, causing mis-detection of actual lane markings and/or improper control.
- High road camber
- Tyre pressure not as per recommended specification
- High vertical slopes
- Rough roads, undulating road surfaces, high occurrence of rumble strips, speed breakers, potholes etc.
- Frequent presence of multiple vehicles in the same lane or moving along lane edges (such as 2-wheeled vehicles and cars occupying the same lane, weaving in and out of the lane along the lane edge or a wide load body of a truck or bus intruding into the lane).

### NOTE

*The ASA system cannot control vehicle position basis vehicles or objects on the side and the driver must monitor the environment and ensure safe steering at all times.*

### WARNING

It is recommended that the system is primarily used on highways with good lane markings, smooth road surface, gradual curves/ gradients/ camber designed for highway speeds and traffic moving at highway speeds with good lane discipline.

### Warning Indicators

When Adaptive Steering Assist is ON but in the standby state, due to any of the pre-conditions not being met, this tell-tale is displayed in the Instrument Cluster, in grey colour.



When the Adaptive Steering Assist feature is ON and active (the feature is actively controlling vehicle position for maintaining lane centre), the above tell-tale is displayed in the Instrument Cluster, in green colour.



## STARTING AND DRIVING

When a system failure of Adaptive Steering Assist is detected, this tell-tale is displayed in the Instrument Cluster, along with visual warning “Adaptive Steering Assist malfunction. Please contact service centre”.



### Warning Indicator in case of Hands OFF situation

ASA feature gives audio and visual warnings in case of hands OFF situation.

If the driver takes their hands off the wheel, the system allows operation for a limited time and then deactivates both Adaptive Cruise Control and Adaptive Steering Assist giving prior warnings as follow:

First instant warning - After taking hands off steering wheel green coloured graphic prompting the driver to hold the steering wheel, accompanied by a single audio alert.

**Keep Hands On Steering Wheel**



Second instant warning - After taking hands off steering wheel amber coloured graphic prompting the driver to hold the steering wheel, accompanied by a continuous audio alert.

**Keep Hands On Steering Wheel**



### High Beam Assist (HBA)

High-beam assist recognizes headlamps of on-coming vehicle & tail lamps of leading vehicle in night condition switching headlights between high and low beam automatically.

#### Prerequisite for activation

1. Front Camera should be fault and blockage free.
2. Front windshield should be clean.
3. HBA is ON in Infotainment Setting
4. Auto light mode is turned ON.
5. High Beam is ON.
6. Vehicle speed is more than around 20 Km/h.

#### User Settings

HBA retains last user settings after every ignition ON.

1. User can turn HBA feature OFF from Infotainment Settings
  2. User can turn HBA feature ON from Infotainment Settings
- In infotainment system user interface,



press the following button sequence to reach HBA user settings page

Go to Home page >> All App >> Settings  
>> Driver Assistance >> Drive assist

### Feature Operation

#### Warning Behavior (When HBA ON)

When HBA feature is active tell-tale or icon shall appear in the instrument cluster panel in green color.



In this case, when an oncoming vehicle is detected, the system switches the High Beam to Low Beam. And when the vehicle is passed, then the Low beam is switched back to High Beam.

If High Beam is turned OFF due to bright streetlights, then once the streetlight location is crossed completely, High Beam may be turned back ON.

#### Feature Failure Warning Behavior

If HBA system fails, then pop up message may appear in the instrument cluster along with tell-tale in amber color.



To resolve this failure the user should try cleaning the windshield camera of any obstruction/blockage. If the issue persists, then turn Ignition OFF to ON. If the issue is still there, then visit service center.

#### Limitations of High Beam Assist

HBA is subject to certain system limitations and may keep High beam OFF for certain limitations and ON for few other limitation conditions.

- Operation at up/down hill.
- Operation at curve.
- If HBA fails due to any reason including vehicle head lamp fault.
- Unrecognized Front headlamp of oncoming vehicle.
- Unrecognized rear taillight.
- Construction area: For temporarily installed reflectors. There is a possibility of false recognition of temporarily installed reflectors.
- In the case of rainy weather: Water is left on the road after rain, and light from the light source is reflected. There is a possibility of misrecognition due to

reflection of the light source due to the moisture of the road during rainy weather.

- After detecting low beam from oncoming vehicle, HBA may give flash from low beam to high beam and again low beam due to any other light sources present around or due to camera detection performance.

#### Disclaimer

- If HBA detects the light sources of oncoming vehicle only for short period, HBA can stay in high beam without switching low beam.
- In the case that obviously appeared light sources of oncoming vehicle over the guardrail are detected, HBA may switch high beam to low beam.
- In the city or in a light source area such as traffic sign, electronic sign board, building light, streetlight, HBA may switch high beam to low beam.
- HBA shall not operate at daytime even though High Beam is on by the auto light sensor.

## STARTING AND DRIVING

- HBA will operate in following weather conditions:
  - The normal nighttime environment in which the light of the front vehicle is detected by the naked eye.

### Traffic Sign Recognition (TSR)

Traffic Sign recognition (TSR) is a feature by which a vehicle is able to recognize the traffic signs available on the road.

### Prerequisite for Activation

Front camera should be fault and blockage free

Front windshield should be clean

Vehicle speed is above 0 km/h

### User Settings

TSR shall be default ON at the start of vehicle.

- User can turn TSR feature ON from Infotainment setting menu.
- User can turn TSR feature OFF from Infotainment setting menu.

In infotainment system user interface, press the following button sequence to reach TSR user settings page

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist

### Feature Operation



Sign is displayed in the instrument cluster panel when TSR recognizes the Speed Limit

Sign.



Sign is displayed in the instrument cluster panel when TSR recognizes the End of speed limit sign/End of restriction sign.



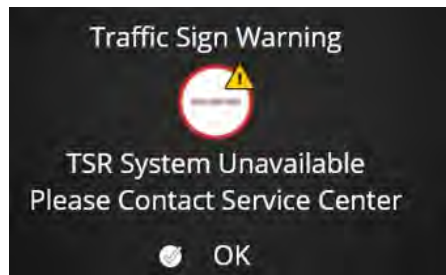
Sign is displayed in the instrument cluster panel when TSR recognizes the No Overtaking

Allowed sign.

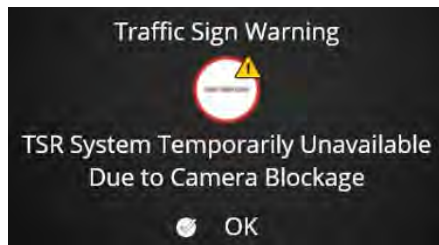
### Failure Warning Behavior

If the system malfunctions, below tell-tale will appear in the instrument cluster panel and a pop up message "TSR System Unavailable. Please Contact Service Center" is shown on Infotainment screen.





If the system Temporary Unavailable due to camera blockage, tell-tale will appear in the instrument cluster panel and a pop up message "TSR System Temporarily Unavailable. Please Contact Service Center" is shown on Infotainment screen.



To resolve this failure the user should try cleaning the windshield camera of any obstruction/blockage. If the issue persists, then turn Ignition OFF to ON. If the issue is still there, then visit service center.

### Limitations of TSR

TSR cannot recognize or it can misrecognize the signs on the road under some conditions. Those conditions are as follows

#### 1. Day coarse conditions

- In the case that the traffic signs cannot be distinguished due to shadow caused by overpass or trees.
- In case of low sunlight, heavy rain and heavy snow.

- In case of traffic sign occluded by obstacle near road such as a tree, vehicle, etc.
  - In case of damaged traffic sign.
  - In case of poor visibility being impossible to recognize traffic signs.
  - In case of faded traffic sign.
  - In case of strong curve road
  - In case of multi-lane traffic signs may not recognized in third lane due to long distance.
- #### 2. Night coarse conditions
- In case of poor illumination to the traffic sign caused by headlamp lighting angle.
  - In case of reflection from the traffic sign.
  - In case of low sunlight, heavy rain and heavy snow.
  - In case of traffic sign occluded by obstacle near road such as a tree, vehicle, etc.
  - In case of damaged traffic sign.
  - In case of poor visibility being im-

## STARTING AND DRIVING

---

possible to recognize traffic signs.

- In case of faded traffic sign.

### **Disclaimer:**

The traffic sign on road which is not as per Vienna convention may not get detected. TSR signs will be detected only when they fall in TSR system detection zone.

Following signs may not be detected in every case or at all:

- Non-standard signs which are not as per Vienna Convention
- Signs with yellow background
- Speed Limit signs ending in 5 like with 15, 25, etc.
- Speed limit signs having extra characters like kmph etc.
- Speed limit signs having extra images like small cars etc.
- Traffic signs will be detected only when they fall in TSR system detection zone within 10 Mtr.

### **Adaptive Cruise Control (ACC) (if equipped)**

Adaptive Cruise Control (ACC) is a comfort feature that allows a vehicle's cruise control system to adapt the host vehicle's speed automatically even in traffic conditions. Feature detects vehicles, which are in the host vehicle's path. If slower moving vehicle is detected the ACC system will slow down the host vehicle speed and maintain appropriate distance from the target vehicle. If the system detects that the forward vehicle is no longer present in the host vehicle's path, the system will accelerate the vehicle back to its set cruise control speed.

The feature identifies the nature of traffic and distance from other vehicles and help to adjust the speed and distance of the host vehicle according to vehicle moving in front.

### **Disclaimer**

- ACC is drive assist feature and it will not avoid collision.
- Please read and understand owner's manual before using ACC feature in

real conditions. Driver should be always careful and attentive while using ACC feature.

- Driver is always responsible when driving vehicle and in all environment conditions even if ACC is performing its control.
- Driver should follow all safety rules and regulations, traffic rules. Its driver responsibility to be alert and driver should have always control on vehicle even in ACC on mode.
- Driver should have safe and proper distance from preceding vehicle and apply brake on time as per real situation if system does not decelerate vehicle or does not maintain speed or distance from preceding vehicle.
- Driver should not validate ACC feature in real road conditions, incorrect way to use ACC may lead to severe accident and damage.
- Driver should always follow ACC alerts or warning given by system.
- ACC may decelerate slowly or may not stop the vehicle in time so driver

should always be attentive and intervene through applying brake if required.

- Always drive in ACC active mode within speed limits.
- ACC should be used in less or moderate traffic road conditions. Use of ACC where pedestrians crossing path is normal may lead to accident.
- ACC may not detect target and decelerate speed when any four-wheeler or two-wheeler cut-in immediately.
- ACC should always use in well-constructed and maintained road conditions.
- ACC should always use on road where clear visible lane marking and signboards are available.

### ACC Switches



The steering wheel switches are used for ACC operations.

**ACC Cruise ON/OFF:** This button is used to ON/OFF the ACC function.



**ACC Cruise RESUME/CANCEL:** This button is used to resume to the initial set speed OR cancel the ACC function temporarily in same IGN cycle.



**ACC SET+:** This button is used to increase the Set speed. If user



presses button for short duration, then set speed will increase by 1kmph and for long press set speed will increase by 10kmph.

**ACC SET-:** This button is used to decrease the Set speed. If user presses button for short duration, then set speed will decrease by 1kmph and for long press set speed will decrease by 10kmph.



**ACC Time Gap:** This button is used to set the desired distance between the host vehicle and the target vehicle.



### Pre-requisites for Activation

- Front Camera and front RADAR are fault/blockage free.
- Front windshield and front bumper should be clean.
- Gear shall be in Drive Mode.
- Driver Seat belt should be buckle.
- Driver Door should be close.
- Autonomous braking feature should be enabled from User settings.
- Park Brake should not be engaged.

## STARTING AND DRIVING

---

- Brake Pedal should not be pressed.
- Accelerator pedal should not be pressed
- Vehicle speed should be within 7 to 180 Km/h.
- No other system failures or degradation of functions related to AEB systems like Braking system, Engine management system, Steering system etc.
- FCW + AEB feature should be active from ADAS setting menu.

### User Settings (Steering Wheel)

- When User presses ACC ON button from Steering wheel, this turns ACC in Standby Mode.
- Once vehicle speed is above 7 km/h and user presses the Resume button on steering wheel, this turns ACC in engaged mode and vehicle speed will set to 30 Km/h.
- When User presses ACC RESUME OR SET+/SET- short or long button from steering wheel, this turns ACC standby mode to ACC active mode un-

less any feature inhibit conditions are true.

- When user presses SET+/- button ACC set speed will increase/decrease.
- When user presses Cancel button from steering wheel, this turns ACC from Engaged to Standby.
- During ACC active follow mode if vehicle stops for more than 5 sec then user has to press resume button to engage ACC again.
- When ACC is in standby mode and user presses ACC Time Gap switch for long duration then ACC will turn to CC standby mode. If user wants to transit from CC to ACC mode again then user need to press same button (ACC Time Gap Switch).
- User can change/set the time gap in active and standby mode by pressing ACC time gap switch.
- When User presses ACC OFF button from Steering wheel, this turns ACC OFF.
- When User Engages ACC at a speed

lower than 30km/h and more than 7 km/h, then ACC speed will be set as 30km/h and vehicle speed will increase to 30km/h if there is no obstacle present in the path.

- When User Engages ACC at a speed higher than 30km/h, then that speed will be set as ACC speed and vehicle will cruise at that speed if no obstacle is present in the path.
- If ACC is ON/Engaged and User Activates the Hill Descent Control function by pressing the HDC button, then ACC will be OFF/Standby. If HDC is already activated and user tries to activate the ACC, then ACC won't be active. HDC function has always higher priority than ACC.
- Overtake Assist Control: During the Follow Control mode, if the driver has will to overtake the target vehicle, OAC function will help to overtake the target vehicle by enhancing the acceleration smoothly. For OAC activation, Turn indicator must be ON and minimum host vehicle speed should be 20km/h.

### ACC Modes

**Active Cruise Mode** – If there is no vehicle in the path of subject vehicle, then it will cruise at a speed equal to ACC set speed.

**Active Follow Mode** – If there is a vehicle in the path of the subject vehicle, then it will maintain a distance from the target vehicle equivalent to the distance set by the user.

**Stop - Hold Mode** – If the preceding vehicle is stopped then host vehicle will also stop. If the preceding vehicle moves then host vehicle will also move without user intervention. Now after 5 sec of stop mode to restart ACC user intervention is required irrespective of preceding vehicle. User can exist from stop hold mode by double pressing ACC resume/Cancel button.

### ACC Telltale Behaviours

**Standby Mode** – During this mode ACC will be ON in standby mode as user presses ACC ON/OFF switch. ACC will not perform any control in standby mode. This telltale will be shown to user in white



color when ACC is in Standby mode.

**Active Mode** – During this mode ACC will be in engaged mode and control for ACC function is performed as per valid target detection. In ACC active mode above telltale will be shown to user in green color



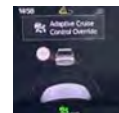
### ACC Display Behaviour

1. Visual Display behavior when ACC is in Active Cruise mode - then only subject vehicle to be shown to user.
2. Visual Display behavior when ACC is in follow mode - then target vehicle will be shown in blue color.
3. Visual Display behavior when ACC is standby mode - then only subject vehicle to be shown to user.

### ACC Warning Behaviour

1. Warning behavior when ACC is in Override

When ACC is Active, driver takes the control of the Vehicle by pressing accelerator. Below popup will appear on the instrument cluster panel.



2. Warning behavior when ACC is Available to Resume.

When Vehicle stopped following a target vehicle and ACC is Active, Driver can resume the vehicle by pressing Resume switch or Acceleration pedal to restart the ACC if the target vehicle moves. Below popup will appear on the instrument cluster panel.

If the target vehicle moves before 5 seconds, then host vehicle will resume itself. No User intervention is required here.



3. Warning behavior when Front Object is disappeared at low speed.

When ACC is active and following a target vehicle moving at low speed (Under 30kmph) and If target vehicle disappears,

## STARTING AND DRIVING

below warning will appear on instrument cluster panel. Here the vehicle speed will increase automatically to the minimum set speed (30kpmh).



4. Warning behaviour when front vehicle apply rapid brake.

When target vehicle is shown in red color, user has to take over ACC system. The situation when ACC de-celeration is insufficient due to rapid braking of preceding vehicle, the user must aware about this case and take control to apply enough braking. ACC will get deactivated and go to standby state when Brake pedal is pressed.

5. Warning behavior when ACC is in Safety check mode.

When ACC is in safety check mode then "Safety Check in Progress. ACC System Will be available in while" message will be shown to user for 5 sec. During this mode ACC will be unavailable to user.

### Failure Warning Behaviour of ACC

1. Telltale behavior when ACC Permanent failure - ACC Permanent unavailable. This below telltale will appear in orange color in instrument cluster panel.



#### *i* NOTE

*Please Visit Nearest TPEM Authorised Service Centre.*

During permanent ACC failure warning message will appear to customer as Adaptive Cruise control System Failure Please contact TPEM Authorised Service Centre for 5 sec of duration. In this case the customer should visit the TPEM Authorised Service Centre and get the issue rectified.

2. Telltale behavior when ACC Temporary failure - ACC Temporarily unavailable due to Radar/Camera blockage. Below telltale will appear in instrument cluster

panel.

#### *i* NOTE

*Please Visit Nearest TPEM Authorised Service Centre to Unblock Radar/Camera.*

During ACC temporary failure warning will appear to customer as Adaptive cruise control temporarily unavailable due to Radar/Camera blockage for 5 sec of duration. In this case the customer should ensure cleaning any dust/mud/blockage from the RADAR and Camera mounting area (Front Bumper and Windshield), then make IGN OFF and then IGN ON again. If this doesn't solve the issue, then customer should visit the TPEM Authorised Service Centre and get the issue rectified.

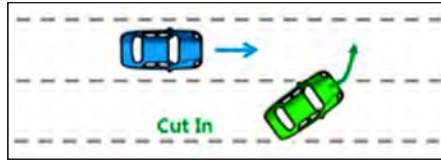
### System Limitation Scenarios

ACC system is subject to system limitations and may be unavailable or degraded performance in following situations.

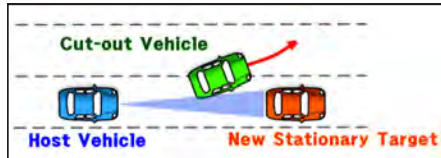
1. Risk of collision in close cut-in - When vehicle is active with ACC and accelerating, if any vehicle comes immediately



from adjacent lane, then there is risk of collision if the distance between subject and target vehicle is small.

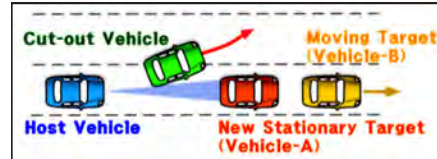


2. Risk of collision in close cut-out vehicle - When Host vehicle ACC is active and it is following a target vehicle, then if the target vehicle changes its lane and a stationary vehicle is present ahead in the host vehicle lane then there is a possibility of collision with the new stationary target vehicle due to backlight glare or other Camera limitation.

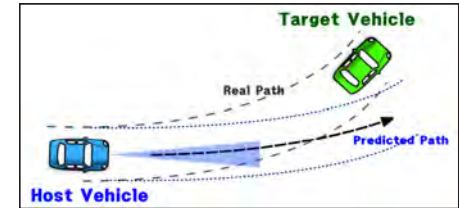
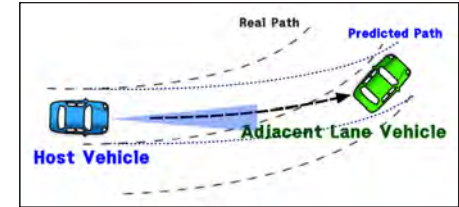


3. Risk of collision ignoring stopped vehicle - When Host vehicle ACC is active

and it is following a target vehicle, then if the target vehicle changes its lane and a stationary vehicle is present and another moving vehicle is present ahead of that in the host vehicle lane then there is a possibility of collision with the new stationary target vehicle due to moving vehicle being recognized as the new target.

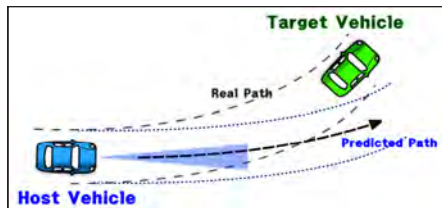


4. Risk of missing lane due to curvature entry/exit situation - ACC system predicts the lane curvature in driving appropriately. If curvature of the road differs from the predicted curvature, then adjacent lane can be selected as the host vehicle lane and target vehicle in the actual host vehicle lane can be missed. Alternately, a vehicle present in the adjacent lane can be detected as target vehicle.

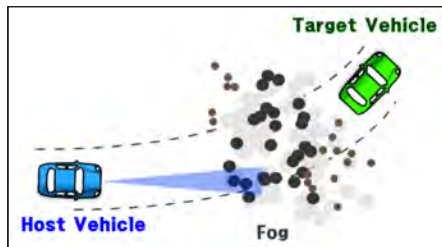


5. Risk of missing lane due to Excessive Curvature Curve - ACC system is designed to work with minimum curvature of around 125m. If the radius of curvature of the road is smaller than 125m, then an inadequate acceleration/deceleration can occur due to difficulty of detecting the inner lane.

## STARTING AND DRIVING

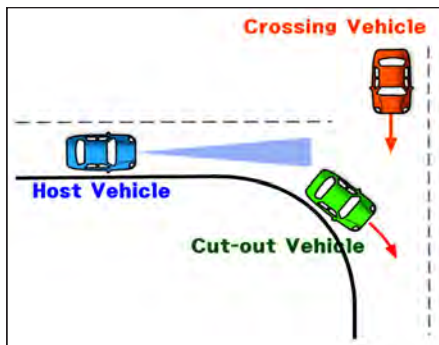


6. Risk of collision due to weather conditions - If the driver uses the ACC in the bad weather conditions (snow, rain, fog, etc.) or during low visibility situation, the sensor limit may cause a collision between the vehicle and pedestrians/Vehicle ahead.

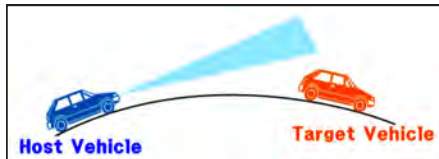


7. Risk of collision due to Intersection situation - If the vehicle ahead cuts out at the intersection, there is a possibility of

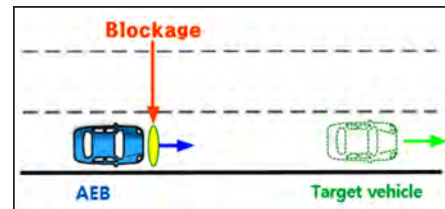
collision with the transverse moving vehicle due to rapid acceleration of host vehicle.



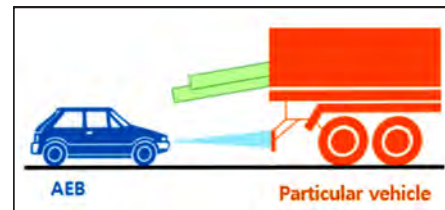
8. Risk of collision due to road gradient change - The target vehicle can be missed when crossing a section of road where the road gradient changes. For e.g. a hill or an underground road.



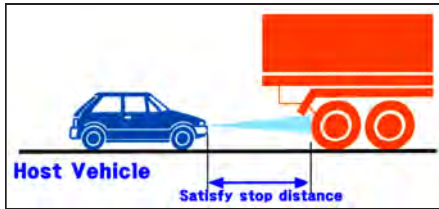
9. Risk of collision due to Sensor Blockage - In case of Sensors (front radar, front camera) contamination, blockage case can occur, and collision of the front vehicle and pedestrians is possible.



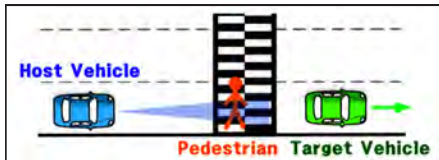
10. Risk of collision due to Particular vehicle - When a specific type of load is loaded or a special vehicle, it can collide if it is not selected as the target vehicle due to the sensor detection limit.



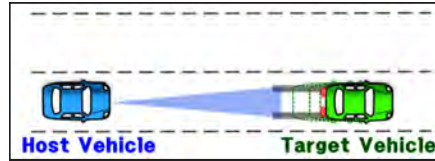
11. Risk of collision due to excessive load of preceding vehicle - If the height of front preceding vehicle is very high, in such case sensor may not recognize the vehicle and collision may occur.



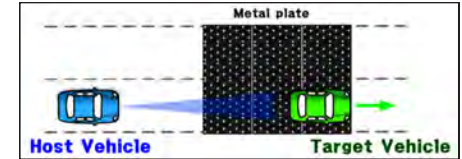
12. Risk of Collision due to inability to recognize the pedestrian - ACC system doesn't control people, especially if a pedestrian appears in front of the host vehicle during the Stop & Go situation, the host vehicle may collide with the pedestrian.



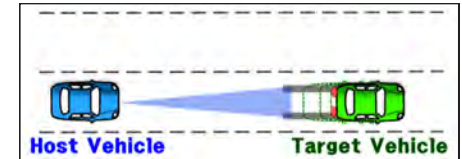
13. Risk of Collision due to excessive braking of the preceding vehicle - When ACC system is active and vehicle is moving, if preceding vehicle applied sudden brake, then collision may occur.



14. Detection performance deterioration due to roadside structures - If a construction section, railroad track, or other metallic object is on the road (e.g., tollgate, subway construction site, tunnel, lane-proximity guardrail, etc.), this may affect the detection performance and front vehicle may not be detected. This may cause the collision.



15. Takeover Request If excessive braking applied by Preceding Vehicle - If the Preceding Vehicle brakes beyond the ACC maximum deceleration limit, Host vehicle might collide with target vehicle. In such case ACC Takeover Request warning displays on cluster to alert the Host vehicle Driver.



## STARTING AND DRIVING

### Intelligent Speed Assistance (isa)

ISA (Intelligent Speed Assist System) is a safety feature that adjust the ACC-set speed based on the speed limit detected via traffic signboards and/or Map pro-vided speed limit. ISA system ensures that vehicle speed does not exceed a safe or legal speed limit while ACC is active.

#### Prerequisite for activation

The front facing camera must be operational and free of obstructions.

Driver must ensure that both the ISA app and the Navigation app are up-dated to the latest version available in the Arcade app store.

Driver must ensure the Traffic Sign Recognition (TSR) system is active.

Driver must ensure that the Adaptive Cruise Control (ACC) system is active in order to utilize the ISA functionality.

Driver must ensure that there are no failures or degradation of functions related to the steering system, braking system, or any other critical systems.

#### NOTE

*This system does not guarantee safety or autonomous driving, so the driver should always pay close attention.*

#### How to use ISA feature

The ISA feature can be activated from infotainment by pressing the following sequence:

- Go to Home page >> All App >> Settings >> Driver Assistance >> Traffic sign recognition and Speed Assist >> Intelligent Speed Assist.
- It is important to ensure that the Adaptive Cruise Control (ACC) mode is active before utilizing the Intelligent Speed Assist (ISA) feature.
- When the ISA system detects that the current road speed limit is lower than the ACC set speed value, the driver will be notified by a blinking this tell-tale indicator on the instrument cluster for 5 seconds, accompanied by a single audible chime alert.



- Upon hearing the chime alert and observing the blinking tell-tale indicator as mentioned above, the driver can press the “SET-” button (while the tell-tale is blinking) on the steering wheel to adjust the ACC set speed to match the detected speed limit.
- When the ACC Set Speed is adapted/adjusted to the detected speed limit, the driver will receive the following pop-up message for a duration of 5 seconds: “ACC set speed adapted to detected speed limit.”
- By observing the above mentioned tell-tale indicator blinking, the driver can press the “SET-” button (while tell-tale is blinking) on the steering wheel to adjust the ACC set speed to match the detected speed limit.
- When ACC Set Speed is adapted / adjusted to detected speed limit, Driver will get below pop up message for 5 sec duration: “ACC set speed changed”.



### Failure Warnings

- When ISA is in a temporary failure state, the tell-tale indicator below will be displayed along with a 5-second pop-up message: "Intelligent speed assist system is temporarily unavailable".



- When ISA is in a permanent failure state, the tell-tale indicator below will be displayed along with a 5-second pop-up message:

- "Intelligent speed assist system failure, please contact the service centre"

### System limitations

ISA feature subject to system limitation and unavailable or degrade performance or give false warning in following situations.

- For Speed limit signs with value lower than 30 Kmph, ISA will not reduce the ACC set Speed. However, Driver will be informed through Over speed warning and ISA prompt about the Over speeding.
- The final functionality of the ISA system depends on the availability of map data and/or Traffic Sign Recognition (TSR).
- TSR Limitations: The Traffic Sign Recognition (TSR) feature relies on a camera sensor and has certain limitations, including the inability to detect three-digit speed limit signs, multi-stacked speed limit signs, and signboards that are not installed according to VINNA conventions (e.g., a white background with numbers in black and red outer circle) or that are incorrectly placed or signboards outside the camera's field of view.
- MAP Data Limitation: The ISA system relies on up-to-date map data to determine speed limits. Inaccuracies or delays in map data updates can lead to the system displaying incorrect speed limit information.
- ISA system will not work in some situation where speed sign boards are present but the MAP data does not provide speed limit information.
- Inclement weather (rain, snow, fog, dust, haze, mist etc.)
- Camera blockage (direct sunlight or headlamp glare on camera, mud/dirt/moisture/ stickers/ debris on wind-shield surface in front of camera etc.)
- The ISA feature may not perform as expected due to temporary events such as road construction, accidents, or adverse weather conditions.
- The performance of ISA can be degraded in areas with poor GPS reception, such as urban canyons, tunnels,

## STARTING AND DRIVING

or areas with dense foliage, as the system requires accurate positioning data.

- ISA systems may occasionally trigger false alarms or fail to detect speed limit changes.

### Cautions

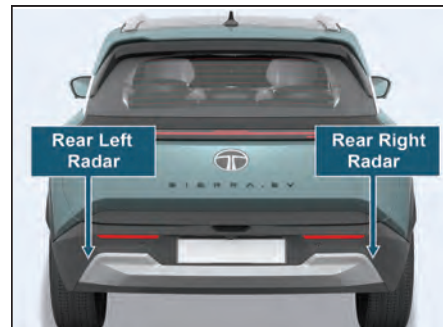
- The ISA system is a driving assist feature designed to enhance driving comfort. However, the driver remains fully responsible for adhering to traffic regulations and maintaining safe control of the vehicle at all times.
- Drivers must remain attentive, monitor their surroundings, and be ready to override the ISA system when necessary to ensure safe operation.
- For accurate and reliable performance, the map data used by the ISA system must be updated frequently. Drivers are responsible for ensuring that the system relies on the most current map information.
- The ISA system may be affected by factors such as outdated map data, complex road layouts, or poor GPS reception. Drivers should be aware of

these limitations and avoid relying solely on the system.

- Weather conditions, road changes, and signal disruptions can impact the performance of the ISA system. Drivers must be prepared to take control whenever the system's functionality is compromised.
- ISA systems may occasionally experience malfunctions or unexpected behaviour. In such events, drivers must take immediate and full control of the vehicle to ensure safety.

### Rear Advanced Driver Assistance System (Rear ADAS)

Rear ADAS features uses two rear corner radars, which will be placed on corners of the rear bumper of the vehicle.



#### *i* NOTE

- *The correct operation of the rear corner radar sensors will be compromised if they are misaligned due to accident damage at the rear of the vehicle.*

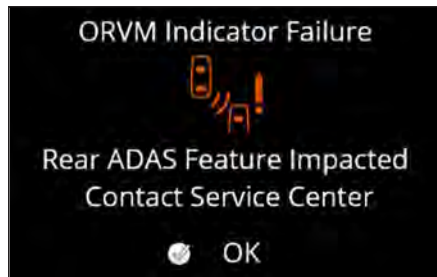
If any damage to rear bumper of the vehicle, it is recommended to get the vehicle inspected by TP&M Authorised Service Centre.



### Rear ADAS Malfunction:



When outside rear view mirror warning light is not working properly, the “ORVM indicator failure. Rear ADAS feature Impacted. Please contact service center” message will appear on the cluster for several seconds.



In this case, it is recommended to get the vehicle inspected by TPEM Authorised Service Centre.

Please make sure the warning indicators in the ORVM's are not obscured by stickers or any other objects.

### Blind Spot Detection (BSD) And Lane Change Alert (LCA)

BSD-LCA detects moving objects present in the adjacent lanes and warns if there are moving objects present in blind spots of the host vehicle or there is a high speed closing vehicle in adjacent lanes. Warning to the driver is given in stages.



### Prerequisite for activation

The following condition shall be satisfied to activate BSD-LCA

1. Rear Corner Radars and other related systems are fault free.

## STARTING AND DRIVING

2. Vehicle shall be in Drive (D) or Neutral (N) Mode.
3. Vehicle speed is above 20 kmph approximately.

### User Settings

1. BSD-LCA will be default ON during start of vehicle i.e. on every Ignition ON the BSD-LCA feature will be enabled even if previously disabled.
2. User can turn ON/OFF BSD-LCA feature using infotainment screen.

In infotainment system user interface, press the following button sequence to reach BSD-LCA user settings page

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist.

### Feature Warning Behavior

When the conditions for BSD-LCA warnings are met for an object in BSD-LCA zone,

1. The warning indicator on the outside rear-view mirror will illuminate as follows:



Level 1 Warning (Object detected in the BSD-LCA zone): The warning light on the ORVM will be continuously ON.

Level 2 Warning (Object detected in the BSD-LCA zone and Turn signal ON-same side as where object is detected): - The warning light on the ORVM will blink.

2. In case of Level 2 warning (Object detected in the zone and Turn signal ON), an audible warning will be given on the respective side. If turn indicator is switched OFF second stage alert will be deactivated.

### NOTE

*Audible warnings are not repeated until the collision threat visual alert disappears.*

### Failure Warning Behavior

When BSD-LCA is not working properly, the “Blind spot detection system failure. Contact service centre” message will appear along with tell-tale on the cluster for several seconds.



When rear corner radar is covered with any material which will influence BSD – LCA working, the “Blind spot detection system temporarily unavailable. Contact service centre” message will appear along with tell-tale on the cluster for several seconds.

If any of these malfunctions occurs, it is recommended to get the vehicle inspected by TPEM Authorised Service Centre.

### Limitations of BSD-LCA



### Limitations of BSD-LCA

- When the hazard-warning indicator is ON, the Level 2 BSD-LCA warning by the turn signal shall not operate.
- BSD-LCA may not function properly, under the following situations:
  - The speed of the other vehicle is very fast that it passes by your vehicle in a very short time.
  - Your vehicle passes by the other vehicle at very high speed.
  - Your vehicle changes lane without giving turn indicators.
  - The vehicle in the next lane moves two lanes away from you, or when the vehicle two lanes away moves to the next lane from you.
  - When your vehicle overtakes other vehicle and relative speed between the two vehicles is low.
  - When your vehicle is braking.
  - When your vehicle is moving in scenarios such as sharp turns, junctions, roundabouts, etc.
- BSD-LCA may not operate properly when driving on a curved road as,

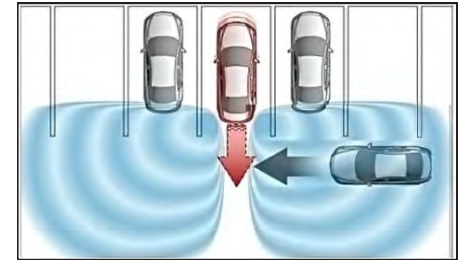
- The function may not detect the vehicle in the next lane.
- The function may recognize a vehicle in the same lane.
- BSD-LCA may not operate properly while driving on the road merges or divides. The function may not detect the vehicle in the next lane.
- BSD-LCA may not operate properly when driving on a slope. The function may not detect the vehicle in the next lane.

### Disclaimer

Since the detecting range of the rear corner radar is determined by a standard road width, therefore, on a narrow road; BSD-LCA may detect other vehicles two lanes over and warn you. On the other hand, on a wide road, BSD-LCA may not be able to detect a vehicle driving in the next lane and may not warn you.

### Rear Cross Traffic Alert (RCTA)

RCTA system warns the driver while reversing out of a parking spot or where there can be a possibility of a collision with a vehicle or an object crossing sideways from behind.



### Prerequisite for activation

The following condition shall be satisfied to activate RCTA

1. Rear Corner Radars AND other related systems are fault free.
2. Ignition shall be in ON state. Engine can be in either Running or in OFF state.
3. The R (Reverse) gear engaged.

## STARTING AND DRIVING

4. Vehicle speed is below 8 km/h approximately.

### User Settings

1. RCTA shall be default ON during start of Vehicle i.e. on every Ignition ON the RCTA feature will be enabled even if previously disabled.
2. User can turn ON/OFF RCTA feature using infotainment screen.

In infotainment system user interface, press the following button sequence to reach RCTA user settings page

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist.

### Feature Warning Behavior

When the conditions for activating the RCTA warning are met for any object in RCTA zone,

1. The warning indicator on the outside rear-view mirror will blink if an object is recognized in RCTA zone.



2. A warning icon (direction specific) will appear on the infotainment screen.



3. At the same time, an audible warning will sound.

### Failure Warning Behavior

When RCTA is not working properly, the "Rear cross traffic alert system failure. Contact service centre" message will appear along with icon on the infotainment screen for several seconds.



When Rear corner radar is covered with any material which will influence RCTA working, the "Rear cross traffic alert sys-

tem temporarily unavailable. Contact service center.” message will appear along with icon on the infotainment screen for several seconds.

If any of these malfunctions occurs, it is recommended to get the vehicle inspected by TP EM Authorised Service Centre.

### Limitations of RCTA

- While reversing near an obstruction (like vehicle or structure): RCTA may not warn the driver when reversing near a vehicle or structure and may not detect the vehicle approaching from the left or right.
- RCTA may provide degraded functionality when the vehicle is on a slope (uphill/downhill) or near it. In this case system may not warn the driver.

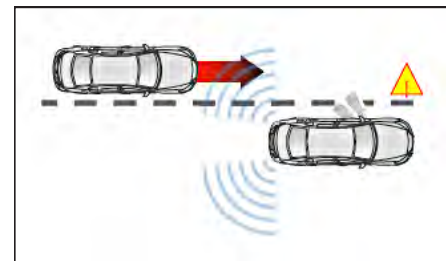
### Disclaimer

- While reversing out of complex parking situations: RCTA may/may not give the warning/may operate unexpectedly in the situations where other vehicles are parking or coming out of parking near your vehicle.

- When your vehicle is parked diagonally:
  - RCTA may provide limited functionality when coming out of diagonal parking space and may not detect vehicle approaching in RCTA zone.
  - RCTA system may operate unexpectedly during complex diagonal parking situations (like slow moving vehicle coming very close, vehicles parked in adjacent very closely etc.)
- RCTA may detect vehicles in front of you while parking reverse into a space with a wall or structure in rear/side area. In this case system may provide unnecessary warning.
- RCTA may not operate properly, or it may operate unexpectedly when the approaching vehicle is very fast or very slow.
- RCTA may operate unexpectedly in case of heavy traffic when multiple other vehicles are approaching from both sides.

## 10. Door Open Alert (doa)

DOA system warns the passengers in the car about the presence of approaching vehicles from behind which may hit the door while opening the respective side door.



DOA

### NOTE

*Door Open Alert is an aid system only. It is the responsibility of the user to open the door with due attention, in a way which is safe for the vehicle, and other road users, so that serious injuries can be avoided.*

## STARTING AND DRIVING

### Prerequisite for activation

The following condition shall be satisfied to activate DOA

1. Rear Corner Radars AND other related systems are fault free.
2. Vehicle speed is below 3 km/h approximately.
3. Ignition is ON and approximately for 3 minutes after ignition is turned OFF, but turns off immediately if the vehicle is locked externally.

### User Settings

1. DOA shall be default ON during start of Vehicle i.e. on every Ignition ON the DOA feature will be enabled even if previously disabled.
2. User can turn ON/OFF DOA feature using infotainment screen.

In infotainment system user interface, press the following button sequence to reach DOA user settings page

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist.

### Feature Warning Behavior

When the conditions for DOA warnings are met for any object in DOA zone,

1. The warning indicator on the outside rear-view mirror will illuminate as follows:



Level 1 Warning (Object detected in the zone): The warning light on the ORVM will be continuously ON.

Level 2 Warning (Object detected in the zone and respective side door opened): The warning light on the ORVM will blink.

2. In case of Level 2 warning (Object detected in the zone and respective side

door opened), an audible warning will be given an indication will be given on the cluster as shown below.



### Failure Warning Behavior



When DOA functionality is not available, the “Door open alert system failure. Contact service centre” message will appear along with icon on the cluster screen for several seconds.

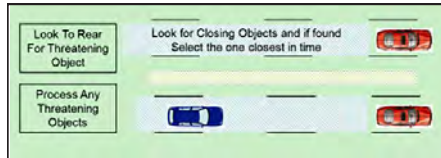
If this malfunction occurs, it is recommended to get the vehicle inspected by TPDM Authorised Service Centre.

### Disclaimer

Door open alert may not operate properly, or it may operate unexpectedly when the approaching vehicle is very fast or very slow or in cases where the approaching vehicle is at certain angles.

### 11. Rear Collision Warning (RCW)

RCW identifies potential collision risks from the back of the vehicle. The system warns the driver of the rear vehicle of identified collision risks by automatically flashing the hazard lights of your vehicle.



#### **NOTE**

*RCW system shall be considered only as an alert function for driver of other vehicle when there is a risk of accident. It is not a reference for starting suitable braking action in such scenario by other vehicles.*

#### Prerequisite for activation

The following condition shall be satisfied to activate RCW

1. Rear Corner Radars AND other related systems are fault free.

2. Vehicle shall be in Drive (D) or Neutral (N) or Park (P) mode. (In neutral vehicle should be in standstill)
3. Vehicle speed is within system operating limit in Drive (D) mode.

#### User Settings

1. RCW shall be default ON during start of Vehicle i.e. on every Ignition ON the RCW feature will be enabled even if previously disabled.
2. User can turn ON/OFF RCW feature using infotainment screen

In infotainment system user interface, press the following button sequence to reach RCW user settings page

Go to Home page >> All App >> Settings >> Driver Assistance >> Drive assist.

#### Feature Warning Behavior

In case of RCW warning, all right and left direction indicators will flash to warn driver of vehicle in behind about a possibility of collision.

#### Failure Warning Behavior

- When RCW is not working properly,

the “Rear Collision Warning system failure. Contact service centre” message will appear on the cluster for several seconds.

- When Rear corner radar is covered with any material which will influence RCW working, the “Rear Collision Warning system temporarily unavailable. Contact service centre” message will appear on the cluster for several seconds.

If any of these malfunctions occurs, it is recommended to get the vehicle inspected by TPEM Authorised Service Centre.

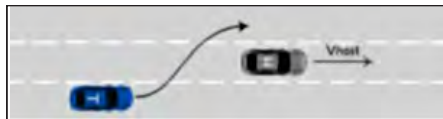
#### Limitations of RCW

- When target is overtaking from left lane to right lane, RCW system may not trigger warning.



- When target is overtaking from right lane to left lane, RCW system may not trigger warning.

## STARTING AND DRIVING



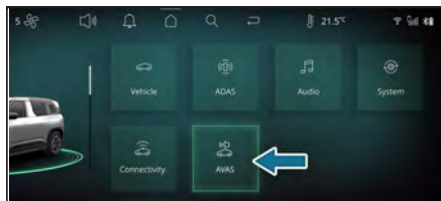
### Disclaimer

- In curved roads, RCW system may trigger warning inconsistently.
- RCW warning may not be triggered in case of lane merges or the rear vehicle is approaching at an angle.
- The system may not operate properly when your vehicle is moving in scenarios such as sharp turns, junctions, roundabouts, etc.
- The system may not operate when overlap of your vehicle and other vehicle is small.

### ACOUSTIC VEHICLE ALERTING SYSTEM (AVAS) (if equipped)

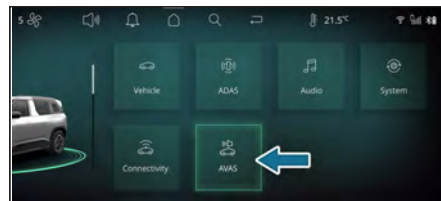
EVs are almost silent while moving, making it challenging for pedestrians to recognize when a car is coming. Sound generators called acoustic vehicle alerting systems (AVAS) are installed in electric vehicles (EVs) to notify/alert the pedestrians.

AVAS unit is placed in front under the bonnet.



### Charging sound

If turned on by user, AVAS informs user with a sound intimating vehicle charging is started & ended.

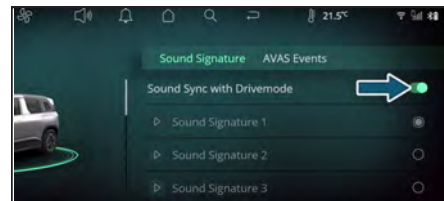


### Lock and Unlock Sound

If turned on by user, AVAS informs user about vehicle lock and unlock in limited scenarios.

### V2X sound

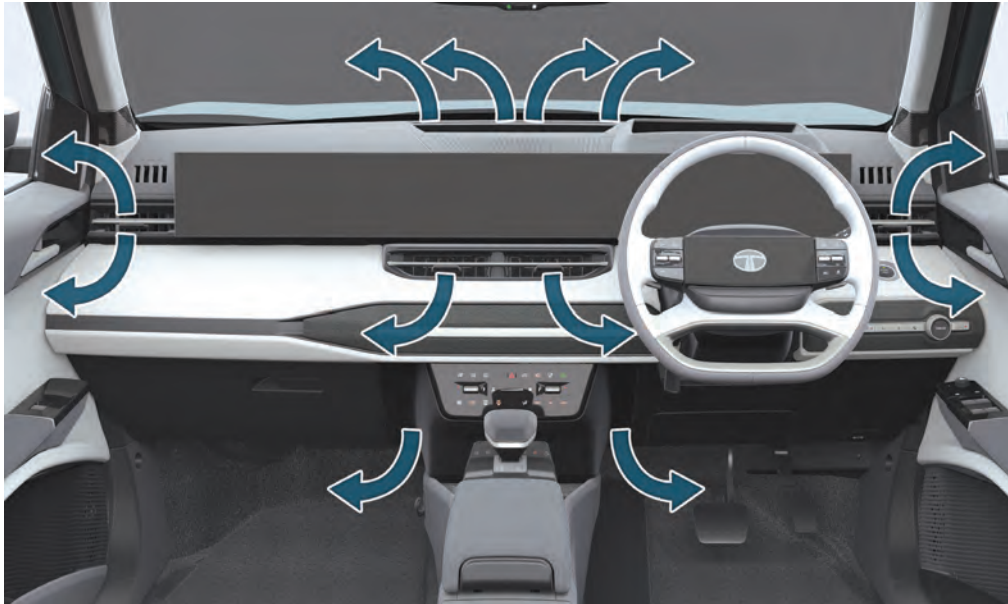
If turned on by user, AVAS informs user with a sound intimating vehicle discharging is started/ended.



### CLIMATE CONTROL

#### Air Distribution

The Climate Control regulates the temperature inside the vehicle and filter the dust particles in cabin based on the user set temperature settings. The air is distributed through the vents in the passenger compartment as shown below:



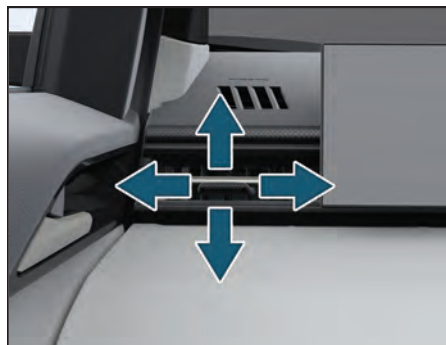


## INTERIOR FEATURES

### Air Vents

#### Dashboard Side And Front Centre Vent

Air vents are available on the dashboard. The direction of air flow can be adjusted using sliders on the respective vents.

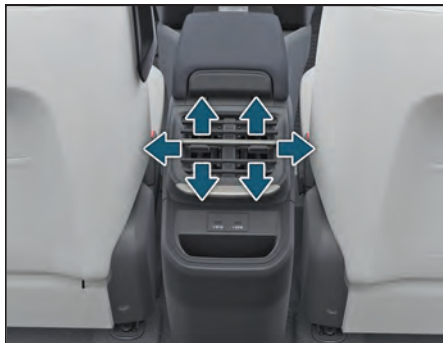


*Dashboard Side Vent*



*Front Center Vent*

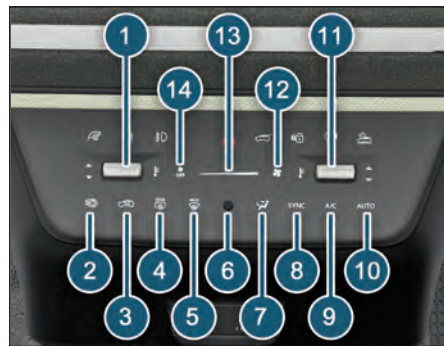
#### Rear Centre Vent



### FULLY AUTOMATIC TEMPERATURE CONTROL (FATC)

FATC system controls the in-cabin temperature of the vehicle automatically and provides maximum passenger convenience regardless of outside weather conditions

#### Option I - Dual Zone



1. Co-driver Temperature toggle switch
2. Xpress Cool
3. Fresh air / Recirculation
4. Rear window demister
5. Maximum defrost



6. In car Sensor
7. Air distribution (mode)
8. SYNC mode
9. AC ON/OFF
10. AUTO Switch
11. Driver temperature toggle switch
12. FAN Speed Increment
13. FAN Slider
14. FAN Speed decrement

### Display Screen



FATC display is shown on main display screen.

Dual zone FATC has separate temperature control switches for driver and co-driver. Also it has fan slider for blower control. FATC functions can be controlled using both the FATC control panel and the touch

screen display.

Whenever the user selects any switch or moves the toggle switch, then the display unit will show the relevant climate information

Also, when the display is not in climate mode then climate information will be displayed on the all-time display available on the top bar and widget.

### 1. Co-driver Temperature control toggle switch



Users can increment & decrement temperature by 0.5 deg c /change in rocker switch position. LO is the minimum possible value of co-driver temperature selected from decrementing push button. HI is the maximum possible value of co-driver temperature selected from incrementing push Button.

User can set co-driver temperature from 18 to 30 deg.

### NOTE

- *The set temperature and vent outlet temperature will not be same during the operation of AC System. It only matches when entire cabin temperature is stabilized.*
- *Foot air vent temperature will be always on warmer side as compared to center/side air vent based on passenger's comfort.*
- *When the ambient temperature is less than the passenger's set temperature, to achieve the same, warm air will come from air outlets and vice versa, this is a normal working condition.*
- *Rear air vent outlet temperature is a combination of both front air vent outlet temperatures.*
- *Driver mode selection is considered as master, however override facility is available for temperature control, blower control, fresh/Recirculation and mode selection.*

## INTERIOR FEATURES

### 2. Xpress Cool



XPRESS Cooling can be turned ON/OFF by selecting XPRESS COOL icon. This helps cabin to reach to comfort temperatures quickly by optimally setting the air conditioning to maximum cooling.

Also, if required, the driver window will roll down to flush the hot air from inside the cabin. Once cabin has been sufficiently flushed, the system will announce to take driver window's roll up which can be taken up using window winding switch.

Driver side window may roll down, if:

- The cabin temperature is more than outside temperature.
- If it is not raining.
- Vehicle Speed is less than 40 kmph.

Further, after sufficiently cooling the cabin, the Xpress cooling function will auto switch off and revert back to customer pre-selected settings.

Xpress cooling functionality is used to improve the HVAC system performance in case of cabin temperature being considerably greater than outside air temperature. The system will be deactivated automatically after 500 sec of continuous operation.

### 2. Fresh Air / Recirculation



When the recirculation switch is turned ON, air from the vehicle's interior is sent throughout the system.

When the recirculation switch is turned OFF, air from outside enters in to the cabin (fresh mode). Whenever discomfort is felt, switch to fresh air mode.

#### NOTE

*The outside air intakes for the climate control systems are at the base of windscreen. Keep this area clear from leaves and other debris.*

Use recirculation mode for faster heating and cooling. However, keeping the system in recirculation mode - particularly when the AC is in OFF - can cause fogging of windows. Use recirculation mode for faster heating and cooling. However, keeping the system in recirculation mode - particularly when the AC is in OFF - can cause fogging of windows.

### 3. Rear Window Demister



Select the rear window demister switch to turn it ON or OFF. The system will be deactivated after 15 mins of continuous operation.

### 4. Maximum Defrost



It directs the main airflow towards windscreen for faster defrosting. (It also overrides any mode selection you may have made).

When you turn off the maximum de-frost, the system returns to its former settings

## **i NOTE**

*For your safety make sure you have a clear view through all the windows before driving*

### 5. In Car Sensor

CCH panel shall contain a sensor that shall be used for measuring the average cabin temperature values.

### 6. Air Distribution (mode)



In AUTO mode, the FATC system will regulate the mode automatically. However, user override is possible with the use of MODE switch to select the desired airflow mode.

Each time you select the MODE switch, the display shows the mode selected.

|  |                                                                              |
|--|------------------------------------------------------------------------------|
|  | Directs air through the center and side air vents                            |
|  | Directs air through the center, side and foot well vents                     |
|  | Directs air through the foot well air vents                                  |
|  | Directs air through the defroster & foot well vents (Default fresh air mode) |
|  | Directs air through the defroster vents (Default fresh air mode)             |

### 7. SYNC Mode (For dual mode only)

It shall be always possible to turn ON the SYNC functionality from the CCH panel. Once CCM detects the SYNC signal, then the driver set temperature shall be copied into Co-driver set temperature. If the SYNC function is OFF, a single touch shall activate SYNC function. If the SYNC function is ON, the single touch shall turn OFF the SYNC function.

### 8. AC ON / OFF

Select the AC ON/OFF switch to turn the air conditioning ON or OFF. The AC icon activated on the display when the AC is

ON.

### 9. Auto Switch

To put the automatic climate control in fully automatic mode:

1. Select the 'AUTO' switch.
2. Set the desired temperature by toggle switch. The display will show all the functions during 'AUTO' mode.
3. The system automatically selects the proper mix of conditioned and/or heated air that will, as quickly as possible, raise or lower the interior temperature to your preference.
4. When you set the temperature to its lower limit (LO) or its upper limit (HI) the system runs at full cooling or heating only. It does not regulate the interior temperature.

## **i NOTE**

*In 'AUTO' mode, the FATC system will regulate the blower speed automatically.*

## INTERIOR FEATURES

### 10. Driver Temperature Control Toggle Switch

Users can increment & decrement temperature by 0.5 deg C / change in rocker switch position. LO is the minimum possible value of driver temperature selected from decrementing Push button. HI is the maximum possible value of driver temperature selected from Incrementing Push Button.

### 11. FAN Speed Increment

It shall always be possible to Increment the blower speed by 1 detent per touch from the CCH panel on the fan speed Increment symbol side with long press on blower increment switch, blower speed will increase by one detent after each 500ms.

### 12. FAN Speed Slider

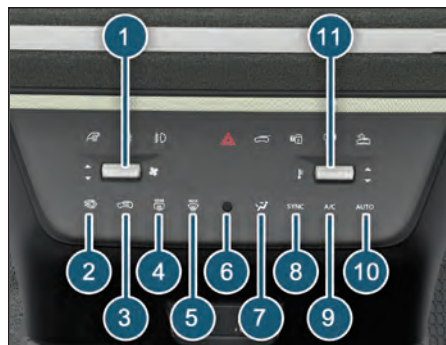
In Dual Zone CCH Panel, Fan Speed can be controlled through the slider. It shall always be possible to increment/decrement the blower speed by 1 detent per touch on blower up press/down press. If the Finger is swiped on the Blower Slider bar from left to right /right to left then the blower will gradually increase up to that position. If

the finger is touched anywhere in the slider, respective blower speed will be set.

### 13. FAN Speed Decrement

It shall always be possible to decrement the blower speed by 1 detent per touch from the CCH panel on the fan speed decrement symbol side. To turn OFF the blower, user needs to press the FAN speed decrement symbol for 3 seconds continuously.

### Option II - Single Zone



1. Blower Speed Control toggle switch
2. Xpress Cool

3. Fresh air / recirculation
  4. Rear window demister
  5. Maximum defrost
  6. In Car sensor
  7. Air distribution (Mode)
  8. OFF Mode
  9. AC ON/OFF
  10. Auto Switch
  11. Temperature Control Toggle Switch
- Display Screen**



FATC display is shown on main display screen.

FATC functions can be controlled using both the FATC control panel and the touch screen display.

Whenever the user selects any switch or moves the toggle switch, then the display

unit will show the relevant climate Information

Also, when the display is not in climate mode then climate information will be displayed on the all-time display available on the top bar and widget.

### 1. Blower Speed Control toggle switch

Move toggle switch up & down to increase & decrease the blower speed.

### 7. OFF Mode

Select the OFF switch to turn the system 'OFF'. OFF will be displayed on the infotainment screen.

### 10. Temperature Control toggle switch

Move the temperature control toggle switch up to increase the temperature of the air. The desired temperature will be increased by steps of 0.5°C. User can select temperature range from 18°C to 30°C. Move the toggle switch down to reduce the temperature.

When you set the temperature to its lower limit (Lo) or its upper limit (Hi), the system runs at full cooling or heating only. It doesn't regulate the interior temperature.

**All other features are similar to Dual Zone Option I.**

## FATC Sensors

### Outside Ambient Temperature (OAT) Sensor



Outside Ambient Temperature (OAT) sensor located under the front bumper grill.

### In-car Sensor On Control Panel

In-car sensor is located on FATC control panel.

#### NOTE

- Do not cover or spill any liquid on sensors.

## INTERIOR FEATURES

- Do not cover sensor, this may cause the sensor to malfunction. This may lead to FATC not functioning to desired level.

### CABIN AIR PURIFICATION (if equipped)

The Climate Control System fitted with advance filter for cabin air purification.

#### NOTE

*Replace the Filter as per Maintenance schedule. More frequent filter replacement are required/ recommended in case of vehicle is driven in heavy dusty conditions. If the vehicle is driven in heavy dusty conditions more frequent filter replacement are required. Replace the filter if you find poor ventilation, cooling or demisting and poor Air Quality Index (AQI).*

### Air Quality Index : (if equipped)

- Climate control system fitted with FATC calculates Air Quality Index (AQI) of cabin using PM2.5 AQI Index.
- FATC System in AUTO Mode automatically sets the blower speed and switches to recirculation air mode to improve AQI inside the cabin.
- The calculated AQI is displayed on dis-

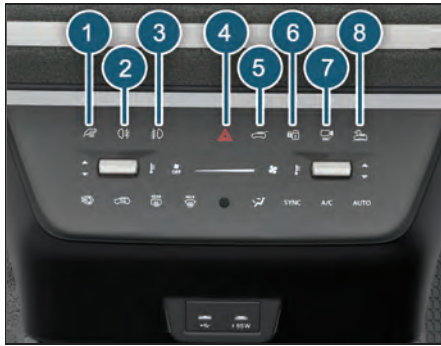
play unit along with severity index.



#### NOTE

- AQI calculation will be effective after 30secs , ignition ON and no value will be displayed during this period.
- If AQI does not improve in some time get sensor and Cabin filter inspected.

### FASCIA SWITCHES



1. Charging gun lock/unlock
2. Rear Fog Lamps (If equipped)
3. Front Fog Lamps (If equipped)
4. Hazard warning switch
5. Tail gate opening
6. Central lock/unlock
7. Surround View System (SVS)
8. TP2 Traction Pro (If equipped)

#### 1. Charger Gun Lock/ Unlock

To release the charger gun, press the switch located on fascia switch. Charging socket inlet is located at Rear LH side of vehicle.

#### NOTE

*Make sure both AC slow charging & DC fast charging is De-energised / off state.*

#### 2. Rear For Lamps (if equipped)

The rear fog lamps are located on the rear bumper. In poor visibility conditions due to fog, snow or rain, the fog lamps make visibility better and make it easier for other road users to see you. It turns to 'ON' when the fog lamp switch is turned on when the ignition is 'ON'.

#### 3. Front Fog Lamps (if equipped)

The front fog lamps are located on the front bumper. In poor visibility conditions due to fog, snow or rain, the fog lamps make visibility better and make it easier for other road users to see you. It turns to 'ON' when the fog lamp switch is turned on when the ignition is 'ON' and when the po-

sition and parking/ head lamp is 'ON'. An indicator on front fog lamp knob will come on when the front fog light is 'ON'.

#### 4. Hazard Warning Switch

Press the hazard warning switch to activate the hazard warning. All the turn signal lamps will flash simultaneously. To turn OFF, press the switch again.

#### 5. Tail Gate Opening

To unlatch the tail gate, press the switch located on fascia switch.

#### 6. Central Lock/unlock

To open the door, press the Lock/un-lock door switch located on the fascia switch.

#### 7. Surround View Camera (SVS)

Press this switch to see the 360° view in the display screen.

#### 8. TP2 Traction Pro (If equipped)

Traction pro 2 feature will assist user in case of uphill as well as downhill. It will also allow user to cruise vehicle at slow speed in rough terrain.

To Activate the TP2 feature in the vehicle, the driver must press the TP2 button on

## INTERIOR FEATURES

---

the fascia when the driver seat belt is buckled and driver door is closed.

After activating TP2 by using the TP2 button, user has to set the target speed with cruise Set+ and Set- buttons and TP2 function will try to maintain the set speed irrespective of variation in terrains.

### Preconditions to activate TP2:

1. Driver seat belt buckled and driver door closed.
2. Parking brake is released
3. Gear in D/R.
4. When in standstill condition, brake pedal pressed and TP2 Button Pressed.
5. When in dynamic condition and vehicle speed < 25 kmph, Press TP2 button and Set + button.

### Deactivation:

1. Driver seat belt unbuckled or driver door opened.
2. Parking brake is applied
3. Transmission in P/N.
4. When driver presses brake and brings

the vehicle to standstill.

5. When vehicle is in D and speed crosses 60 Kmph.
6. When vehicle is in R and speed crosses 25 Kmph.

The TP2 feature can control the vehicle from 6kmph till 25kmph above which the TP2 features goes to standby and if the driver accelerates the vehicle greater than 60kmph the TP2 feature is deactivated post which the TP2 button must be pressed again when the vehicle speed is less than 25 kmph to activate the TP2 feature.

TP2 Set Speed will be indicated in upper right corner of cluster with Green tell-tale indicating active intervention state of the function.

The driver can set the speed at which he wants to cruise by using the Cruise Button on steering wheel.

The Driver can also control the set speed by pressing the brake pedal or accelerator pedal. As soon as the driver releases the Brake/ Accelerator pedal that vehicle speed is taken as the set speed.

The TP2 standby lamp in the instrument cluster lights up when the TP2 feature is turned ON by the user but hasn't got activated.



The TP2 Green lamp in the instrument cluster lights up when the TP2 feature activated and actively controls the vehicle, the lamp may also blink based on the actual feature state.



The TP2 malfunction lamp in the instrument cluster lights up after every ignition ON. It goes OFF after a few seconds if the system is healthy.





## WARNING

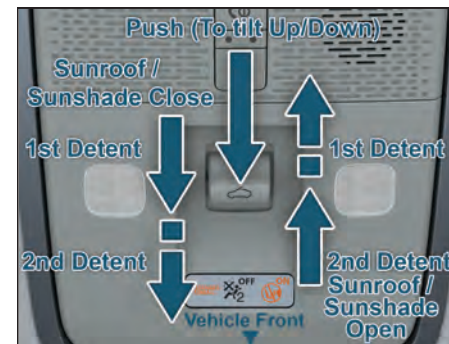
When the TP2 feature gets activated and actively controls while driving, the TP2 ON lamp blinks on the cluster indicating the TP2 feature is active and is actively controlling the vehicle speed. If TP2 is faulty, The TP2 malfunction lamp remains ON in the Instrument cluster, have TP2 checked immediately at a TPEM Authorised Service Centre.

## POWER SUNROOF (if equipped)

It brings natural light and fresh air into passenger compartment. The cabin becomes more illuminated and bright which gives pleasant feeling while driving and makes driving experience more enjoyable and relaxed.

Sunroof allows air to flow evenly from the roof which is quieter and less intrusive than wind blowing through a side window.

Sunroof can be operated by Manual Switch, Voice Command and by Rain Detection/Vehicle Lock.



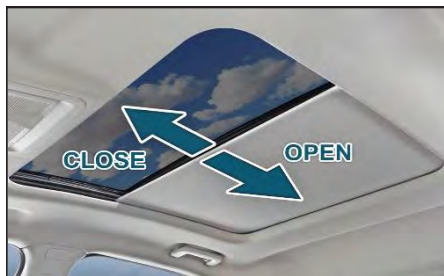
Sunroof switch is used to open, close, tilt up & tilt down the sunroof as and when required with switch intended operation.

1. Push the switch away from the windshield to open the sunroof. It has two detents
  - 1st detent : First click [Express operation] only sunshade will get open
  - 2nd detent : Second Click [Sunroof will get open]
  - Long Press : Sunshade and Sun-

## INTERIOR FEATURES

- roof both will get open respectively.
2. Push the switch towards the windshield to close the sunroof. It has two detents
    - 1st detent : First click [Express operation] [only sunroof will get close]
    - 2nd detent: Second Click [Sunshade will get closed]
    - Long Press: Sunroof and Sunshade both will get close respectively.
  3. Press at the centre of the switch for -
    - Tilt up / down: while sunroof is closed and sunshade should be in open position.
    - Stop at desired position: while sunroof and sunshade is in moving condition.

### Sunshade Open / Close Position



### Sunroof Open / Close Position



### Sunroof Voice Command

Ensure vehicle is in IGN ON/running condition.

Enable voice recognition via steering wheel switch or TATA Assist icon from the infotainment screen.



System will prompted with "How can I help?"

Give the "sunroof open/close" command. Sunroof will be opened/closed.



### **WARNING**

- Speak the commands / Instructions in a neutral English accent for best results.
- Do not take long pauses (greater than 1 second) while speaking the words in a command. Speak the words of the command at a constant rate.
- Avoid varying your pitch and volume while speaking the commands. Speak clearly and loudly at a reasonable speed.

- Ensure that there is no noise disturbance when you speak the commands like, other passengers in the vehicle are talking or there is lot of wind noise. Disturbance from external sound sources may result in poor voice recognition.
- Always face forward while speaking your commands as the voice recognition quality is best in this orientation.

### **Sunroof Closure On Auto Detection Of Rain/vehicle Lock**

For User Convenience / Protection of vehicle, sunroof will automatically close under following conditions:

- **Rain Detection:** When sunroof is open and rain is detected (based on Wiper speed is slow/high upon raining), then Sunroof will close automatically
- **Vehicle Locking:** The sun roof will close automatically when ignition is off and vehicle is locked from out through driver door manual key or by remote key.

### **NOTE**

- *Combi Switch should be in auto mode to close sunroof with rain sensor.*
- *Never deliberately use your body parts to test the automatic reversal function. The sunroof glass may reverse direction, but there is a risk of injury.*
- *Do not open sunroof during/after rain, water may damage interior of vehicle.*

## INTERIOR FEATURES

### Automatic Reversal / Anti-pinch Function



If the sunroof senses any obstacle while it is closing then it will reverse its direction and opens the sunroof so that trapped object will get released easily. The auto reverse function may not work if very thin or soft object is caught between the sunroof assemblies. Anti-Pinch/ Automatic reversal is a safety feature however to override it and operate sunroof manually, press sunroof close switch within 10 seconds of auto-reversal completion and hold it till sunroof is fully closed.

#### **WARNING**

Never try pinching of any part of your body intentionally to activate the Automatic reversal function. The Automatic reversal function may not work if something gets stuck just before the sunroof fully closes.

#### **WARNING**

- Even though the sunroof can be operated when the ignition key is in the ON position (the vehicle is stationary position), operating the sunroof repeatedly with the ignition turned OFF will run down the battery.
- When a desired sunroof operation is completed, release the switch. If you keep pressing the switch, it could cause a malfunction. Especially in winter, never operate the sunroof if moving areas are iced. Wait until the areas are de-iced.
- Ensure your head, hands, arms, or other body parts or objects are out

of the way before operating the sunroof. Body parts or objects may get pinched causing injuries or vehicle damage.

- Dust accumulated between the sunroof and roof panel can make noise or cause any damage. Open the sunroof and remove dust regularly using a clean cloth.
- Avoid sitting on top of the sunroof, it may cause injury or vehicle damage.
- Do not open sunroof immediately after vehicle wash, accumulated water might go inside sunroof sealing area/ inside vehicle.
- Do not allow passengers to lean out of an open sunroof whilst the vehicle is in motion. Injuries may occur from objects such as tree branches.
- Safety of the vehicle occupants must be observed at all times. Do not allow limbs to be placed in the moving path of the sunroof at any time, injury may occur.

- High Pressure wash Jet Flow should not be directed on Sunroof sealing area around periphery of glass. Doing so may lead to water leakage inside cabin.
- The sunroof motor incorporates thermal protection to ensure safe operation. If the motor temperature rises beyond safe limits, the sunroof movement will be stopped to avoid damage. It will be temporarily disabled and will require approx. 30 minutes of cooling time before it can be used again.

### **Initializing The Power Sunroof (ignition / Vehicle On Condition)**

A. In the event of a power failure or fuse dead or battery disconnection when the sunroof is in motion, then sunroof will require initialization when the power is re-stored.

B. In the event of Sunroof first click (Express operation) not working.

C. In the event of Sunroof, not closing fully or partially closing.

### **Initializing Procedure for condition (A & B) only sunroof - Glass panel**

1. Turn ON the ignition.
2. Close the sunroof by pressing 1st detent switch. After closing completely still keep it pressed for 1-2 seconds until click sound comes from Sun-roof.

The Initializing process is completed. Check if Sunroof open/close operation is working, if not then repeat step 1 & 2.

### **Initializing Procedure for condition (C) only Sunroof-Glass Panel**

1. Close the sunroof fully by pressing 1st Detent and keep the switch pressed for 10 seconds.
2. The Re-initializing process is completed. Check the sunroof is closing completely, if not repeat step 1 and 2.

### **Initializing Procedure for condition (A & B) only Sunshade**

1. Turn ON the ignition.
2. Close the sunroof by pressing 2nd detent switch. After closing completely, still keep it pressed for 1-2 seconds until click

sound comes from Sun-roof.

The Initializing process is completed. Check if Sunroof open/close operation is working, if not then repeat step 1 & 2

### **Initializing Procedure for condition (C) only Sunshade**

1. Close the sunshade fully by pressing 2nd Detent and keep the switch pressed for 10 seconds.
2. The Re-initializing process is completed. Check the sunshade is closing completely, if not repeat step 1 and 2.

### **Power Sunroof - Self-learning Procedure**

In the event of Sunroof glass panel / Sunshade automatically reversing after pressing detent switch respectively.

#### **Sunroof Glass self-learning:**

Keep the sunroof glass closed, press and hold 1st Detent switch until completion of the self-learning process.

2. During this time the sunroof glass will automatically close, pause for 5 seconds, open partially then close fully.

## INTERIOR FEATURES

3. This indicates that the sunroof glass self-learning process is completed.

### Sunshade self-learning:

1. After the sunroof glass self-learning is completed, Keep the sunroof closed, press and hold 2nd Detent until completion of the self-learning process.

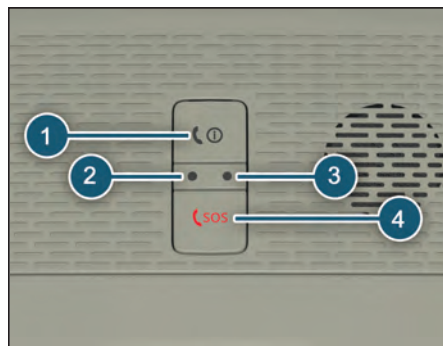
2. During this time the sunshade will automatically close, pause for 5 seconds, open partially then close fully.

3. This indicates that the sunshade self-learning process is completed.

### NOTE

*In case malfunction of any of the above function, please contact TPEM Authorised Service Centre.*

## E-call And B-call Switch



1. B-Call Switch: B-Call (Breakdown Assistance) will connect you to a TPEM Roadside assistance for Towing. Not for ambulance service.
2. Red LED Indication: Red LED indicates the fault or failure in B-Call/E-Call functionality.
3. Green LED Indication: Green LED indicates the status of ongoing B-Call or E-Call.
4. E-Call Switch (Emergency Call or SOS Switch): E-Call will connect you to tow-

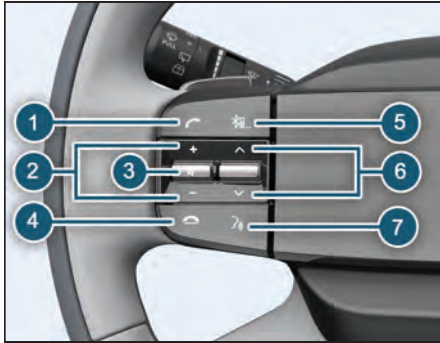
ing and ambulance services.

### NOTE

*\*Subject to mobile network, connectivity and location mentioned.*

### STEERING MOUNTED CONTROLS

#### Steering Mounted Controls (LHS)



##### 1. Phone Receive

Press above switch to accept incoming call when a cell phone is connected via Bluetooth.

##### 2. Volume

Press above switch to increase or decrease volume of music system / radio.

##### 3. Mute

For mute, press this switch. The system mutes/ pauses the currently played audio.

Press above switch to reject or hang up a phone call. It is also used to mute the volume of music system/radio

##### 4. Phone Reject

Press the switch to reject or hang up a phone call.

##### 5. Source

Press above switch to select the required source in the infotainment system i.e. USB, AM, FM and Bluetooth.

##### 6. Seek Forward/backward

This will function in the two modes. Radio Mode - will change radio channels. Media Mode - will change sound tracks.

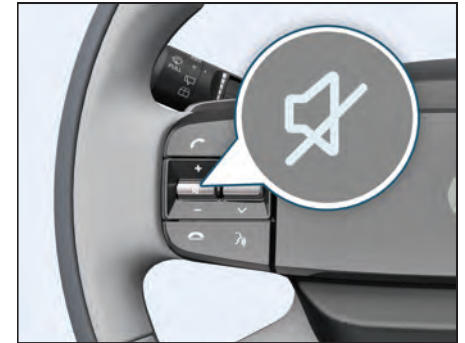
##### 7. Push to talk

For Voice Recognition, press this switch. The system mutes/ pauses the currently played audio and you will hear a beep sound to indicate the activation of the voice recognition feature. The system displays the voice recognition screen on Infotainment to indicate activation of the feature.

### Master /force Restart Process

If your infotainment system touch screen becomes unresponsive or shows some unusual behavior, then you can restart it to potentially resolve the issue. Follow some basic steps given below and you can restart the system.

To restart the infotainment system



1. Park the vehicle.
2. Hold the Steering wheel Mute button (long press) (as shown in above image) for about 10 secs. Now hold the Steering wheel source



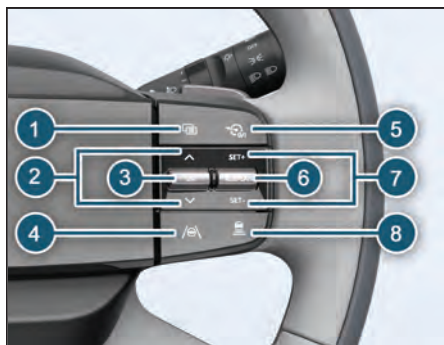
## INTERIOR FEATURES

- Now hold the Steering wheel source change button (long Press) for more than 10 sec) and release as soon as display's goes blank.

### NOTE

- It is preferable to do one Ignition OFF to ON cycle after Master/Force restart to synchronize vehicle settings with the TATA Infotainment System.
- If the reboot does not work or master/force restarts are required on a weekly or daily basis, vehicle shall be taken to dealership. There, the dealer can update your firmware or inspect the system for hard-ware problem.
- Force/Master restart keeps the stored data, such as call history, text message information, and previously paired phones as it is.

### Steering Mounted Controls (RHS)



#### 1. Pagination

Press the switch to enter in to cluster screen.

#### 2. Page Up/down

If cluster screen is selected, with Up/Down switch you can access the sub-menu screens of a main menu.

#### 3. Selection Ok

Push the OK button to access/select the sub menu screens of a main menu item.

#### 4. Lane Keep Assist (LKA) (if equipped)

Press the switch to activate the Lane Keep Assist function.

#### 5. Cruise ON/OFF

Press the switch to turn ON/OFF the cruise function.

#### 6. RES/CAN

To resume a previously set speed, push the RES button and release. This switch is also used to cancel /deactivate (CAN) cruise control system without erasing the set speed from memory.

#### 7. Set+ / Set-

Accelerate the vehicle to desired speed, Press the SET + to select the required cruise speed. When the cruise control is set, you can increase or decrease the speed by pushing SET+/ SET- buttons respectively.

#### 8. Adaptive Cruise Control (acc) (if Equipped)

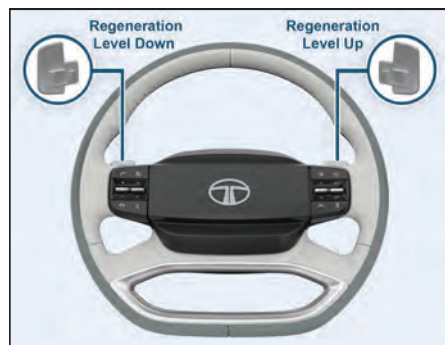
Press the switch to activate the adaptive cruise control function.



## REGENERATION SWITCH (if equipped)

Regeneration is equipped in Harrier EV, which converts the available kinetic energy during decelerating or braking condition of the vehicle to electrical energy which can charge the high voltage battery. Regeneration level for the vehicle can be controlled manually or can be turned off.

The Regeneration switch equipped behind the steering wheel assembly at both the sides Level Up and Level Down



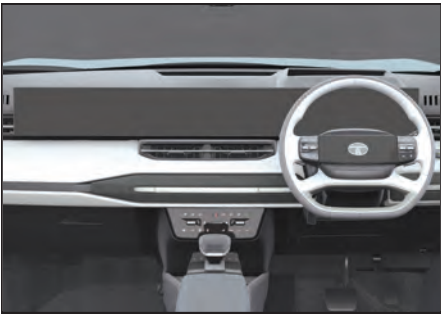
- By using the regeneration switch Pull type) driver can Up/Down regeneration level.
- Level down switch can be used to reduce regeneration level (3 →2 →1→OFF)
- Level Up switch can be used to increase the regeneration level (OFF→1 →2 →3)
- Default regeneration level on the vehicle when vehicle is cranked will be Level 1.

| Drive Modes | Default Regeneration      |
|-------------|---------------------------|
| OFF         | No Regeneration           |
| Level 1     | Minimum Regeneration      |
| Level 2     | Intermediate Regeneration |
| Level 3     | Maximum Regeneration      |

| Drive Modes | Default Regeneration |
|-------------|----------------------|
| Sport       | Level 1              |
| City        | Level 1              |
| Eco         | Level 1              |

INTERIOR FEATURES

INFOTAINMENT SYSTEM DISPLAY  
Option I



Option II



Voice Commands In Multiple Languages (if Equipped)

Speak your own language while driving and your car will do the rest.



Silent Mode:



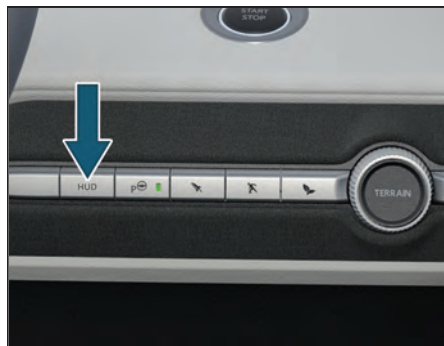
The silent mode feature allows user to selectively enable or disable non-regulatory

chimes and voice alerts based on their preferences and driving conditions.



If silent mode toggle is enabled, then audio warning and voice alert shall not be played.

### HEAD UP DISPLAY (HUD) (If equipped)



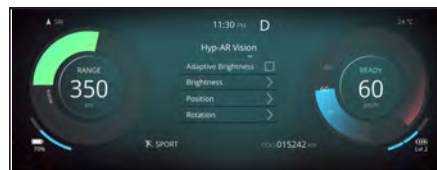
The AR-HUD (Augmented Reality - Head Up Display)

Projects virtual images on windshield in driver's field of view, enabling the driver to remain focused on the road ahead. The driver gets all the important information such as speed, warning signals and navigation without looking down at the instrument cluster or the secondary display.

#### Activation

##### Through HUD Switch

- HUD can be activated through HUD switch available on the right side of dashboard.
- Long press HUD switch to turn ON/OFF the display.
- Short press HUD switch to invoke the HUD adjustment settings in IPC.
- Right side steering wheel switches can be used to access the HUD settings.



- **Adaptive Brightness:** Enabling auto brightness adjusts illumination of AR-HUD display according to light sensor.
- **Brightness:** Brightness can be increased or decreased according to the user choice manually.
- **Position:** Position can be adjusted

manually (Up or down) according to user feasibility.

- **Rotation:** Rotation can be adjusted manually (right or left tilting) according to user with UP and Down switches on steering wheel.

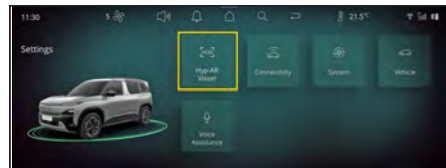
#### NOTE

*For a better experience during night-time, the user is requested to turn on the Position Lamp*

## INTERIOR FEATURES

### Through Infotainment Screen:

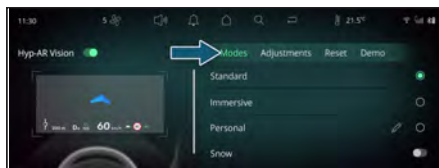
1. To access the settings, select AR-HUD icon from the app drawer on infotainment.



2. AR-HUD can be turned ON/OFF by the toggle button.



3. The mode (Theme) of the display can be changed from Mode's selection.



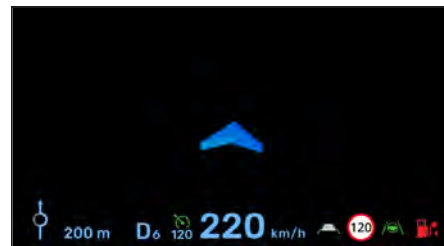
- **Standard:** This mode includes all static features and augmented navigation. In AR only navigation is visible. This mode offers a limited view to the user.
- **Immersive:** This mode includes all static information along with all augmented information. This mode offers a rich view for maximum awareness.
- **Personal:** Personal mode includes all static information. AR feature can be turned OFF or ON based on user selection by selecting the edit icon. This mode offers customization according to user needs.

### NOTE

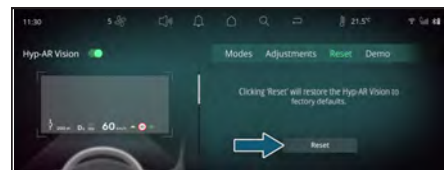
*On clicking "i" button the graphic in AR area will appear for 5 seconds and then*

*disappear.*

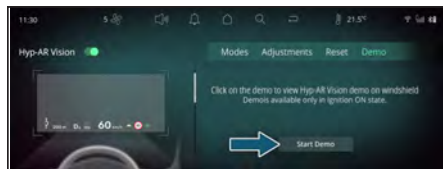
- **Snow:** Snow mode will change color theme of current mode. It improves visibility in rainy or foggy (lower visibility) conditions.



4. Current settings of AR-HUD can be reset by selecting the reset button. This will change the settings to default state.

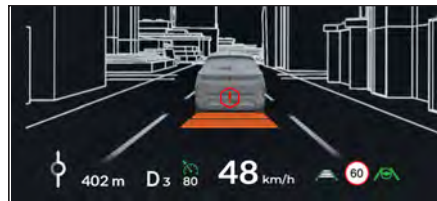


- Demo mode can be turned ON from the Demo mode tab. It can be used to experience the visual elements and interactive features of Hyp-AR vision on the windshield without real-time driving conditions.



### NOTE

*Keep the vehicle in Ignition ON condition to access the demo mode.*



### Do's And Don'ts

- Avoid using chemical solutions for cleaning the Hyp-AR Vision top surface.
- Avoid scratches on top surface of Hyp-AR Vision. To avoid permanent damage, wipe the protective glass using a gentle, lint-free cloth.
- Avoid using damp hands on Hyp-AR Vision.
- For clear vision, do not place any objects or items on the Hyp-AR Vision top surface.
- A dedicated windshield is used for the Hyp-AR Vision functionality. While replacing the windshield, user is requested to use only the dedicated windshield suitable for Hyp-AR Vision

from TPEM Authorised Service Centre.

- Avoid using polarized sunglasses while using Hyp-AR Vision. It may affect the visibility of features.
- Under extreme changes in sunlight, the display may temporarily and gradually disappear and then recover. This is a normal phenomenon.

### WARNING

- Do not operate Hyp-AR Vision settings while driving.
- Do not continue to see Hyp-AR Vision while driving, this could distract you from the road in front of you and may cause an accident.
- If the user feels distracted by continuous use of the Hyp-AR Vision, user can turn it OFF.

### NOTE

- Adjust the display brightness and height properly not to obstruct the driving.*

## INTERIOR FEATURES

- *In the following conditions, you may not see the display properly:*
  - *Sitting posture is not correct*
  - *Object covering Hyp-AR Vision*
  - *Driving on bright Sunny day or rainy day*
  - *Wearing a polarize/ inadequate glasses*
  - *Any object/ vehicle coming in front of vehicle*
- *In case of any malfunction in Hyp-AR Vision, contact the TPEM Authorised Service Centre.*

### HUD Features

#### Augmented Features:

Augmented Navigation: Display AR arrows for easy navigation.



Pedestrian Detection: It appears when there is pedestrian detected to prevent collision.



Other Object Detection: Object Detection warns about obstacles around the vehicle to prevent collisions.



Vehicle Detection: Blue line indicates car,

truck, two-wheeler under the vehicle before possible collision.



Adaptive Cruise Control: This warning appears while setting the level from the front target vehicle.



Forward Collision Warning/Autonomous Emergency Braking: FCW and AEB System display all types of moving or non-moving objects on the vehicle path and issue a warning before a possible collision.



Blind Spot Detection: BSD warns whenever there is object in the blind spot of vehicle.

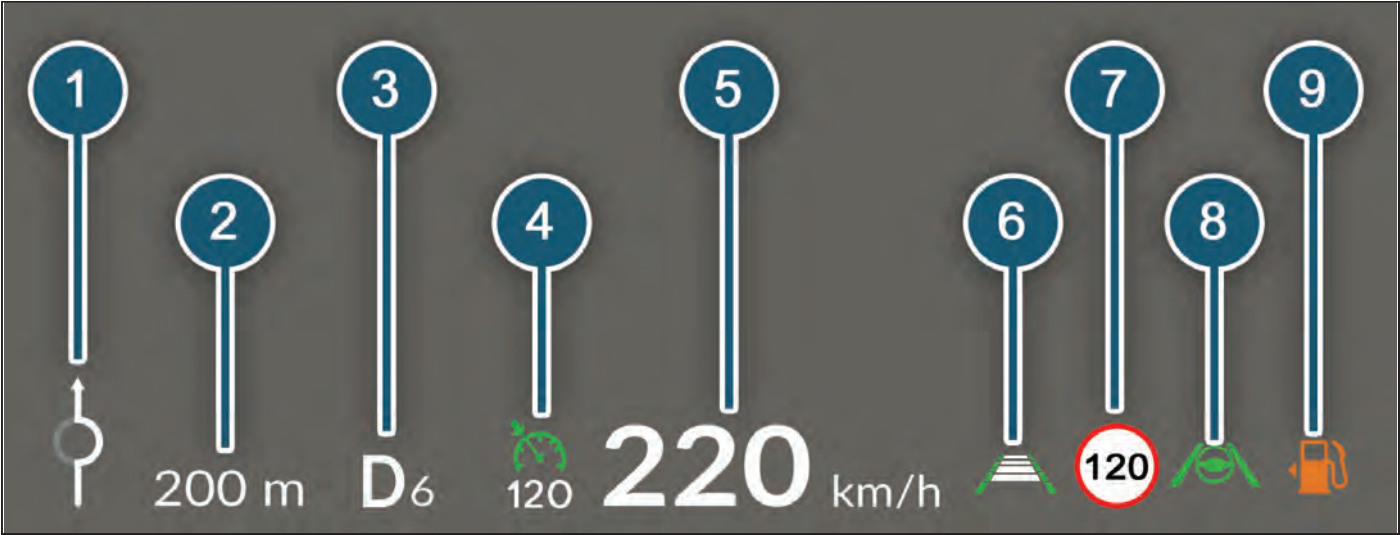


Lane Departure Warning: Warns the driver through visual alerts if the vehicle unintentionally drifts out of its lane. Lane Keep Assist: It helps maintain the vehicle's position within the lane.



# INTERIOR FEATURES

Static Features:



| Sr.no. | Feature                  | Feature Description                                                                                               |
|--------|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1      | Traffic Sign Recognition | The Traffic Sign Recognition System should detect the various traffic signs and give visual displays accordingly. |
| 2      | Adaptive Cruise Control  | ACC set speed indicates level for the front target vehicle.                                                       |
| 3      | Low Charge               |                                                                                                                   |



## INTERIOR FEATURES

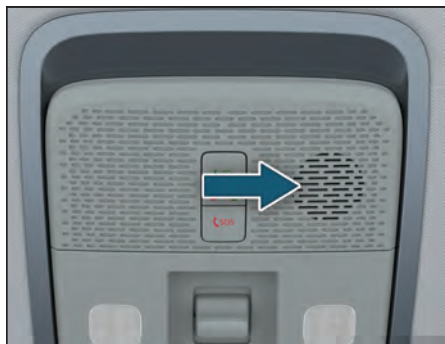
| Sr.no. | Feature                 | Feature Description                                                                                                         |
|--------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|        |                         | When the vehicle's battery charge level is low, the low Charge warning telltale turns ON.                                   |
| 4      | Lane Keep Assist        | This warning helps to maintain the vehicle within the lane.                                                                 |
| 5      | Gear Display            | Current gear engaged by the transmission will be displayed.                                                                 |
| 6      | Adaptive Cruise Control | This symbol comes when the adaptive cruise function is activated in the vehicle.                                            |
| 7      | Speed and Speed Limiter | The Speedometer Indicates the vehicle speed in km/h. Vehicle speed color will change according to the speed of the vehicle. |
| 8      | Turn by Turn Indicator  | It provides step-by-step directions for easy navigation.                                                                    |
| 9      | Turn by Turn Distance   | Display distance for next maneuver.                                                                                         |
| 10     | Destination ETA         | Time to reach the destination is updated here.                                                                              |

### Troubleshooting

| Sr.no. | Problem Observed           | Possible Cause (s)                                                                                      | Action To Be Taken                                                                                                         |
|--------|----------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1      | The HUD display is too dim | 1. The vehicle's ambient light sensor is obstructed or covered.<br>2. The brightness setting is too low | 1. Check for and remove any objects that are obstructing or placed near the HUD.<br>2. Adjust the settings of the display. |
| 2      | (System is) unresponsive   | System malfunction                                                                                      | Turn off the Vehicle and then restart it.                                                                                  |

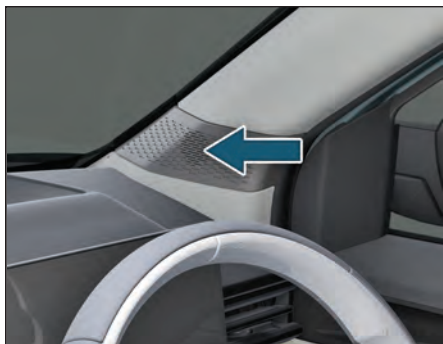
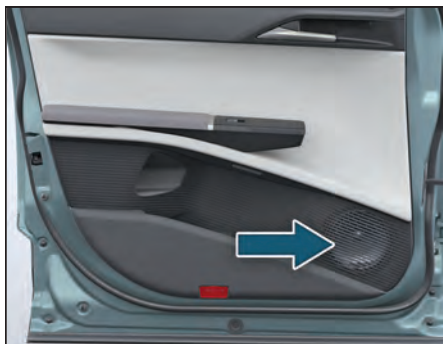
## INTERIOR FEATURES

### MIC



Mic is provided on the roof near the roof lamp.

### SPEAKERS (if equipped)



Speakers are available in models with infotainment system. Provisions are given for music system and speakers on versions without the infotainment system.

### USB PORT (if equipped)

#### Front Multimedia C + C Charger



C type media USB port is used to connect your portable digital music players, pen drives etc. for playing music tracks through the vehicle's music system.

C type USB port is used for fast charging of mobiles/ Tablets/ Laptops/ iPad/ iPhone etc.

### Rear Multimedia C + C Type Charger



C type USB port is used for fast charging of mobiles/ Tablets/ Laptops/ iPad/ iPhone etc.

### POWER SOCKET (if equipped)



12V power socket is available in luggage compartment on the right side.

## INTERIOR FEATURES

### LAMPS

#### Roof Lamp

##### Front Roof Lamp

Interior roof lighting lamp is provided on the roof.



The lamp can be turned ON by pushing the lens of the lamp.

Push it again to turn OFF the lamp.

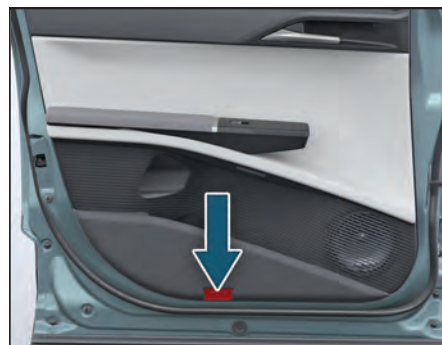
#### Boot Lamp



Boot lamp is provided in the rear luggage compartment to illuminate the luggage area.

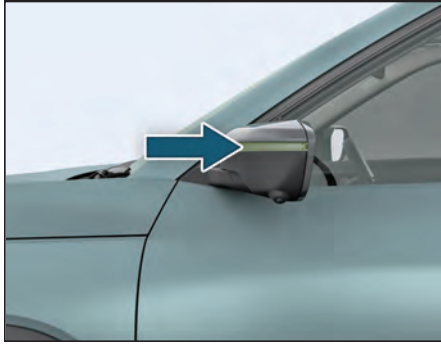
Whenever a door or tailgate is opened, it will turn ON.

#### Puddle Lamp



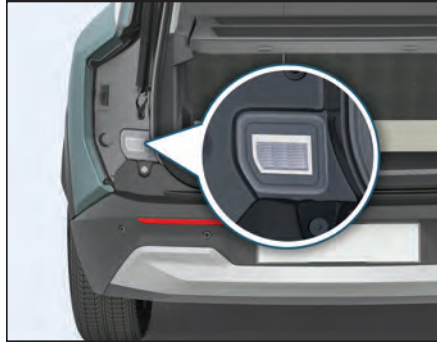
Puddle / Ajar Lamp is provided on driver, front passenger & rear doors trim. It will be ON only when particular door is in open condition.

### Side Indicator Lamp / Orvm



It is provided on outer part of ORVM. It will turn ON when the turn signal indicator switch is ON and Hazard Warning indicator switch is ON.

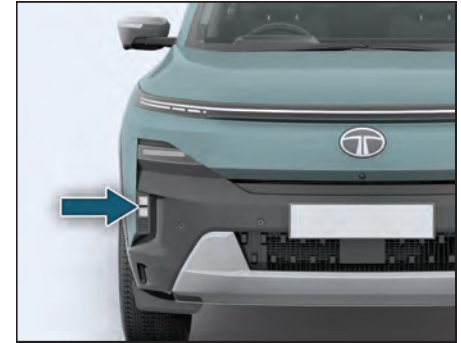
### Auxiliary Tail Lamp



This feature enhances safety during loading and unloading operations, roadside stops, or any situation where the vehicle is stationary with the tailgate open. The auxiliary tail lamp helps reduce the risk of rear-end collisions by providing an additional light source to alert other road users to the vehicle's presence.

### Front Fog Lamp (if equipped)

Front fog lamp is located on front bumper. When visibility is poor due to fog, snow or rain, use the fog lamps to improve visibility as well as making it easier for other road users to see you.



For switching 'ON' the Fog lamps, either the Head Lamp or Parking lamp must be 'ON'.

Operate the front fog lamp switch provided on fascia to 'ON' or 'OFF' the fog lamp.

## INTERIOR FEATURES

### Cornering Feature

For front fog lamp cornering feature the Head Lamp must be 'ON'.

As you turn steering wheel to left or right, the corresponding fog lamp will automatically get 'ON'.

### Rear Fog Lamp (if Equipped)



Rear Fog Lamp is provided on rear bumper to improve visibility in adverse weather conditions to vehicles behind you.

For switching 'ON' the rear fog lamps, front fog lamp must be 'ON'.

Operate the rear fog lamp switch provided

on fascia to 'ON' or 'OFF' the fog lamp.

An indicator on the switch will be illuminated when the rear fog light is 'ON'.

### Lamp Condensation / Fogging Condition

Condensation is a natural phenomenon in lamp. This occurs mainly because of atmospheric condition/weather change.

During normal condensation, thin film of mist is visible on the inside surface of the exterior lens. Generally, this condition is considered normal and will be eliminated by turning on the respective lamp with ignition ON or during normal driving conditions.

By doing this if the condensation has begun to clear after the drying time it indicates that the lamp sealing has not been breached and will eventually clear. The lamp must not be replaced.

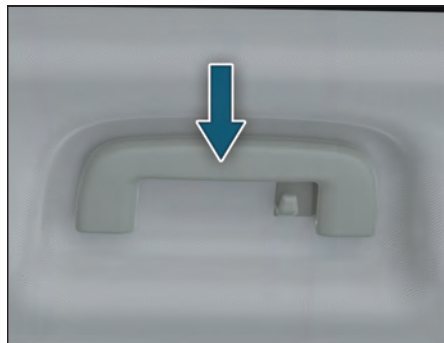
#### *i* NOTE

- All Exterior lamp fogging / condensation is natural occurrence and respective lamp assembly

*replacement will not be necessary to resolve the issue.*

- High-pressure washer jet direct on vent system of lamp are not recommended, there might be possibility of water ingress causing for heavy fogging
- Presence of condensation / mist in nonfunctional area is normal and acceptable, no action is recommended.

### ROOF GRAB HANDLE



Grab handles are installed on the roof for all seats except for the driver's seat. These help the passengers to position themselves comfortably during the journey.

### MOOD LIGHTS (if equipped)



#### Ambient Light Function

Ambient lighting comforts user by illuminating the vehicle interiors at defined locations. Single colored LEDs are fixed at various locations of the vehicle interiors.

#### Turning Ambient Lighting ON and OFF:

- Ambient lights turn ON in themes and color options (as per vehicle applicability) whenever parking light is turned to ON.
- Ambient lights turn OFF whenever parking light is turned to OFF.

#### Ambient Lights (entry/exit):

- Ambient lights turn ON in themes and color options (as per vehicle applicability) whenever roof lamp made active by removing the ignition key from key slot and opening any door.

- Once the opened door is closed, the ambient lights dim off after approximately 25 seconds.
- If door is left open, the ambient lights will turn OFF after set battery saver time.

#### Level Brightness Control

- Drag the slider to the right or left to adjust the brightness.

INTERIOR FEATURES

VEHICLE TELEMATICS (if equipped)

Car is equipped with iRA - Connected car Technology which offers a host of features to the users through the “TPEM iRA.ev” Mobile Application (APP). The Vehicle is equipped with an Electronic Control unit which monitors & records the data from various vehicle systems like Airbag, Brake, Battery & other electrical systems. This data is then processed & used for providing the connected Car features.

Vehicle Telemetry

This includes the periodic transmission of data from other vehicle ECUs & Electronic systems like BCM, EMS,ABS, Airbag, VCU, BMS etc. along with the geographical location of the vehicle.

Event based recording

This includes data generated during specific events like vehicle collision, intrusion etc.

The Data collected through Connected Car module is used by TATA Passenger Electric Mobility Limited for various pur-

poses, including, but not limited to, providing connected car features through mobile APP.

- Evaluation of Vehicle performance.
- Research & improvement of current & future vehicle designs.
- Troubleshooting & diagnostics of the vehicle.

TPEM does not disclose the data recorded from your vehicle to any third party except:

- After obtaining a written consent from the Car Owner.
- Upon request from Law enforcing agencies and regulatory bodies.
- Used for research purpose without the Personal Verifiable information (anonymized).
- Used as defence of TPEM in a Law-suit.

*network availability & vehicle telematics network availability.*

*i* NOTE

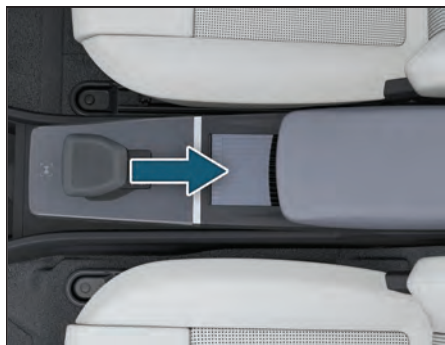
*Feature operations & data collection through TPEM iRA mobile app effectiveness is dependent on User mobile*





## INTERIOR FEATURES

### Location Of Wireless Charger

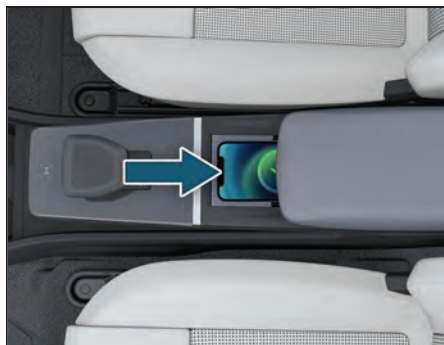


### Function of Each Component of WPC

- **Wireless Power Charging (WPC) ECU:** Generates power up to 15W and transfer power wirelessly by magnetic induction.
- **Anti-skid Mat:** Holds the position of smart phone placed on it against any jerk and acts a charging contact surface for the smart phone.
- **Cooling FAN:** It is provided to keep charging surface temperature within ambient temperature range.

- **FAN Cover:** It has ducts to direct FAN air on WPC ECU surface.
- **Infotainment unit:** It is status display unit to display the status of wireless power charger. Status symbol and text display is displayed on Infotainment unit.

### Functions Of WPC System



### Conditions to Charge Phone Properly

1. Keep the charging surface clear of any metal objects (coins, credit cards, smart cards, keys etc.)
2. Place the smart phone on the charging

area marked for positioning the phone, for best results place the smart phone at the center of the charging pad.

3. Charge the smartphone without its cover or not a thick cover otherwise it would halt the wireless charging
4. Ensure that the phone is placed with display facing upwards and charging area touching the charging pad surface.
5. Turn ON the ignition to start the charging.

**INFORMATION**

- The wireless charging function is supported to charge smart phones which are Qi compatible. Certain features may not function as not supported by the smart phone manufacturer and not a malfunction of the wireless charging.
- Wireless charging stops with Ignition OFF.
- Wireless charging stops when the smart phone is not completely in touch with the charging pad surface or not positioned correctly on charging pad

**Do's and Don'ts** **WARNING**

If any metal object such as coin is located between wireless charging pad and phone back, the charging may get disrupted. Also, metal object may heat up.

**Do's:**

1. Please ensure that the phone is compatible to the charging standard "Qi".
2. If any metal object found on charging pad remove it immediately.
3. In case of water/Liquid spillage on charging pad, dry out the pad surface area properly & clean the ECU surface area by removing Anti-skid mat. Do not charge the smart phone until surface is completely dry.
4. The smart phone may become hot while getting charged. Please be cautious about the high temperature while picking up the smartphone from the charging pad.

5. Wireless charger works on principle of magnetic induction, i.e. it converts electrical energy into magnetic energy to transfer energy from charging pad to phone. Please maintain safe distance from the charger most of the time as it may cause irritation to sensory organs or active implants if implemented in the body. Please consult medical specialist in case implant organ in the body of the user
6. Always turn ON the engine while using this feature to avoid vehicle battery drain issue.
7. Always keep charging pad clean and dust free.
8. Vehicle AC may turn ON during wireless mobile charging for efficient use of this feature.

**Don'ts:**

1. Do not use metal smart phone covers as it would halt the wireless charging function. The wireless charging may not function properly when there is a heavy & thick accessory cover on the smart phone.

## INTERIOR FEATURES

---

2. Do not place smart phone up-side down on charging pad or do not miss aligned mobile phone on charging pad in such case smartphone charging will not happen.
3. Do not keep any metal objects like coins, smart keys, electronic cards e.g. credit card, debit card, smartcard from the charging pad as it may disrupt the charging process and/or may damage the card.
4. Do not keep any liquid (e.g. water, cold drink, and sanitizer), flammable object on antiskid mat.
5. Do not cover the wireless charger with a cloth or other object while charging. It may heat up the device and reduce the charging efficiency.
6. Do not disassemble, modify or remove the wireless charger & do not apply force or impact to the wireless.

### Information

1. Small noise may be heard when a smart phone which does not support wireless charging or any foreign object is placed on the charging pad. This small sound is due to the vehicle discerning compatibility of the object placed on the charging pad. It does not affect the vehicle performance or the smartphone in any way.
2. For certain cellular phones with their own protection, the wireless charging speed may decrease and the wireless charging may stop.
3. When the interior temperature of the wireless charger rises above a set temperature, the wireless charging will cease to charging function. After the interior temperature drops below set threshold, the wireless charging function will resume.
4. When the mobile phone temperature rises above a set cut off threshold, the wireless charging will cease to charging function due to mobile phone stops demand power from wireless charger.

After the mobile phone temperature drops below threshold, the wireless charging function will resume. Mobile temperature cut off threshold is much lower than WPC temperature cut off threshold.

5. When charging certain smart phones, the charging full message on head unit may not display when the smart phone is fully charged. It depends on smart phone manufacturer.
6. Smartphone of some manufacturers may display messages on weak current. This is due to particular characteristic of smartphone and does not imply a malfunction on wireless charger. Smaller smartphone users (ex. iPhone) may face intermittent charging issues due to its smaller size. (To avoid this, place the smartphone at center of the charging pad). Small mobile phones may not be able to charge in every position on charging pad.
7. The wireless charger may not operate correctly when the vehicle is near a TV tower, electric power plant, gas station,

large display, airport, or other facility that generates strong radio waves or electrical noise.

### WPC ECU in Standby Mode

The infotainment system displays no warning message/ indication in this mode.

This mode represents that the charging function is halted and not functional. The charging function could halt because of below reasons like:

1. Phone is not properly aligned with the charging pad or not positioned correctly on pad wireless charger in standby mode
2. Phone is kept in upside down position wireless charger in standby mode
3. Phone is fully charged, and phone does not demand power wireless charger in standby mode

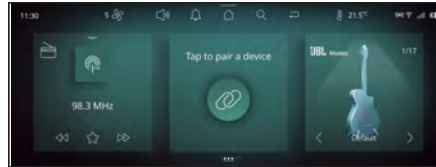
### Smart Phone Charging ON Mode

When the smart phone is placed correctly and the conditions are favorable to perform the function of wireless charging, the infotainment system shows following messages. After Popup, The charging symbol

stays ON until the phone is fully charged.



*Charging Mode ON - Popup*



*Charging Mode ON – All Time Display*

### Metal Object Detection Mode

The charging gets interrupted/stopped due to metal object placed on the charging pad. The infotainment head unit displays following message. Check if there are any foreign objects between the smart phone and the charging pad, please clean if so. Smartphone shall be lifted for removing foreign objects and place it back on charging pad.



*Metal Objects*

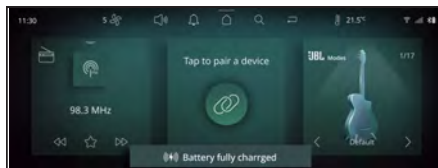
## INTERIOR FEATURES

### NOTE

- *Delay in restarting of mobile charging will be observed if foreign objects are removed without lifting smart phone.*
- *After removing the foreign objects, if smart phone do not resume charging immediately lift the phone and place on charging pad to start the charging.*
- *In case, Phone overheat, remove and keep it after sometime.*

### Smart Phone Battery is Full/ Charging Completed

The smart phone fully charged status is indicated on the infotainment display screen with following message. This Indication depends on phone profile whether it communicate the battery full charge status to WPC System.



Charging Complete Indication - Popup



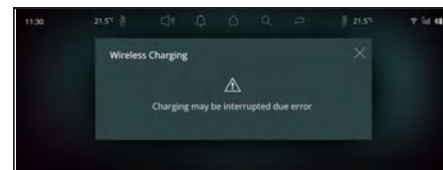
Charging Complete Indication - All Time Display

### WPC System Error Mode

The error in the WPCF wireless power charger with FAN, system may cause the error message to get displayed on the infotainment screen.

Some of errors that can occur while charging which halt/interrupt charging can be covered with this indication are:

- WPCF internal fault which lead to permanent failure in charger functionality
- WPC Fan Stuck / Jam is detected
- Coil Failure
- High Temperature of WPC device (70 Degree C)
- Memory failure



System Error Mode- Popup

### **i NOTE**

*Avoid charging your smartphone and go to the nearest service center if an error message appears on the head unit.*

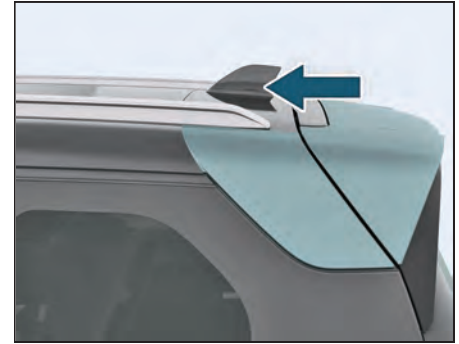
### **ROOF RAIL (AESTHETIC) (if equipped)**



### **i NOTE**

*Do not apply load or mount roof rack on roof rails.*

### **SHARK FIN ANTENNA**



Shark FIN antenna is provided on the roof at rear end.

## INTERIOR FEATURES

### WELCOME AND GOOD BYE STRATEGY

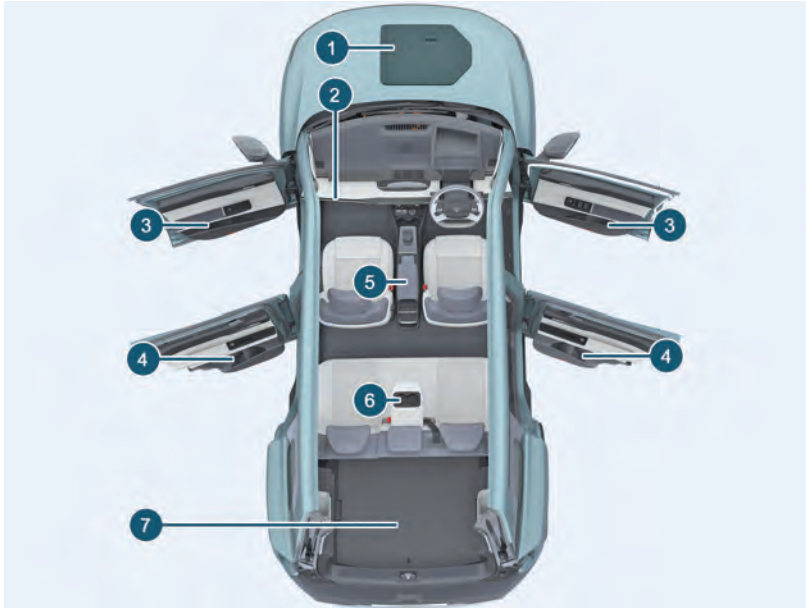
| S.n. | Function            | Vehicle Condition | Key Inputs | Lamp Animation                                                                                                                              |
|------|---------------------|-------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Welcome Animation   | Lock              | Unlock     | A single flash of all direction turn indicators followed with a welcome animation of three cycles. Total animation time is approx. 4.5 Sec. |
| 2    | Goodbye Animation   | Unlock            | Lock       | Two flashes of all direction turn indicators followed with a goodbye animation of three cycles. Total animation time is approx. 4.5 Sec     |
| 3    | Second input Unlock | Lock              | Unlock     | Single flash of all direction turn indicators.                                                                                              |
| 4    | Second input Lock   | Lock              | Unlock     | Four flashes of all direction turn indicators.                                                                                              |

#### For Animation

| S.n. | Function                  | Animation                                                                         | Lamp Animation                                                                                                                              |
|------|---------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Buffering Animation       |  | A single flash of all direction turn indicators followed with a welcome animation of three cycles. Total animation time is approx. 4.5 Sec. |
| 2    | Charging Animation        |  | Two flashes of all direction turn indicators followed with a goodbye animation of three cycles. Total animation time is approx. 4.5 Sec     |
| 3    | Charging Error            |  | Single flash of all direction turn indicators.                                                                                              |
| 4    | Battery Charge state 100% |  | Four flashes of all direction turn indicators.                                                                                              |



### STORAGE COMPARTMENT



1. Frunk
2. Glove Box
3. Utility Pockets on Front Doors
4. Utility Pockets on Rear Doors
5. Center Console Below Arm Rest
6. Foldable Arm Rest/ Cup Holder
7. Luggage Compartment

## STOWAGE AREA

### GLOVE BOX



#### Opening And Closing

**To open-** Pull the knob and open the glove box flap.

**To close -** Lift glove box flap upward until it engages.

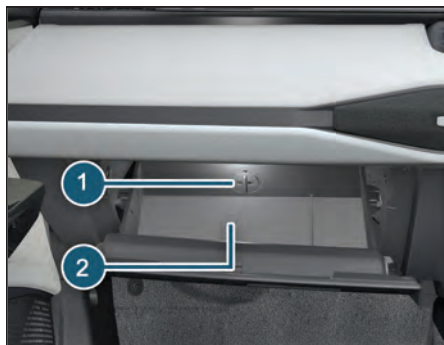
#### Glove Box Illumination

The glove box lamp illuminates when the glove box flap is opened.

#### NOTE

*Make sure that glove box flap is closed while driving.*

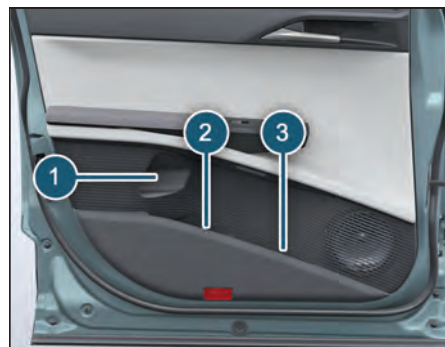
#### Stowage Detail



Following items can be stored in glove box.

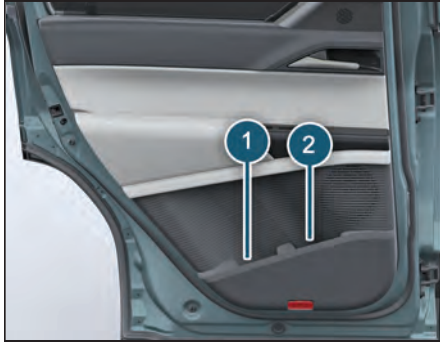
1. Owner's manual and other vehicle document
2. First aid kit, Visiting card, Pen, Receipts etc.

### UTILITY POCKETS ON FRONT DOORS



Utility pockets are provided on front doors and they can be used to keep following items.

1. Suitable Umbrella
2. Suitable Water Bottle
3. Magazine / Paper / Books

**UTILITY POCKETS ON REAR DOORS**

Utility pockets are available on rear doors and they can be used to keep following items.

1. Suitable Umbrella
2. Suitable Water Bottle

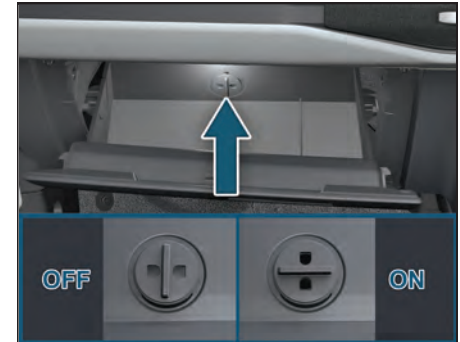
***i* NOTE**

*Remove the water from umbrella and fold it properly before storing it in umbrella holder.*

**CENTER CONSOLE****Stowage Below Arm Rest**

Stowage below arm rest

Stowage compartment is provided below the arm rest for keeping cell phones, iPod's, chargers etc.

**Cooling Facility**

Cooling facility is provided below arm rest.

## STOWAGE AREA

### Cup Holder



Space for cup holder is provided in center console.

### Stowage For Rear Passenger (if equipped)



Stowage for rear passenger is provided on rear side of floor console between the front passenger seats. It can be used to keep mobile charger, mobile and small items like wallet, Power bank etc.

### FOLDABLE ARM REST (if equipped)



A foldable arm rest has been provided in the rear seat. It also has two cup holders, which can be accessed by opening the cover. When not required, fold the armrest back into the seat.

#### NOTE

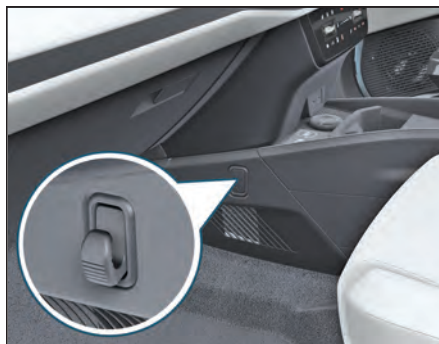
*Remove all items and cups before folding the cup holder. Use cups, containers, bottles of right size and which have lids. The content could otherwise spill.*

**HOOKS (if equipped)****Coat Hook**

Coat hangers are provided for rear passenger on both grab handles.

***i* NOTE**

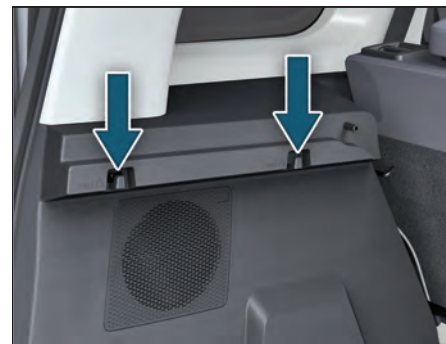
- The coat hook is not designed to carry heavy objects or luggage items.
- Do not hang hard, sharp-edged or fragile objects on the coat hook.

**Collapsible Hook**

Carrier hook is provided for hanging small carry bags etc.

***i* NOTE**

*Do not use these hooks for securing luggage like using nets etc.*

**LUGGAGE COMPARTMENT**

Store the luggage in luggage compartment. You can keep suitcase, bags, etc.

** WARNING**

- Distribute the items of luggage as evenly as possible.
- Position heavy loads towards rear seat and low down in the trunk as possible.
- Do not allow occupants to travel in the luggage compartment.

## STOWAGE AREA

- Do not place anything on luggage cover as it could obstruct driver's rear view. Also in case of an accident or sudden braking, it could cause an injury to occupants.

### STOWAGE BACK OF DRIVE SEAT



Stowage for the rear passenger is available back of driver seat. It can be used to keep phone and small items.

### FRUNK



Frunk is an extra storage provided in your vehicle under bonnet.

#### Opening the Frunk

- Open the bonnet and lock with stay rod.
- Press the opening knob provided on frunk cover. Open it fully and it shall remain open.



### **NOTE**

*If the hinge and latch are not used correctly, it may get damaged and need to be replaced.*

### **Closing the Frunk**

1. Before closing the frunk cover ensure that there are no obstructions to it.
2. Push down the frunk cover and gently press it until it latches against the frunk latch.

### **NOTE**

*Before opening the frunk, please ensure that the bonnet is fully open and locked with stay bar.*



### **CAUTION**

- Do not overload the frunk, it may get damaged.
- Maximum allowable frunk load for AWD: 17Kg and RWD: 28Kg. Do not store fragile objects in the frunk.

- Always ensure to keep the frunk cover closed while driving as stored items might come out of frunk and may get damaged.
- Do not store water, liquids, or frozen items in the frunk, as this may damage the vehicle's driving system.

### **CAUTION**

- Do not store liquid in frunk, if so leakage may damage electrical system in vehicle.
- Do not force press the frunk cover or place heavy objects on it, it may get damaged.
- Do not wash the frunk with jet spray. Clean it with wet cloth.

### **WARNING**

- Never place/keep animals, baby, toddlers etc inside/outside the frunk, it can cause serious injury.

## STOWAGE AREA

---

- Never try to enter inside frunk, it can cause serious injury.
- Do not store flammable/explosive items in frunk as it may catch fire if vehicle is exposed to high temperatures for long duration.



### EMERGENCY EQUIPMENT

You should be familiar with the location of the emergency equipment provided in the vehicle and how to use it.

Do a check of this equipment periodically and make sure that they are in proper working condition and stowed at their locations.

#### First Aid Kit

The first aid kit is kept inside the glove box compartment.

The kit contains items that can be used in case of minor injuries only.

#### NOTE

*Examine contents of the first aid kit periodically and replenish consumed or expired items.*

#### NOTE

*The jack should be used only to change wheels. It is important to read the instructions in this section before attempting to use the jack.*

#### Tool Kit

Tool kit is provided in the Tool kit bag in Luggage Compartment.

#### Advance Warning Triangle

An advance warning triangle is kept in the luggage compartment. Use advance warning triangle to warn the approaching traffic in case of vehicle break-down or during emergency, where your vehicle could become a potential traffic hazard.



Keep the warning triangle at an approximate distance of 50-150m behind your vehicle in the same lane of traffic. The

reflecting side of the triangle should face the oncoming traffic and it should be free from any obstacles. Remove the advance warning triangle carefully from the bag and assemble. Refer instructions given on the bag.

Press Hazard Warning Switch given on fascia. When you press the hazard warning switch, all turn signal lamps will start to blink.

#### NOTE

*After using the warning triangle tie it firmly and keep it inside the bag to avoid rattling noise.*

#### NOTE

*The hazard warning lamps can operate even if the ignition is switched off.*

## EMERGENCY AND BREAKDOWN

### PUNCTURE REPAIR KIT

#### WARNING

Compliance to below instructions is vital to ensure vehicle safety and personal safety. Non-compliance may result in serious injury or death. Damage to tire will affect vehicle handling and lead to loss of overall vehicle control.

#### Introduction

- The tire puncture repair kit seals most tire punctures to restore temporary mobility.
- Recommended to use only for passenger car ground tubeless tires only and vehicle tire inflation pressure up to 300kPa (3 bar /43psi).
- The system consists of a compressor and a sealant, and serves to effectively and conveniently seal punctures in car tires caused, for example, by nails or similar foreign objects with a diameter of up to ¼" (6 mm).
- Depending on the type and extent of tire damage, some tires can only be partially sealed or not sealed at all.
- Loss of tire pressure can affect vehicle handling and vehicle control.
- Drive with caution and avoid making sudden steering or driving maneuvers, especially if the vehicle is heavily loaded or you are towing a trailer.
- The system will provide you with an emergency temporary repair, enabling you to continue your journey to the next vehicle or tire dealer, or to drive a maximum distance of 200 Kms.
- Do not exceed a maximum speed of 80 km/h.
- Keep the Puncture repair Kit out of the reach of children.
- If used for other than its intended purpose, the tire puncture repair Kit may cause severe accident or injury due to the fact that compressed air can act as an explosive or propellant.
- Park your vehicle at the roadside so that you do not obstruct the flow of traffic and you are able to use the Puncture repair Kit without being in danger.
- Engage the hand brake, even if you have parked on a level road, to ensure that the vehicle will not move.
- Do not attempt to remove foreign objects like nails or screws penetrating the tire leave them as it is.
- Always ensure the vehicle is running during the tire puncture repair kit is in use, but not if the vehicle is in an enclosed or poorly ventilated area.
- Never leave the tire puncture repair kit unattended while in use.
- Do not keep the compressor operating for more than 10 minutes otherwise there is a risk of it overheating.
- Replace the sealant bottle with a new one before the expiry date is reached (see bottle label). In case that the sealant is expired the functionality cannot be fully guaranteed. Only use original tire puncture repair kit bottles which are pressure resistant.

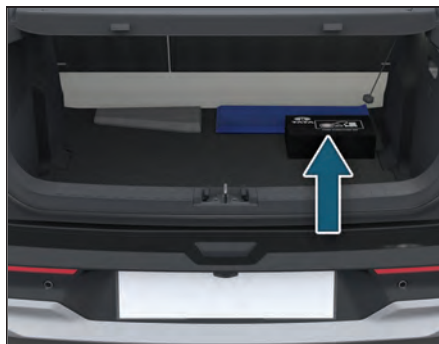
### WARNING

Do not use the Puncture repair Kit if the tire has already been damaged as a result of being driven underinflated. Do not try to seal damage other than that located within the visible tread of the tire. Do not try to seal damage to the tire's sidewall.

### WARNING

TPMS/iTPMS (if equipped) functionality to be checked by TPEM Authorised Service Centre, if any error occurs due to the use of tire puncture repair kit.

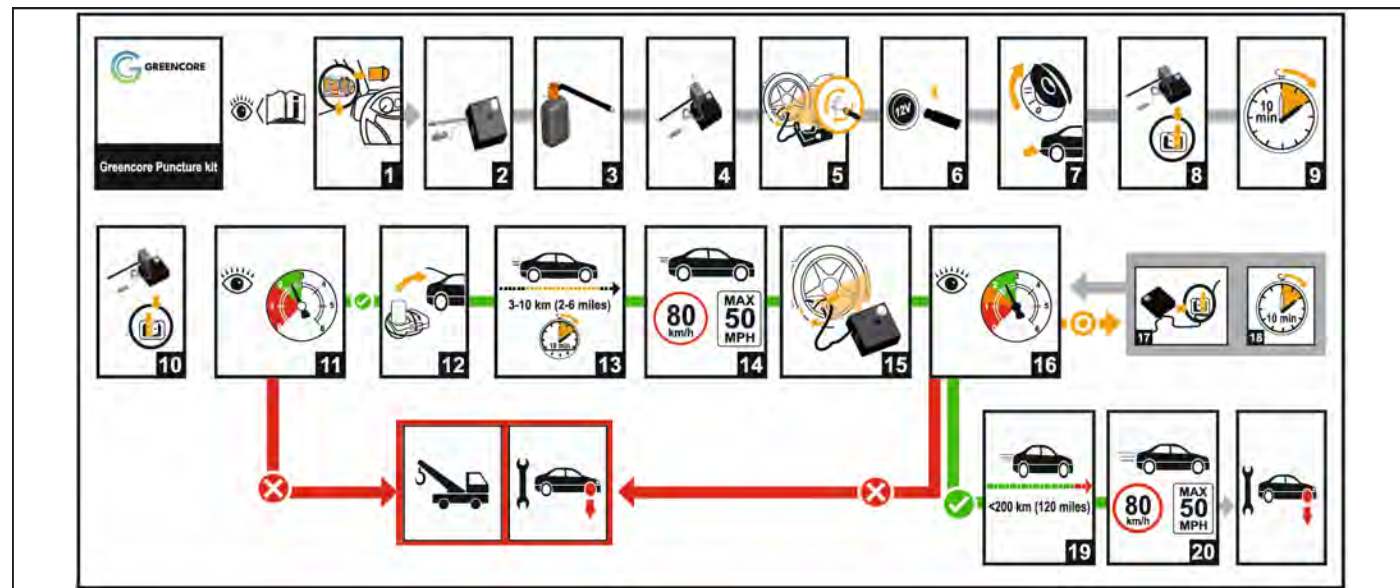
### Location In Vehicle



*In Luggage compartment*

## EMERGENCY AND BREAKDOWN

### Steps



### How To Proceed In The Event Of A Tire Puncture

First pump the tire sealant and air into the tire (see Step 1). Immediately thereafter, drive a short distance (3-10 km) in order to distribute the sealant in the tire. After that check the tire pressure and pump more air into the tire if necessary (see Step 2). Then you can proceed to drive with caution for a maximum distance of 200 kms and at a maximum speed of 80 km/h.

#### Step 1: Pumping The Tyre Sealant And Air Into The Tyre

1. Take out the hose and power plug with cable out of the tyre puncture repair kit casing. Remove the white cap of T type connector of Sealant Bottle.
2. Connect the hose pipe of the compressor with T type connector of sealant bottle. Ensure it is tightly Fitted.
3. Avoid skin contact with the sealant which contains natural rubber latex. Do not open pressure "air release" valve. Use enclosed protective gloves.
4. Install sealant Bottle by rotating clock-

wise firmly against the bottle holding grooves on the compressor. Remove the white cap of the hose pipe of the sealant bottle.

5. Insert power plug into the 12 volt power socket connection.
6. Start the vehicle in idling (only if the vehicle is outdoors or in a well ventilated area).
7. Press compressor switch to ON Important: When pumping in the sealant through the tyre valve, the pressure may rise up to 500 KPa (5 bar. 73 psi) but will drop again after about 30 seconds
8. Inflate the tyre to an inflation pressure of minimum 180kPa, (1.8 bar/26 psi) and a maximum of 300kPa (3 bar/43 psi).
9. Remove the sealant bottle from the compressor grooves & tight the white caps on the hose pipe of the sealant bottle as well as T -type connector of the sealant bottle. This avoids unexpected leakage of sealant residue and Lock it.

10. Make sure the puncture repair kit stored safely, but it's still easily accessible, in the vehicle.
11. The compressor will be needed again when you check the tyre pressure.

#### WARNING

- Ensure pump should not be ON for more than 10 min as it may heat up and stop working.
- Check the sidewall of the tyre prior to inflation. If there are any cracks, bumps or similar damage, do not attempt to inflate the tyre. Do not stand directly beside the tyre while the compressor is pumping. Watch the sidewall of the tyre. If any cracks, bumps or similar damage appear, turn off the compressor and let the air out by means of the pressure "air release" valve. In this case, do not continue to use the tyre.

#### NOTE

## EMERGENCY AND BREAKDOWN

*When pumping in the sealant through the tyre valve, the pressure may rise up to 500 kPa (5 bar/73 psi) but will drop again after about 30 seconds.*

### WARNING

Need to drain fluid from tyre before repair.

### Step 2: Checking The Tyre Pressure

1. Once a tyre inflation pressure of at least 180kPa (1.8 bar/26 psi) has been reached. Switch the compressor to "0" in order to read the actual tyre pressure from the pressure gauge.
2. Pull the power plug from the 12 volt power socket connection.
3. Slowly unscrew the hose from the tyre valve (sealant residues may escape from the hose) and put the protective cap back onto the hose.
4. Leave the bottle in the holder. This avoids unexpected leakage of sealant residue.
5. Make sure the Puncture repair Kit, the

cap of the bottle and the orange cap are stored safely, but are still easily accessible, in the vehicle.

6. The kit will be needed again when you check the tyre pressure.
7. Start and drive for about 3-10 km so that the sealant can seal the damaged area. Do not drive for more than 10 min and not faster than 80 km/h.
8. Stop the vehicle after driving about 3-10 km. Check and where necessary, adjust the pressure of the damaged tyre. Remove the protective cap from the end of the hose. Screw the hose firmly onto the valve of the damaged tyre. Read the tyre pressure from the pressure gauge.
9. If the pressure of the sealant filled tyre is 130kPa (1.3 bar/19 psi) or more, it must now be adjusted to the pressure specified for your vehicle (Refer tyre pressure sticker pasted on driver door side).
10. Deflate the tyre to the specified pressure using the pressure "air release" valve. Rest of the remaining sealant in

the hose might leak out when opening pressure "air release" valve or taking off the protective cap of the hose. Please use protective glove for safety purpose.

11. Once you have inflated the tyre to its correct tyre pressure, switch off the compressor, pull the plug out of the socket, unscrew the hose, fasten the tyre valve cap and put back on the protective cap of the hose.
12. Leave the bottle in the holder and store the Puncture repair Kit away safely in the vehicle trunk
13. Drive to the nearest workshop to get the damaged tyre repaired and if the tyre repair is not possible it should be removed from the car. Before the tyre is removed from the rim, inform your tyre dealer that the tyre contains sealant.

### WARNING

If heavy vibrations, unsteady steering behavior or noises should occur while driving, reduce your speed and drive

with caution to a place where it is safe for you to stop the vehicle. Recheck the tyre and its pressure. If the tyre pressure is less than 130kPa (1.3bar, 19 psi) or if there are any visible cracks, bumps or similar damage on the side wall, do not continue to use the tyre.

### **WARNING**

After using the sealant you may drive no faster than 80 km/h and the damaged tyre must be replaced as quickly as possible (within a maximum driving distance of 200 km).

New sealant and replacement parts can be purchased from your authorized repair shop or dealer. Sealant bottles can be disposed with house-hold waste.

### For Normal Tire - Checking / Inflation Of Tire Pressure Follow The Below Process



- Remove the puncture repair kit from the luggage area.
- Insert the power plug into the 12V power socket connection and start the vehicle in idling.
- Remove the plug from the tire valve and screw the inflator hose into the tire valve
- Press the switch to “I” present on the inflator and the motor will start to inflate.
- As specified pressure is achieved then switch to “O” present on the inflator and compressor will turn off.
- Check the tire pressure again. If tire pressure is too high, deflate the tire to

the specified pressure using the pressure “air release” valve.

- Remove the inflator hose from the tire valve and plug the tire valve safely.
- Remove the power plug 12V from the power socket and assemble it properly and keep the unit in luggage space again for next use.

### **NOTE**

*Remember that emergency road-side tire repair kits only provide temporary mobility. You should consult a tire specialist for advice.*

### **NOTE**

*New sealant and replacement parts can be purchased from your authorized repair shop or dealer. Sealant bottles can be disposed with household waste.*

## EMERGENCY AND BREAKDOWN

### TOWING

#### Guidelines (do's & Don'ts)

When towing a break down vehicle, certain precautions and procedures must be taken to prevent damage to the vehicle and/or components. Failure to use standard towing precautionary measures when lifting or towing a break down vehicle could result in an unsafe operating condition.

To correctly tow and prevent accidental damage to your vehicle, take help of a TPEM Authorised Service Centre or a commercial tow-truck service.

#### NOTE

*Make sure that the parking brake is released, vehicle is in neutral and steering wheel is unlocked. The power steering functions only when vehicle is running. Hence, during towing the steering efforts will be more.*

#### WARNING

Do not jump start the vehicle, since it is an EV. If the 12V battery is completely

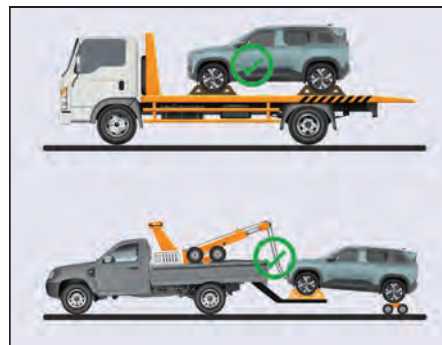
discharged, contact the nearest TPEM Authorised Service Centre

#### WARNING

- Do not get under your vehicle after it has been lifted by a tow truck.
- For towing a vehicle, the best way is to use a wrecker. Alternatively use a rigid tow bar.
- Switch 'ON' the hazard warning indicators of both the vehicles to warn other road users.
- Limit the speed to 20-30 kmph.
- In case of brake failure, use the parking brake to control the vehicle.
- Fasten the tow rope or tow bar at the towing eyes. Otherwise, the vehicle could be damaged.
- When towing, pull away slowly and smoothly. If the tractive power is too high, the vehicles could be damaged.

### Recommended Towing

In case of break down, we recommend that your vehicle be towed with the driving wheels off the ground or place the vehicle on a flatbed truck as shown.



#### NOTE

*Second option can be used if towing dollies are available otherwise flatbed towing is recommended.*

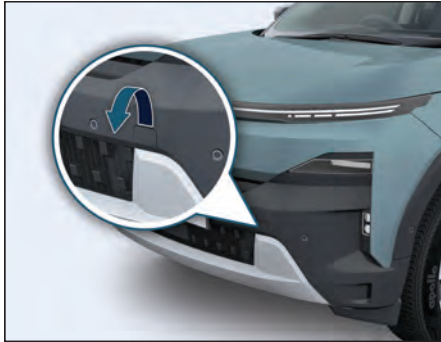


### **WARNING**

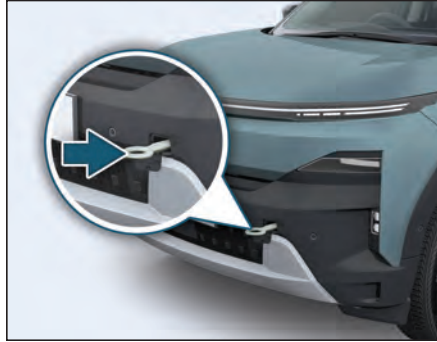
Do not tow your vehicle with the rear wheels on the ground, as this may cause serious damage to the motor.

#### **Tow Hook Fitment**

- Open the tow hook cover provided on the front bumper by pulling it downward.



- Assemble the tow hook as shown in image.

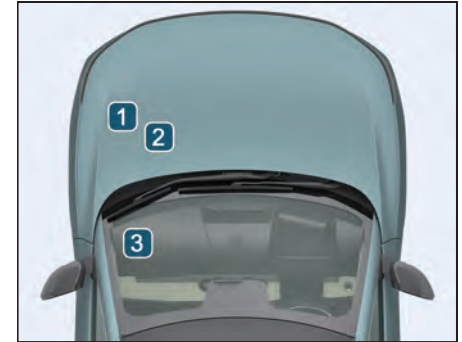


- After towing, remove the towing hook and press fit the cover properly.
- Place the towing hook in the vehicle tool kit.

#### **FUSES**

Your vehicle has fuse boxes at three locations.

The vehicles electrical circuits have fuses to protect the wiring from short circuits or sustained overload.



1. Battery Mounted Fuse Box.
2. Motor Compartment Fuse Box.
3. Cabin Compartment Fuse Box.

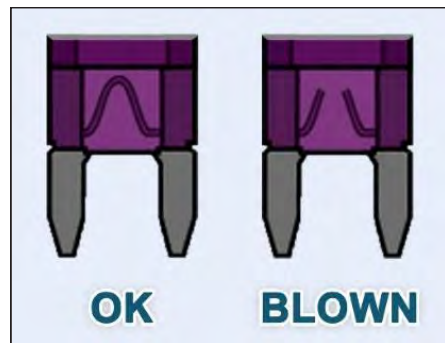
#### **Checking And Replacing Fuses**

If any electrical unit in your vehicle is not functioning, check the fuses first.

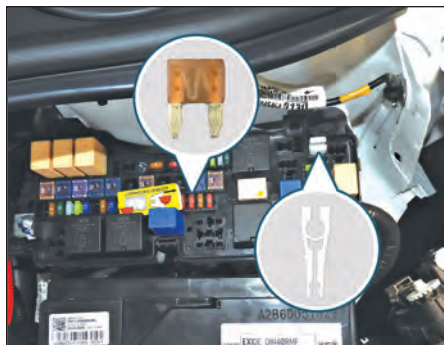
## EMERGENCY AND BREAKDOWN

Please follow the steps below that will guide you to check and replace them.

- Apply parking brake
- Switch off all electrical accessories.
- Turn the ignition key to the 'LOCK' position.
- In the fuse box, identify the defective fuse from its melted wire.



- Remove the defective fuse by “fuse puller”. The fuse puller and spare fuses are provided in the motor compartment fuse box.



- Defective fuses must be replaced with fuses of same rating, which you can recognize by color and value. Fuses of same rating, which you can recognize by color and value.

### *i* NOTE

*Always make sure that the spare fuses are added.*

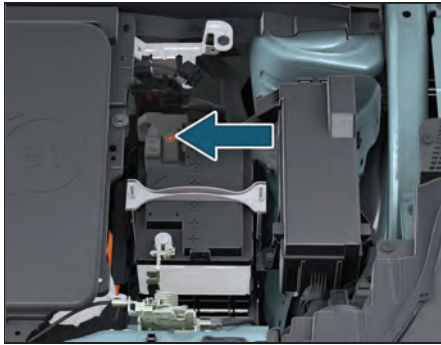
- Make sure that all other fuses are pressed firmly in position.
- If a newly inserted fuse also blows, have the cause traced and rectified at

nearest TPEM Authorised Service Centre immediately.

### WARNING

- If you manipulate or bridge a faulty fuse or if you replace it with a fuse with higher amperage, the electric cables could be overloaded. This could result in a fire. There is a risk of an accident and injury.
- Always replace faulty fuses with the specified new fuses having the correct amperage.

### 1. Battery Mounted Fuse Box



#### **WARNING**

If fuse box cover is removed for any reason, it should be refitted properly in its original position.

### 2. Motor Compartment Fuse Box



#### **NOTE**

*The fuse box layout is for reference purpose only. Please refer the sticker provided inside the fuse box cover.*



| Motor Compartment Fuse Box |             |                             |
|----------------------------|-------------|-----------------------------|
| Fuse No                    | Fuse Rating | Description                 |
| F1                         | 100A        | INTERIOR FUSEBOX SUPPLY - I |
| F2                         | 125A        | EPAS                        |
| F3                         | 60A         | INTERIOR FUSEBOX IGN/ACC    |
| F4                         | 60A         | IVBAC BATT PUMP             |
| F5                         | -           | -                           |
| F6                         | 60A         | BLDC MOTOR                  |
| F7                         | -           | -                           |
| F8                         | 60A         | EPAS                        |
| F9                         | 5A          | VCU KP SUPPLY BATT          |
| F10                        | -           | -                           |
| F11                        | -           | -                           |
| F12                        | 5A          | BRAKE SW                    |
| F13                        | -           | -                           |
| F14                        | 15A         | HORN HIGH / LOW             |
| F15                        | -           | -                           |
| F16                        | -           | -                           |
| F17                        | 100A        | INTERIOR FUSE BOX SUPPLY II |
| F18                        | -           | -                           |
| F19                        | 60A         | EV POWERTRAIN               |

## EMERGENCY AND BREAKDOWN

| Motor Compartment Fuse Box |             |                   |
|----------------------------|-------------|-------------------|
| Fuse No                    | Fuse Rating | Description       |
| F20                        | 30A         | IGNITION LOAD     |
| F21                        | -           | -                 |
| F22                        | 30A         | VCU RELAY FUSE    |
| F23                        | -           | -                 |
| F24                        | -           | -                 |
| F25                        | -           | -                 |
| F26                        | 10A         | BMS POWER BATT    |
| F27                        | 15A         | HEADLAMP LH       |
| F28                        | 20A         | FRONT WIPER MOTOR |
| F29                        | 15A         | HEADLAMP RH       |
| F30                        | 10A         | AVAS              |
| F31                        | -           | -                 |
| F32                        | -           | -                 |
| F33                        | 5A          | PT RELAY CONTROL  |
| F34                        | -           | -                 |
| F35                        | -           | -                 |
| F36                        | 10A         | PT OBD            |
| F37                        | 30A         | VCU               |
| F38                        | -           | -                 |
| F39                        | -           | -                 |

## EMERGENCY AND BREAKDOWN

| Motor Compartment Fuse Box |             |                            |
|----------------------------|-------------|----------------------------|
| Fuse No                    | Fuse Rating | Description                |
| F40                        | 10A         | COMPRESSOR BAT             |
| F41                        | -           | -                          |
| F42                        | 5A          | E DRIVE BATTERY RELAY COIL |
| F43                        | -           | -                          |
| F44                        | 5A          | EVAPORATED SOLENOID        |
| F45                        | 5A          | FRONT E DRIVE BATT         |
| F46                        | 30A         | TRACTION COOLANT PUMP      |
| F47                        | 10A         | 3 IN 1 COMBO               |
| F48                        | -           | —                          |
| F49                        | -           | -                          |
| F50                        | 5A          | E-TXV                      |
| F51                        | 5A          | REAR E-DRIVE BATT          |
| F52                        | 30A         | HV BATT COOLANT PUMP BATT  |
| F53                        | -           | -                          |
| F54                        | -           | -                          |
| F55                        | -           | -                          |
| F56                        | 5A          | IVBAC IGN                  |
| F57                        | 5A          | FRONT PDC                  |
| F58                        | 5A          | IGNITION SUPPLY            |
| F59                        | -           | -                          |

## EMERGENCY AND BREAKDOWN

| Motor Compartment Fuse Box |             |                     |
|----------------------------|-------------|---------------------|
| Fuse No                    | Fuse Rating | Description         |
| F60                        | -           | -                   |
| F61                        | 5A          | HEAD LAMP LEVELLING |
| F62                        | -           | -                   |

| Motor Compartment Relay Fuse Box |             |                             |
|----------------------------------|-------------|-----------------------------|
| Fuse No                          | Fuse Rating | Description                 |
| R1                               | 20A         | HORN                        |
| R2                               | 20A         | WIPER ON /OFF               |
| R3                               | 20A         | WIPER SPEED SLOW / FAST     |
| R4                               | -           | -                           |
| R5                               | -           | -                           |
| R6                               | -           | -                           |
| R7                               | 20A         | FRONT WIPER MOTOR           |
| R8                               | 20A         | REAR E DRIVE                |
| R9                               | 20A         | FRONT E DRIVE               |
| R10                              | 40A         | RATCTION COOLING PUMP RELAY |
| R11                              | -           | -                           |
| R12                              | 40A         | COOLING PUMP BATT RELAY     |
| R13                              | 40A         | VCU                         |
| R14                              | -           | -                           |
| R15                              | 40A         | IGNITION                    |



## EMERGENCY AND BREAKDOWN

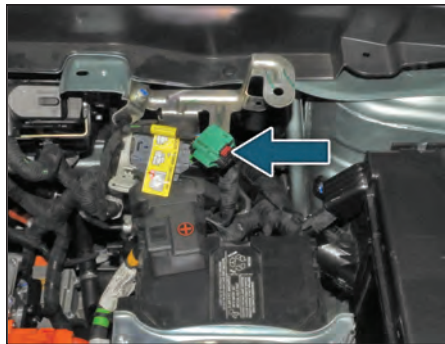
| Motor Compartment Relay Fuse Box |             |                        |
|----------------------------------|-------------|------------------------|
| Fuse No                          | Fuse Rating | Description            |
| R16                              | -           | -                      |
| R17                              | -           | -                      |
| R18                              | -           | -                      |
| R19                              | -           | -                      |
| R20                              | -           | -                      |
| R21                              | 20A         | AV BATT KP ALIVE RELAY |

## EMERGENCY AND BREAKDOWN

### Motor Compartment High Voltage Service Disconnect

During the service of vehicle, the technician or service operator can be prone to electric shock with Low Voltage.

So to avoid this electric shock hazard the high voltage should be disconnected before servicing the vehicle.



Below steps to be followed to isolate High voltage from vehicle and components:-

- Technician or service operator should locate the motor compartment which is located under the bonnet.

- Without touching any other high voltage components he should remove the HV MSD short link connector.
- HV MSD connector is connected with green color connector parked near to 3in1 combo unit over a bracket.
- HV MSD short link connector labeled with yellow color (As per picture).



- After successfully completion of service HV MSD short link must be reconnected with green connector same as original condition.

### 3. Cabin Compartment Fuse Box Cover Removal Procedure

Fuse box is located inside the cover below steering column. To access the fuse box, remove cover as per procedure given below.

1. Fuse box cover is mounted on dash board with the help of lugs at the top and bottom of the cover from inside.



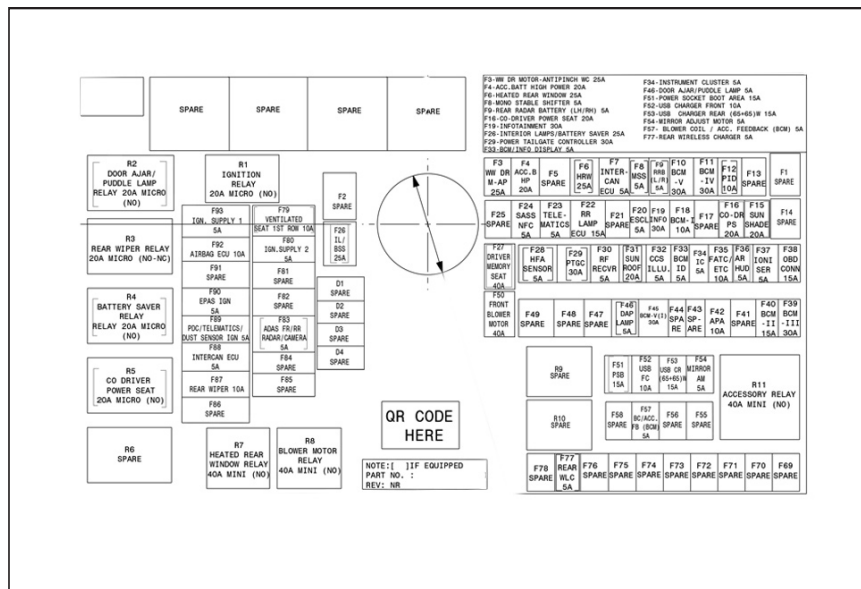
2. To remove the cover, gently pull the cover from upper side that the lugs get disengaged.

### **Re-fitment Procedure**

Align bottom lugs and push upper part with respective slots on dashboard and press the cover firmly.

# EMERGENCY AND BREAKDOWN

## Fuses - Cabin Compartment



Note: Please refer fuse box sticker on vehicle for more clarity.

## EMERGENCY AND BREAKDOWN

| Cabin Compartment Fuse Box Details |               |                            |
|------------------------------------|---------------|----------------------------|
| Fuse No                            | Ratings (Amp) | Description                |
| F1                                 | -             | -                          |
| F2                                 | -             | -                          |
| F3                                 | 25A           | WW DR MOTOR - ANTIPINCH WC |
| F4                                 | 20A           | ACC BATT HIGH POWER        |
| F5                                 | 20A           | -                          |
| F6                                 | 25A           | HEATED REAR WINDOW         |
| F7                                 | 5A            | INTERCAN ECU               |
| F8                                 | 5A            | MONOSTABLE SHIFTER         |
| F9                                 | 5A            | REAR RADAR BATT (LH/RH)    |
| F10                                | 30A           | BCM - V                    |
| F11                                | 30A           | CBM -IV                    |
| F12                                | 10A           | PID                        |
| F13                                | -             | -                          |
| F14                                | -             | -                          |
| F15                                | 10A           | SUNSHADE                   |
| F16                                | 10A           | CO-DRIVER POWER SEAT       |
| F17                                | -             | -                          |
| F18                                | 10A           | BCM - I                    |
| F19                                | 30A           | INFOTAINMNET               |
| F20                                | 5A            | ESCL                       |

## EMERGENCY AND BREAKDOWN

| Cabin Compartment Fuse Box Details |                  |                                |
|------------------------------------|------------------|--------------------------------|
| Fuse No                            | Ratings<br>(Amp) | Description                    |
| F21                                | -                | -                              |
| F22                                | 15A              | REAR LAMP ECU                  |
| F23                                | 5A               | TELEMATICS                     |
| F24                                | 5A               | SASS NFC                       |
| F25                                | -                | -                              |
| F26                                | 25A              | INTERIOR LAMPS / BETTERY SAVER |
| F27                                | 40A              | DRIVER MEMORY SEAT             |
| F28                                | 5A               | HFA SENSOR                     |
| F29                                | 30A              | POWER TAILGATE CONTROLLER      |
| F30                                | 5A               | RF RECEIVER                    |
| F31                                | 20A              | SUNROOF                        |
| F32                                | 5A               | CCS ILLUMINATION               |
| F33                                | 5A               | BCM / INFO DISPLAY             |
| F34                                | 5A               | INSTRUMENT CLUSTER             |
| F35                                | 10A              | FATC / ETC                     |
| F36                                | 5A               | AR HUD (HEAD UP DISPLAY)       |
| F37                                | 5A               | IONISER                        |
| F38                                | 15A              | OBD                            |
| F39                                | 30A              | BCM - III                      |
| F40                                | 5A               | BCM - II                       |

## EMERGENCY AND BREAKDOWN

| Cabin Compartment Fuse Box Details |                  |                                  |
|------------------------------------|------------------|----------------------------------|
| Fuse No                            | Ratings<br>(Amp) | Description                      |
| F41                                | -                | -                                |
| F42                                | 10A              | AUTONOMOUS PARK ASSIST (APA)     |
| F43                                | -                | -                                |
| F44                                |                  |                                  |
| F45                                | A                | BCM-V                            |
| F46                                | 5A               | DOOR AJAR / PUDDLE LAMP          |
| F47                                | -                | -                                |
| F48                                | -                | -                                |
| F49                                | -                | -                                |
| F50                                | 40A              | FRONT BLOWER MOTOR               |
| F51                                | 15A              | POWER SOCKET - BOOT AREA         |
| F52                                | 10A              | USB CHARGER FRONT                |
| F53                                | 15A              | USB CHARGER REAR (65+65)         |
| F54                                | 5A               | MIRROR ADJUST MOTORS             |
| F55                                | -                | -                                |
| F56                                | -                | -                                |
| F57                                | 5A               | BLOWER COIL / ACC FEEDBACK (BCM) |
| F58                                | -                | -                                |
| F69                                | -                | -                                |
| F70                                | -                | -                                |

## EMERGENCY AND BREAKDOWN

| Cabin Compartment Fuse Box Details |                  |                                  |
|------------------------------------|------------------|----------------------------------|
| Fuse No                            | Ratings<br>(Amp) | Description                      |
| F71                                | -                | -                                |
| F72                                | -                | -                                |
| F73                                | -                | -                                |
|                                    | -                | -                                |
| F75                                | -                | -                                |
| F76                                | -                | -                                |
| F77                                | 5A               | FRONT WIRELESS CHARGER           |
| F78                                | -                | -                                |
| F79                                | 10A              | VENTILATED SEAT 1ST ROW          |
| F80                                | 5A               | IGNITION SUPPLY 2                |
| F81                                | 5A               | ADAS_RR RADAR IGN                |
| F82                                | -                | -                                |
| F83                                | 5A               | ADAS_FR RADAR / CAMERA IGN       |
| F84                                | -                | -                                |
| F85                                | -                | -                                |
| F86                                | -                | -                                |
| F87                                | 10A              | REAR WIPER                       |
| F88                                | 5A               | INTERCAN ECU                     |
| F89                                | 5A               | PDC / TELEMATIC /DUST SENSOR IGN |
| F90                                | 5A               | EPAS IGN                         |



## EMERGENCY AND BREAKDOWN

| Cabin Compartment Fuse Box Details |                  |                   |
|------------------------------------|------------------|-------------------|
| Fuse No                            | Ratings<br>(Amp) | Description       |
| F91                                | -                | -                 |
| F92                                | 10A              | AIRBAG ECU        |
| F93                                | F93              | IGNITION SUPPLY 1 |
| F35                                | 10A              | FATC / ETC        |

## EMERGENCY AND BREAKDOWN

### If An Accident Occurs...

- If your vehicle is drivable, park your vehicle off the road; rotate the gear selector knob to "N" and apply the parking brake.
- If not drivable do not try to start the vehicle. Rotate the gear selector knob to N and apply the parking brake.
- Roll down the windows and open the door locks if possible. If the 'Ready' message does not come in the instrument cluster, do not try to switch ON the supply by pressing the Start/Stop button.
- If there is no electrical supply, at-least try to unlock single door manually.
- If the vehicle 'Ready' message flashes in the instrument cluster, press the Start/Stop button to turn off the supply, and ensure 'Ready' message goes off to verify the high-voltage system is disconnected.
- De-latch the bonnet from inside the cabin by pulling the lever to open it.
- If the lever is not reachable, do not

spend time to de-latch the bonnet.

- Come out of the vehicle and move the smart key at least 2 meters away from the vehicle to avoid any accidental restart or activation of high voltage systems.
- Try to evacuate the occupants from inside of the vehicle.
- Secure vehicle by barricading it, without touching the vehicle.
- Inform the TPEM Roadside-Assistance immediately.
- Do not touch the vehicle. Keep a safe distance.

### WARNING

- Do not touch electric wires that may become exposed from inside or outside the vehicle, high voltage electric wires (orange), connectors and any exposed electric components and devices. Doing so may result in electric shock and lead to injuries or even death.

- If you observe any coolant leaks and rupture in refrigerant lines, do not drive the vehicle and contact TPEM Roadside Assistance.
- If the vehicle switches off after an accident, come out of the vehicle immediately without touching any metal parts.
- Leaks or damage to the Lithium Iron Phosphate battery may result in a fire. If you discover them, contact emergency services immediately. Never touch the fluid leaked inside or outside the vehicle. If the fluid contacts with your skin or eyes, wash it off immediately with a large amount of water or saline solution and receive immediate medical attention to help avoid serious injury.
- If water enters inside the vehicle: If your vehicle is flooded or if water has soaked the carpets, you should not try to start the vehicle. Never touch the high voltage cables, connectors and package modules, be

cause an electric shock may occur causing injury or death. (High voltage components are orange in colour)

- If a submersion in water occurs: Do not touch your vehicle, if the vehicle has been submerged in water. The high voltage battery may cause shock or may catch fire. Immediately contact the authorities and advise them of the condition of your vehicle and that an electric vehicle is involved.

If a small scale fire occurs, use a fire extinguisher (C, ABC, BC) that is meant for electrical fires. If it is impossible to extinguish the fire in the early stage, remain a safe distance from the vehicle and immediately call the authorities. Also, advise them that an electric vehicle is involved.

When approaching a high voltage vehicle in a situation of fire, rescue or recovery, follow the standard rule:

- Always assume the high-voltage system is live in the vehicle.

- Only High Voltage System trained personnel with necessary high voltage PPEs (hand gloves, electrical safety shoes, etc..) should access and analyse the EV after all occupants are safely evacuated.

### Emergency Shut Off System

When vehicle detects any fault in HV system, it activates the emergency shut OFF for safety purpose. Even if the gear knob is in Drive mode, the system may shut-OFF suddenly. In this case, contact the nearest TPEM Authorised Service Centre to rectify the issue.

### In Case Of Emergency Situations

If the vehicle stalls at a crossroad or crossing, rotate the rotary knob to N (Neutral) position and then push the vehicle to a safe place

### If The Vehicle Stalls While Driving

- Reduce your speed gradually, keeping a straight line. Move cautiously off the road to a safe place.
- Turn on the hazard lamps.
- Try to start the vehicle again. If your vehicle will not start, contact TPEM Authorised Service Centre or seek other qualified assistance.
- Since this vehicle runs on electric power, it generates little sound. Beware of your driving environment and

## EMERGENCY AND BREAKDOWN

---

drive safely.

- After you park the vehicle or while you are waiting at a traffic light, check whether there are kids or obstacles around the vehicle.
- Check if there is something behind the vehicle when driving in reverse. Pedestrians may not hear the sound of the vehicle.

**BULB SPECIFICATION**

| Sn. | Description    | Rating  | Type | Qty. |
|-----|----------------|---------|------|------|
| 1   | REAR BOOT LAMP | 12V, 5W | W5W  | 1    |
| 2   | GLOVE BOX LAMP | 12V, 5W | W5W  | 1    |

## EMERGENCY AND BREAKDOWN

---

### 24 X 7 ROADSIDE ASSISTANCE

**Dear Customer,**

It is our responsibility and our endeavor to ensure that you have our complete service backup if ever, wherever and whenever you need the same. When you have a road network that spans wide area, the probability of a breakdown happening within hailing distance of a TPEM Authorised Service Centre is very low.

It is precisely for this reason, we have tied up with TVS AA, who will provide breakdown assistance including towing to the nearest TPEM Authorised Service Centre through their Authorized Service Providers (ASP).

The 24X7 On Roadside Assistance Program shall be automatically available to your vehicle for the duration of Warranty period. The program shall also be available, if you avail the same post warranty.

### Standard Procedure When Calling For Roadside Assistance In Case Of A Breakdown

- Dial the toll free help line number – **1800 209 8282**
- Identify your vehicle with the Vehicle chassis number that is available in the Owner's Manual.
- Explain your exact location with landmarks and tell us about the problem you face with the vehicle.
- Park your vehicle on the edge of the road, open the bonnet and put on the hazard warning signal.
- Place the advance warning triangle supplied with the vehicle approx. 3 m from the vehicle in the direction of on-coming traffic.

### Coverage Under 24 X 7 Roadside Assistance Program

I. The **24x7 Roadside Assistance** Program Service covers the following services on your vehicle during warranty period.

- Wheel change through spare wheel.

- Re-opening the vehicle in cases of key lock out.
- Rectification of electrical problems related to battery, fuses etc.
- On spot repairs for complaints repairable at site. ^
- Vehicle to vehicle towing or winching & towing for non-accident cases up to the nearest TPEM Authorised Dealer/Service Centre. Towing charges at actual cost beyond the same to be paid to the ASP in cash/online. (Any ferry or toll charges levied in relation to the vehicle being towed to be paid by the customers in actuals in cash/online).
- In the event of major accident ambulance assistance will be provided if needed (If equipped).



II. The **24x7 Roadside Assistance** Program coverage on availing the 24X7 policy, post warranty is upto maximum of 6 instance of assistance in one year for both the plans- Basic and Premium. In the premium plan, this includes 2 instances of towing up to the nearest TPEM Authorised Service Centre.

### Exclusions

**24 X 7 Roadside Assistance Program does not apply to**

- Cost of parts consumables and labor

for such repairs not covered under warranty\*. These charges are to be settled with ASP in cash/online.

- Toll or ferry charges paid by ASP in reaching to the breakdown site to be settled with ASP in actuals in cash/online.
- Cases involving accident, fire, theft, vandalism, riots, lightening, earthquake, windstorm, hail, tsunami, unusual weather conditions, other acts of God, flood, etc.
- Vehicles that are unattended, un-registered, impounded or abandoned.
- Breakdown/defects caused by misuse, abuse, negligence, alterations or modifications made to the vehicle.
- Lack of maintenance as per the maintenance schedule as detailed in the owner's manual.
- Cases involving racing, rallies, vehicle testing or practice for such events.

### Disclaimer

- The Service is not available in Lakshadweep. \*\*The reach time is indica-

tive & the actual reach time will be conveyed by the call center at the time of breakdown call.

- The reach time can vary depending on the traffic density & time of the day.
- The reach time indicated does not account for delays due to but not limited to acts of God, laws, rules & regulations for time being in force, orders of statutory or Govt. authorities, industrial disputes, inclement weather, heavy down pour, floods, storms, natural calamities, road blocks due to accidents, general strife and law & order conditions viz. fire, arson, riots, strikes, terrorist attacks, war etc.
- ^ On spot repairs at breakdown site shall depend on nature of complaints & will be as per the discretion of the ASP.
- \* The decision for free of charge repairs will be as per the warranty policy & procedures of TATA PASSENGER ELECTRIC MOBILITY LTD. and as per the interpretation of the same by ASP. You will be duly informed by the ASP &

## EMERGENCY AND BREAKDOWN

---

call center for the change applicable if any.

- All charges wherever applicable need to be settled directly with the ASP.

### Exclusion of Liabilities

- It is understood that TPEM shall be under no liability whatsoever in respect of any loss or damage arising directly or indirectly out of any delay in or non-delivery of, defect/deficiency in service/parts provided by ASP.
- In case vehicle cannot be repaired on-site, customers are advised to use the towing facility for taking their vehicle to the nearest TPEM Authorised Service Centre only. In no condition shall the vehicle be towed to any TPEM Authorised Service Centre. TPEM will not be responsible for any repairs carried out in such unauthorized workshop.
- Customer are advised to take acknowledgment from the ASP for the list of accessories/extra fittings and other belongings in the vehicle as well as the current condition related to dents/scratches breakages of parts/fit-

ments of the vehicle at the time of ASP taking possession of the vehicle & to verify these items when delivery is taken back by them, Claim for loss of or damage to items, if any should be taken up with ASP directly. TPEM shall not be responsible for any such claims, damages/loss or any deficiency of service of the ASP.

- Vehicles will be handled, repaired & towed as per the customer's risk & TPEM shall not be liable for any damages / claims as a result of the same.
- Services entitled to the customers can be refused or cancelled on account of abusive behavior, fraudulent representation, malicious intent and refusal to pay the charges for any charges related services and spare parts during service or on previous occasions on part of the customer.
- On site repairs may be temporary in nature. The completion of repairs does not certify the road worthiness of the vehicle. The customer is advised to ensure temporary repairs carried out on-

site is followed by permanent repairs at a TPEM Authorised Service Centre at the earliest. Terms and conditions and service coverage, exclusions etc. are subject to change without notice.

### E-call and B-call Services (if applicable)

1. Emergency Hard switch button is present on roof console to operate the E-Call / B-Call feature (Emergency Assistance / Breakdown Assistance).
2. E-Call will be automatically triggered in case of vehicle crash/air bag is deployed in the event of accident.
3. On pressing emergency call button, a pop up message appears on HU stating: „Dialing emergency number in 5 seconds” & have a cancel button on the message box, so that user can cancel connecting to emergency number in case it was done unknowingly.
4. In case of breakdown, B-Call can be triggered by pressing B-call switch or by selecting “Tata Assist” option from infotainment.



5. Once call is connected, user can cancel the call using hard press button.
6. B-Call for breakdown assistance can be triggered by pressing B-Call switch provided in the vehicle.

### **i NOTE**

- *\*Subject to mobile network, connectivity and location mentioned.*
- *The connected services offered through the iRA.ev application are dependent on the e-sim (provided by third-party telecom vendor) embedded inside the telematics unit installed inside your vehicle.*
- *If the e-sim is not activated or renewed with an active subscription plan within a time period of 12 months from date of sale or the expiry of previous active plan, the e-sim is at a risk of being deactivated.*
- *Once de-activated, connected services cannot be availed on the same e-sim and in the event, the customer is desirous of availing connected*

*services in the future, the entire telematics unit would need to be replaced.*

- *The expense of telematics replacement is to be borne by the customer alone, with no recourse to the Company. Accordingly to avoid this outcome, we strongly urge you to keep the iRA.ev subscription active at all times by paying the subscription fee.*
- *For any assistance or queries, please reach out to [irasupport@tatamotors.com](mailto:irasupport@tatamotors.com)*

### **Response Time \*\* For The B-call (roadside Assistance Program)**

|                                             |                 |
|---------------------------------------------|-----------------|
| Reach time within city (Listed 54 cities)   | 45 Minutes      |
| Reach time On State and National High- ways | 90 Minutes      |
| Reach time on remote location               | 120 Minutes +/- |

## EMERGENCY AND BREAKDOWN

---

| Response Time ** For The B-call<br>(roadside Assistance Program)                                          |                              |
|-----------------------------------------------------------------------------------------------------------|------------------------------|
| Within City Limits                                                                                        | 60Minutes                    |
| Highways and<br>other places                                                                              | 120 Minutes +/-              |
| States of North<br>East, J&K and Hi-<br>machal                                                            | Same Day (Within<br>24 Hrs.) |
| ** The response time will depend on the<br>location, terrain, traffic density and the<br>time of the day. |                              |

**E-call And B-call Led (near E-call And B-call Switch) Indications**

| SN | Parameters                          | System Health Led- Red                 | Call Status Led- Green                   |
|----|-------------------------------------|----------------------------------------|------------------------------------------|
| 1  | During IGN ON                       | LED remains ON for 3 Sec               | LED remains ON for 3 Sec                 |
| 2  | E-Call/ B-Call Subscription Active  | OFF                                    | OFF                                      |
| 3  | E-Call/ B-Call Subscription Expired | Slow Blinking (0.5 Sec ON & 4 Sec OFF) | OFF                                      |
| 4  | During Call Initiation              | OFF                                    | Fast Blinking (0.5 Sec ON & 0.5 Sec OFF) |
| 5  | Call Connected                      | OFF                                    | Permanent ON                             |
| 6  | System Failure                      | ON                                     | OFF                                      |

If RED light is continuously ON near E-Call (SOS) switch then E-Call/B-Call will not work. User has to contact nearest TPEM Authorised Service Centre.

Call back is available during E-call only.

In case of call transfer from B-Call to E-Call, Call back case will not work. If RED light is continuously ON near E-Call (SOS) switch then E-Call/B-Call will not work. User has to contact nearest TPEM Authorised Service Centre. Call back is available during E-call only. In case of call transfer from B-Call to E-Call, Call back case will not work.

**List of below cities coverage for E-Call support**

Ambulance services will be provided only in the cities mentioned under the below mentioned categories. Allianz shall provide an ambulance with basic life support or advanced life support based on the availability at the time of emergency call received from the breakdown location. The criteria to choose the kind of ambulance shall be as per the requirement of the customer, distance from the breakdown location & the nearest hospital & subject to availability. The judgement of Allianz in all cases will be final.

## EMERGENCY AND BREAKDOWN

---

### Category A:

Ambulance services will be given under municipal limits of these cities varying from 15-25 Km radius.

|           |              |            |             |             |
|-----------|--------------|------------|-------------|-------------|
| Hyderabad | Vadodara     | Coimbatore | Madurai     | Noida       |
| Bangalore | Kochi        | Vijayawada | Kolhapur    | Jaipur      |
| Mumbai    | Trivandrum   | Vizag      | Mysore      | Jodhpur     |
| Kolkata   | Indore       | Raipur     | Aurangabad  | Chandigarh  |
| Chennai   | Nagpur       | Surat      | Kozikhode   | Patna       |
| Delhi     | Pune         | Rajkot     | Warangal    | Amritsar    |
| Gurgaon   | Bhubaneshwar | Bhopal     | Cuttack     | Navi Mumbai |
| Faridabad | Lucknow      | Ranchi     | Goa         | Thane       |
| Ghaziabad | Kanpur       | Nashik     | Pondicherry |             |
| Ahmedabad | Varanasi     | Mangalore  | Guntur      |             |

### Category B:

Ambulance services will be given under municipal limits of these within 15 Km radius.

|            |                  |            |                |
|------------|------------------|------------|----------------|
| Adilabad   | Dahod            | Khandwa    | Rajnandgaon    |
| Adityapur  | Dakshin Dinajpur | Kharagpur  | Ramanathapuram |
| Agartala   | Daman            | Kishanjang | Rampur         |
| Agra       | Dankuni          | Koch Bihar | Raniganj       |
| Ahmednagar | Darbhanga        | Kollam     | Ratlam         |
| Aizawl     | Darjeeling       | Korba      | Rewa           |
| Ajmer      | Davanagere       | Kota       | Rewari         |

## EMERGENCY AND BREAKDOWN

|              |             |                 |              |
|--------------|-------------|-----------------|--------------|
| Akola        | Dehradun    | Kottayam        | Rishikesh    |
| Alappuzha    | Deoghar     | Kovilpatti      | Rohtak       |
| Aligarh      | Deoria      | Kozhikode       | Roorkee      |
| Allahabad    | Dhanbad     | Krishnagiri     | Rourkela     |
| Ambala       | Dhar        | Kurnool         | Rudrapur     |
| Ambikapur    | Dharamshala | Kurukshetra     | Sagar        |
| Ambur        | Dharbhanga  | Kutch           | Saharsa      |
| Amethi       | Dharmapuri  | Lakhimpur Kheri | Salem        |
| Amravati     | Dharmavaram | Lalitpur        | Samastipur   |
| Amreli       | Dharwad     | Latur           | Sambalpur    |
| Anand        | Dhaulpur    | Lonavala        | Sangli       |
| Anantapur    | Dhule       | Ludhiana        | Sangrur      |
| Ankleshwar   | Dibrugarh   | Maharajanj      | Satara       |
| Arakkonam    | Dindigul    | Mahbubnagar     | Satna        |
| Ariyalur     | Dispur      | Mahesana        | Secunderabad |
| Arrah        | Durg        | Mainpuri        | Sehore       |
| Asansol      | Durgapur    | Malda           | Shahdol      |
| Auraiya      | Eluru       | Maldah          | Shahjahanpur |
| Aurangabad   | Ernakulam   | Manaparai       | Shajapur     |
| Azamgarh     | Erode       | Mandsaur        | Shikohabad   |
| Badshahnagar | Etah        | Manmad          | Shillong     |
| Bagbera      | Etawah      | Mansa           | Shimla       |

## EMERGENCY AND BREAKDOWN

|             |             |               |                |
|-------------|-------------|---------------|----------------|
| Baghpat     | Faizabad    | Mapusa        | Shimoga        |
| Bahadurgarh | Farrukhabad | Margao        | Sholinghur     |
| Bahraich    | Fatehpur    | Mathura       | Sholingur      |
| Balaghat    | Ferozpur    | Mau           | Siddharthnagar |
| Balasore    | Firozabad   | Mavelikara    | Sikar          |
| Baleshwar   | Firozpur    | Medak         | Siliguri       |
| Ballia      | Gadwal      | Medinipur     | Sirsa          |
| Baloda      | Gandhidham  | Meerut        | Sitapur        |
| Balrampur   | Gandhinagar | Mehsana       | Sivaganga      |
| Bambolim    | Gaya        | Miryalaguda   | Siwan          |
| Banda       | Ghazipur    | Mirzapur      | Solan          |
| Bankura     | Giridih     | Moga          | Solapur        |
| Barasat     | Godhra      | Moradabad     | Sonbhadra      |
| Bareilly    | Gonda       | Morbi         | Sonipat        |
| Barwani     | Gondia      | Morena        | Sri City       |
| Basti       | Gorakhpur   | Motihari      | Sri Ganganagar |
| Bathinda    | Gudivada    | Muktsar       | Srirangam      |
| Begusarai   | Gulbarga    | Murshidabad   | Sullurupeta    |
| Belgaum     | Guna        | Muzaffarnagar | Supaul         |
| Bellary     | Gurdaspur   | Muzaffarpur   | Suryapet       |
| Bengaluru   | Guwahati    | Nadia         | Tadepalligudem |
| Betul       | Gwalior     | Nagapattinam  | Tamaram        |

## EMERGENCY AND BREAKDOWN

|              |                        |                            |                    |
|--------------|------------------------|----------------------------|--------------------|
| Bhadohi      | Hajipur                | Nagercoil                  | Taregana           |
| Bhadrak      | Haora                  | Nainital                   | Tatanagar          |
| Bhagalpur    | Hapur                  | Nalgonda                   | Tenkasi            |
| Bharatpur    | Hardoi                 | Namakkal                   | Thalassery         |
| Bharuch      | Haridwar               | Nanded                     | Thanjavur          |
| Bhatinda     | Hassan                 | Nandyal                    | The Nilgiris       |
| Bhavnagar    | Hathras                | Navsari                    | Theni              |
| Bhilai       | Hazaribagh             | Nawanshahr                 | Thiruchirapalli    |
| Bhilwara     | Hingoli                | Nellore                    | Thiruvalla         |
| Bhind        | Hisar                  | New Farakka                | Thiruvallur        |
| Bhiwani      | Hooghly South          | Nizamabad                  | Thiruvananthapuram |
| Bhuj         | Hoshiarpur             | North Twenty four Parganas | Thiruvarur         |
| Bhusawal     | Hosur                  | Ooty                       | Thrissur           |
| Bidhan Nagar | Hubli                  | Osmanabad                  | Thrivandram        |
| Biharsharif  | Imphal                 | Palamapur                  | Tindivanam         |
| Bijapur      | Jabalpur               | Palampur city              | Tiruchirappalli    |
| Bijnor       | Jaisalmer              | Palanpur                   | Tirunelveli        |
| Bikaner      | Jalandhar              | Pali                       | Tirupati           |
| Bilaspur     | Jalgaon (Chalisingaon) | Panaji                     | Tirupattur         |
| Birbhum      | Jalgaon                | Panipat                    | Tiruppur           |
| Bokaro       | Jalpaiguri             | Panjim                     | Tirur              |
| Bolpur       | Jammu                  | Panna                      | Tiruvanamalai      |

## EMERGENCY AND BREAKDOWN

|                  |                     |                |                  |
|------------------|---------------------|----------------|------------------|
| Brahmapur        | Jamnagar            | Pantnagar      | Toothukudi       |
| Bulandshahr      | Jamshedpur          | Parappanangadi | Tumkuru          |
| Buxar            | Jaunpur             | Paravur        | Tundla           |
| Calicut          | Jehanabad           | Parbhani       | Udaipur          |
| Chaibasa         | Jhansi              | Pathankot      | Udupi            |
| Chakradharpur    | Jhunjhunu           | Patiala        | Ujjain           |
| Chakulia         | Jind                | Perambalur     | Unnao            |
| Champa           | Jorhat              | Phagwara       | Vaikom           |
| Chandauli        | Junagadh            | Phaphund       | Valsad           |
| Chandil          | Jyotiba Phule Nagar | Pithapuram     | Vaniyambadi      |
| Channapatna      | Kaithal             | Pollachi       | Vapi             |
| Chapra           | Kancheepuram        | Ponda          | Varkala Sivagiri |
| Chengalpattu     | Kanniyakumari       | Port Blair     | Vellore          |
| Chhatarpur       | Kannur              | Prayagraj      | Viluppuram       |
| Chhindwara       | Kanpur Nagar        | Puducherry     | Virudhampet      |
| Chhota Gobindpur | Kanyakumari         | Pudukkottai    | Virudhunagar     |
| Chitradurga      | Karimnagar          | Puri           | Visakhapatnam    |
| Chitrakoot       | Karnal              | Purna          | Wardha           |
| Chittoor         | Katni               | Purnia         | Yavatmal         |
| Churu            | Katpadi             | Rae Bareli     | Yelahanka        |
| Cooch Behar      | Kaushambi           | Raigar         | Yemmiganur       |
| Cuddalore        | Kayamkulam          | Raigarh        | Yeswanthpur      |



## EMERGENCY AND BREAKDOWN

|          |         |             |  |
|----------|---------|-------------|--|
| Cuddapah | Khammam | Rajahmundry |  |
|----------|---------|-------------|--|

### Category C:

For Outside city Limit cases & rest of the cities – Services will be aligned on best effort basis utilizing the service network from the nearest city or publicly available network.

AWP's liability for the provision of Assistance Services will cease once the emergency services have arrived at the breakdown location or on completion of towing or upon arrival of the Covered Vehicle at Authorised TATA Dealer or TPEM Authorised Service Centre.



### LUBRICANT SPECIFICATIONS

Use following genuine fluids, coolants and lubricants recommended for optimum performance of your vehicle.

| Item                                                                 | Specification                        | Company                                 | Brand                                  | Qty.                                                                               |
|----------------------------------------------------------------------|--------------------------------------|-----------------------------------------|----------------------------------------|------------------------------------------------------------------------------------|
| Coolant (Pre mixed)<br>Anti freeze agent + soft<br>water 40:60 ratio | Class II/ JIS K2234<br>TATA SS7700S1 | SUNSTAR CCI                             | Golden Cruiser LLC<br>2200 NP          | BCS 9.8 L (75kWh)<br>4.5 L (63kWh)<br>TCS for RWD - 6 L<br>TCS for AWD - 6.5 L     |
|                                                                      |                                      | IOCL                                    | TPEM GENUINE<br>COOLANT KOOL PLUS      |                                                                                    |
| Gear box Oil                                                         | BOT 350 M3                           | Castrol                                 | Castrol                                | Front 1.1 +/- 0.1 L<br>(1100 +/- 100 ml)<br>Rear 0.6 +/- 0.02 L<br>(600 +/- 20 ml) |
| Brake Fluid                                                          | SAE J 1704, DOT 4                    | PETRONAS Lubricants<br>(India) Pvt Ltd. | Brake fluid (DOT4)                     | 0.850 L (850 ml)                                                                   |
| Refrigerant                                                          | R-134a                               | —                                       | —                                      | 670 +/- 20 gm                                                                      |
| AC Compressor Oil                                                    | POE RB100EV                          | ENEOS                                   | RB 100EV / Equivalent<br>grade POE Oil | 150 +/- 15 gm                                                                      |
| Sunroof Grease                                                       | MULTEMP 2C194                        | —                                       | —                                      | As required                                                                        |

Note - AC compressor oil RL 68H, Emkarate can be used for top up in case of unavailability of RB 100EV.

## TECHNICAL SPECIFICATION

### VEHICLE SPECIFICATIONS

| Parameter          | 63 kWh                                                        | 75 kWh                                                    |
|--------------------|---------------------------------------------------------------|-----------------------------------------------------------|
| Powertrain         |                                                               |                                                           |
| Electric motor     | Front E-drive-Induction; Rear E-drive-PMSM                    |                                                           |
| Nominal voltage    | 403.2 V                                                       |                                                           |
| Maximum power, kW  | RWD :Rear - 175 kW                                            | RWD :Rear - 149 kW<br>AWD : Front - 103 kW, Rear - 149 kW |
| Maximum torque, Nm | RWD - 315 Nm (Rear only)                                      | RWD - 315 Nm (Rear only)<br>AWD - 504Nm (Boost mode only) |
| Gear Box           |                                                               |                                                           |
| Model and Type     | Electric Vehicle Transaxle                                    |                                                           |
| No. Of gears       | Single speed, 1 Forward Drive + 1 Reverse Drive               |                                                           |
| Steering           |                                                               |                                                           |
| Type               | Electric Power Assisted Steering System                       |                                                           |
| Brakes             |                                                               |                                                           |
| Brakes             | Front (Disc); Rear (Disc)                                     |                                                           |
| Parking brake      | Electronic Parking Brake                                      |                                                           |
| Shock Absorber     |                                                               |                                                           |
| Front and Rear     | MacPherson strut Damper twin tube with gas filled             |                                                           |
| Suspension         |                                                               |                                                           |
| Front              | Independent, lower wishbone, Mcpherson strut with coil spring |                                                           |

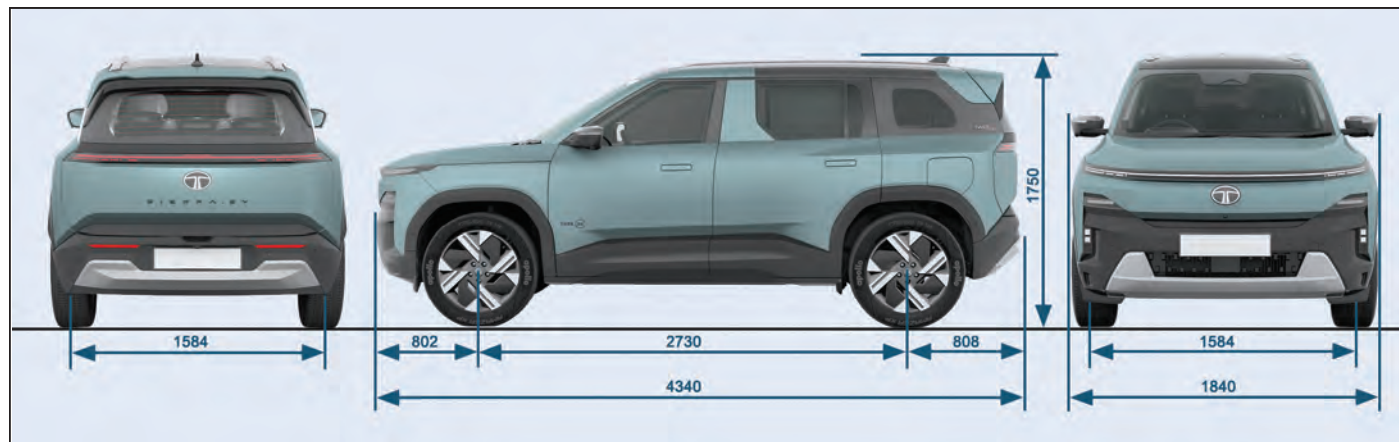
## TECHNICAL SPECIFICATION

| Parameter                      | 63 kWh                                                                                             | 75 kWh                         |
|--------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------|
| Rear                           | Independent Multi link suspension with coil spring and shock absorber                              |                                |
| Wheels & Tyre                  |                                                                                                    |                                |
| Tyres                          | Option I - 225/60 R18 100 H (Radial / Tubeless)<br>Option II - 225/55 R19 99 H (Radial / Tubeless) |                                |
| Wheel rims                     | Option I: 7J x 18 Alloy wheel Rim<br>Option II: 7J x 19 Alloy wheel Rim                            |                                |
| Cab / Body                     |                                                                                                    |                                |
| Type                           | Steel Monocoque Body                                                                               |                                |
| Electrical System              |                                                                                                    |                                |
| System voltage                 | 12 Volts                                                                                           |                                |
| Auxiliary battery              | 12V DIN 47.5 Ah                                                                                    |                                |
| Main Chassis Dimension (in mm) |                                                                                                    |                                |
| Wheel base, mm                 | 2730                                                                                               |                                |
| Track front, mm                | 1584                                                                                               |                                |
| Track rear, mm                 | 1584                                                                                               |                                |
| Overall length, mm             | 4340                                                                                               |                                |
| Overall height, mm (unladen)   | 1690                                                                                               |                                |
| Max. Width, mm                 | 1841                                                                                               |                                |
| Ground clearance, mm (laden)   | RWD : 172 , AWD : 165                                                                              |                                |
| Performance                    |                                                                                                    |                                |
| Max. Speed                     | AWD: 180 Kmph<br>RWD: 160 Kmph                                                                     | AWD: 180 Kmph<br>RWD: 160 Kmph |

## TECHNICAL SPECIFICATION

| Parameter                                                       | 63 kWh              | 75 kWh                                     |
|-----------------------------------------------------------------|---------------------|--------------------------------------------|
| Max. Recommended Gradeability                                   | RWD: 32%            | AWD: 48%<br>RWD: 32%                       |
| Minimum Turning Circle Dia. in meter as per IS:12222            | 10.64 m             |                                            |
| Minimum Turning Clearance circle dia. in meters as per IS:12222 | 11.24 m             |                                            |
| Weight                                                          |                     |                                            |
| Gross vehicle weight (Laden), kg                                | RWD: 2312 - 2361 kg | RWD: 2443 - 2469 kg<br>AWD: 2540 - 2555 kg |
| Kerb weight (unladen), kg                                       | RWD: 1900 - 1967 kg | RWD: 2018 - 2044 kg<br>AWD: 2100 - 2130 kg |

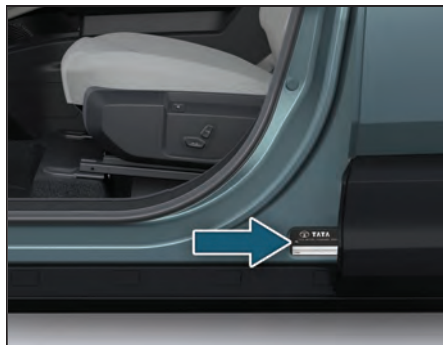
## VEHICLE DIMENSIONS



**NOTE:** Dimensions are in mm unladen condition

## TECHNICAL SPECIFICATION

### AGGREGATE IDENTIFICATION NUMBERS



VIN plate near co-driver seat



Chassis No. punching below driver seat



### SERVICE INSTRUCTIONS

The **TATA SIERRA EV** has been manufactured to give you economical and trouble free performance. To achieve this, please follow the instructions as stated.

Your vehicle is entitled to three free services (labour only). The free service coupons are attached to the sales invoice. Please present these coupons to the servicing dealer while availing free services.

**1st free service** - At 1500 kms. OR 2 months, whichever is earlier.

**2nd free service** - At 5000 kms. OR 6 months, whichever is earlier.

**3rd free service** - At 10,000 kms. OR 12 months, whichever is earlier.

All services other than free services are chargeable.

Servicing of the vehicle can be done at any TPEM Authorised Dealer Workshop or TPEM Authorised Service Centre.

Warranty claims can be settled by any TPEM Authorised Dealer Workshop or TPEM Authorised Service Centre.

## MAINTENANCE AND CARE

### SERVICE SCHEDULE

| S.N | Periodic Maintenance Schedule                                                                                    |        | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |
|-----|------------------------------------------------------------------------------------------------------------------|--------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|
|     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer)<br>Or Months Which Ever Comes Earlier. | Km     | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |
|     |                                                                                                                  | Months | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |

**Note:** i → inspect (check, clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T → Tighten To Specified Torque ; R → replace

| GENERAL |                                                                                                                             |                             |
|---------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1       | Wash the vehicle & Clean Condenser Fins                                                                                     | To be done at every service |
| 2       | Check & Top up Fluids (If required): Transaxle oil, Coolant, Brake Fluid, 12V Battery Electrolyte, Wind Screen washer fluid | To be done at every service |
| 3       | Check HV Battery box under the vehicle for cracks/leakages                                                                  | To be done at every service |
| 4       | Check and clean (if required) HV Battery breather plug for mud/dust/external particle accumulation.                         | To be done at every service |
| 5       | Check all the HV cables for looseness, cuts, wear & tear.                                                                   | To be done at every service |

## MAINTENANCE AND CARE

| S.N                                                                                                                                 | Periodic Maintenance Schedule                                                                                                                               |                                                            | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|--|
|                                                                                                                                     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer) Or Months Which Ever Comes Earlier.                                               | Km                                                         | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |  |
|                                                                                                                                     |                                                                                                                                                             | Months                                                     | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |  |
| Note: i→ inspect (check,clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T→ Tighten To Specified Torque ; R→ replace |                                                                                                                                                             |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 6                                                                                                                                   | Clean Charging socket area.<br>Also clean dust around hinge area of Charging Port Housing lid                                                               | To be done at every service                                |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 7                                                                                                                                   | Inspect Battery cooling system (BCS) and Traction cooling system (TCS)                                                                                      | To be done at every service                                |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 8                                                                                                                                   | Coolant - Battery cooling system (BCS) and Traction cooling system (TCS)                                                                                    | Replace Every 60,000 km/ 36 Months whichever comes earlier |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 9                                                                                                                                   | Using TDS diagnostic tool,Capture generate report & keep it in record,check and capture all DTC's. Check VCU, BMS & CDC software are latest, update if olde | To be done at every service                                |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 10                                                                                                                                  | Check all door latches & striker operations,apply grease if required.                                                                                       |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |

## MAINTENANCE AND CARE

| S.N                                                                                                                                 | Periodic Maintenance Schedule                                                                                                                                                                                                                                                                                                                                                                                                           |        | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|--|
|                                                                                                                                     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer)<br>Or Months Which Ever Comes Earlier.                                                                                                                                                                                                                                                                                                                        | Km     | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |  |
|                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                         | Months | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |  |
| Note: i→ inspect (check,clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T→ Tighten To Specified Torque ; R→ replace |                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 11                                                                                                                                  | Check condition of rubber bushes/parts in lower control arms, front and rear coil spring seats, front & rear bump stoppers, anti-roll bar links, rear twist beam, rubber boots/dust cover/bellow in rack & pin ion, steering and suspension ball joints, steering column. Replace if necessary.(First at 5K / 6M then at every service). For severe usage, above checks to be done at every 5,000 km or after every severe usage event. |        | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
| 12                                                                                                                                  | Check for tightening torque of all bolts & nuts. For severe usage, above checks to                                                                                                                                                                                                                                                                                                                                                      | T      | T            | T     | T      | T            | T      | T      | T      | T      | T      | T      | T      | T       | T       | T       | T       | T       | T       |  |

## MAINTENANCE AND CARE

| S.N                                                                                                                                 | Periodic Maintenance Schedule                                                                                    |                                                            | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|--|
|                                                                                                                                     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer)<br>Or Months Which Ever Comes Earlier. | Km                                                         | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |  |
|                                                                                                                                     |                                                                                                                  | Months                                                     | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |  |
| Note: i→ inspect (check,clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T→ Tighten To Specified Torque ; R→ replace |                                                                                                                  |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|                                                                                                                                     | be done at every 5000Km or after every severe event.                                                             |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|                                                                                                                                     | BRAKES                                                                                                           |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 13                                                                                                                                  | Check front & rear brake pads. Replace if necessary                                                              |                                                            |              | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
| 14                                                                                                                                  | Replace brake fluid and check brake system components for leakages                                               | Replace Every 40,000 km/ 24 Months whichever comes earlier |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 15                                                                                                                                  | EPB operation                                                                                                    |                                                            | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
|                                                                                                                                     | WHEELS & TYRES                                                                                                   |                                                            |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 16                                                                                                                                  | Wheel alignment & balancing,Tyre rotation.                                                                       |                                                            |              | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
| 17                                                                                                                                  | Check Tyre pressure, condition of all tyres                                                                      |                                                            |              | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |

## MAINTENANCE AND CARE

| S.N                                                                                                                                 | Periodic Maintenance Schedule                                                                                     |                             | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|--|
|                                                                                                                                     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer)<br>Or Months Which Ever Comes Earlier.  | Km                          | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |  |
|                                                                                                                                     |                                                                                                                   | Months                      | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |  |
| Note: i→ inspect (check,clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T→ Tighten To Specified Torque ; R→ replace |                                                                                                                   |                             |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|                                                                                                                                     | TRANSAXLE                                                                                                         |                             |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 18                                                                                                                                  | Front E-Drive Oil change interval (First at 7.5K/6M). Rear E-Drive to be only visually inspected for oil leakage. |                             | R            |       |        | R            |        |        | R      |        |        | R      |        |         | R       |         |         |         | R       |  |
| 19                                                                                                                                  | Front & Rear E-Drive visual inspection for oil leakage/Seepage                                                    | I                           | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
|                                                                                                                                     | ELECTRICAL                                                                                                        |                             |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 20                                                                                                                                  | Check specific gravity of battery electrolyte for 12V Aux. battery                                                |                             | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
| 21                                                                                                                                  | Headlamp focussing                                                                                                |                             | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
|                                                                                                                                     | A.C. SYSTEM                                                                                                       |                             |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 22                                                                                                                                  | Inspect Air-conditioning / HVAC System for satisfactory performance                                               | To be done at every service |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |

## MAINTENANCE AND CARE

| S.N                                                                                                                                 | Periodic Maintenance Schedule                                                                                    |        | Free Service |       |        | Paid Service |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|--------------|-------|--------|--------------|--------|--------|--------|--------|--------|--------|--------|---------|---------|---------|---------|---------|---------|--|
|                                                                                                                                     | Operations:<br>1)service Intervals To Be Followed Based On Km (kilometer)<br>Or Months Which Ever Comes Earlier. | Km     | 1,500        | 5,000 | 10,000 | 20,000       | 30,000 | 40,000 | 50,000 | 60,000 | 70,000 | 80,000 | 90,000 | 100,000 | 110,000 | 120,000 | 130,000 | 140,000 | 150,000 |  |
|                                                                                                                                     |                                                                                                                  | Months | 2            | 6     | 12     | 24           | 36     | 48     | 60     | 72     | 84     | 96     | 108    | 120     | 132     | 144     | 156     | 168     | 180     |  |
| Note: i→ inspect (check,clean, Correct, Top Up, Adjust, Repair, Replace As Necessary) ; T→ Tighten To Specified Torque ; R→ replace |                                                                                                                  |        |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 23                                                                                                                                  | Replace Pollen filter (if applicable)                                                                            |        | I            | R     | R      | R            | R      | R      | R      | R      | R      | R      | R      | R       | R       | R       | R       | R       | R       |  |
| 24                                                                                                                                  | Replace Combi PM2.5 filter (if applicable)                                                                       |        | I            | R     | R      | R            | R      | R      | R      | R      | R      | R      | R      | R       | R       | R       | R       | R       | R       |  |
|                                                                                                                                     | SUNROOF (if equipped)                                                                                            |        |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 25                                                                                                                                  | Clean guide rails and drain holes                                                                                |        |              | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |
|                                                                                                                                     | EXTERIOR TRIM                                                                                                    |        |              |       |        |              |        |        |        |        |        |        |        |         |         |         |         |         |         |  |
| 26                                                                                                                                  | Grease application on the hinge arm at the tappet area, charging port                                            |        | I            | I     | I      | I            | I      | I      | I      | I      | I      | I      | I      | I       | I       | I       | I       | I       | I       |  |

### Note:

Beyond 1,50,000 carry out the same services at the same intervals of 10,000 kms respectively.

### Vehicle operating in Severe Driving Conditions are as below -

Under severe driving conditions additional maintenance is required:

A: Repeatedly driving short distance of less than 7 km.

## MAINTENANCE AND CARE

B: Extensive low speed driving for long distances.

C: Driving on rough, sandy, dusty, muddy, unpaved, graveled or salt-spread roads.

D: Frequently driving in heavy traffic area & in stop & go condition

E: Frequently operating in mountainous area, uphill, downhill

F: Driving in conditions such as patrolling, Taxi, Pickup Van, Vehicle Towing, trailer towing

| Item                                 | Interval                | Condition |   |    |    |    |    |
|--------------------------------------|-------------------------|-----------|---|----|----|----|----|
|                                      |                         | A         | B | C  | D  | E  | F  |
| Check and adjust wheel alignment     | Inspect every 10,000 km | I         | I | I  | I  | I  | I  |
| A/C Filter                           | Inspect every 7500 Km   | I         | I | NA | NA | NA | NA |
| Gear Oil                             | Inspect every 7500 km   | I         | I | I  | I  | I  | I  |
| Sunroof- Guide rails and drain holes | Inspect every 7500 km   | I         | I | I  | I  | I  | I  |

### NOTE

- In Free service, labor charges are free, any parts and consumables will be chargeable.
- Filters, Fluid, Consumables, Grease, Components, wheel alignment, Wheel Balancing and weights etc will be replaced on chargeable basis.
- Check the coolant level and top up as required, replace if it is contaminated /discolored or there is sludge formation.
- Periodic maintenance Schedule is for reference. TPEM reserves the rights to change the Periodic Maintenance Schedule.
- As the High Voltage (HV) battery is covered under a lifetime warranty\*\* for the first owner, it is mandatory that the vehicle be serviced exclusively at TPEM Authorised Service Centres in accordance with the periodic maintenance schedule outlined in the table above, up to 150,000 Kms or 180 months, whichever occurs first. Subsequent to this period, the vehicle must continue to be serviced at intervals of every 10,000 Kms or 12 months, whichever occurs earlier.

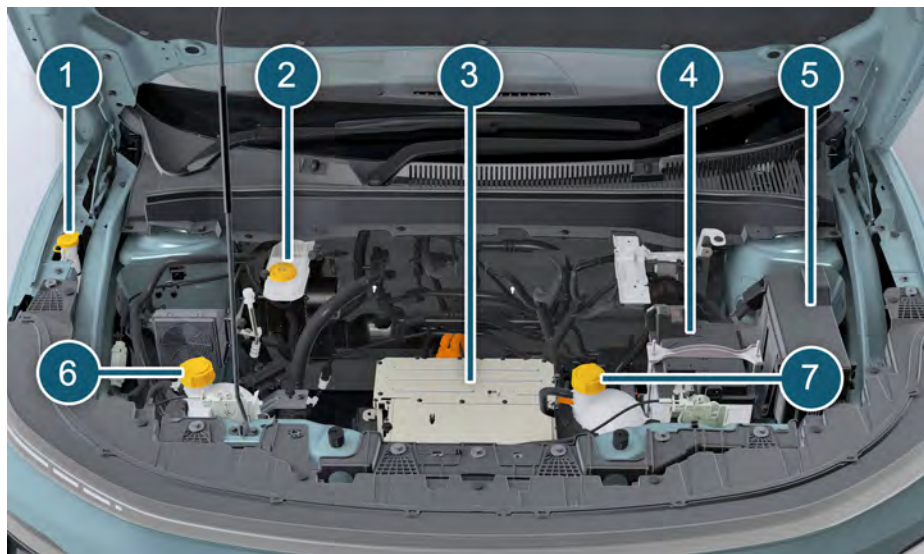


*\*\*Refer "High Voltage Battery Warranty: Terms and Conditions" section in owner's manual for details.*

## MAINTENANCE AND CARE

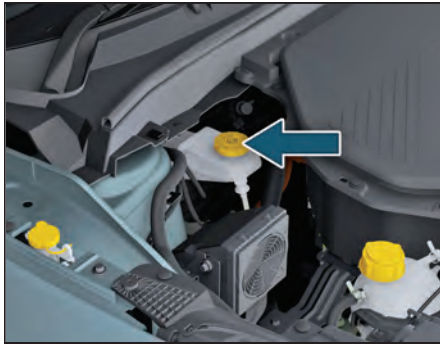
---

### MOTOR COMPARTMENT



1. Windshield Washer Container
2. Brake Fluid Reservoir
3. Motor (Below Frunk)
4. Low Voltage Battery
5. Motor Compartment Fuse Box
6. Coolant Tank (TCS)
7. Coolant Tank (BCS)

## BRAKE FLUID LEVEL



The level of the brake fluid should be between the 'MIN' and 'MAX' marks provided on the side of the brake fluid container. If the level falls below the 'MIN' mark, add recommended brake fluid.

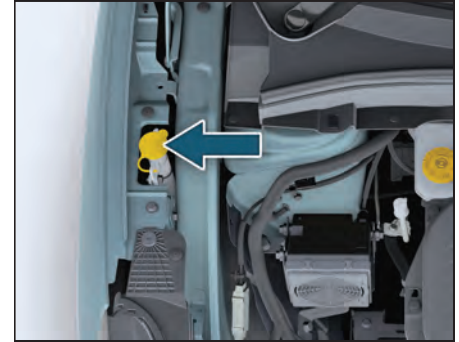
### **i** NOTE

- Do not allow brake fluid to make contact with the skin or eyes.
- Do not allow brake fluid to splash or spill on the paint surface as it will damage the paint. In case of

*spillage, wipe it off immediately.*

*For more clarity about location of Brake Fluid Container and filling cap, please refer respective motor Compartment*

## WINDSHIELD WASHER FLUID LEVEL



Examine if there is washer fluid in the tank. Fill it if necessary. Use a good quality fluid, diluted with water as necessary.

### **i** NOTE

- Do not use detergent or any other additive in the windshield washer reservoir. This can severely impair visibility when sprayed on the windshield, and can also damage your vehicle's paint.

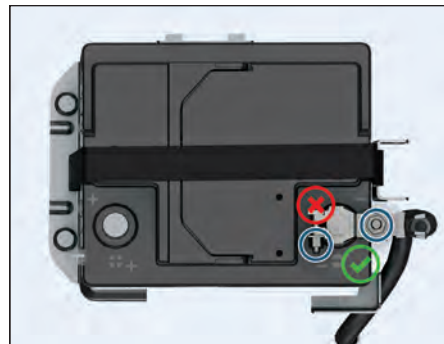
## MAINTENANCE AND CARE

- Do not operate washer motor with no fluid in washer tank, washer motor will be damaged.

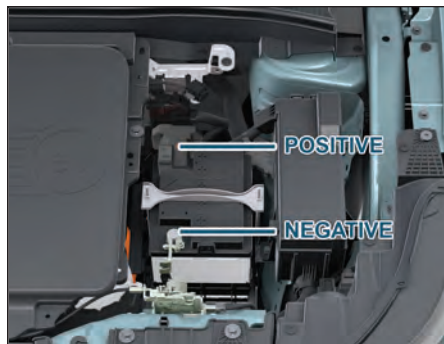
***For more clarity about location of Wind-shield Washer Container and filling cap, please refer image of the respective motor Compartment.***

### 12V BATTERY

- Examine the battery for electrolyte level against the marking on the battery outer case.
- Use a proper spanner to loosen and remove cables from the terminals.
- Always disconnect the negative (-ve) terminal first and reconnect it last.
- If your vehicle is equipped with Battery Sensor, then disconnect only the Sensor output cable.
- Ensure battery is securely mounted.
- If you need to connect the battery to charger, remove both the terminal to prevent damage to the vehicle electrical system.
- Apply terminal coats/spray/petroleum jelly to prevent battery terminal from oxidation.
- Refer the below Battery Sensor image for do's and don'ts.



***For location of battery, please refer image of the respective motor Compartment.***



## *i* NOTE

Use only authorized Battery recommended by TPEM. Use of any other unauthorized Battery will result into Intelligent Alternator Control (IAC) function deterioration.

## *i* NOTE

- During normal operation, the battery generates gas which is explosive in nature. A spark or open flame can cause the battery to explode causing very serious injuries.
- Keep all sparks, open flames and smoking materials away from the battery.
- The battery contains sulphuric acid (electrolyte) which is poisonous and highly corrosive in nature. Getting electrolyte in your eyes or on the skin can cause severe burns. Wear protective clothing and a face shield or have a skilled technician to do the battery maintenance.

## TYRES



|   |                       |                             |
|---|-----------------------|-----------------------------|
| 1 | Under inflation       | Excessive side tread wear   |
| 2 | Correct tyre pressure | Uniform wear                |
| 3 | Over inflation        | Excessive center tread wear |

## MAINTENANCE AND CARE

### Inflation

Check for inflation and condition of your vehicle tyres periodically.

Check the pressure in the tyres when they are cold.

Keeping the tyres properly inflated gives you the best combination of riding comfort, handling, tyre life and better energy efficiency.

Over inflation of tyres makes the vehicle ride bumpy and harsh. Tyres are more prone to uneven wear and damage from road hazards.

Under inflated tyres reduce your comfort in vehicle handling and are prone to failures due to high temperature. They also cause uneven wear and more energy consumption.

#### *i* NOTE

*Every time you check inflation pressure, you should also examine tyres for uneven wear, damage and trapping of foreign objects in the treads and wear.*

### Recommended Tyre Pressures

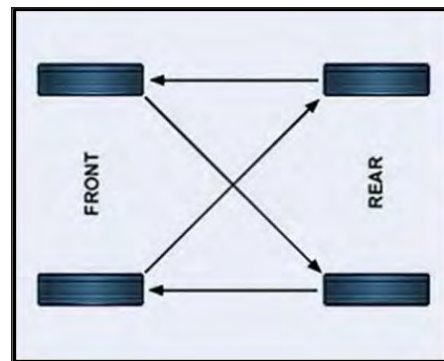


#### *i* NOTE

*Kindly refer Tyre pressure as indicated on tyre pressure sticker provided on vehicle (near driver seat )*

### Tyre Rotation

To help increase tyre life rotated at specified intervals or earlier depending on the operation of vehicle. The illustrations shows how to rotate tyres without spare wheel.



#### *i* NOTE

- *Do not use spare wheel for tyre rotation, in case of temporary spare wheel used.*

- *Two or more temporary tyres should not be used on one vehicle.*
- *Tyre pressure to be checked every 15 days.*
- *Tyre pressure of temporary wheel is to be checked at least once in a month.*

### Wheel Alignment And Balancing

#### Alignment

Incorrect wheel alignment causes excessive and uneven tyre wear. Check wheel alignment at specified intervals.

#### Balancing

Wheels of your vehicle are balanced for better ride comfort and longer tyre life. Balancing needs to be done whenever tyre is removed from rim.

#### WARNING

If the vehicle vibrates abnormally on a smooth road, have the wheel balanced done immediately.

### Special Care For Tubeless Tyres

- When you remove the tyre and install it back on the rim, take precautions not to damage tyre bead. Use tyre removal and assembly machines. Damage or cut on tyre bead may cause gradual loss of air and deflation of tyre.
- Do not scratch the inner surface of tubeless tyre with metallic or sharp object. Tubeless tyres are coated with impermeable layer of rubber from the inner surface which holds the air in the tyre. Removal of this layer due to scratching may cause gradual loss of air and deflation.
- If wheel rim gets damaged in service, get the wheel rim repaired/ replaced immediately. Running the vehicle with damaged rim may cause deflation of tyre and subsequent dislodging of tyre from rim.
- Keep the recommended inflation pressure. Over-inflation, in particular, may cause puncture or bursting of tyre.

#### NOTE

- *Life and wear pattern of tyres depends on various parameters like tyre pressure, wheel alignment, wheel balancing, tyre rotation, etc. It also largely depends on vehicle speed, load carried, usage, driving habits, road conditions, tyre quality, etc.*
- *In case fault is suspected to be due to poor quality of tyres, the same may be taken up with concerned tyre manufacturer.*

### Tyre Equipment

Summer tires have a tread designed to provide superior performance on dry pavement. However, the performance of these tyres will be substantially reduced in wet conditions. If you operate your vehicle on wet roads, use all season tyres for all four wheels

### Special Winter Equipment

It is recommended that the following items be carried in the vehicle during winter:

## MAINTENANCE AND CARE

- A scraper and stiff-bristled brush to remove ice and snow from the windows and wiper blades.
- A shovel to dig the vehicle out of snow-drifts.
- Extra windshield-washer fluid to refill the reservoir tank

### Tread Wear Indicator

Tread wear indicator (TWI) a ribs of approx. 1.6 mm height moulded inside tread grooves which help to identify worn-out tyres. This is available on multiple location around the tyre and it can be shows by triangle  making (supplier-specific symbol) next to tread pattern.

When the tread is worn up to TWI height, tyre must be replaced.



## VEHICLE PARKING FOR LONG DURATION

Following care is to be taken:

1. Park the vehicle in covered, dry and if possible well-ventilated premises. Ensure parking brake engaged.
2. Remove the battery terminal cables (first remove the cable from the negative terminal). Ensure that battery is fully charged.
3. Use wheel chocks to prevent movement of the car.
4. Clean and protect the painted parts using protective wax.
5. Clean and protect the shiny metal parts using commercially available special compounds.
6. Sprinkle talcum powder on the rubber windscreen wiper and lift them off the glass.
7. Slightly open the windows.
8. Cover the vehicle with a cloth or perforated plastic sheet. Do not use sheets of imperforated plastic as they do not allow moisture on the vehicle body to



## MAINTENANCE AND CARE

- evaporate.
9. Inflate the tyres to 0.5 bar above the normal specified pressure and check it at regular intervals.
  10. Check the battery charge every six weeks.
  11. Avoid parking vehicle below 20% SoC continuously for two weeks (<14 days)
  12. Ideal condition to park vehicle for longer duration (> 14 days) is with 40% to 60% SoC (Disconnect the Aux battery – Ve terminal for longer duration parking)
  13. After the resting period the vehicle must be charged to 100% using Slow/AC Charging before use.
  14. During the resting period, the user may choose either of the following options to prevent discharge of low voltage battery.
  15. Periodically (weekly once) user may switch on the remote Air conditioning for 20-30 mins. This wakes up both the high voltage and low voltage systems and prevents low voltage battery from

getting discharged.

16. If possible, user may choose to disconnect the negative terminal of the low voltage battery. This results in complete vehicle sleep and minimum loss of charge for both low voltage and high voltage batteries.

### SMART KEY BATTERY REPLACEMENT

#### Procedure

1. Open rear key cover by small tool.
2. Remove battery from battery slot by small tool and replace it with new battery.
3. Ensure that the “+” symbol on the battery is facing upwards. The correct polarity is shown on the battery cover.
4. Close the rear key cover.
5. Make sure that the key cover is intact properly.



## MAINTENANCE AND CARE

### NOTE

- Use CR 2032 battery only.
- An inappropriately disposed battery can be harmful to the environment and human health. Dispose of the battery according to your local law(s) and regulations.

### REMOTE KEY BATTERY REPLACEMENT (if equipped)

Rotate the back side of remote key and remove the rear cover.

Insert the coin in coin slot, rotate and remove the next back cover.

Battery will be visible now.

Remove the battery and insert the new one. Observe the positive terminal marking to be facing up-side.

Place the back cover and rotate using coin. Then place the outer cover and rotate.1. Rotate the back side of remote key and remove the rear cover.2. Insert the coin in coin slot, rotate and remove the next back cover.3. Battery will be visible now.4. Remove the battery and insert the new one. Observe the positive terminal marking to be facing up-side.

Place the back cover and rotate using coin. Then place the outer cover and rotate.

### NOTE

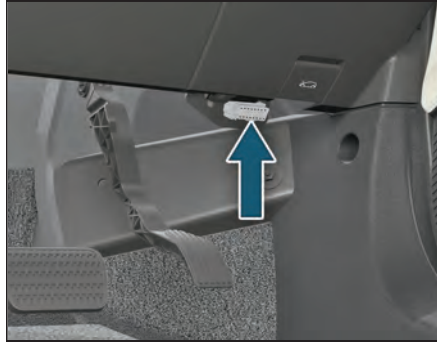
- Use 3V, CR 2450 battery only.
- An inappropriately disposed battery can be harmful to the environment and human health. Dispose of the battery according to your local law(s) and regulations.



### ON BOARD DIAGNOSTIC (OBD II) SYSTEM

The OBD system also has a diagnostic connector that can be interfaced with appropriate diagnostic tools, which makes it possible to read the fault codes stored in the Electronic Control Unit, together with a series of specific parameters for Motor operation and Diagnosis. This check can also be carried out by the traffic police. To access the diagnostic connector, open the cockpit fuse box cover, which is located on RH side below the steering wheel.

On board diagnostic located in cabin compartment fuse box. (refer below image)



### DO IT YOURSELF

#### Daily Checks

- Tyres for unusual wear, cracks or damage and embedded foreign material such as nails, stones, etc.
- Traces of fluid and oil below vehicle.
- There is sufficient charging for the trip.
- Windshield, windows, mirrors, lights, and reflectors are clean and unobstructed.
- All lamps, wipers, wiper blades and horn for proper operation.
- All switches, gauges and tell-tales are working properly.
- All doors, motor compartment and tail gate are securely closed and latched.
- Tool kit, jack & handle, warning triangle, owner's manual, first aid kit and vehicle documents are available and stored at their locations.

#### NOTE

*Water dripping below the car is normal.  
This is due to the usage of air condition*

## MAINTENANCE AND CARE

ing system.

### Weekly Checks

- Coolant level
- Brake fluid level
- Windshield washer fluid level
- Battery electrolyte level

### NOTE

*Tyre pressure always be measured in cold conditions. Do a check of tyre pressure and condition after every 15 days, including the spare tyre.*

### Brake Hoses And Lines

Visually check for proper installation, chafing, cracks, deterioration and any leakage. Replace any deteriorated or damaged parts immediately.

### Replacing The Components of Your EV

Since the electrical components of vehicle are not user serviceable, it is recommended that you approach your nearest

TPEM Authorised Service Centre to replace any electrical components of the car.

## CAR CARE

Your vehicle is subjected to many external influences such as climate, road conditions, industrial pollution and proximity to the sea. These conditions demand regular care of the vehicle body. Dirt, insects, bird droppings, oil, grease and stone chippings should be removed as soon as possible.

### Washing

Following these tips while washing your vehicle.

1. Always wash your vehicle in shade and the surface is at room temperature.
2. Wash with mild vehicle wash soap like 'Car Shampoo' and use a soft bristle brush, sponge or soft cloth and rinse it frequently while washing to avoid scratches.
3. When cleaning ensure that drain holes of lower panels and rocker panel are free from mud, slug etc to make way for entrapped water inside it to avoid corrosion.
4. To avoid scratches, please wear soft

gloves. Remove finger rings, nails, wrist watch while washing.

5. To remove stubborn stains and contaminants like tar, use turpentine or cleaners like 'Stain remover' which are safe for paint surfaces.
6. Avoid substances like petrol, diesel, kerosene, benzene, thinner, acids or other solvents that cause damage to vehicle interior, exterior and paint.
7. Dry your vehicle thoroughly to prevent any damp spots.
8. Rinse all surfaces thoroughly to prevent any traces of soap and other cleaners as this may lead to the formation of stains on the painted surface later.
9. During washing of the vehicle, do not apply water jet OR pressure water from pipe directly on any rubber material or Seal surface. This is to avoid damage to Rubber sealing parts in the sunroof system or in the vehicle. The damage to the sunroof sealing or any other seal in the vehicle will cause water ingress inside the vehicle.



### NOTE

- *Avoid parking the car under trees without proper cover, it will reduce the amount of bird droppings, tree sap and pollen contact on paint surface. Regularly remove the twigs, leaves and vegetation near the windshield areas, to avoid water stagnation.*
- *Always close the sunroof while washing the vehicle.*

### WARNING

Do not direct high pressure washer fluid/ water jets (Pressure above 0.5 bar) at electrical devices and connector during washing. This is to prevent malfunction / failure of electrical system due to water ingress.

After drying the vehicle, inspect it for chips and scratches that could allow corrosion to start. Apply touch up paint where necessary.

### Cleaning of Carpets

Vacuum clean the carpet regularly to remove dirt. Dirt will make the carpet wear out faster. Periodically, shampoo the carpet to keep it looking new.

Use carpet cleaners (preferably foam type). Follow the instructions that come with the cleaner. Apply it with a sponge or soft brush. Keep the carpeting as dry as possible by not adding water to the foam.

## MAINTENANCE AND CARE

### NOTE

*Avoid wiping of painted surface in dry condition as it may leave scratches on the painted surface.*

### Cleaning of Windows, Front and Rear Glasses

Clean the windows inside and outside with commercially available glass cleaners.

This will remove the haze that builds up on the inside of windows. Use a soft cloth or paper towels to clean all glass and plastic surfaces.

### Waxing

Waxing and polishing is recommended to maintain the gloss and wetlook appearance of your paint finish.

1. Use good quality polish and wax for your vehicle.
2. Re-wax your vehicle when the water does not slip off the surface but collects over the surface in patches.

### Polishing

Polishes and cleaners can restore shine to the painted surface that has oxidized and become dull. They normally contain mild abrasives and solvents that remove the top layer of the finish coat. Polish your vehicle, if the finish does not regain its original shine after using wax.

### Interior Fabric Cleaning Tips

1. Stains should be treated immediately. If left for a long time, they can leave a permanent mark.
2. Cleaning the stains immediately is important especially for stains, which contain artificial colors in the stain creating liquid or semisolid substance. The colorant may leave a stain if kept for longer time.
3. Stain should not be removed by rubbing. As far as possible, try to blot or lift the stain with cloth or plastic spatula and then clean the remaining stain with cloth or sponge.
4. If the stain has dried, then gently brush off the material and then press with

damp cloth or sponge till it disappears.

5. Do not use household detergents to clean the fabric.
6. Always use clean cotton cloth for cleaning.

### Paint Care

Following guidelines will help you to protect your vehicle from corrosion effectively.

### NOTE

*Avoid Spillage or Direct contact of Air freshener liquid/chemicals with painted plastic parts. These chemicals may cause damage to paint like blisters, peel off, wrinkles etc.*

### Proper Cleaning

In order to protect your vehicle from corrosion it is recommended that you wash your vehicle thoroughly and frequently in case:

- There is a heavy accumulation of dirt and mud especially on the underbody.
- It is driven in areas having high atmospheric pollution due to smoke, soot, dust, iron dust and other chemical pol-

lutants.

- It is driven in coastal areas.
- The underbody must be thoroughly pressure washed after every three months.
- In addition to regularly washing your car, the following precautions need to be taken.

#### **Periodic Inspection**

- Regularly inspect your vehicle for any damage in the paint film such as deep scratches and immediately get them repaired from an authorized service outlet, as these defects tend to accelerate corrosion.
- Inspect mud liners for damages
- Keep all drain holes clear from clogging.

#### **Proper Parking**

- Always park your vehicle in shade to protect it from harsh sunlight or in a well-ventilated garage so that there is no dampness on any part of the vehicle.

#### **Wiper Care**

- To prevent damage to the wipers or windshield, do not operate the wipers when the windshield is dry.
- To prevent damage to the wiper arms and other components, do not attempt to move the wipers manually.

#### **NOTE**

*We strongly recommended to avoid applying any external coating solution on vehicle glazing / glasses, especially on Front & Rear Windscreen Glass. This may affect the Wiper performance & lead to poor visibility while using Wipers in wet condition/Rainy season.*

#### **Special Care**

Illuminated Steering Wheel and Fascia Switch Panel

- Always use dry and soft cloth for cleaning,
- Avoid using extremely hot or cold water, as rapid temperature changes can cause the plastic components to become brittle. Do not use high-pressure water for cleaning.
- Please use a soft, clean cloth dampened with normal water to clean the steering wheel.
- Do not use carwax, window cleaner, glass coatings or chemical solvents such as strong detergents containing high alkaline or volatile organic solvents (gasoline, acetone etc). This may damage the driver airbag parts.
- Do not apply shiner, sanitizer, petrol, soap solutions, strong detergents, foam based cleaners or any other liquids etc. as these may cause damage to the surface.
- Do not use any sharp or other objects

## MAINTENANCE AND CARE

---

which can create scratch on illuminated surface.

- When cleaning the horn pad, do not hard press it.
- Use microfiber cloth on high gloss surfaces.
- Do not apply any protective film on the driver Airbag from un-authorised sources.



### VALUE ADDED SERVICES

**Why are Corrosion Protection Waxes necessary?**

**Corrosion Is Caused By:**

Water/salt water acid rain & atmospheric fallouts.

**Critical Areas Are:**

Cavities: joints, crevices, spot welds, underbody

- Corrosion is the most important factor when we talk about the vehicle life. If you treat your car you can prolong the life.
- It is very dangerous to drive around in a corroded vehicle.
- The corrosion creeps onto the vehicle from the inside and from the outside. The most dangerous kind of corrosion is often not discovered until it is too late.

**Benefits of Anti - Rust Treatment:**

- A professionally applied range of world class products offering real value to the new and used vehicle customer.
- The treatment has been developed to withstand the harshest environmental and climatic conditions (rust. Pollutants, stone and gravel impact, etc.)
- Insulate cabin space from external noises.
- Expensive tin work and Denting / Painting avoided.
- Higher resale value for the vehicle.
- Higher safety – Uncorroded vehicle
- 10 free checkups available

**MAINTENANCE AND CARE**

---

These treatments are available in all Authorised workshops. The Dealer Service Marketing Executive will explain to you the benefits and terms and conditions of this treatment.

I / We have been explained the Benefits, Terms and conditions and the prices of these treatments by the Dealer Service Marketing Executive.

☐ I wish to avail / ☐ Do not wish to avail extended warrant policy.

**Customer's Signature**

**Dealer's Signature**

**VEHICLE INTERIOR ENRICHMENT****Why To Protect Your New Car's Fabric Interior?**

- Someone will spoil your vehicle's fabric carpet or seats.
- A significant detractor from your vehicle's resale value.
- A permanent stain on your vehicle's interior fabric.

**Vehicle Interior Gets Affected From:**

Drink Spills - Food Stains - Mud - Ultraviolet Rays - Pets - Traffic

Benefits: Vehicle Interior Enrichment

- Removal of medium stains and dirt from all interior parts of the car i.e., carpet, upholstery and roof lining.
- Cleaning of windshield and all windows (inside and outside).
- Dressing of all internal plastics (e.g.: door pad trims) and rubber parts.
- The treatment involves cleaning and dressing of all parts of the exposed interiors.
- Specialised protection for seat fabric from liquid spills.

This treatment is available in all authorized workshops. The Dealer Service Marketing Executive will explain to you the benefits and terms and conditions of this treatment.

I / We have been explained the Terms and conditions, Coverage and Owner's responsibility by the Dealer Service Marketing Executive.

☐ I wish to avail / ☐ Do not wish to avail extended warrant policy.

**Customer's Signature**

**Dealer's Signature**

## **MAINTENANCE AND CARE**

---

### **VEHICLE EXTERIOR ENRICHMENT**

#### **Why Vehicles Are Painted?**

- For Corrosion protection of the metal surfaces.
- Ease of application from other corrosion protection treatments.
- Cheaper than other corrosion protection methods eg. Galvanizing, anodizing.
- For decoration and identification.

#### **Various Environmental Hazards Affecting Paints**

Environmental hazards: damage your vehicle's finish.

Even as your new vehicle rolls off the assembly line, the paint is not protected.

#### **The Enemy**

Ultraviolet Rays, Pollution, Tree Sap, Bird Droppings, Car Wash Chemicals, Road Salt, Acid Rain.

#### **Benefits:** Vehicle Exterior Enrichment

- Removal of medium scratches, orange peel, oxidation, dust nibs etc. & swirl marks from painted surface.
- Restoration of original gloss levels, UV protection after gloss is restored.
- Cleaning & dressing of tyres, Bumpers and all exterior plastic moldings/trim.

This treatment is available in all authorized workshops. The Dealer Service Marketing Executive will explain to you the benefits and terms and conditions of this treatment.

---

**VEHICLE WARRANTY - TERMS AND CONDITIONS**

We WARRANT each **TATA SIERRA EV** vehicle and parts thereof manufactured by us to be free from defect in material and workmanship subject to the following terms and conditions:

1. This warranty shall be for a period of **36 months from the date of sale of the car or a mileage of 1,25,000 km whichever occurs earlier. The warranty on the Battery and Motor shall be for a period of 96 months from the date of sale of car or a mileage of 1,60,000 KMs whichever occur earlier.**
2. Our obligation under this warranty shall be limited to repairing or replacing, free of charge, such parts of the car which, in our opinion, are defective, on the car being brought to us or to our dealers within the period. The parts so repaired or replaced shall also be warranted for quality and workmanship but such warranty shall be co-terminus with this original warranty.
3. Any part which is found to be defective and is replaced by us under the warranty shall be our property.
4. As for such parts as Tyres, Batteries, Audio and / or Video equipment (if any), etc. not manufactured by us but supplied by other parties, this warranty shall not apply, but buyers of the car shall be entitled to, so far as permissible by law, all such rights as we may have against such parties under their warranties in respect of such parts.
5. This warranty shall not apply if the car or any part thereof is repaired or altered otherwise than in accordance with our standard repair procedure or by any person other than from our sales or service establishments, our authorized dealers, service centres or service points in any way so as, in our judgment which shall be final and binding, to affect its reliability, nor shall it apply if, in our opinion which shall be final and binding, the car is subjected to misuse, negligence, improper or inadequate maintenance or accident or loading in excess of such carrying capacity as certified by us, or such services as prescribed in our Owner's Manual are not carried out by the buyer through our sales or service establishments, our authorized dealers, service centres or service points.
6. **This warranty shall not apply to the replacement of normal wear parts, including without limitation, drive belts, hoses, wiper blades, fuses, clutch disc, brake shoes or brake pads, cables and all rubber parts (except oil seal and glass run).**
7. This warranty shall not cover any inherent normal deterioration of the car or any of its parts arising from the actual use of the car or any damage due to negligent or improper operation or storage of the car.
8. This warranty shall not apply to normal maintenance services like oils & fluid changes, head lamps focusing, fastener retightening, center hub cap/wheel cover. wheel balancing and alignment, tyre rotation, adjustment of valve clearance, ignition timing and consumables like bulbs, air and gas leaks in case

## WARRANTY

---

of air conditioned cars.

9. This warranty shall not apply to any damage or deterioration caused by environmental pollution or bird droppings. Slight irregularities not recognized as affecting the function or quality of the vehicle or parts, such as slight noise or vibration, defects appearing only under particular or irregular operations are items considered characteristics of the vehicle.
10. This warranty shall be null and void if the car is subjected to abnormal use such as rallying, racing or participation in any other competitive sport. This warranty shall not apply to any repair or replacements as a result of accident or collision.
11. This warranty is expressly in lieu of all warranties, whether by law or otherwise, expressed or implied, and all other obligations or liabilities on our part and we neither assume, nor authorize any person to assume on our behalf, any other liability arising from the sale of the car or any agreement in relation thereto.
12. The buyer shall have no other rights except those set out above and have, in particular, no right to repudiate the sale, or any agreement or to claim any reduction in the purchase price of the car, or to demand any damages or compensation for losses, incidental or indirect, or inconvenience or consequential damages, loss of car, or loss of time, or otherwise, incurred or accrued.
13. Any claim arising from this warranty shall be recognized only if it is notified in writing to us or to our authorized dealer with-

out any delay soon after such defects as covered & ascertained under this warranty.

**14. This warranty is fully transferable to subsequent vehicle owner. Only unexpired remaining period of warranty applies.**

15. We reserve our rights to make any change or modification in design of the car or its parts or to introduce any improvement therein or to incorporate in the car any additional part or accessory at any time without incurring any obligation to incorporate the same in the cars previously sold.

## HIGH VOLTAGE BATTERY WARRANTY - TERMS AND CONDITIONS

TATA Passenger Electric Mobility Limited offers high voltage battery warranty covering lifetime of your vehicle, subject to the following terms and conditions:

It is clarified that notwithstanding what is stated under [VEHICLE WARRANTY section] of the owner's manual of SIERRA.EV, the "Lifetime HV Battery Warranty" terms and conditions stated hereunder shall be applicable to the warranty coverage for the HV Battery and shall prevail over any other conflicting provision in the owner's manual.

**Coverage of HV Battery Warranty offered is as below:**

| Lifetime Warranty First Registration |             | Standard Warranty Second Registration Onwards*** (whichever Occurs First) |         |
|--------------------------------------|-------------|---------------------------------------------------------------------------|---------|
| Years                                | Km          | Years                                                                     | Km      |
| Lifetime*                            | Unlimited** | 10^                                                                       | 2 Lacs^ |

\* The term 'Lifetime' refers to the period of fifteen years from the first date of registration of the vehicle at the local regional transport office, in accordance with the Motor Vehicles Act, 1988, and is restricted for the period of use during the first registration only.

\*\* The term 'Unlimited' is on the assumption of fair usage by a customer for personal use of the vehicle and therefore excludes extra-ordinary usage or usage outside the ordinary course as

would be reasonably expected for vehicle not being used for commercial / trade purposes.

\*\*\* It is incumbent on every second (or subsequent) owner to inform TATA PASSENGER ELECTRIC MOBILITY LIMITED of the vehicle ownership transfer for HV Battery Warranty to apply. The HV Battery Warranty stated above shall not be available to the second or subsequent owner in absence of such notification.

^ For every sale of vehicle second registration onwards, the date of first registration and kilometer coverage starting from such date of first registration, shall be considered as the baseline for calculating the said term of 8 years and said coverage of 1,60,000 Kms. Respectively.

### Battery Energy Capacity offered under HV Battery Warranty coverage

- In the event the HV Battery falls below minimum 70% of its State of Health ("SOH") of nominal energy at any time during the warranty period, then TPEM shall either repair / re-place the HV Battery to achieve the following:
  - In case of repair / replacement of HV battery pack or cells during the warranty period (to be decided at sole discretion of TATA PASSENGER ELECTRIC MOBILITY LIMITED), TPEML will bring the energy capacity of HV battery at a value equal to steady state SOH before battery repair / replacement or at 80% SOH, whichever is higher.
- It is clarified that the HV Battery, just like all other Lithium-Iron Phosphate or other heavy material batteries, shall undergo

## WARRANTY

---

gradual power and energy loss over the course of time and use. Unless such gradual loss results in a depreciation of the SOH value for the HV Battery below said threshold of 70%, the HV Battery Warranty being offered herein shall not apply.

**Applicability of Lifetime HV Battery Warranty is exclusively applicable under the following conditions:**

**The Lifetime HV Battery warranty is exclusively applicable under the following conditions**

1. Owner of vehicle is first registrant / first registered user of the vehicle
2. Registrations in the name of private individuals.
3. Registrations made under the company perk car scheme for employees, including subsequent transfers of the vehicle to the same employee.

*[It is clarified that in the event the perk car is sold or transferred to any third party (not being the aforementioned employee), such third party shall be treated as second or subsequent owner and the specifications of the standard HV battery warranty as applicable to second registration onwards, shall apply]*

**Vehicles excluded from Lifetime HV Battery Warranty**

1. This warranty does not extend to non-private registrations, including vehicles used for display, demonstration, test drives, courtesy purposes, commercial operations, fleet management, taxi services, or other company vehicle uses. In each such instance, the HV battery warranty shall be limited to a

duration of eight (8) years or a mileage of 2,00,000 kilometers from the date of first registration of the vehicle, whichever occurs first.

2. Private registrations held in the name of a company, corporate body, firm, trust, or similar entities, where the vehicle is being used for commercial / trade purposes

**General Terms and Conditions for HV Battery Warranty**

The warranty coverage listed above is applicable for vehicles meeting the following criteria, subject to compliance with the terms and conditions of the warranty coverage:

1. In the regular course, the vehicle must be serviced or repaired at TPEM Authorised Service Centre, following the recommended maintenance and servicing schedule.
2. Upon discovery of any defect, it is incumbent that the vehicle be promptly brought to TPEM Authorised Service Centre alone for application of this HV Battery Warranty.
3. Any use of lubricants or oils (if applicable) or other fluids, spare parts, components or accessories, made by a person or facility, during repair, alteration or modification or maintenance, who is authorized or certified by TPEM to do so.
4. Prompt and timely installation of all vehicle's software updates after notification that there is an update available.
5. The vehicle must maintain an active & uninterrupted IRA.ev connection throughout the entire ownership period, which the owner is informed and aware is mandatory for monitoring the



HV Battery health of connected vehicles. IRA.ev subscription maintenance is crucial for telemetry data to be transferred back to TPEM to allow visibility of the last steady state of the HV Battery for application of the HV Battery Warranty.

6. Vehicle shall not have more than 2 warnings related to EV charging pattern between two service intervals.
7. Please refer “BATTERY AND COMPONENT” and “CHARGING” section of owner’s manual of your vehicle for important information on how to maximize the life and capacity of the Battery.
8. Servicing or opening of the Battery pack should not be by a non- TPEM Authorised Service Centre personnel;
9. If the vehicle stopped due to over-discharge of Battery, then it must be plugged for charge within 24 hours from time of stop-page.
10. Charging port recommended by TPEM is always used without default, and during off-roading activity for SIERRA.EV QWD, it is recommended to fit the TATA Genuine accessory off-roading kit (soon to be made available for separate purchase) into the vehicle and to ensure the vehicle insurance covers off-roading activity.

### Exclusions:

The warranty is for inherent defects in the HV Battery and does not extend to any damage or defects to the HV Battery arising from:

- Customers’ own acts or omissions.
- External or physical damage or tampering, to both the HV Battery pack as well as the HV Battery container or tray. (For clarity, while undertaking any warranty service, if it is determined (as per the sole discretion of TPEM) that the HV Battery case / container / tray has suffered such denting, scratching, defiling, diminishing, abrading, chafing or other such similar damage owing to which the physical conditions and properties of the HV Battery case / container / tray cannot be restored or maintained, then, the HV Battery case / container / tray may be replaced at a chargeable basis to be paid for by the vehicle owner, and the same shall not be covered as part of the HV Battery Warranty.
- Accidents or collisions or damage arising from flying trajectory of gravel or other objects capable of causing, puncture, dents or damage to the HV Battery Pack.
- Damaging the Battery, or intentionally attempting, either by physical means, programming, or other methods, to extend (other than as specified in your Owner’s Manual) or reduce the life of the Battery;
- Exposing the Battery to direct flame; or Flooding the Battery or Immersion of HV battery or vehicle in flood or salt water over and above the maximum permitted exposure limit under IP67 (whether caused voluntarily or inadvertently).
- The vehicle / HV battery is subjected to any form of unusual stress, including stress due to improper charging racing; over-

## WARRANTY

---

loading; or other misuse; abuse; or negligent use etc.

### **Limitation of Liability**

Except as provided in this warranty document and to the maximum extent permitted by law, in no event shall TPEML (or its Affiliates or any of its, or its Affiliates' directors, officers, employees, advisors or agents) be liable for any consequential, incidental, special or punitive damages (including with-out limitation of loss of profits, harm to goodwill or business reputation or loss of opportunity, or damages arising from de-lay in services), incurred by or claimed against the owner of the vehicle, or any third-party (in rem) which arise from or out of or in connection with the purchase, installation, use, defect, repair, replacement, storage, or otherwise handling of the HV Battery or HV Battery case, whether such claim is based on contract, warranty, negligence, strict liability, or any other the-ory.

It is further clarified that TPEML's aggregate liability to the customer shall be repair or replacement of the HV Battery Pack in accordance with the terms and conditions of this HV Battery Warranty, which shall constitute the Customer's (or any third-party's) sole remedy against TPEML.

### **Force Majeure**

The Customer agrees that the operation of this HV Battery Warranty and TPEML's obligations hereunder shall stand fully suspended and excused (without it resulting in or being interpreted as a breach by TPEML) for the entire duration of any Force Majeure Event. For the purpose of this HV Battery Warranty docu-

ment, a Force Majeure Event shall include, without limitation, natural catastrophes (such as flooding, fires, earth-quakes, lightning, extreme weather, rampant outbreak of dis-ease pandemics, epidemics, or illness and all other like-kind occurrences), disaster, crime, terrorism, piracy, war, sedition, rioting, shortages, labor strikes, lock-outs, extreme price fluctuations of necessary goods or services, restrictions on import-export of materials (including without limitation, essential elements and precious minerals necessary for the body-building of the HV Battery), or any other changes in law or regulation, and governmental acts or omissions, among other events, beyond TPEML's reasonable control.

For any other details in connection with the Terms and Conditions or for any other matters please visit our Tata.ev website (<https://ev.tatamotors.com/>)

**EXTENDED WARRANTY**

TATA PASSENGER ELECTRIC MOBILITY LIMITED recommends the purchase of its extended warranty program.

**Coverage** - Mechanical + Electrical

**Benefits**

- Insures you against unforeseen break down repair bills.
- Documentation is simple and hassle free.
- Near cashless & speedy claim

**Term**

36 + 12 months or 1,25,000 kms whichever occurs first

OR

36 + 24 months or 1,25,000 kms whichever occurs first

Extended Warranty is available in the dealership from where you have purchased your vehicle. We strongly recommend purchase of Extended Warranty at time of purchase of your vehicle. Extended Warranty can be availed until warranty period from date of purchase of vehicle. The Dealer Service Marketing Executive shall explain to you the Terms and conditions, Coverage and Owner's responsibility.

**Note**

- The extended warranty comes into force once the manufacturer's warranty expires.
- It is more restrictive as by the time it comes into force the vehicle is already 36 months old.

**What Is Covered?**

- Mechanical / Electrical break down as defined in this warranty and confirmed by the dealer within the stipulated terms and conditions.



## EXTENDED WARRANTY

---

- TPEM dealer shall either repair or replace any part found to be defective with a new part or an equivalent at no cost to the owner for parts or labour.
- Such defective parts which have been replaced will become property of TATA PASSENGER ELECTRIC MOBILITY LIMITED.

### What Is Not Covered?

Please refer the Extended Warranty Booklet for details of the exclusion list. Soft copy can be available with the dealer.

### Owner's Responsibility

- Proper use, maintenance and care of the vehicle in accordance with the instructions contained in the Owner's Manual and Service Booklet. The records of the same to be ensured in Owner's Manual.
- Retention of maintenance service bills.

I / We have been explained the Terms and conditions, Coverage and Owner's responsibility by the Dealer Service Marketing Executive

☐ I wish to avail / ☐ Do not wish to avail extended warranty policy.

**Customer's Signature**

**Dealer's Signature**